

Enquiry into Plants

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1.1.1

BOOK I. OF THE PARTS OF PLANTS AND THEIR COMPOSITION.

In considering the distinctive characters of plants and their nature generally one must take into account their parts, their qualities, the ways in which their life originates, and the course which it follows in each case: (conduct and activities we do not find in them, as we do in animals). Now the differences in the way in which their life originates, in their qualities and in their life-history are comparatively easy to observe and are simpler, while those shewn in their 'parts' present more complexity. Indeed it has not even been satisfactorily determined what ought and what ought not to be called 'parts,' and some difficulty is involved in making the distinction.

1.1.2

Now it appears that by a 'part,' seeing that it is something which belongs to the plant's characteristic nature, we mean something which is permanent either absolutely or when once it has appeared (like those parts of animals which remain for a time undeveloped) — permanent, that is, unless it be lost by disease, age or mutilation. However some of the parts of plants are such that their existence is limited to a year, for instance, flower, 'catkin,' leaf, fruit, in fact all those parts which are antecedent to the fruit or else appear along with it. Also the new shoot itself must be included with these; for trees always make fresh growth every year alike in the parts above ground and in those which pertain to the roots. So that if one sets these down as 'parts,' the number of parts will be indeterminate and constantly changing; if on the other hand these are not to be called 'parts,' the result will be that things which are essential if the plant is to reach its perfection, and which are its conspicuous features, are nevertheless not 'parts'; for any plant always appears to be, as indeed it is, more comely and more perfect when it makes new growth, blooms, and bears fruit. Such, we may say, are the difficulties involved in defining a 'part.'

1.1.3

But perhaps we should not expect to find in plants a complete correspondence with animals in regard to those things which concern reproduction any more than in other respects; and so we should reckon as 'parts' even those things to which the plant gives birth, for instance their fruits, although we do not so reckon the unborn young of animals. (However, if such a product seems fairest to the eye, because the plant is then in its prime, we can draw no inference from this in support of our argument, since even among animals those that are with young are at their best.)

1.1.4

Again many plants shed their parts every year, even as stags shed their horns, birds which hibernate their feathers, four-footed beasts their hair: so that it is not strange that the parts of plants should not be permanent, especially as what thus occurs in animals and the shedding of leaves in plants are analogous processes.

1.1.5

In like manner the parts concerned with reproduction are not permanent in plants; for even in animals there are things which are separated from the parent when the young is born, and there are other things- which are cleansed away, as though neither of these belonged to the animal's essential nature. And so too it appears to be with the growth of plants; for of course growth leads up to reproduction as the completion of the process.

1.1.6

And in general, as we have said, we must not assume that in all respects there is complete correspondence between plants and animals. And that is why the number also of parts is indeterminate; for a plant has the power of growth in all its parts, inasmuch as it has life in all its parts. Wherefore we should assume the truth to be as I have said, not only in regard

to the matters now before us, but in view also of those which will come before us presently; for it is waste of time to take great pains to make comparisons where that is impossible, and in so doing we may lose sight also of our proper subject of enquiry. The enquiry into plants, to put it generally, may either take account of the external parts and the form of the plant generally, or else of their internal parts: the latter method corresponds to the study of animals by dissection.

1.1.7

Further we must consider which parts belong to all plants alike, which are peculiar to some one kind, and which of those which belong to all alike are themselves alike in all cases; for instance, leaves roots bark. And again, if in some cases analogy ought to be considered (for instance, an analogy presented by animals), we must keep this also in view; and in that case we must of course make the closest resemblances and the most perfectly developed examples our standard; and, finally, the ways in which the parts of plants are affected must be compared to the corresponding effects in the case of animals, so far as one can in any given case find an analogy for comparison. So let these definitions stand.

1.1.8

Now the differences in regard to parts, to take a general view, are of three kinds: either one plant may possess them and another not (for instance, leaves and fruit), or in one plant they may be unlike in appearance or size to those of another, or, thirdly, they may be differently arranged. Now the unlikeness between them is seen in form, colour, closeness of arrangement or its opposite, roughness or its opposite, and the other qualities; and again there are the various differences of flavour. The inequality is seen in excess or defect as to number or size, or, to speak generally, all the above-mentioned differences too are included under excess and defect: for the 'more' and the 'less' are the same thing as excess and defect, whereas

‘differently arranged’ implies a difference of position; for instance, the fruit may be above or below the leaves, and, as to position on the tree itself, the fruit may grow on the apex of it or on the side branches, and in some cases even on the trunk, as in the sycamore; while some plants again even bear their fruit underground, for instance arakhidna and the plant called in Egypt uingon; again in some plants the fruit has a stalk, in some it has none. There is a like difference in the floral organs: in some cases they actually surround the fruit, in others they are differently placed: in fact it is in regard to the fruit, the leaves, and the shoots that the question of position has to be considered.

1.1.9

Or again there are differences as to symmetry: in some cases the arrangement is irregular, while the branches of the silver-fir are arranged opposite one another; and in some cases the branches are at equal distances apart, and correspond in number, as where they are in three rows.

1.1.10

Wherefore the differences between plants must be observed in these particulars, since taken together they shew forth the general character of each plant.

1.1.11

But, before we attempt to speak about each, we must make a list of the parts themselves. Now the primary and most important parts, which are also common to most, are these — root, stem, branch, twig; these are the parts into which we might divide the plant, regarding them as members, corresponding to the members of animals: for each of these is distinct in character from the rest, and together they make up the whole.

1.1.12

The root is that by which the plant draws its nourishment, the stem that to which it is conducted. And by the ‘stem’ I mean that part which grows above ground and is single; for that is the

part which occurs most generally both in annuals and in long-lived plants; and in the case of trees it is called the 'trunk.' By 'branches' I mean the parts which split off from the stem and are called by some 'boughs.'- By 'twig' I mean the growth which springs from the branch regarded as a single whole, and especially such an annual growth.

1.1.13

Now these parts belong more particularly to trees. The stem however, as has been said, is more general, though not all plants possess even this, for instance, some herbaceous plants are stemless; others again have it, not permanently, but as an annual growth, including some whose roots live beyond the year. In fact your plant is a thing various and manifold, and so it is difficult to describe in general terms: in proof whereof we have the fact that we cannot here seize on any universal character which is common to all, as a mouth and a stomach are common to all animals; whereas in plants some characters are the same in all, merely in the sense that all have analogous characters, while others correspond otherwise. For not all plants have root, stem, branch, twig, leaf, flower or fruit, or again bark, core, fibres or veins; for instance, fungi and truffles; and yet these and such like characters belong to a plant's essential nature. However, as has been said, these characters belong especially to trees, and our classification of characters belongs more particularly to these; and it is right to make these the standard in treating of the others.

1.1.14

Trees moreover shew forth fairly well the other features also which distinguish plants; for they exhibit differences in the number or fewness of these which they possess, as to the closeness or openness of their growth, as to their being single or divided, and in other like respects. Moreover each of the characters mentioned is not 'composed of like parts'; by which I mean that though any given part of the root or trunk is

composed of the same elements as the whole, yet the part so taken is not itself called 'trunk,' but 'a portion of a trunk.' The case is the same with the members of an animal's body; to wit, any part of the leg or arm is composed of the same elements as the whole, yet it does not bear the same name (as it does in the case of flesh or bone); it has no special name. Nor again have subdivisions of any of those other organic parts which are uniform special names, subdivisions of all such being nameless. But the subdivisions of those parts which are compound have names, as have those of the foot, hand, and head, for instance, toe, finger, nose or eye. Such then are the largest parts of the plant.

1.2.1

Again there are the things of which such parts are composed, namely bark, wood, and core (in the case of those plants which have it), and these are all 'composed of like parts.' Further there are the things which are even prior to these, from which they are derived — sap, fibre, veins, flesh: for these are elementary substances — unless one should prefer to call them the active principles of the elements; and they are common to all the parts of the plant. Thus the essence and entire material of plants consist in these.

1.2.2

Again there are other as it were annual parts, which help towards the production of the fruit, as leaf, flower, stalk (that is, the part by which the leaf and the fruit are attached to the plant) - and again tendril, 'catkin' (in those plants that have them). And in all cases there is the seed which belongs to the fruit: by 'fruit' is meant the seed or seeds, together with the seed-vessel. Besides these there are in some cases peculiar parts, such as the gall in the oak, or the tendril in the vine.

1.2.3

In the case of trees we may thus distinguish the annual parts, while it is plain that in annual plants all the parts are annual: for

the end of their being is attained when the fruit is produced. And with those plants which bear fruit annually, those which take two years (such as celery and certain others) and those which have fruit on them for a longer time — with all these the stem will correspond to the plant's length of life: for plants develop a stem at whatever time they are about to bear seed, seeing that the stem exists for the sake of the seed.

1.2.4

Let this suffice for the definition of these parts: and now we must endeavour to say what each of the parts just mentioned is, giving a general and typical description.

1.2.5

The sap is obvious: some call it simply in all cases 'juice,' as does Menestor among others: others, in the case of some plants give it no special name, while in some they call it 'juice,' and in others 'gum.' Fibre and 'veins' have no special names in relation to plants, but, because of the resemblance, borrow the names of the corresponding parts of animals. — It may be however that, not only these things, but the world of plants generally, exhibits also other differences as compared with animals: for, as we have said,² the world of plants is manifold. However, since it is by the help of the better known that we must pursue the unknown, and better known are the things which are larger and plainer to our senses, it is clear that it is right to speak of these things in the way indicated: for then in dealing with the less known things we shall be making these better known things our standard, and shall ask how far and in what manner comparison is possible in each case. And when we have taken the parts, we must next take the differences which they exhibit, for thus will their essential nature become plain, and at the same time the general differences between one kind of plant and another.

1.2.6

Now the nature of the most important parts has been indicated already, that is, such parts as the root, the stem, and the rest:

their functions and the reasons for which each of them exists will be set forth presently. For we must endeavour to state of what these, as well as the rest, are composed, starting from their elementary constituents.

1.2.7

First come moisture and warmth: for every plant, like every animal, has a certain amount of moisture and warmth which essentially belong to it; and, if these fall short, age and decay, while, if they fail altogether, death and withering ensue. Now in most plants the moisture has no special name, but in some it has such a name, as has been said -: and this also holds good of animals: for it is only the moisture of those which have blood which has received a name; wherefore we distinguish animals by the presence or absence of blood, calling some 'animals with blood,' others 'bloodless.' Moisture then is one essential 'part,' and so is warmth, which is closely connected with it.

1.2.8

There are also other internal characters, which in themselves have no special name, but, because of their resemblance, have names analogous to those of the parts of animals. Thus plants have what corresponds to muscle; and this quasi-muscle is continuous, fissile, long: moreover no other growth starts from it either branching from the side or in continuation of it. Again plants have veins: these in other respects resemble the 'muscle,' but they are longer and thicker, and have side-growths and contain moisture. Then there are wood and flesh: for some plants have flesh, some wood. Wood is fissile, while flesh can be broken up in any direction, like earth and things made of earth: it is intermediate between fibre and veins, its nature being clearly seen especially in the outer covering' of seed-vessels. Bark and core are properly so called, yet they too must be defined. Bark then is the outside, and is separable from the substance which it covers. Core is that which forms the middle of the wood, being third in order from the bark, and

corresponding to the marrow in bones. Some call this part the 'heart,' others call it 'heart-wood': some again call only the inner part of the core itself the 'heart,' while others distinguish this as the 'marrow.'

1.2.9

Here then we have a fairly complete list of the 'parts,' and those last named are composed of the first 'parts'; wood is made of fibre and sap, and in some cases of flesh also; for the flesh hardens and turns to wood, for instance in palms ferula and in other plants in which a turning to wood takes place, as in the roots of radishes. Core is made of moisture and flesh: bark in some cases of all three constituents, as in the oak black poplar and pear; while the bark of the vine is made of sap and fibre, and that of the cork-oak of flesh and sap. Moreover out of these constituents are made the most important parts,- those which mentioned first, and which may be called 'members': however not all of them are made of the same constituents, nor in the same proportion, but the constituents are combined in various ways.

1.2.10

Having now, we may say, taken all the parts, we must endeavour to give the differences between them and the essential characters of trees and plants taken as wholes.

1.3.1

Now since our study becomes more illuminating if we distinguish different kinds, it is well to follow this plan where it is possible. The first and most important classes, those which comprise all or nearly all plants, are tree, shrub, under-shrub, herb.

1.3.2

A tree is a thing which springs from the root with a single stem, having knots and several branches, and it cannot easily be uprooted; for instance, olive fig vine. 'A shrub is a thing which rises from the root with many branches; for instance, bramble

Christ's thorn. An under-shrub is a thing which rises from the root with many stems as well as many branches; for instance, savory-rue. A herb is a thing which comes up from the root with its leaves and has no main stem, and the seed is borne on the stem; for instance, corn and pot-herbs.

1.3.3

These definitions however must be taken and accepted as applying generally and on the whole. For in the case of some plants it might seem that our definitions overlap; and some under cultivation appear to become different and depart from their essential nature, for instance, mallow when it grows tall and becomes tree-like. For this comes to pass in no long time, not more than six or seven months, so that in length and thickness the plant becomes as great as a spear, and men accordingly use it as a walking-stick, and after a longer period the result of cultivation is proportionately greater. So too is it with the beets; they also increase in stature under cultivation, and so still more do chaste-tree Christ's thorn ivy, so that, as is generally admitted, these become trees, and yet they belong to the class of shrubs. On the other hand the myrtle, unless it is pruned, turns into a shrub, and so does filbert: indeed this last appears to bear better and more abundant fruit, if one leaves a good many of its branches untouched, since it is by nature like a shrub. Again neither the apple nor the pomegranate nor the pear would seem to be a tree of a single stem, nor indeed any of the trees which have side stems from the roots, but they acquire the character of a tree when the other stems are removed. However some trees men even leave with their numerous stems because of their slenderness, for instance, the pomegranate and the apple, and they leave the stems of the olive and the fig cut short. Indeed it might be suggested that we should classify in some cases simply by size, and in some cases by comparative robustness or length of life. For of under-shrubs and those of the pot-herb class some have only one stem and come as it were

to have the character of a tree, such as cabbaore and rue: wherefore some call these 'tree-herbs'; and in fact all or most of the pot-herb class, when they have been long in the ground, acquire a sort of branches, and the whole plant comes to have a tree-like shape, though it is shorter lived than a tree.

1.3.4

For these reasons then, as we are saying, one must not make a too precise definition; we should make our definitions typical. For we must make our distinctions too on the same principle, as those between wild and cultivated plants, fruitbearing and fruitless, flowering and flowerless, evergreen and deciduous. Thus the distinction between wild and cultivated seems to be due simply to cultivation, since, as Hippon remarks, any plant may be either wild or cultivated according as it receives or does not receive attention.

1.3.5

Again the distinctions between fruitless and fruitbearing, flowering and flowerless, seem to be due to position and the climate of the district. And so too with the distinction between deciduous and evergreen, thus they say that in the district of Elephantine neither vines nor figs lose their leaves.

1.3.6

Nevertheless we are bound to use such distinctions. For there is a certain common character alike in trees, shrubs, under-shrubs, and herbs. Wherefore, when one mentions the causes also, one must take account of all alike, not giving separate definitions for each class, it being reasonable to suppose that the causes too are common to all. .

d in fact there seems to be some natural difference from the first in the case of wild and cultivated, seeing that some plants cannot live under the conditions of those grown in cultivated ground, and do not submit to cultivation at all, but deteriorate under it; for instance, silver-fir, fir, holly, and in general those which affect cold snowy country; and the same

is also true of some of the under-shrubs and herbs, such as caper and lupin. Now in using the terms 'cultivated' and 'wild' we must make these on the one hand our standard, and on the other that which is in the truest sense 'cultivated.' Now Man, if he is not the only thing to which this name is strictly appropriate, is at least that to which it most applies.

1.4.1

Again the differences, both between the plants' wholes and between their parts, may be seen in the appearance itself of the plant. I mean differences such as those in size, hardness, smoothness or their opposites, as seen in bark, leaves, and the other parts; also, in general, differences as to comeliness or its opposite and as to the production of good or of inferior fruit. For the wild kinds appear to bear more fruit, for instance, the wild pear and wild olive, but the cultivated plants better fruit, having even flavours which are sweeter and pleasanter and in general better blended, if one may so say.

1.4.2

These then as has been said, are differences of natural character, as it were, and still more so are those between fruitless and fruitful, deciduous and evergreen plants, and the like. But with all the differences in all these cases we must take into account the locality,² and indeed it is hardly possible to do otherwise. Such differences would seem to give us a kind of division into classes, for instance, between that of aquatic plants and that of plants of the dry land, corresponding to the division which we make in the case of animals. For there are some plants which cannot live except in wet; and again these are distinguished from one another by their fondness for different kinds of wetness; so that some grow in marshes, others in lakes, others in rivers, others even in the sea, smaller ones in our own sea, larger ones in the Red Sea. Some again, one may say, are lovers of very wet places, or plants of the marshes, such as the willow and the plane. Others again cannot live at all in

water, but seek out dry places; and of the smaller sorts there are some that prefer the shore.

1.4.3

However, if one should wish to be precise, one would find — that even of these some are impartial and as it were amphibious, such as tamarisk, willow, alder, and that others even of those which are admitted to be plants of the dry land sometimes live in the sea, as palm, squill, asphodel. But to consider all these exceptions and, in general, to consider in such a manner is not the right way to proceed. For in such matters too nature certainly does not thus go by any hard and fast law. Our distinctions therefore and the study of plants in general must be understood accordingly. To return — these plants as well as all others will be found to differ, as has been said, both in the shape of the whole and in the differences between the parts, either as to having or not having certain parts, or as to having a greater or less number of parts, or as to having them differently arranged, or because of other differences such as we have already mentioned. And it is perhaps also proper to take into account the situation in which each plant naturally grows or does not grow. For this is an important distinction, and specially characteristic of plants, because they are united to the ground and not free from it like animals.

1.5.1

Next we must try to give the differences as to particular parts, in the first instance speaking broadly of those of a general character, and then of special differences between individual kinds; and after that we must take a wider range, making as it were a fresh survey.

1.5.2

Some plants grow straight up and have tall stems, as silver-fir, fir, cypress; some are by comparison crooked and have short stems, as willow, fig, pomegranate; and there are like differences as to degree of thickness. Again some have a single

stem, others many stems; and this difference corresponds — more or less to that between those which have sidegrowths and those which have none, or that between those which have many branches and those which have few, such as the date-palm. And in these very instances we have also differences in strength thickness and the like. Again some have thin bark, such as bay and lime; others have a thick bark, such as the oak. And again some have smooth bark, as apple and fig; others rough bark, as ‘wild oak’ (Valonia oak) cork-oak and date-palm. However all plants when young have smoother bark, which gets rougher as they get older; and some have cracked bark, as the vine; and in some cases it readily drops off, as in andrachne, apple, and arbutus. And again of some the bark is fleshy, as in cork-oak, oak, poplar; while in others it is fibrous and not fleshy; and this applies alike to trees shrubs and annual plants, for instance to vines, reeds and wheat. Again in some the bark has more than one layer, as in lime, silver-fir, vine, Spanish broom, onions; while in some it consists of only as that of the silver-fir, while others are rather breakable, such as the wood of the olive. Again some are without knots, — as the stems of elder, others have knots, as those of fir and silver-fir.

1.5.3

Now such differences also must be ascribed to the essential character of the plant: for the reason why the wood of silver-fir is easily split is that the grain is straight, while the reason why olive-wood is easily broken is that it is crooked and hard. Limewood and some other Moods on the other hand are easily bent because their sap is viscid. Boxwood and ebony are heavy because the grain is close, and oak because it contains mineral matter. In like manner the other peculiarities too can in some way be referred to the essential character.

1.6.1

Again there are differences in the ‘core’: in the first place according as plants have any or have none, as some say is the

case with elder among other things; and in the second place there are differences between those which have it, since in different plants it is respectively fleshy, woody, or membranous; fleshy, as in vine, fig, apple, pomegranate, elder, ferula; woody, as in Aleppo pine, silver-fir, fir; in the lastnamed especially so, because it is resinous.- Harder again and closer than these is the core of dog-wood kermes-oak, oak, laburnum, mulberry, ebony, nettletree.

1.6.2

The cores in themselves also differ in colour; for that of ebony and oak is black, and in fact in the oak it is called 'oak-black'; and in all these the core is harder and more brittle than the ordinary wood; and for this reason the core of these trees can not be bent. Again the core differs in closeness of texture. A membranous core is not common in trees, if indeed it is found at all; but it is found in shrubby plants and woody plants generally, as in reed ferula and the like. Again in some the core is large and conspicuous, as in kermes-oak, oak and the other trees mentioned above; while in others it is less conspicuous, as in olive and box. For in these trees one cannot find it isolated, but, as some say, it is not found in the middle of the stem, being diffused throughout, so that it has no separate place; and for this reason some trees might be thought to have no core at all; in fact in the date-palm the wood is alike throughout.

1.6.3

Again plants differ in their roots, some having many long roots, as fig oak plane; for the roots of these, if they have room, run to any length. Others again have few roots, as pomegranate and apple, others a single root, as silver-fir and fir; these have a single root in the sense that they have one long one which runs deep, and a number of small ones branching from this. Even in some of those which have more than a single root the middle root is the largest and goes deep, for instance, in the almond; in the olive this central root is small, while the others are larger

and, as it were, spread out crabwise. Again the roots of some are mostly stout, of some of various degrees of stoutness, as those of bay and olive; and of some they are all slender, as those of the vine. Roots also differ in degree of smoothness and in density. For the roots of all plants are less dense than the parts above ground, but the density varies in different kinds, as also does the woodiness. Some are fibrous, as those of the silver-fir, some fleshier, as those of the oak, some are as it were branched and tassel-like, as those of the olive; and this is because they have a large number of fine small roots close together; for all in fact produce these from their large roots, but they are not so closely matted nor so numerous in some cases as in others.

1.6.4

Again some plants are deeprooting, as the oak, and some have surface roots, as olive, pomegranate, apple, cypress. Again some roots are straight and uniform, others crooked and crossing one another. For this comes to pass not merely on account of the situation because they cannot find a straight course; it may also belong to the natural character of the plant, as in the bay and the olive; while the fig and such like become crooked because they can not find a straight course.

1.6.5

All roots have core, just as the stems and branches do, which is to be expected, as all these parts are made of the same materials. Some roots again have side-growths shooting upwards, as those of the vine find pomegranate, while some have no side-growth, as those of silver-fir, cypress and fir. The same differences are found in under-shrubs and herbaceous plants and the rest, except that some have no roots at all, as truffle, mushroom, bullfist 'thunder-truffle.' Others have numerous roots, as wheat, one-seeded wheat, barley and all plants of like nature, for instance, - Some have few roots, as leguminous plants. And in general most of the potherbs have single roots, as cabbage, beet, celery, monk's rhubarb; but some

have large side-roots, as celery and beet, and in proportion to their size these root deeper than trees. Again of some the roots are fleshy, as in radish, turnip, cuckoo-pint, crocus; of some they are woody, as in rocket and basil. And so with most wild plants, except those whose roots are to start with numerous and much divided, as those of wheat barley and the plant specially called 'grass.' For in annual and herbaceous plants this is the difference between the roots: — Some are more numerous and uniform and much divided to start with, but the others have one or two specially large roots and others springing from them.

1.6.6

To speak generally, the differences in roots are more numerous in shrubby plants and pot-herbs; — for some are woody, as those of basil, some fleshy, as those of beet, and still more those of cuckoo-pint asphodel and crocus; some again are made, as it were, of bark and flesh, as those of radishes and turnips; some have joints, as those of reeds and dog's tooth grass and of am-thing of a reedy character; and these roots alone, or more than any others, resemble the parts above ground; they are in fact like reeds fastened in the ground by their fine roots. Some again have scales or a kind of bark, as those of squill and purse-tassels, and also of onion and things like these. In all these it is possible to strip off a coat.

1.6.7

Now all such plants, seem, as it were, to have two kinds of root; and so, in the opinion of some, this is true generally of all plants which have a solid 'head' and send out roots from it downwards. These have, that is to say this fleshy or bark-like root. The squill, as well as the roots which grow from this. For these roots not only differ in degree of stoutness, like those of trees and pot-herbs; they are of quite distinct classes. This is at once quite evident in cuckoo-pint and galingale, the root being in the one case thick smooth and fleshy, in the other thin and fibrous.

Wherefore we might question if such roots should be called roots ‘; inasmuch as they are under ground they would seem to be roots, but, inasmuch as they are of opposite character to other roots, they Mould not. For your root gets slenderer as it gets longer and tapers continuously to a point; but the so-called root of squill purse-tassels and cuckoo-pint does just the opposite.

1.6.8

Again, while the others send out roots at the sides, this is not the case with squill and pursetassels, nor yet with garlic and onion. In general in these plants the roots which are attached to the ‘head’ in the middle appear to be real roots and receive nourishment,’ and this ‘head’ is, as it were, an embryo or fruit; Therefore those who call such plants ‘plants which reproduce themselves underground’ give a fair account of them. In other kinds of plants there is nothing of this sort. But a difficult question is raised, since here the ‘root’ has a character which goes beyond what one associates with roots. For it is not right to call al that which is underground ‘root,’ since in that case the stalk of purse-tassels and that of long onion and in general any part which is underground would be a root, and so would the truffle, the plant which some call puff-ball, the uingon, and all other underground plants. Whereas none of these is a root; for we must base our definition on natural function and not on position.

1.6.9

However it may be that this is a true account and yet that such things are roots no less; but in that case we distinguish two different kinds of root, one being of this character and the other of the other, and the one getting its nourishment from the other; though the fleshy roots too themselves seem to draw nourishment. At all events men invert the roots of cuckoo-pint before it shoots, and so they become larger by being prevented from pushing through to make a shoot. For it is evident that the

nature of all such plants is to turn downwards for choice; for the stems and the upper parts generally are short and weak, while the underground parts are large numerous and strong, and that, not only in the instances given, but in reeds dog's-tooth grass and in general in all plants of a reedy character and those like them. Those too which resemble ferula have large fleshy roots.

1.6.10

Many herbaceous plants likewise have such roots, as colchicum crocus and the plant called 'partridge-plant'; for this too has thick roots which are more numerous than its leaves. (It is called the 'partridge-plant' because partridges roll in it and grub it up.) So too with the plant called in Egypt uingon; for its leaves are large- and its shoots short, while the root is long and is, as it were, the fruit. It is an excellent thing and is eaten; men gather it when the river goes down by turning the clods. But the plants which afford the most conspicuous instances and shew the greatest difference as compared with others are silphium and the plant called magydaris; the character of both of these and of all such plants is especially shewn in their roots. Such is the account to be given of these plants.

1.6.11

Again some roots would seem to shew a greater difference than those mentioned, for instance, those of arakhidna, and of a plant which resembles arakos. For both of these bear a fruit underground which is as large as the fruit above ground, and this arakos has one thick root, namely, the one which runs deep, while the others which bear the 'fruit' are slenderer and branch in many directions at the tip. It is specially fond of sandy ground. Neither of these plants has a leaf nor anything resembling a leaf, but they bear, as it were, two kinds of fruit instead, which seems surprising. So many then are the differences shewn in the characters and functions of roots.

1.7.1

The roots of all plants seem to grow earlier than the parts above ground (for growth does take place downwards). But no root goes down further than the sun reaches, since it is the heat which induces growth. Nevertheless the nature of the sod, if it is light open and porous, contributes greatly to deep rooting, and still more to the formation of long roots; for in such soils growth goes further and is more vigorous. This is evident in cultivated plants. For, provided that they have water, they run on, one may say, wherever it may be, whenever the ground is unoccupied and there is no obstacle. For instance the plane-tree by the watercourse in the Lyceum when it was still young sent out its roots a distance of thirty-three cubits, having both room and nourishment.

1.7.2

The fig would seem, one may say, to have the longest roots, and in general plants which have wood of loose texture and straight roots would seem to have these longer. Also young plants, provided that they have reached their prime, root deeper and have longer roots than old ones; for the roots decay along with the rest of the plant's body. And in all cases alike the juices of plants are more powerful in the roots than in other parts, while in some cases they are extremely powerful; wherefore the roots are bitter in some plants whose fruits are sweet; some roots again are medicinal, and some are fragrant, as those of the iris.

1.7.3

The character and function of the roots of the 'Indian fig' (banyan) are peculiar, for this plant sends out roots from the shoots till it has a hold on the ground and roots again; and so there comes to be a continuous circle of roots round the tree, not connected with the main stem but at a distance from it.

1.7.4

Something similar to this, but even more surprising, occurs in those plants which emit roots from their leaves, as they say does a certain herb — which grows about Opus, which is also

sweet to taste. The peculiarity again of lupins is less surprising, namely that, if the seed is dropped where the ground is thickly overgrown, it pushes its root through to the earth and germinates because of its vigour. But we have said enough for study of the differences between roots.

1.8.1

One may take it that the following are the differences between trees: — Some have knots, more or less, others are more or less without them, whether from their natural character or because of their position. But, when I say ‘without knots,’ I do not mean that they have no knots at all (there is no tree like that, but, if it is true of any plants, it is only of other kinds, such as rush bulrush galingale and plants of the lake side generally) but that they have few knots. Now this is the natural character of elder bay fig and all smooth-barked trees, and in general of those whose wood is hollow or of a loose texture. Olive fir and wild olive have knots; and some of these grow in thickly shaded windless and wet places, some in sunny positions exposed to storms and winds, where the soil is light and dry; for the number of knots varies between trees of the same kind. And in general mountain trees have more knots than those of the plain, and those that grow in dry spots than those that grow in marshes.

1.8.2

Again the way in which they are planted makes a difference in this respect; those trees that grow close together are knotless and erect, those that grow far apart have more knots and a more crooked growth; for it happens that the one class are in shade, the others in full sun. Again the ‘male’ trees have more knots than the ‘female’ in those trees in which both forms are found, as cypress, silver-fir, hop-hornbeam, cornelian cherry — for there is a kind called ‘female cornelian cherry’ (cornel) — and wild trees have more knots than trees in cultivation: this is true both in general and when we compare those of the same

kind, as the wild and cultivated forms of olive fig and pear. All these have more knots in the wild state; and in general those of closer growth have this character more than those of open growth; for in fact the 'male' plants are of closer growth, and so are the wild ones; except that in some cases, as in box and nettle-tree, owing to the closer growth there are no knots at all, or only a few.

1.8.3

Again the knots of some trees are irregular and set at haphazard, while those of others are regular, alike in their distance apart and in their number, as has been said; wherefore also they are called trees with regular knots.' For of some the knots are, i\s it were, at even distances, while in others the distance between them is greater at the thick end of the stem. And this proportion holds throughout. This is especially evident in the wild olive and in reeds — in which the joint corresponds to the knot in trees. Again some knots are opposite one another, as those of the wild olive, while others are set at random. Again some trees have double knots, some treble, some more at the same point; some have as many as five. — In the silver-fir both the knots and the smaller branches are set at right angles, as if they were stuck in, but in other trees they are not so. And that is why the silver-fir is such a strong tree. Most peculiar are the knots of the apple, for they are like the faces of wild animals; there is one large knot, and a number of small ones round it. Again some knots are blind, others productive; by 'blind' I mean those from which there is no growth. These come to be so either by nature or by mutilation, according as either the knot' is not free and so the shoot does not make its way out, or, a bough having been cut off, the place is mutilated, for example by burning. Such knots occur more commonly in the thicker boughs, and in some cases in the stem also. And in general, wherever one chops or cuts part of the stem or bough, a knot is formed, as though one thing were made thereby into two and a

fresh growing point produced, the cause being the mutilation or some other such reason; for the effect of such a blow cannot of course be ascribed to nature.

1.8.4

Again in all trees the branches always seem to have more knots, because the intermediate parts have not yet developed, just as the newly formed branches of the fig are the roughest,' and in the vine the highest shoots. (For to the knot in other trees correspond the 'eye' in the vine, the joint in the reed) In some trees again there occurs, as it were, a diseased formation of small shoots,- as in elm, oak, and especially in the plane; and this is universal if they grow in rough, waterless or windy spots. Apart from any such cause this affection occurs near the ground in what one may call the 'head' of the trunk, when the tree is getting old.

1.8.5

Some trees again have what are called by some 'excrecences' (or something corresponding), as the olive; for this name belongs most properly to that tree, and it seems most liable to the affection; and some call it 'stump,' some Arotone, others have a different name for it. It does not occur, or only occurs to a less extent, in straight young trees, which have a single root and no side-growths. To the olive also, both wild and cultivated, are peculiar certain thickenings in the stem.

1.9.1

Now those trees which grow chiefly or only in the direction of their height are such as silver-fir date-palm cypress, and in general those which have a single stem and not many roots or branches (the date-palm, it may be added, has no side-growths at all). And trees like these have also similar growth downwards. Some however divide from the first, such as apple; some have many branches, and their greater mass of growth high up, as the pomegranate: however training position and cultivation chiefly contribute to all of these characters. In proof of which

we have the fact that the same trees which, when growing close together, are tall and slender, when grown farther apart become stouter and shorter; and if we from the first let the branches grow freely, the tree becomes short, whereas, if we prune them, it becomes tall, — for instance, the vine. This too is enough for proof that even some potherbs acquire the form of a tree, as we said of mallow and beet. Indeed all things grow well in congenial places. . . . For even among those of the same kind those which grow in congenial places have less knots, and are taller and more comely: thus the silver-fir in Macedon is superior to other silver-firs, such as that of Parnassus. Not only is this true of all these, but in general the wild woodland is more beautiful and vigorous on the north side of the mountain than on the south.

1.9.2

Again some trees are evergreen, some deciduous. Of cultivated trees, olive date-palm bay myrtle a kind of fir and cypress are evergreen, and among wild trees silver-fir, fir, Phoenician cedar, yew, odorous cedar, the tree which the Arcadians call ‘cork-oak’ (holm-oak), mock-privet, prickly cedar, ‘wild pine’, tamarisk, box, kermes-oak, holly, alaternus, cotoneaster, hybrid arbutus (all of which grow about Olympus) andrachne, arbutus, terebinth, ‘wild bay’ (oleander). Andrachne and arbutus seem to cast their lower leaves, but to keep those at the end of the twigs perennially, and to be always adding leafy twigs. These are the trees which are evergreen.

1.9.3

Of shrubby plants these are evergreen: — ivy, bramble, buckthorn, reed, kedris (juniper) — for there is a small kind of kedros so called which does not grow into a tree. Among under-shrubs and herbaceous plants there are rue, cabbage, rose, gilliflower, southernwood, sweet marjoram, tufted thyme, marjoram, celery, alexanders, poppy, and a good many more kinds of wild plants. However some of these too, while

evergreen as to their top growths, shed their other leaves, as marjoram and celery; for rue too is injuriously affected and changes its character.

1.9.4

And all the evergreen plants in the other classes too have narrower leaves and a certain glossiness and fragrance. Some moreover which are not evergreen by nature become so because of their position, as was said about the plants at Elephantine and Memphis, while lower down the Nile in the Delta there is but a very short period in which they are not making new leaves. It is said that in Crete in the district of Gortyna there is a plane near a certain spring which does not lose its leaves; (indeed the story is that it was under this tree that Zeus lay with Europa), while all the other plants in the neighbourhood shed their leaves. At Sybaris there is an oak within sight of the city which does not shed its leaves, and they say that it does not come into leaf along with the others, but only after the rising of the dog-star. It is said that in Cyprus too there is a plane which has the same peculiarity.

1.9.5

The fall of the leaves in all cases takes place in autumn or later, but it occurs later in some trees than in others, and even extends into the winter. However the fall of the leaf does not correspond to the growth of new leaves (in which case those that come into leaf earlier would lose their leaves earlier), but some (such as the almond) which are early in coming into leaf are not earlier than the rest in losing their leaves, but are even comparatively late.

1.9.6

Others again, such as the mulberry, come into leaf late, but are hardly at all later than the others in shedding their leaves. It appears also that position and a moist situation conduce to keeping the leaves late; for those which grow in dry places, and in general where the soil is light, shed their leaves earlier, and

the older trees earlier than young ones. Some even cast their leaves before the fruit is ripe, as the late kinds of fig and pear.

1.9.7

In those which are evergreen the shedding and withering of leaves take place by degrees; for it is not the same leaves which always persist, but fresh ones are growing while the old ones wither away. This happens chiefly about the summer solstice. Whether in some cases it occurs even after the rising of Arcturus or at a quite different season is matter for enquiry. So much for the shedding of leaves.

1.10.1

Now, while the leaves of all other trees are all alike in each tree, those of the abele ivy — and of the plant called kroton (castor-oil plant) are unlike one another and of different forms. The young leaves in these are round, the old ones angular, and eventually all the leaves assume that form. On the other hand in the ivy, when it is young, the leaves are somewhat angular, but when it is older, they become rounder: for in this plant too a change of form takes place. There is a peculiarity special to the olive lime elm and abele: their leaves appear to invert the upper surface after the summer solstice, and by this men know that the solstice is past. Now all leaves differ as to their upper and under surfaces; and in most trees the upper surfaces are greener and smoother, as they have the fibres and veins in the under surfaces, even as the human hand has its 'lines,' but even the upper surface of the leaf of the olive is sometimes whiter and less smooth. So all or most leaves display their upper surfaces, and it is these surfaces which are exposed to the light. Again most leaves turn towards the sun; wherefore also it is not easy to say which surface is next to the twig; for, while the way in which the upper surface is presented seems rather to make the under surface closer to it, yet nature desires equally that the upper surface should lie the nearer, and this is specially seen in the turning back of the leaf towards the sun. One may observe

this in trees whose leaves are crowded and opposite, such as those of myrtle.

1.10.2

Some think that the nourishment too is conveyed to the upper surface through the under surface, because this surface always contains moisture and is downy, but they are mistaken. It may be that this is not due to the trees' special character, but to their not getting an equal amount of sunshine, though the nourishment conveyed through the veins or fibres is the same in both cases. That it should be conveyed from one side to the other is improbable, when there are no passages for it nor thickness for it to pass through. However it belongs to another part of the enquiry to discuss the means by which nourishment is conveyed.

1.10.3

Again there are various other differences between leaves; some trees are broad-leaved, as vine fig and plane, some narrow-leaved, as olive, pomegranate, myrtle. Some have, as it were, spinous' leaves, as fir, Aleppo pine, prickly cedar; some, as it were, fleshy leaves; and this is because their leaves are of fleshy substance, as cypress, tamarisk, apple, among under-shrubs kneoros, and stoibe, and among herbaceous plants house-leek, and hulwort. This plant is good against moth in clothes. For the leaves of beet and cabbage are fleshy in another way, as are those of the various plants called rue; for their fleshy character is seen in the flat instead of in the round. Among shrubby plants the tamarisk has fleshy leaves. Some again have reedy leaves, as date-palm, doum-palm and such like. But, generally speaking, the leaves of these end in a point; for reeds galingale sedge and the leaves of other marsh plants are of this character. The leaves of all these are compounded of two parts, and the middle is like a keel, placed where in — other leaves is a large passage dividing the two halves. Leaves differ also in their shapes; some are round, as those of pear, some

rather oblong, as those of the apple; some come to a sharp point and have spinous projections at the side, as those of smilax. So far I have spoken of undivided leaves; but some are divided and like a saw, as those of silver-fir and of fern. To a certain extent those of the vine are also divided, while those of the fig one might compare to a crow's foot. Some leaves again have notches, as those of elm filbert and oak, others have spinous projections both at the tip and at the edges, as those of kermes oak, oak, smilax, bramble, Christ's thorn and others. The leaf of fir, Aleppo pine, silver-fir and also of prickly cedar and kedris (juniper) has a spinous point at the tip. Among other trees there is none that we know which has spines for leaves altogether, but it is so with other woody plants, as akorna, drypis, pine thistle and almost all the plants which belong to that class. For in all these spines, as it were, take the place of leaves, and, if one is not to reckon these plants. It is peculiar to pot-herbs to have hollow leaves, as in onion and horn-onion.

1.10.4

To sum up, the differences between leaves are shewn in size, number, shape, hollowness, in breadth, roughness and their opposites, and in the presence or absence of spinous projections; also as to their attachment, according to the part from which they spring or the means by which they are attached; the part from which they spring being the root or a branch or the stalk or a twig, while the means by which they are attached may be a leaf-stalk, or they may be attached directly; and there may be several leaves attached by the same leaf-stalk. Further some leaves are fruit-bearing, enclosing the fruit between them, as the Alexandrian laurel, which has its fruit attached to the leaves. These are all the differences in leaves stated somewhat generally, and this is a fairly complete list of examples.

1.10.5

Leaves are composed some of fibre bark and flesh, as those of the fig and vine, some, as it were, of fibre alone, as those of reeds and corn. But moisture is common to all, for it is found both in leaves and in the other annual parts, leaf-stalk, flower, fruit and so forth but more especially in the parts which are not annual; in fact no part is without it. Again it appears that some leaf-stalks are composed only of fibre, as those of corn and reeds, some of the same materials as the stalks.

1.10.6

Of flowers some are composed of bark veins and flesh, some of flesh only, as those in the middle of cuckoo-pint.'

1.10.7

So too with fruits; some are made of flesh and fibre, some of flesh alone, and some of skin also. And moisture is necessarily found in these also. The fruit of plums and cucumbers is made of flesh and fibre, that of mulberries and pomegranates of fibre and skin. The materials are differently distributed in different fruits, but of nearly all the outside is bark, the inside flesh, and this in some cases includes a stone.)

1.11.1

Last in all plants comes the seed. This possesses in itself natural moisture and warmth, and, if these fail, the seeds are sterile, like eggs in the like case. In some plants the seed comes immediately inside the envelope, as in date filbert almond (however, as in the case of the date, there may be more than one covering). In some cases again there is flesh and a stone between the envelope and the seed, as in olive plum and other fruits. Some seeds again are enclosed in a pod, some in a husk, some in a vessel, and some are completely naked.

1.11.2

Enclosed in a pod are not only the seeds of annual plants, as leguminous plants, and of considerable numbers of wild plants, but also those of certain trees, as the carob-tree (which some call the 'Egyptian fig'), Judas-tree, and the koloitium of the

Liparae islands. In a husk are enclosed the seeds of some annuals, as wheat and millet; and in like manner some plants have their seeds in a vessel, some have them naked. In a vessel are those of the poppy and plants of the poppy kind; (the case of sesame however is somewhat peculiar), while many pot-herbs have their seeds naked, as dill coriander- anise cummin fennel and many others. No tree has naked seeds, but either they are enclosed in flesh or in shells, which are sometimes of leathery nature, as the acorn and the sweet chestnut, sometimes woody, as almond and nut. Moreover no tree has its seeds in a vessel, unless one reckons a cone as a vessel, because it can be separated from the fruits.

1.11.3

The actual seeds are in some cases fleshy in themselves, as all those which resemble nuts or acorns; in some cases the fleshy part is contained in a stone, as in olive bay and others. The seeds in some plants again merely consist of a stone, or at least are of stone-like character, and are, as it were, 'dry'; for instance those of plants like safflower, millet and many pot-herbs. Most obviously of this character are those of the date, for they contain no cavity, but are throughout dry; — not but what there must be even in them some moisture and warmth, as we have said.

Further seeds differ in that in some cases they are massed together, in others they are separated and arranged in rows, as those of the gourd and bottle-gourd, and of some trees, such as the citron. Again of those that are massed together some differ in being contained in a single case, as those of pomegranate pear apple vine and fig; others in being closely associated together, yet not contained in a single case, as, among annuals, those which are in an ear — unless one regards the ear as a case. In that case the grape-cluster and other clustering fruits will come under the description, as well as all those plants which on

account of good feeding or excellence of soil bear their fruits massed together, as they say the olive does in Syria and elsewhere.

1.11.5

But this too seems to be a point of difference, that some grow massed together from a single stalk and a single attachment, as has been said in the case of plants with clusters or ears whose seeds do not grow contained in one common case; while others grow otherwise. For in these instances, if one takes each seed or case separately, it has its own special point of attachment, for instance each grape or pomegranate, or again each grain of wheat or barley. This would seem to be least of all the case with the seeds of apples and pears, since these touch one another and are enclosed in a sort of skin-like membrane, outside which is the fruit-case. However each of these too has its own peculiar point of attachment and character; this is most obvious in the separation of the pomegranate seeds, for the stone is attached to each, and the connexion is not, as in figs, obscured by the moisture. For here too there is a difference, although in both cases the seeds are enclosed in a sort of fleshy substance, as well as in the case which encloses this and the other parts of the fruit. For in the pomegranate the stones have this moist fleshy substance enclosing each separate stone; but in the case of fig-seeds, as well as in that of grape stones and other plants which have the same arrangement, the same pulp is common to all. However one might find more such differences, and one should not ignore the most important of them, namely those which specially belong to the plant's natural character.

1.12.1

The differences in taste, shape, and form as a whole are tolerably evident to all, so that they do not need explanation; except that it should be stated that the case containing the fruit is never right-lined in shape and never has angles. Of tastes some are like wine, as those of vine mulberry and myrtle: some

are like olive-oil, as, besides olive itself, bay, hazel, almond, fir, Aleppo pine, silver fir; some like honey, as fig, date, chestnut; some are pungent, as marjoram, savory cress mustard; some are bitter, as wormwood, centaury. Some also are remarkably fragrant, as anise and juniper; of some the smell would seem to be insipid,- as in plums; of others sharp, as in pomegranates and some kinds of apples. But the smells even of those in this class must in all cases be called winelike, though they differ in different kinds, on which matter we must speak more precisely, when we come to speak of flavours,- reckoning up the different kinds themselves, and stating what differences there are between them, and what is the natural character and property of each.

1.12.2

Now the sap of the trees themselves assumes different kinds of tastes as was said; sometimes it is milky, as that of the fig and poppy, sometimes like pitch, as in silver-fir, fir and the conifers; sometimes it is insipid, as in vine pear and apple, as well as such pot-herbs as cucumber gourd lettuce; while others' again have a certain pungency, such as the juice of thyme and savory; others have a fragrance, such as the juices of celery, dill, fennel, and the like. To speak generally, all saps correspond to the special character of the several trees, one might almost add, to that of each plant. For every plant has a certain temperament and composition of its own, which plainly belongs in a special sense to the fruits of each. And in most of these is seen a sort of correspondence with the character of the plant as a whole, which is not however exact nor obvious; it is chiefly in the fruitcases that it is seen, and that is why it is the character of the flavour which becomes more complete and matures into something separate and distinct; in fact we must consider the one as 'matter,' the other — as 'form' or specific character.

1.12.3

Again the seeds themselves and the coats containing them have different flavours. And, to speak generally, all parts of trees and plants, as root stem branch leaf fruit, have a certain relationship to the character of the whole, even if there is variation in scents and tastes, so that of the parts of the same plant some are fragrant and sweet to the taste, while others are entirely scentless and tasteless.

1.12.4

For in some plants the flowers are more fragrant than the leaves, in others on the contrary it is rather the leaves and twigs which are fragrant, as in those used for garlands. In others again it is the fruits; in others it is neither of these parts, but, in some few cases, the root or some part of it. And so too with the flavours. Some leaves and some fruit-pulps are, and some are not good for food. Most peculiar is the case of the lime: the leaves of this are sweet, and many animals eat them, but the fruit no creature eats, (for, as to the contrary case, it would not be at all surprising that the leaves should not be eaten, while the fruits were eaten not only by us but by other animals). But concerning this and other such matters we must endeavour to consider the causes on some other occasion.

1.13.1

For the present let so much be clear, that in all the parts of plants there are numerous differences shewn in a variety of ways. Thus of flowers some are downy, as that of the vine, mulberry and ivy, some are 'leafy,' as in almond, apple, pear, plum. Again some of these flowers are conspicuous, while that of the olive, though it is 'leafy,' is inconspicuous. Again it is in annual and herbaceous plants alike that we find some leafy, some downy. All plants again have flowers either of two colours or of one; most of the flowers of trees are of one colour and white, that of the pomegranate being almost the only one which is red, while that of some almonds is reddish. The flower of no other cultivated trees is gay nor of two colours, though it may

be so with some uncultivated trees, as with the flower of silverfir, for its flower is of saffron colour; and so with the flowers of those trees by the ocean which have, they say, the colour of roses.

1.13.2

However, among annuals, most are of this character — their flowers are two-coloured and twofold. I mean by ‘twofold’ that the plant has another flower inside the flower, in the middle, as with rose, lily, violet. Some flowers again consist of a single ‘leaf,’ having merely an indication of more, as that of bindweed. For in the flower of this the separate ‘leaves’ are not distinct; nor is it so in the lower part of the narcissus,’ but there are angular projections from the edges. And the flower of the olive is nearly of the same character.

1.13.3

But there are also differences in the way of growth and the position of the flower; some plants have it close above the fruit, as vine and olive; in the latter, when the flowers drop off, they are seen to have a hole through them, and this men take for a sign — whether the tree has blossomed well; for if the flower is burnt up or sodden, it sheds the fruit along with itself, and so there is no hole through it. The majority of flowers have the fruit-case in the middle of them, or, it may be, the flower is on the top of the fruit-case, as in pomegranate apple pear plum and myrtle, and among under-shrubs, in the rose and in many of the coronary plants. For these have their seeds below, beneath the flower, and this is most obvious in the rose because of the size of the seed-vessel. In some cases again the flower is on top of the actual seeds, as in pine-thistle safflower and all thistle-like plants; for these have a flower attached to each seed. So too with some herbaceous plants, as anthemion, and among pot-herbs, with cucumber gourd and bottle-gourd; all these have their flowers attached on top of the fruits, and the flowers persist for a long time while the fruits are developing.

1.13.4

In some other plants the attachment is peculiar, as in ivy and mulberry; in these the flower is closely attached to the whole fruit-case; it is not however set above it, nor in a seed-vessel that envelops each separately, but it occurs in the middle part of the structure — except that in some cases it is not easily recognised because it is downy.

1.13.5

Again some flowers are sterile, as in cucumbers those which grow at the ends of the shoot, and that is why men pluck them off, for they hinder the growth of the cucumber. And they say that in the citron those flowers which have a kind of distaff growing in the middle are fruitful, but those that have it not are sterile. And we must consider whether it occurs also in any other flowering plants that they produce sterile flowers, whether apart from the fertile flowers or not. For some kinds of vine and pomegranate certainly are unable to mature their fruit, and do not produce anything beyond the flower.

1.13.6

(The flower of the pomegranate is produced abundantly and is solid: in general appearance it is a substantial structure with a flat top, like the flower of the rose; but, as seen from below, the inferior part of the flower is different-looking, being like a little two-eared jar turned on one side and having its rim indented.)

1.13.7

Some say that even of plants of the same kind some specimens flower while others do not; for instance that the ‘male’ date-palm flowers but the ‘female’ does not, but exhibits its fruit without any antecedent flower. Such is the difference which we find between plants of the same kind; and the like may be said in general of those which cannot mature their fruit. And it is plain from what has been said that flowers shew many differences of character.

1.14.1

Again as to the production of fruit trees differ in the following respects. Some bear on their new shoots, some on last year's wood, some on both. Fig and vine bear on their new shoots; on last year's wood olive pomegranate apple almond pear myrtle and almost all such trees. And, if any of these does happen to conceive and to produce flowers on its new shoots, (for this does occur in some cases, as with myrtle, and especially, one may say, in the growth which is made after the rising of Arcturus) it can not bring them to perfection, but they perish half-formed. Some apples again of the twice-bearing kinds and certain other fruit-trees bear both on last year's wood and on the new shoots; and so does the olynihos, which ripens its fruit as well as bearing figs on the new shoots.

1.14.2

Most peculiar is the growth of fruit direct from the stem, as in the sycamore; for this, they say, bears fruit on the stem. Others say that it bears both in this way and also on the branches, like the carob; for the latter bears on the branches too, though not abundantly: (the name carob is given to the tree which produces what are called 'Egyptian figs'). Again some trees, and some plants in general, produce fruit at the top, others at the sides, others in both ways. But bearing fruit at the top is less common in trees than in other plants, as among grains in those which have an ear, among shrubby plants in heath privet chaste tree and certain others, and among pot-herbs in those with a bulbous root. Among plants which bear both on the top and at the sides are certain trees and certain potherbs, as blite, orach, cabbage. I say trees, since the olive does this too in a way, and they say that, when it bears at the top, it is a sign of fruitfulness. The date-palm too bears at the top, in a sense, but this tree also has its leaves and shoots at the top; indeed it is in the top that its whole activity is seen. Thus we must endeavour to study in the light of the instances mentioned the differences seen in the various parts of the plant.

1.14.3

But there appear to be the following differences which affect the plant's whole being: some are cultivated, some wild; some fruitful, some barren; some evergreen, some deciduous, as was said, while some again have no leaves at all; some are flowering plants, some flowerless; some are early, some late in producing their shoots and fruits; and there are other differences similar to these. Now it may be said that such differences are seen in the parts, or at least that particular parts are concerned in them. But the special, and in a way the most important distinction is one which may be seen in animals too, namely, that some are of the water, some of the land. For of plants too there is a class that cannot grow except in moisture, while others will indeed grow on dry land, but they lose their character and are inferior. Again of all trees, one might almost say, and of all plants there are several forms to each kind; for hardly any kind contains but a single form.

1.14.4

But the plants which are called respectively cultivated and wild shew this difference in the clearest and most emphatic way, for instance the cultivated and wild forms of fig olive and pear. In each of these pairs there are differences in fruit and leaves, and in their forms and parts generally. But most of the wild kinds have no names and few know about them, while most of the cultivated kinds have received names — and they are more commonly observed; I mean such plants as vine, fig, pomegranate, apple, pear, bay, myrtle and so forth; for, as many people make use of them, they are led also to study the differences.

1.14.5

But there is this peculiarity as to the two classes respectively; in the wild kinds men find only or chiefly the distinction of male 'and' female,' while in the cultivated sorts they recognise a number of distinguishing features. In the former case it is easy

to mark and count up the different forms, in the latter it is harder because the points of difference are numerous. However we have said enough for study of the differences between parts and between general characters. We must now speak of the methods of growth, for this subject comes naturally after what has been said.

2.1.1

BOOK II. OF PROPAGATION, ESPECIALLY OF TREES.

The ways in which trees and plants in general originate are these: — spontaneous growth, growth from seed, from a root, from a piece torn off, from a branch or twig, from the trunk itself; or again from small pieces into which the wood is cut up (for some trees can be produced even in this manner). Of these methods spontaneous growth comes first, one may say, but growth from seed or root would seem most natural; indeed these methods too may be called spontaneous; wherefore they are found even in wild kinds, while the remaining methods depend on human skill or at least on human choice.

2.1.2

However all plants start in one or other of these ways, and most of them in more than one. Thus the olive is grown in all the ways mentioned, except from a twig; for an olive-twigg will not grow if it is set in the ground, as a fig or pomegranate will grow from their young shoots. Not but what some say that cases have been known in which, when a stake of olive-wood was planted to support ivy, it actually lived along with it and became a tree; but such an instance is a rare exception, while the other methods of growth are in most cases the natural ones. The fig grows in all the ways mentioned, except from root-stock and cleft wood; apple and pear grow also from branches, but rarely. However it appears that most, if not practically all, trees may grow from branches, if these are smooth, young and vigorous. But the other methods, one may say, are more natural, and we

must reckon what may occasionally occur as a mere possibility.

2.1.3

In fact there are quite few plants which grow and are brought into being more easily from the upper parts, as the *viae* is grown from branches; for this, though it cannot be grown from the head, yet can be grown from the branch, as can all similar trees and under-shrubs, for instance, as it appears, rue, gilliflower, bergamot-mint, tufted thyme, calamint. So the commonest ways of growth with all plants are from a piece torn off or from seed; for all plants that have seeds grow also from seed. And they say that the bay too grows from a piece torn off, if one takes off the young shoots and plants them; but it is necessary that the piece torn off should have part of the root or stock attached to it. However the pomegranate and 'spring apple' will grow even without this, and a slip of almond grows if it is planted. The olive grows, one may say, in more ways than any other plant; it grows from a piece of the trunk or of the stock, from the root, from a twig, and from a stake, as has been said. Of other plants the myrtle also can be propagated in several ways; for this too grows from pieces of wood and also from pieces of the stock. It is necessary however with this, as with the olive, to cut up the wood into pieces not less than a span long and not to strip off the bark.

2.1.4

Trees then grow and come into being in the abovementioned ways; for as to methods of grafting and inoculation, these are, as it were, combinations of different kinds of trees; or at all events these are methods of growth of a quite different class and must be treated of at a later stage.

2.2.1

Of under-shrubs and herbaceous plants the greater part grow from seed or a root, and some in both ways; some of them also grow from cuttings, as has been said, while roses and lilies grow from pieces of the stems, as also does dog's-tooth grass. Lilies

and roses also grow when the whole stem is set. Most peculiar is the method of growth from an exudation; for it appears that the lily grows in this way too, when the exudation that has been produced has dried up. They say the same of alexanders, for this too produces an exudation. There is a certain reed also which grows if one cuts it in lengths from joint to joint and sets them sideways, burying it in dung and soil. Again they say that plants which have a bulbous root are peculiar in their way of growing from the root.

2.2.2

The capacity for growth being shewn in so many ways, most trees, as was said before, originate in several ways; but some come only from seed, as silver fir, fir, Aleppo pine, and in general all those that bear cones: also the date-palm, except that in Babylon it may be that, as some say, they take cuttings from it. The cypress in most regions grows from seed, but in Crete from the trunk also, for instance in the hill country about Tarra; for there grows the cypress which they clip, and when cut it shoots in every possible way, from the part which has been cut, from the ground, from the middle, and from the upper parts; and occasionally, but rarely, it shoots from the roots also.

2.2.3

About the oak accounts differ; some say it only grows from seed, some from the root also, but not vigorously, others again that it grows from the trunk itself, when this is cut. But no tree grows from a piece torn off or from a root except those which make side-growths.

2.2.4

However in all the trees which have several methods of originating the quickest method and that which promotes the most vigorous growth is from a piece torn off, or still better from a sucker, if this is taken from the root. And, while all the trees which are propagated thus or by some kind of slip seem to be alike in their fruits to the original tree, those raised from the

fruit, where this method of growing is also possible, are nearly all inferior, while some quite lose the character of their kind, as vine apple fig pomegranate pear. As for the fig, no cultivated kind is raised from its seed, but either the ordinary wild fig or some wild kind is the result, and this often differs in colour from the parent; a black fig gives a white, and conversely. Again the seed of an excellent vine produces a degenerate result, which is often of quite a different kind; and at times this is not a cultivated kind at all, but a wild one of such a character that it does not ripen its fruit; with others again the result is that the seedlings do not even mature fruit, but only get as far as flowering.

2.2.5

Again the stones of the olive give a wild olive, and the seeds of a sweet pomegranate give a degenerate kind, while the stoneless kind gives a hard sort and often an acid fruit. So also is it with seedlings of pears and apples; pears give a poor sort of wild pears, apples produce an inferior kind which is acid instead of sweet; quince produces wild quince. Almond again raised from seed is inferior in taste and in being hard instead of soft; and this is why men bid us graft on to the almond, even when it is fully grown, or, failing that, frequently plant the offsets.

2.2.6

The oak also deteriorates from seed; at least many persons having raised trees from acorns of the oak at Pyrrha could not produce one like the parent tree. On the other hand they say that bay and myrtle sometimes improve by seeding, though usually they degenerate and do not even keep their colour, but red fruit gives black — as happened with the tree in Antandros; and frequently seed of a ‘female’ cypress produces a male tree. The datepalm seems to be about the most constant of these trees, when raised from seed, and also the ‘cone-bearing pine’ (stone-pine) and the ‘lice-bearing pine.’ So much for degeneration in cultivated trees; among wild kinds it is plain

that more in proportion degenerate from seed, since the parent trees are stronger. For the contrary would be very strange, seeing that degenerate forms are found even in cultivated trees, and among these only in those which are raised from seed. (As a general rule these are degenerate, though men may in some cases effect a change by cultivation).

2.2.7

Again differences in situation and climate affect the result. In some places, as at Philippi, the soil seems to produce plants which resemble their parent; on the other hand a few kinds in some few places seem to undergo a change, so that wild seed gives a cultivated form, or a poor form one actually better. We have heard that this occurs, but only with the pomegranate, in Egypt and Cilicia; in Egypt a tree of the acid kind both from seeds and from cuttings produces one whose fruit has a sort of sweet taste, while about Soli in Cilicia near the river Pinaros (where the battle with Darius was fought) all those pomegranates raised from seed are without stones.

2.2.8

If anyone were to plant our palm at Babylon, it is reasonable to expect that it would become fruitful and like the palms of that country. And so would it be with any other country which has fruits that are congenial to that particular locality; for the locality is more important than cultivation and tendance. A proof of this is the fact that things transplanted thence become unfruitful, and in some cases refuse to grow altogether.

2.2.9

There are also modifications due to feeding and attention of other kinds, which cause the wild to become cultivated, or again cause some cultivated kinds to go wild, such as pomegranate and almond. Some say that wheat has been known to be produced from barley, and barley from wheat, or again both growing on the same stool; but these accounts should be taken as fabulous. Anyhow those things which do change in this

manner do so spontaneously,- and the alteration is due to a change of position (as we said happens with pomegranates in Egypt and Cilicia), and not to any particular method of cultivation.

2.2.10

So too is it when fruit-bearing trees become unfruitful, for instance the persion when moved from Egypt, the date-palm when planted in Hellas, or the tree which is called 'poplar' in Crete,' if anyone should transplant it. Some again say that the sorb becomes unfruitful if it comes into a very warm position, since it is by nature cold-loving. It is reasonable to suppose that both results follow because the natural circumstances are reversed, seeing that some things entirely refuse to grow when their place is changed. Such are the modifications due to position.

2.2.11

As to those due to method of culture, the changes which occur in things grown from seed are as was said; (for with things so grown also the changes are of all kinds). Under cultivation the pomegranate and the almond change character, the pomegranate if it receives pig-manure and a great deal of river water, the almond if one inserts a peg and removes for some time the gum which exudes and gives the other attention required. In like manner plainly some wild things become cultivated and some cultivated things become wild: for the one kind of change is due to cultivation, the other to neglect: — however it might be said that this is not a change but a natural development towards a better or an inferior form; (for that it is not possible to make a wild olive pear or fig into a cultivated olive pear or fig). As to that indeed which is said to occur in the case of the wild olive, that if the tree is transplanted with its topgrowth entirely cut off, it produces 'coarse olives,' this is no very great change. However it can make no difference which way one takes this.

2.3.1

Apart from these changes it is said that in such plants there is a spontaneous kind of change, sometimes of the fruit, sometimes of the tree itself as a whole, and soothsayers call such changes portents. For instance, an acid pomegranate, it is said, may produce sweet fruit, and conversely; and again, in general, the tree itself sometimes undergoes a change, so that it becomes sweet instead of acid, or the reverse happens. And the change to sweet is considered a worse portent. Again a wild fig may turn into a cultivated one, or the contrary change take place; and the latter is a worse portent. So again a cultivated olive may turn into a wild one, or conversely, but the latter change is rare. So again a white fig may change into a black one. and conversely; and similar changes occur in the vine.

2.3.2

Now these changes they interpret as miraculous and contrary to nature; but they do not even feel any surprise at the ordinary changes, for instance, when the 'smoky' vine, as it is called, produces alike white grapes instead of black or black grapes instead of white. Of such changes the soothsayers take no account, any more than they do of those instances in which the soil produces a natural change, as was said of the pomegranate in Egypt. But it is surprising when such a change occurs in our own country, because there are only one or two instances and these separated by wide intervals of time. However, if such changes occur, it is natural that the variation should be rather in the trait than in the tree as a whole. In fact the following irregularity also occurs in fruits; a fig-tree has been known to produce its figs from behind the leaves, pomegranate and vines from the stem, while the vine has been known to bear fruit without leaves. The olive again has been known to lose its leaves and yet produce its fruit; this is said to have happened to Thettalos, son of Pisistratus. This may be due to inclement weather; and some changes, which seem to be abnormal, but are

not really so, are due to other accidental causes; for instance, there was an olive that, after being completely burnt down, sprang up again entire, the tree and all its branches. And in Boeotia an olive whose young shoots had been eaten off by locusts grew again: in this case however the shoots had, so to speak, only been shed. But after all such phenomena are perhaps far from strange, since the cause in each case is obvious; rather is it strange that trees should bear fruit not at the places where it naturally forms, or else fruit which does not belong to the character of the tree. And most surprising of all is it when, as has been said, there is a change in the entire character of the tree. Such are the changes which occur in trees.

2.4.1

Of other plants it appears that bergamot-mint turns into cultivated mint, unless it is fixed by special attention; and this is why men frequently transplant it; so too wheat turns into darnel. Now in trees such changes, if they occur, are spontaneous, but in annual plants they are deliberately brought about: for instance, one-seeded wheat and rice-wheat change into wheat, if bruised before they are sown; and this does not happen at once, but in the third year. This change resembles that produced in the seeds by difference of soil; for these grains vary according to the soil, and the change takes about the same time as that which occurs in one-seeded wheat. Again wild wheats and barleys also with tendance and cultivation change in a like period.

2.4.2

These changes appear to be due to change of soil and cultivation, and in some cases the change is due to both, in others to cultivation alone; for instance, in order that pulses may not become uncookable, men bid one moisten the seed in nitre for a night and sow it in dry ground the next day. To make lentils vigorous they plant the seeds in dung; to make chick-peas large they bid one moisten the seed while still in the

pod, before sowing. Also the time of sowing makes differences which conduce to digestibility and harmlessness: thus, if one sows vetches in spring, they become quite harmless and are not indigestible like those sown in autumn.

2.4.3

Again in pot-herbs change is produced by cultivation; for instance, they say that, if celery seed is trodden and rolled in after sowing, it comes up curly; it also varies from change of soil, like other things. Such variations are common to all; we must now consider whether a tree, like animals, becomes unproductive from mutilation or removal of a part. At all events it does not appear that division is an injury, as it were, which affects the amount of fruit produced; either the whole tree perishes, or else, if it survives, it bears fruit. Old age however is a cause which in all plants puts an end to life.

2.4.4

It would seem more surprising if the following changes occurred in animals naturally and frequently; some animals do indeed seem to change according to the seasons, for instance, the hawk the hoopoe and other similar birds. So also changes in the nature of the ground produce changes in animals, for instance, the water-snake changes into a viper, if the marshes dry up. Most obvious are certain changes in regard to the way in which animals are produced, and such changes run through a series of creatures; thus a caterpillar changes into a chrysalis, and this in turn into the perfect insect; and the like occurs in a number of other cases. But there is hardly anything abnormal in this, nor is the change in plants, which is the subject of our enquiry, analogous to it. That kind of change occurs in trees and in all woodland plants generally, as was said before, and its effect is that, when a change of the required character occurs in the climatic conditions, a spontaneous change in the way of growth ensues. These instances must suffice for investigation of the ways in which plants are produced or modified.

2.5.1

Since however methods of cultivation and tendance largely contribute, and, before these, methods of planting, and cause great differences, of these too we must speak.

2.5.2

And first of methods of planting: as to the seasons, we have already stated at what seasons one should plant. Further we are told that the plants chosen should be the best possible, and should be taken from soil resembling that in which you are going to plant them, or else inferior; also the holes should be dug as long as possible beforehand, and should always be deeper than the original holes, even for those whose roots do not run very deep.

2.5.3

Some say that no root goes down further than a foot and a half, and accordingly they blame those who plant deeper. However there are many instances in which it appears that what they say does not hold good: a plant which is naturally deep-rooting pushes much deeper if it finds either a deep mass of soil or a position which favours such growth or again the kind of ground which favours it. In fact, a man once said that when he was transplanting a fir which he had uprooted with levers, he found that it had a root more than eight cubits long, though the whole of it had not been removed, but it was broken off.

2.5.4

The slips for planting should be taken, if possible, with roots attached, or, failing that, from the lower rather than from the higher parts of the tree, except in the case of the vine; those that have roots should be set upright, while in the case of those which have none about a handsbreadth or rather more of the slip should be buried. Some say that part even of those which have roots should be buried, and that the position should be the same as that of the tree from which the slip was taken, facing north or east or south, as the case may be. With those plants

with which it is possible, shoots from the boughs should also, they say, be planted, some being set on the trees themselves, as with olive pear apple and fig, but in other cases, as in that of the vine, they must be set separately, for that the vine cannot be grafted on itself.

2.5.5

If the slips cannot be taken with root or stock attached, as with the olive, they say that one must split the wood at the lower end and plant with a stone on top; and the fig and other trees must be treated in like manner with the olive. The fig is also propagated by sharpening a stout shoot and driving it in with a hammer, till only a small piece of it is left above ground, and then piling sand above so as to earth it up; and they say that the plants thus raised grow finer up to a certain age.

2.5.6

Similar is the method used with vines, when they are propagated by the 'peg' method; for the peg makes a passage for that sort of shoot on account of its weakness; and in the same manner men plant the pomegranate and other trees. The fig progresses more quickly and is less eaten by grubs, if the cutting is set in a squill-bulb; in fact anything so planted is vigorous and grows faster. All those trees which are propagated by pieces cut from the stem should be planted with the cut part downwards,' and the pieces cut off should not be less than a handsbreadth in length, as Mas said, and the bark should be left on. From such pieces new shoots grow, and as they grow, one should keep on heaping up earth about them, till the tree becomes strong. This kind of propagation is peculiar to the olive and myrtle, while the others are more or less common to all trees.

2.5.7

The fig is better than any other tree at striking roots, and will, more than any other tree, grow by any method of propagation. We are told that, in planting the pomegranate, myrtle or bay, one should set two trees close together, not further than nine

feet apart, apples a little further, pears and wild pears still further, almonds and figs further still, and in like manner the olive. Again the distance apart must be regulated by the nature of the ground, being less — in hilly parts than in low ground.

2.5.8

Most important of all, one may say, is it to assign to each the suitable soil; for then is the tree most vigorous. Speaking generally, they say that low ground is most suitable for the olive fig and vine, and the lower slopes of hills for fruit trees. Nor should one fail to note what soil suits each variety even of those closely related. There is the greatest difference, one may say, between the different kinds of vine: for they say that there are as many kinds of vine as there are of soil. If they are planted as their nature requires, they turn out well, if otherwise, they are unfruitful. And these remarks apply almost equally to all trees.

2.6.1

The method of propagating date-palms is peculiar and exceptional, as also is their subsequent cultivation. They plant several seeds together, putting two below and two above, which are fastened on; but all face downwards. For germination starts not, as some say, from the 'reverse' or hollow side, but from the part which is uppermost; wherefore in joining on the seeds which are placed above one must not cover up the points from which the growth is to come; and these can be recognised by experts. And the reason why they set several together is that a plant that grows from one only is weak. The roots which grow from these seeds become entangled together and so do the first shoots from the very start, so that they combine to make a single stem.

2.6.2

Such is the method of growing from the fruits. But propagation is also possible from the tree itself, by taking off the top, which contains the 'head.' They take off about two cubits' length, and, splitting it, set the moist end.- It likes a soil which contains salt;

wherefore, where such soil is not available, the growers sprinkle salt about it; and this must not be done about the actual roots: one must keep the salt some way off and sprinkle about a gallon. To shew that it seeks such a soil they offer the following proof; wherever date-palms grow abundantly, the soil is salt, both in Babylon, they say, where the tree is indigenous, in Libya in Egypt and in Phoenicia; while in Coele-Syria, where are most palms, only in three districts, they say, where the soil is salt, are dates produced which can be stored; those that grow in other districts do not keep, but rot, though when fresh they are sweet and men use them at that stage.

2.6.3

The tree is likewise very fond of irrigation; as to dung there is a difference of opinion: some say that the date-palm does not like it, but that it is most injurious, others that it gladly accepts it and makes good growth thereby, but plenty of water should be given, after manuring, as the Rhodians use. This then is matter for enquiry; it may be that there are two distinct methods of cultivation, and that dung, if accompanied by watering, is beneficial, though without it it is harmful. — When the tree is a year old, they transplant it and give plenty of salt, and this treatment is repeated when it is two years old, for it delights greatly in being transplanted.

2.6.4

Most transplant in the spring, but the people of Babylon about the rising of the dog-star, and this is the time when most people propagate it, since it then germinates and grows more quickly. As long as it is young, they do not touch it, except that they tie up the foliage, so that it may grow straight and the slender branches may not hang down. At a later stage they prune it, when it is more vigorous and has become a stout tree, leaving the slender branches only about a handsbreadth long. So long as it is young, it produces its fruit without a stone, but later on the fruit has a stone.

2.6.5

However some say that the people of Syria use no cultivation, except cutting out wood and watering, also that the date-palm requires spring water rather than water from the skies; and that such water is abundant in the valley in which are the palm-groves. And they add that the Syrians say that this valley extends through Arabia to the Red Sea, and that many profess to have visited it, and that it is in the lowest part of it that the date-palms grow. Now both accounts may be true, for it is not strange that in different soils the methods of cultivation should differ, like the trees themselves.

2.6.6

There are several kinds of palm. To begin with, to take first the most important difference; — some are fruitful and some not; and it is from this latter kind that the people of Babylon make their beds and other furniture. Again of the fruitful trees some are ‘male,’ others ‘female’; and these differ from one another in that the ‘male’ first bears a flower on the spathe, while the ‘female’ at once bears a small fruit. Again there are various differences in the fruits themselves; some have no stones, others soft stones; as to colour, some are white, some black, some yellow; and in general they say that there is not less variety of colour and even of kind than in figs; also that they differ in size and shape, some being round like apples and of such a size that four of them make up a cubit in length, ... while others are small, no bigger than chick-peas; and that there is also much difference in flavour.

2.6.7

The best kind alike in size and in quality, whether of the white or black variety, is that which in either form is called the royal palm ‘; but this, they say, is rare; it grows hardly anywhere except in the park of the ancient Bagoas, near Babylon. In Cyprus there is a peculiar kind of palm which does not ripen its fruit, though, when it is unripe, it is very sweet and luscious,

and this lusciousness is of a peculiar kind. Some palms again differ not merely in their fruits but in the character of the tree itself as to stature and general shape; for instead of being large and tall they are low growing; but these are more fruitful than the others, and they begin to bear as soon as they are three years old; this kind too is common in Cyprus. Again in Syria and Egypt there are palms which bear when they are four or five years old, at which age they are the height of a man.

2.6.8

There is yet another kind in Cyprus, which has broader leaves and a much larger fruit of peculiar shape; in size it is as large as a pomegranate, in shape it is long; it is not however juicy like others, but like a pomegranate, so that men do not swallow it, but chew it and then spit it out. Thus, as has been said, there are many kinds. The only dates that will keep, they say, are those which grow in the Valley of Syria, while those that grow in Egypt Cyprus and elsewhere are used when fresh.

2.6.9

The palm, speaking generally, has a single and simple stem; however there are some with two stems, as in Egypt, which make a fork, as it were; the length of the stem up to the point where it divides is as much as five cubits, and the two branches of the fork are about equal in length. They say that the palms in Crete more often than not have this double stem, and that some of them have three stems; and that in Lapaia one with five heads has been known. It is after all not surprising that in more fertile soils such instances should be commoner, and in general that more kinds and more variation should be found under such conditions. There is another kind which is said to be abundant in Ethiopia, called the doum-palm; this is a shrubby tree, not having a single stem but several, which sometimes are joined together up to a certain point; and the leaf-stalks are not long, only the length of a cubit, but they are plain, and the leafage is borne only at the tip. The leaf is broad and, as it were, made up

of at least two leaflets. This tree is fair to look upon, and its fruit in shape size and flavour differs from the date, being rounder larger and pleasanter to the taste, though not so luscious. It ripens in three years, so that there is always fruit on the tree, as the new fruit overtakes that of last year. And they make bread out of it. These reports then call for enquiry.

2.6.10

The dwarf-palm, as it is called, is a distinct kind, having nothing but its name in common with other palms. For if the head is removed, it survives, and, if it is cut down, it shoots again from the roots. It differs too in the fruit and leaves; for the leaf is broad and flexible, and so they weave their baskets and mats out of it. It is common in Crete and still more so in Sicily. However in these matters we have said more than our purpose required.

2.6.11

To return to the other trees: — in propagating them they set the cuttings upside down, as with vine-shoots. Some however say that that makes no difference, and least of all in propagating the vine; while others contend that the pomegranate thus propagated has a bushier growth and shades the fruit better, and also that it is then less apt to shed the flower. This also occurs, they say, with the fig; when it is set upside down, it does not shed its fruit, and it makes a more accessible tree; and it does not shed its fruit, even if one breaks off the top as it begins to grow.

2.6.12

Thus we have given a general sketch of what we find about methods of propagation, and of the ways in which these trees are reproduced.

2.7.1

As to cultivation and tendance some requirements apply equally to all trees, some are peculiar to one. Those which apply equally to all are spadework watering and manuring, and

moreover pruning and removal of dead wood. But different trees differ in the degree. Some love moisture and manure, some not so much, as the cypress, which is fond neither of manure nor of water, but actually dies, they say, if it is overwatered when young. But the pomegranate and vine are water-loving. The fig grows more vigorously if it is watered, but then its fruit is inferior, except in the case of the Laconian variety, which is water-loving.

2.7.2

All trees require pruning; for they are improved by removal of the dead wood, which is, as it were, a foreign body, and prevents growth and nourishment. Wherefore when the (tree) — becomes old, they cut off all its boughs: for then the tree breaks afresh. Androtion says that the myrtle and olive need more pruning than any other trees; for the smaller you leave them, the better they will grow, and they will bear better fruit. But the vine of course needs pruning even more; for it is in the case of this tree more necessary for promoting both growth and fruitfulness. However, speaking generally, both this and other kinds of tendance must be suited to the particular natural character in each case.

2.7.3

Androtion further says that the olive the myrtle and the pomegranate require the most pungent manure and the heaviest watering, as well as the most thorough pruning, for that then they do not get ‘softwood’ nor any disease underground; but when the tree is old, he adds, one should cut off the boughs, and then attend to the stem as though it were a tree just planted. Thus treated they say that the myrtle and olive are longer lived and very’ robust. These statements might be a subject ^{2.7.4} for further enquiry, or, if not all of them, at least what is stated of the ‘softwood.’

2.7.5

Manure does not suit all alike, nor is the same manure equally good for all. Some need it pungent, some less so, some need it quite light. The most pungent is human dung: thus Chartodras says that this is the best, pig-manure being second to it, goat-manure third, fourth that of sheep, fifth that of oxen, and sixth that of beasts of burden. Litter manure is of different kinds and is applied in various ways: some kinds are weaker, some stronger.

2.7.6

Spade-work is held to be beneficial to all trees, and also hoeing for the smaller ones, as they then become more vigorous. Even dust is thought to fertilise some things and make them flourish, for instance the grape; wherefore they often put dust to the roots of the vine. Some also dig in dust about the figs in places where it is deficient. In Megara, when the etesian winds are past, they cover the cucumber and gourd plants with dust by raking, and so make the fruits sweeter and tenderer by not watering. On this point there is general agreement. But some say that dust should not be put to the vine, and that it should not be meddled with at all when the grape is turning, or, if at all, only when it has turned black. Some again say that even then nothing should be done except to pluck up the weeds. So on this point there is a difference of opinion.

2.7.7

If a tree does not bear fruit but inclines to a leafy growth, they split that part of the stem which is underground and insert a stone corresponding to the crack thus made, and then, they say, it will bear. The same result follows, if one cuts off some of the roots, and accordingly they thus treat the surface roots of the vine when it runs to leaf. In the case of figs, in addition to root-pruning, they also sprinkle ashes about the tree, and make gashes in the stems, and then, they say, it bears better. Into the almond tree they drive an iron peg, and, having thus made a hole, insert in its place a peg of oak-wood and bury it in the

earth, and some call this ‘punishing’ the tree, since its luxuriance is thus chastened.

2.7.8

Some do the same with the pear and with other trees. In Arcadia they have a similar process which is called ‘correcting’ the sorb (for that tree is common in that country). And they say that under this treatment those trees that would not bear do so, and those that would not ripen their fruit now ripen them well. It is also said that the almond becomes sweet, instead of bitter, if one digs round the stem and, having bored a hole about a palmsbreadth, allows the gum which exudes from all sides to flow down into it and collect. The object of this would be alike to make the tree bear and to improve the fruit.

2.8.1

Trees which are apt to shed their fruit before ripening it are almond, apple, pomegranate, pear and, above all, fig and date-palm; and men try to find the suitable remedies for this. This is the reason for the process called ‘caprification’; gall insects come out of the wild figs which are hanging there, eat the tops of the cultivated figs and so make them swell. The shedding of the fruit differs according to the soil: in Italy they say that it does not occur, and so they do not use caprification, nor is it practised in places which face north nor in those with light soils, as at Phalykos in the Megarid, nor in certain parts of the district of Corinth. Also conditions as to wind make a difference; the fruit is shed more with northerly than with southerly winds, and this also happens more if the winds are cold and frequent.- Moreover the character of the tree itself makes a difference; for some kinds, such as the Laconian and other such kinds, shed their early figs but not the later ones. Wherefore caprification is not practised with these. Such are the changes to which the fig is subject in respect of locality kind and climatic conditions.

2.8.2

Now the gall-insects come, as has been said, out of the wild fig, and they are engendered from the seeds. The proof given of this is that, when they come out, there are no seeds left in the fruit; and most of them in coming out leave a leg or a wing behind. There is another kind of gall-insect which is called kentrines; these insects are sluggish, like drones, they kill those of the other kind who are entering the figs, and they themselves die in the fruit. The black kind of wild fig which grows in rocky places is most commended for caprification, as these figs contain numerous seeds.

2.8.3

A fig which has been subject to caprification is known by being red and parti-coloured and stout, while one which has not been so treated is pale and sickly. The treatment is applied to the trees which need it, after rain. The wild figs are most plentiful and most potent where there is most dust. And they say that hylwort also, when it fruits freely, and the ‘gallbags’ of the elm are used for caprification. For certain little creatures are engendered in these also. When the knips is found in figs, it eats the gall-insects. It is to prevent this, it is said, that they nail up the crabs; for the knips then turns its attention to these. Such are the ways of assisting the fig trees.

2.8.4

With dates it is helpful to bring the male to the female; for it is the male which causes the fruit to persist and ripen, and this process some call, by analogy, ‘the use of the wild fruit.’ The process is thus performed: when the male palm is in flower, they at once cut off the spathe on which the flower is, just as it is, and shake the bloom with the flower and the dust over the fruit of the female, and, if this is done to it, it retains the fruit and does not shed it. In the case both of the fig and of the date it appears that the ‘male’ renders aid to the ‘female,’ — for the fruit-bearing tree is called ‘female’ — but while in the latter case there is a union of the two sexes, in the former the result is

brought about somewhat differently.

BOOK III. OF WILD TREES.

3.1.1

Now that we have spoken of cultivated trees, we must in like manner speak of wild ones, noting in what respects they agree with or differ from cultivated trees, and whether in any respects their character is altogether peculiar to themselves.

3.1.2

Now the ways in which they come into being are fairly simple; they all grow either from seed or from a root. But the reason of this is not that they could not possibly grow in any other way, but merely perhaps that no one even tries to plant them otherwise; whereas they might grow from slips, if they were provided with a suitable position and received the fitting kind of tendance, as may be said even now of the trees of woodland and marsh, such as plane willow abele black poplar and elm; all these and other similar trees grow very quickly and well when they are planted from pieces torn off, so that they survive, even if at the time of shifting they are already tall and as big as trees. Most of these are simply planted by being set firmly, for instance, the abele and the black poplar.

3.1.3

Such is the way in which these originate as well as from seed or from roots; the others grow only in these two ways — while some of them, such as silver-fir, fir, and Aleppo pine grow only from seed. All those that have seed and fruit, even if they grow from a root, will grow from seed too; for they say that even those which, like elm and willow, appear to have no fruit reproduce themselves. For proof they give the fact that many such trees come up at a distance from the roots of the original tree, whatever the position may be; and further, they have observed a thing which occasionally happens; for instance,

when at Pheneos in Arcadia the water which had collected in the plain since the underground channels were blocked burst forth, where there were willows growing near the inundated region, the next year after it had dried up they say that willows grew again; and where there had been elms, elms grew, even as, where there had been firs and silverfirs, these trees reappeared — as if the former trees followed the example of the latter.

3.1.4

But the willow is said to shed its fruit early, before it is completely matured and ripened; and so the poet not unfittingly calls it the willow which loses its fruit.

3.1.5

That the elm also reproduces itself the following is taken to be a proof: when the fruit is carried by the winds to neighbouring spots, they say that young trees grow from it. Something similar to this appears to be what happens in the case of certain under-shrubs and herbaceous plants; though they have no visible seed, but some of them only a sort of down, and others only a flower, such as thyme, young plants nevertheless grow from these. As for the plane, it obviously has seeds, and seedlings grow from them. This is evident in various ways, and here is a very strong proof — a plane-tree has before now been seen which came up in a brass pot.

3.1.6

Such we must suppose are the ways in which wild trees originate, apart from the spontaneous ways of which natural philosophers tell. Anaxagoras says that the air contains the seeds of all things, and that these, carried down by the rain, produce the plants; while Diogenes says that this happens when water decomposes and mixes in some sort with earth. Kleidemos maintains that plants are made of the same elements as animals, but that they fall short of being animals in proportion as their composition is less pure and as they are colder. And there are other philosophers also who speak of

spontaneous generation.

3.1.7

But this kind of generation is somehow beyond the ken of our senses. There are other admitted and observable kinds, as when a river in flood gets over its banks or has altogether changed its course, even as the Nesos in the district of Abdera often alters its course, and in so doing causes such a growth of forest in that region that by the third year it casts a thick shade. The same result ensues when heavy rains prevail for a long time; during these too many plants shoot up. Now, as the flooding of a river, it would appear, conveys seeds of fruits of trees, and, as they say, irrigation channels convey the seeds of herbaceous plants, so heavy rain acts in the same way; for it brings down many of the seeds with it, and at the same time causes a sort of decomposition of the earth and of the water. In fact, the mere mixture of earth with water in Egypt seems to produce a kind of vegetation. And in some places, if the ground is merely lightly worked and stirred, the plants native to the district immediately spring up; for instance, the cypress in Crete. And something similar to this occurs even in smaller plants; as soon as the earth is stirred, wherever it may be, a sort of vegetation comes up. And in partly saturated soil, if you break up the ground, they say that caltrop appears. Now these ways of origination are due to the change which takes place in the soil, whether there were seeds in it already, or whether the soil itself somehow produces the result. And the latter explanation is perhaps not strange, seeing that the moist element is also locked up in the soil. Again, in some places they say that after rain a more singular abundance of vegetation has been known to spring up; for instance, at Cyrene, after a heavy pitchy shower had fallen: for it was under these circumstances that there sprang up the wood which is near the town, though till then it did not exist. They say also that silphium has been known to appear from some such cause, where there was none

before. Such are the ways in which these kinds of generation come about.

3.2.1

All trees are either fruit-bearing or without fruit, either evergreen or deciduous, either flowering or flowerless; for certain distinctions apply to all trees alike, whether cultivated or wild. To wild trees, as compared with cultivated ones, belong the special properties of fruiting late, of greater vigour, of abundance of fruit, produced if not matured; for they ripen their fruit later, and in general their time of flowering and making growth is later; also they are more vigorous in growth, and so, though they produce more fruit, they ripen it less; if this is not universally true, at least it holds good of the wild olive and pear as compared with the cultivated forms of these trees. This is generally true with few exceptions, as in the cornelian cherry and sorb; for the wild forms of these, they say, ripen their fruit better, and it is sweeter than in the cultivated forms. And the rule also does not hold good of anything which does not admit of cultivation, whether it be a tree or one of the smaller plants, as silphium caper and, among leguminous plants, the lupin; these one might say are specially wild in their character. For, as with animals which do not submit to domestication, so a plant which does not submit to cultivation may be called wild in its essential character. However Hippon declares that of every plant there exists both a cultivated and a wild form, and that 'cultivated' simply means that the plant has received attention, while 'wild' means that it has not; but though he is partly right, he is partly wrong. It is true that any plant deteriorates by neglect and so becomes wild; but it is not true that every plant may be improved by attention, as has been said. Wherefore we must make our distinction and call some things wild, others cultivated — the latter class corresponding to those animals which live with man and can be tamed.

3.2.2

But perhaps it does not matter which way this should be put. Any tree which runs wild deteriorates in its fruits, and itself becomes dwarfed in leaves branches bark and appearance generally; for under cultivation these parts, as well as the whole growth of the tree, become closer, more compact and harder; which indicates that the difference between cultivated and wild is chiefly shown in these respects. And so those trees which show these characteristics under cultivation they say are really wild, for instance fir cypress, or at least the 'male' kind, hazel and chestnut.

3.2.3

Moreover these wild forms are distinguished by having greater liking for cold and for hilly country: for that too is regarded as a means of recognising wild trees and wild plants generally, whether it is so regarded in itself or as being only incidentally a distinguishing mark.

3.2.4

So the definition of wild kinds, whether it should be thus made or otherwise, perhaps makes no difference for our present purpose. But it is certainly true, speaking broadly and generally, that the wild trees are more to be found in hilly country, and that the greater part of them flourish more in such regions, with the exception of those which love water or grow by river sides or in woods; these and such-like trees are rather trees of the plain. However on great mountains, such as Parnassus, Cyllene, the Pierian Olympus and the Mysian Olympus, and such regions anywhere else, all kinds grow, because of the diversity of positions afforded them. For such mountains offer positions which are marshy, wet, dry, deep-soiled or rocky; they have also their meadow land here and there, and in fact almost every variety of soil; again they present positions which lie low and are sheltered, as well as others which are lofty and exposed to wind; so that they can bear all sorts, even those which belong to the plains.

3.2.5

Yet it is not strange that there should be some mountains which do not thus bear all things, but have a more special kind of vegetation to a great extent if not entirely; for instance the range of Ida in Crete; for there the cypress grows; or the hills of Cilicia and Syria, on which the Syrian cedar grows, or certain parts of Syria, where the terebinth grows. For it is the differences of soil which give a special character to the vegetation. However the word 'special' is used here in a somewhat extended sense.)

3.3.1

The following trees are peculiar to mountain country and do not grow in the plains; let us take Macedonia as an example. Silver-fir, fir, 'wild pine,' lime, zygia, Valonia oak, box, andrachne, yew, Phoenician cedar, terebinth, wild fig, alaternus, hybrid arbutus, hazel, chestnut, kermes-oak. The following grow also in the plain: tamarisk, elm, abele, willow, black poplar, cornelian cherry, cornel, alder, oak, lakare (bird-cherry), wild pear, apple, hop-hornbeam, holly, manna-ash, Christ's thorn, cotoneaster, maple, which when it grows in the mountains, is called zygia, when in the plain, gleinos: others however, classify differently and make maple and zygia distinct trees.

3.3.2

All those trees which are common to both hill and plain are taller and finer in appearance when they grow in the plain; but the mountain forms are better as to producing serviceable timber and fruits, with the exception of wild pear, pear, and apple; these are in the plain better in fruit and also in timber; for in the hills they grow small with many knots and much spinous wood. But even on the mountains all trees grow fairer and are more vigorous when they have secured a suitable position; and, to speak generally, those which grow on the level parts of the mountains are specially fair and vigorous; next to

these come those which grow on the lower parts and in the hollows; while those that grow on the heights are of the poorest quality, except any that are naturally cold-loving. But even these shew some variation in different positions, of which we must speak later; for the present we must in our distinctions in each case take account only of the differences already mentioned.

3.3.3

Now among wild trees those are evergreen which were mentioned before, silver-fir, fir, 'wild pine,' box, andrachne, yew, Phoenician cedar, terebinth, alaternus, hybrid arbutus, bay, phellodrys' (holm-oak), holly, cotoneaster, kermes-oak, tamarisk; but all the others shed their leaves, unless it be that in certain places they keep them exceptionally, as was said of the plane and oak in Crete and in any other place which is altogether favourable to luxuriant growth.

3.3.4

Most trees are fruit-bearing, but about willow, black poplar, and elm men hold different opinions, as was said; and some, as the Arcadians, say that only the black poplar is without fruit, but that all the other mountain trees bear fruit. However in Crete there are a number of black poplars which bear fruit; there is one at the mouth of the Idaean cave, in which the dedicatory offerings are hung, and there is another small one not far off, and there are quite a number about a spring called the Lizard's Spring about twelve furlongs off. There are also some in the hill-country of Ida in the district called Kindria, and in the mountains about Praisia. Others again, as the Macedonians, say that the elm is the only tree of this class which bears fruit.

3.3.5

Again the character of the position makes a great difference as to fruit-bearing, as in the case of the persea and the date-palm. The persea of Egypt bears fruit, and so it does wherever it grows in the neighbouring districts, but in Rhodes it only gets as far as

flowering. The date-palm in the neighbourhood of Babylon is marvellously fruitful; in Hellas it does not even ripen its fruit, and in some places it does not even produce any.

3.3.6

The same may be said of various other trees: in fact even of smaller herbaceous plants and bushes some are fruitful, others not, although the latter are growing in the same place as the former, or quite near it. Take for instance the centaury in Elea; where it grows in hill-country, it is fruitful; where it grows in the plain, it bears no fruit, but only flowers; and where it grows in deep valleys, it does not even flower, unless it be scantily. Any way it appears that, even of other plants which are of the same kind and all go by the same name, one will be without fruit, while another bears fruit; for instance, one kermes-oak will be fruitful, another not; and the same is true of the alder, though both produce flowers. And, generally speaking, all those of any given kind which are called 'male' trees are without fruit, and that though some of these, they say, produce many flowers, some few, some none at all. On the other hand they say that in some cases it is only the 'males' that bear fruit, but that, in spite of this, the trees grow from the flowers, (just as in the case of fruit-bearing trees they grow from the fruit). And they add that in both cases, the crop of seedlings which comes up is sometimes so thick that the woodmen cannot get through except by clearing a way.

3.3.7

There is also a doubt about the flower of some trees, as we said. Some think that the oak bears flowers, and also the filbert, the chestnut, and even the fir and Aleppo pine; some however think that none of these has a flower, but that, — resembling and corresponding to the wild figs which drop off prematurely, we have in the nuts the catkin, in the oak the oak-moss, in the pine the 'flowering tuft'. The people of Macedonia say that these trees also produce no flowers — Phoenician cedar, beech, aia

(holm-oak), maple. Others distinguish two kinds of Phoenician cedar, of which one bears flowers but bears no fruit, while the other, though it has no flower, bears a fruit which shows itself at once — just as wild figs produce their abortive fruit. However that may be, it is a fact that this is the only tree which keeps its fruit for two years. These matters then need enquiry.

3.4.1

Now the budding of wild trees occurs in some cases at the same time as that of the cultivated forms, but in some cases somewhat, and in some a good deal later; but in all cases it is during the spring season. But there is greater diversity in the time of fruiting; as we said before, the times of ripening do not correspond to those of budding, but there are wide differences. For even in the case of those trees which are somewhat late in fruiting, — which some say take a year to ripen their fruit — such as Phoenician cedar and kermes-oak, the budding nevertheless takes place in the spring. Again there are differences of time between individual trees of the same kind, according to the locality; those in the marshes bud earliest, as the Macedonians say, second to them those in the plains, and latest those in the mountains.

3.4.2

Again of particular trees some wild ones bud along with the cultivated forms, as andrachne and hybrid arbutus; and the wild pear is a little later than the cultivated. Some again bud both before zephyr begins to blow, and immediately after it has been blowing. Before it come cornelian cherry and cornel, after it bay and alder; a little before the spring equinox come lime zygia Valonia oak fig. Hazel oak and elder are also early in budding, and still more those trees which seem to have no fruit and to grow in groves, the aspen elm willow black poplar; and the plane is a little later than these. The others which bud when the spring is, as it were, becoming established, are such as wild fig alaternus cotoneaster Christ's thorn terebinth hazel chestnut.

The apple is late in budding, latest of all generally are ipsos (cork-oak) aria (holm-oak) tetragonia odorous cedar yew. Such are the times of budding. The flowering times in general follow in proportion; but they present some irregularity, and so in still more cases and to a greater extent do the times at which the fruit is matured. The cornelian cherry produces its fruit about the summer solstice; the early kind, that is to say, and this tree is about the earliest of all. The late form, which some call 'female cornelian cherry' (cornel), fruits quite at the end of autumn. The fruit of this kind is inedible and its wood is weak and spongy; that is what the difference between the two kinds amounts to. The terebinth produces its fruit about the time of wheat-harvest or a little later, manna-ash and maple in summer; alder, hazel, and a certain kind of wild pear in autumn; oak and chestnut later still, about the setting of the Pleiad; and in like manner alaternus, kermes-oak, Christ's-thorn, cotoneaster, after the setting of the Pleiad; aria (holm-oak) when winter is beginning, apple with the first cold weather, wild pear late — in winter. Andrachne and hybrid arbutus first ripen their fruit when the grape is turning, and again when winter is beginning; for these trees appear to bear twice. As for silver-fir and yew, they flower a little before the solstice; (the flower of the silverfir is yellow and otherwise pretty); they bear their fruit after the setting of the Pleiad. Fir and Aleppo pine are a little earlier in budding, about fifteen days, but produce their fruit after the setting of the Pleiad, though proportionately earlier than silver-fir and yew.

3.4.3

In these trees then the difference of time is not considerable; the greatest difference is shewn in Phoenician cedar holly and kermes-oak; for Phoenician cedar appears to keep its fruit for a year, the new fruit overtaking that of last year; and, according to some, it does not ripen it at all; wherefore men gather it unripe and keep it, whereas if it is left on the tree, it shrivels up.

The Arcadians say that the kermes-oak also takes a year to perfect its fruit; for it ripens last year's fruit at the same time that the new fruit appears on it; the result of which is that such trees always have fruit on them. They say also that holly loses its fruit owing to the winter. Lime and box are very late in fruiting, (lime has a fruit which no animal can eat, and so have cornel and box. Ivy, Phoenician cedar, fir, and andrachne are late fruiting) though, according to the Arcadians, still later than these and almost latest of all are tetragonia, odorous cedar and yew. Such then are the differences as to the time of shedding and ripening their fruit between wild as compared with cultivated trees, and likewise as compared with one another.

3.5.1

Now most trees, when they have once begun to bud, make their budding and their growth continuously, but with fir silver-fir and oak there are intervals. They make three fresh starts in growth and produce three separate sets of buds; wherefore also they lose their bark thrice a year. For every tree loses its bark when it is budding. This first happens in mid-spring at the very beginning of the month Thargelion,⁸ on Mount Ida within about fifteen days of that time; later, after an interval of about thirty days or rather more, the tree puts on fresh buds which start from the head of the knobby growth which formed at the first budding-time; and it makes its budding partly on the top of this, partly all round it laterally, - using the knob formed at the first budding as a sort of joint, just as in the case of the first budding. This happens about the end of the month Skirrophorion.

3.5.2

(It is only at the time of this second budding that the galls also are produced, both the white and the black; the liquid forming them is mostly produced in quantity at night, and, after swelling for one day — except the part which is of resinous character — it hardens if it is caught by the heat, and so cannot grow any more; otherwise it would have grown greater in bulk;

wherefore in some trees the formation is not larger than a bean. The black gall is for several days of a pale green colour; then it swells and sometimes attains the size of an apple.)

3.5.3

Then, after an interval of about fifteen days, the tree for the third time puts on buds in the month Hekatombaion; but this growth continues for fewer days than on either of the previous occasions, perhaps for six or seven at most. However the formation of the buds is as before and takes place in the same manner. After this period there is no increase in length, but the only increase is in thickness.

3.5.4

The periods of budding can be seen in all trees, but especially in fir and silver-fir, because the joints of these are in a regular series and have the knots at even distances. It is then the season also for cutting the timber, because the bark is being shed; for at other times the bark is not easy to strip off, and moreover, if it is stripped off, the wood turns black and is inferior in appearance; for as to its utility this makes no difference, though the wood is stronger if it is cut after the ripening of the fruit.

3.5.5

Now what has been said is peculiar to the above-mentioned trees. But the buddings which take place at the rising of the dog-star and at that of Arcturus after the spring budding are common to nearly all, though they may be most clearly seen in cultivated trees, and, among these, especially in fig, vine, pomegranate, and in general in all those that are luxuriant in growth or are growing in rich soil. Accordingly they say that the budding at the rising of Arcturus is most considerable in Thessaly and Macedonia; for it also happens that the autumn in these countries is a fair and a long season; so that the mildness of the climate also contributes. Indeed it is for this reason, one may say, that in Egypt too the trees are always budding, or at least that the process is only suspended for quite a short time.

3.5.6

Now the facts as to the later buddings apply, as has been said, to all trees alike; but those which belong to the intervals after the first period of budding are peculiar to those mentioned above. Peculiar to some also is the growth of what are called winter buds,' for instance in the abovementioned trees; silver-fir, fir, and oak have them, and also lime, hazel, chestnut, and Aleppo pine. These are found in the oak before the leaf-buds grow, when the spring season is just beginning. This growth consists of a sort of leaf-like formation, which occurs between the first swelling of the leaf-buds and the time when they burst into leaf. In the sorb it occurs in the autumn after the shedding of the leaves, and has from the first a glistening look,' as though swelling had taken place, just as if it were about to burst into leaves; and it persists through the winter till the spring. The filbert after casting its fruit produces its clustering growth, which is as large as a good-sized grub: several of these grow from one stalk, and some call them catkins. Each of these is made up of small processes arranged like scales, and resembles the cone of the fir, so that its appearance is not unlike that of a young green fir-cone, except that it is longer and almost of the same thickness throughout. This grows through the winter (when spring comes, the scale-like processes open and turn yellow); it grows to the length of three fingers, but, when in spring the leaves are shooting, it falls off, and the cup-like fruit-cases of the nut are formed, closed all down the stalk and coiresponding in number to the flowers; and in each of these is a single nut. The case of the lime and of any other tree that produces winter-buds needs further consideration.

3.6.1

Some trees are quick-growing, some slow. Quick-growing are those which grow by the waterside, as elm, plane, aspen, black poplar, willow; (however some dispute about the last-named, and consider it a slow grower) and of fruit-bearing trees,

silver-fir, fir, oak. Quickest growing of all are . . . yew lakara (bird-cherry) Valonia oak, Phoenician cedar, maple, hop-hornbeam, zygia, manna-ash, alder, Aleppo pine, andrachne, cornelian cherry, box, wild pear. But silver-fir, fir and Aleppo pine bear fruit from the very first, whatever size they have attained.

3.6.2

While the growth and budding of most trees are irregular as regards the position in which the buds appear, the growth and budding of the silver-fir follow a regular rule, and its development afterwards is also in a regular sequence. For, when the trunk first divides, then again from the divided trunk the second division takes place in like manner, and so the tree goes on with each fresh formation of buds. In other trees not even the knots are opposite to one another, except in some few cases, as wild olive and others.' Here too we find a difference in the manner of growth which belongs to all trees alike, both cultivated and wild: in some cases the growth is from the top of the shoots and also from the sidebuds,' as in pear pomegranate fig myrtle and the majority of trees, one may say: in some cases the growth is not from the top, but only from the sidebuds, and the already existing part is pushed out further, as is the whole trunk with the upper branches. This occurs in the walnut and in the filbert as well as in other trees. In all such trees the buds end in a single leaf; wherefore it is reasonable that they should not make fresh buds and growth from this point, as they have no point of departure. (To a certain extent the growth of corn is similar; for it also regularly increases by pushing forward of the already existing part, even if the leaves are mutilated, as in corn which is bitten down by animals. Corn however does not make side-growths, as some leguminous plants do.) Here then we may find a difference which occurs both in the making of buds and in the making of fresh growth.

3.6.3

Some say that wild trees are not deep rooting, because they all grow from seed; but this is not a very accurate statement. For it is possible that, when they are well established, they may send their roots down far; in fact even most pot-herbs do this, though these are not so strong as trees, and are undoubtedly grown from seed planted in the ground. The kermes-oak however seems to be the deepest rooting of wild trees; silver-fir and fir are only moderately so, and shallowest are joint-fir plum bullace (which is a sort of wild plum). The last two also have few roots, while joint-fir has many. Trees which do not root deep, and especially silverfir and fir, are liable to be rooted up by winds.

3.6.4

So the Arcadians say. But the people who live near Mount Ida say that the silver fir is deeper rooting than the oak, and has straighter roots, though they are fewer. Also that those which have the deepest roots are plum and filbert, the latter having strong slender roots, the former having many; but they add that both trees must be well established to acquire these characters; also that plum is very tenacious of life. Maple, they say, has shallow roots and few of them; but manna-ash has more and they are thickly matted and run deep. Phoenician cedar and prickly cedar, they say, have shallow roots, those of alder are slender and 'plain,' — as also are those of beech; for this too has few roots, and they are near the surface. Sorb, they say, has its roots near the surface, but they are strong and thick and hard to kill, though not very numerous. Such are the trees which are or are not deep-rooting.

3.7.1

Almost all trees shoot from the side if the trunk is cut down, unless the roots have previously been injured; but fir and silver-fir wither away completely from the roots within the year, if merely the top has been cut off. And there is a peculiar thing about the silver-fir; when it is topped or broken off short

by wind or some other cause affecting the smooth part of the trunk — for up to a certain height the trunk is smooth knotless and plain (and so suitable for making a ship's mast), — a certain amount of new growth forms round it, which does not however grow much vertically: and this is called by some amphauxis and by others amphiphyta; it is black in colour and exceedingly hard, and the Arcadians make their mixing-bowls out of it; the thickness is in proportion to the tree, according as that is more or less vigorous and sappy, or again according to its thickness. There is this peculiarity too in the silver-fir in the same connexion; when, after taking off all the branches, one cuts off the top, it soon dies; yet, when one takes off the lower parts, those about the smooth portion of the trunk, what is left survives, and it is on this part that the amphauxis forms. And plainly the reason why the tree survives is that it is sappy and green because it has no side-growths. Now this is peculiar to the silver-fir.

3.7.2

Now, while other trees bear merely their own fruit and the obvious parts which form annually, to wit, leaf, flower and bud, some bear also catkins or tendrils, and some produce other things as well, for instance the elm its 'cluster' and the familiar baglike thing, the fig both the immature figs which drop off and (in some kinds) the untimely figs — though perhaps in a sense these should be reckoned as fruit. Again filbert produces its catkin, kermes-oak its scarlet 'berry,' and bay its 'cluster.' The fruit-bearing sort of bay also produces this, or at all events one kind certainly does so; however the sterile kind, which some call the 'male,' produces it in greater quantity. The fir again bears its 'tuft,' which drops off.

3.7.3

The oak however bears more things besides its fruit than any other tree; as the small gall and its other black resinous gall. Again it has another growth, like a mulberry in shape, but hard

and difficult to break; this however is not common. It has also another growth like the penis in shape, which, when it is further developed, makes a hard prominence and has a hole through it. This to a certain extent resembles also a bull's head, but, when split open, it contains inside a thing shaped like the stone of an olive.' The oak also produces what some call the 'ball'; this is a soft woolly spherical object enclosing a small stone which is harder,' and men use it for their lamps; for it burns well, as does the black gall. The oak also produces another hairy ball, which is generally useless, but in the spring season it is covered with a juice which is like honey both to touch and taste.

3.7.4

Further the oak produces right inside the axil of the branches another ball with no stalk or else a hollow one; this is peculiar and of various colours: for the knobs which arise on it are whitish or black and spotted, while the part between these is brilliant scarlet; but, when it is opened, it is black and rotten. It also occasionally produces a small stone which more or less resembles pumice-stone; also, less commonly, there is a leaf-like ball, which is oblong and of close texture. Further the oak produces on the rib of the leaf a white transparent ball, which is watery, when it is young; and this sometimes contains flies: but as it develops, it becomes hard, like a small smooth gall.

3.7.5

Such are the growths which the oak produces as well as its fruit. For as for the fungi which grow from the roots or beside them, these occur also in other trees. So too with the oak-mistletoe; for this grows on other trees also. However, apart from that, the oak, as was said, produces more things than any other tree; and all the more so if, as Hesiod says, it produces honey and even bees; however, the truth appears to be that this honey-like juice comes from the air and settles on this more than on other trees. They say also that, when the oak is burnt, nitre is produced

from it. Such are the things peculiar to the oak.

3.8.1

Taking, as was said, all trees according to their kinds, we find a number of differences. Common to them all is that by which men distinguish the 'male' and the 'female,' the latter being fruitbearing, the former barren in some kinds. In those kinds in which both forms are fruit-bearing the 'female' has fairer and more abundant fruit; however some call these the 'male' trees — for there are those who actually thus invert the names. This difference is of the same character as that which distinguishes the cultivated from the wild tree, while other differences distinguish different forms of the same kind; and these we must discuss, at the same time indicating the peculiar forms, where these are not obvious and easy to recognise.

3.8.2

Take then the various kinds of oak; for in this tree men recognise more differences than in any other. Some simply speak of a cultivated and a wild kind, not recognising any distinction made by the sweetness of the fruit; (for sweetest is that of the kind called Valonia oak, and this they make the wild kind), but distinguishing the cultivated kind by its growing more commonly on tilled land and having smoother timber, while the Valonia oak has rough wood and grows in mountain districts. Thus some make four kinds, others five. They also in some cases vary as to the names assigned; thus the kind which bears sweet fruit is called by some *hemeris*, by others 'true oak.' So too with other kinds. However, to take the classification given by the people of Mount Ida, these are the kinds: *hemeris* (gall-oak), *aigilops* (Turkey-oak), 'broad-leaved' oak (scrub oak), Valonia oak, sea-bark oak, which some call 'straight-barked' oak. All these bear fruit; but the fruits of Valonia oak are the sweetest, as has been said; second to these those of *hemeris* (gall-oak), third those of the 'broad-leaved' oak (scrub oak), fourth sea-bark oak, and last *aigilops* (Turkey oak), whose fruits

are very bitter. However the fruit is not always sweet in the kinds specified as such; sometimes it is bitter, that of the Valonia oak for instance. There are also differences in the size shape and colour of the acorns. Those of Valonia oak and sea-bark oak are peculiar; in both of these kinds on what are called the 'male' trees the acorns become stony at one end or the other; in one kind this hardening takes place in the end which is attached to the cup, in the other in the flesh itself. Wherefore, when the cups are taken off, we find a cavity like the visceral cavities in animals.

3.8.3

There are also differences in leaves trunk timber and general appearance. Hemeris (gall-oak) is not straight-growing nor smooth nor tall, for its growth is very leafy and twisted, with many side-branches, so that it makes a low much-branched tree; its timber is strong, but not so strong as that of the Valonia oak, for that is the strongest and the least liable to rot. This kind too is not straight-growing, even less so than the hemeris (gall-oak), but the trunk is very thick, so that the whole appearance is stunted; for in growth this kind too is very leafy and not erect. The aigilops (Turkey oak) is the straightest growing and also the tallest and smoothest, and its wood, cut lengthways, is the strongest. It does not grow on tilled land, or very rarely.

3.8.4

The 'broad-leaved' oak (scrub oak) comes second as to straightness of growth and length of timber to be got from it, but for use in building it is the worst next after the sea-bark oak, and it is even poor wood for burning and making charcoal, as is also that of the sea-bark oak, and next after this kind it is the most worm-eaten. For the sea-bark oak has a thick trunk, but it is generally spongy and hollow when it is thick; wherefore it is useless for building. Moreover it rots very quickly, for the tree contains much moisture; and that is why it also becomes

hollow; and some say that it is the only oak which has no heart. And some of the Aeolians say that these are the only oaks which are struck by lightning, although they are not lofty; nor do they use the wood for their sacrifices. Such then are the differences as to timber and general appearance.

3.8.5

All the kinds produce galls, but only hemeris (gall-oak) produces one which is of use for tanning hides. That of aigilops (Turkey-oak) and that of the 'broad-leaved' oak (scrub oak) are in appearance like that of hemeris (gall-oak), but smoother and useless. This also produces the other gall, the black kind, with which they dye wool. The substance which some call tree-moss and which resembles rags is borne only by the aigilops (Turkey-oak); it is grey and rough and hangs down for a cubit's length, like a long shred of linen. This grows from the bark and not from the knob whence the acorn starts; nor does it grow from an eye, but from the side of the upper boughs. The sea-bark oak also produces this, but it is blackish and short.

3.8.6

Thus the people of Mount Ida distinguish. But the people of Macedonia make four kinds, 'true-oak,' or the oak which bears the sweet acorns, 'broadleaved' oak (scrub oak), or that which bears the bitter ones, Valonia oak, or that which bears the round ones, and aspris (Turkey-oak); the lastnamed some say is altogether without fruit, some say it bears poor fruit, so that no animal eats it except the pig, and only he when he can get no others, and that after eating it the pig mostly gets an affection of the head. The wood is also wretched; when hewn with the axe it is altogether useless, for it breaks in pieces and falls asunder; if it is not hewn with the axe it is better, wherefore they so use it. It is even wretched for burning and for making charcoal; for the charcoal is entirely useless except to the smith, because it springs about and emits sparks. But for use in the smithy it is more serviceable than the other kinds, since, as it goes out

when it ceases to be blown, little of it is consumed. The wood of the sea-bark oak is only useful for wheel-axles and the like purposes. Such are the varieties of the oak which men make out.

3.9.1

The differences between other trees are fewer; for the most part men distinguish them merely according as they are 'male' or 'female,' as has been said, except in a few cases including the fir; for in this tree they distinguish the wild and the cultivated kinds, and make two wild kinds, calling one the 'fir of Ida' (Corsican pine) the other the 'fir of the sea-shore' (Aleppo pine); of these the former is straighter and taller and has thicker leaves, while in the latter the leaves are slenderer and weaker, and the bark is smoother and useful for tanning hides, which the other is not. Moreover the cone of the seaside kind is round and soon splits open, while that of the Idaean kind is longer and green and does not open so much, as being of wilder character. The timber of the seaside kind is stronger, — for one must note such differences also between trees of the same kind, since it is by their use that the different characters are recognised.

3.9.2

The Idaean kind is, as we have said, of straighter- and stouter growth, and moreover the tree is altogether more full of pitch, and its pitch is blacker sweeter thinner and more fragrant when it is fresh; though, when it is boiled, it turns out inferior, because it contains so much watery matter. However it appears that the kinds which these people distinguish by special names are distinguished by others merely as 'male' and 'female.' The people of Macedonia say that there is also a kind of fir which bears no fruit whatever, in which the male' (Aleppo pine) is shorter and has harder leaves, while the 'female' (Corsican pine) is taller and has glistening delicate leaves which are more pendent. Moreover the timber of the 'male' kind has much heart-wood, is tough, and warps in joinery work, while that of the 'female' is easy to work, does not warp, and is softer.

3.9.3

This distinction between 'male' and 'female' may, according to the woodmen, be said to be common to all trees. Any wood of a 'male' tree, when one comes to cut it with the axe, gives shorter lengths, is more twisted, harder to work, and darker in colour; while the 'female' gives better lengths. For it is the 'female' fir which contains what is called the aegis; this is the heart of the tree; the reason being that it is less resinous, less soaked with pitch, smoother, and of straighter grain. This aegis is found in the larger trees, when, as they have fallen down, the white outside part has decayed; when this has been stripped off and the core left, it is cut out of this with the axe; and it is of a good colour with fine fibre. However the substance which the torch-cutters of Mount Ida call the 'fig,' which forms in the fir and is redder in colour than the resin, is found more in the 'male' trees; it has an evil smell, not like the smell of resin, nor will it burn, but it leaps away from the fire.

3.9.4

Such are the kinds of fir which they make out, the cultivated and the wild, the latter including the 'male' and the 'female' and also the kind which bears no fruit. However the Arcadians say that neither the sterile kind nor the cultivated is a fir, but a pine; for, they say, the trunk closely resembles the pine and has its slenderness, its stature, and the same kind of wood for purposes of joinery, the trunk of the fir being thicker smoother and taller; moreover that the fir has many leaves, which are glossy massed together and pendent, while in the pine and in the above-mentioned cone-bearing tree the leaves are few and drier and stiffer; though in both the leaves are hair-like. Also, they say, the pitch of this tree is more like that of the pine; for in the pine too it is scanty and bitter, as in this other cone-bearing tree, but in the fir it is fragrant and abundant. Now the pine is rare in Arcadia, but common in Elis. The Arcadians then dispute altogether the nomenclature.

3.9.5

The pine appears to differ also from the fir in being glossier and having finer leaves, while it is smaller in stature and does not grow so straight; also in bearing a smaller cone, which is stiffer and has a more pitchy kernel, while its wood is whiter, more like that of the silver-fir, and wholly free from pitch. And there is another great difference between it and the fir; the fir, if it is burnt down to the roots, does not shoot up again, while the pine, according to some, will do so; for instance this happened in Lesbos, when the pine-forest of Pyrrha was burnt. The people of Ida say that the fir is liable to a kind of disease; — when not only the heart but the outer part of the trunk becomes glutted with pitch, the tree then is as it were choked. This happens of its own accord through the excessive luxuriance of the tree, as one may conjecture; for it all turns into pitch-glutted wood. This then is an affection peculiar to the fir.

3.9.6

The silver-fir is either ‘male’ or ‘female,’ and has differences in its leaves; those of the ‘male’ are sharper more needle-like and more bent; wherefore the whole tree has a more compact appearance. There are also differences in the wood, that of the ‘female’ being whiter softer and easier to work, while the whole trunk is longer; that of the ‘male’ is less of a uniform colour thicker and harder, has more heart-wood, and is altogether inferior in appearance. In the cone of the ‘male’ are a few seeds at the apex, while that of the ‘female,’ according to what the Macedonians said, contains none at all. The foliage is feathered and the height disproportionate so that the general appearance of the tree is dome-like, and closely resembles the Boeotian peasant’s hat; and it is so dense that neither snow nor rain penetrates it. And in general the tree has a handsome appearance; for its growth is somewhat peculiar, as has been said, compared with the others, it being the only one which is regular, and in stature it is large, much taller than the fir.

3.9.7

There is also not a little difference in the wood: that of the silver-fir is fibrous soft and light, that of the fir is resinous heavy and more fleshy. The fir has more knots, but the silver-fir harder ones; indeed they may be said to be harder than those of any tree, though the wood otherwise is softer. And in general the knots of silver-fir and fir are of the closest and most solid texture and almost transparent: in colour they are like resin-glutted wood, and quite different from the rest of the wood; and this is especially so — in the silver-fir. And just as the fir has its aegis, so the silver-fir has what is called its white ‘centre,’ which answers, as it were, to the aegis of the fir, except that it is white, while the other is bright-coloured because it is glutted with pitch. It becomes close white and good in trees which are of some age, but it is seldom found in good condition, while the ordinary form of it is abundant and is used to make painters’ boards and ordinary writing tablets, superior ones being made of the better form.

3.9.8

However the Arcadians call both substances aegis, alike that of the fir and the corresponding part of the silver-fir, and say that, though the silver-fir produces more, that of the fir is better; for that, though that of the silver-fir is abundant smooth and close, that of the fir, though scanty, is compacter, stronger and fairer in general. The Arcadians then appear to differ as to the names which they give. Such are the differences in the silver-fir as compared with the fir, and there is also that of having the amphauxis, which we mentioned before.

3.10.1

The beech presents no differences, there being but one kind. It is a straight-growing smooth and unbranched tree, and in thickness and height is about equal to the silver-fir, which it also resembles in other respects; the wood is of a fair colour strong and of good grain, the bark smooth and thick, the leaf

undivided, longer than a pear-leaf, spinous at the tip, the roots neither numerous nor running deep; the fruit is smooth like an acorn, enclosed in a shell, which is however without prickles and smooth, not spinous,- like the chestnut, though in sweetness and flavour it resembles it. In mountain country it also grows white and has timber which is useful for many purposes, for making carts beds chairs and tables, and for shipbuilding; While the tree of the plains is black and useless for these purposes; but the fruit is much the same in both.

3.10.2

The yew has also but one kind, is straightgrowing, grows readily, and is like the silver-fir, except that it is not so tall and is more branched. Its leaf is also like that of the silver-fir, but glossier and less stiff. As to the wood, in the Arcadian yew it is black or red, in that of Ida bright yellow and like prickly cedar; wherefore they say that dealers practise deceit, selling it for that wood: for that it is all heart, when the bark is stripped off; its bark also resembles that of prickly cedar in roughness and colour, its roots are few slender and shallow. The tree is rare about Ida, but common in Macedonia and Arcadia; it bears a round fruit a little larger than a bean, which is red in colour and soft; and they say that, if beasts of burden eat of the leaves they die, while ruminants take no hurt. Even men sometimes eat the fruit, which is sweet and harmless. The oslrys (hop-hornbeam), which some call oslrya, has also but one kind: it is like the beech in growth and bark; Its leaves are in shape like a pear's, except that they are much longer, come to a sharp point, are larger, and have many fibres, which branch out like ribs from a large straight one- in the middle, and are thick; also the leaves are wrinkled along the fibres and have a finely serrated edge; the wood is hard colourless and whitish; the fruit is small oblong and yellow like barley; it has shallow roots; it loves water and is found in ravines. It is said to be unlucky to bring it into the house, since, wherever it is, it is supposed to cause a painful

death or painful labour in giving birth.

3.10.3

The lime has both 'male' and 'female' forms, which differ in their general appearance, in that of the wood, and in being respectively fruit-bearing and sterile. The wood of the 'male' tree is hard yellow more branched closer, and also more fragrant; that of the 'female' is whiter. The bark of the 'male' is thicker, and, when stripped off, is unbending because of its hardness; that of the 'female' is thinner and flexible; men make their writingcases out of it. The 'male' has neither fruit nor flower, but the 'female' has both flower and fruit; the flower is cup-shaped, and appears alongside of the stalk of the leaf, or alongside of next year's winter-bud on a separate stalk; it is green, when in the cup-like stage, but brownish as it opens; it appears at the same time as in the cultivated trees. The fruit is rounded oblong as large as a bean, resembling the fruit of the ivy; when mature, it has five angular projections, as it were, made by projecting fibres which meet in a point; the immature fruit is less articulated. When the mature fruit is pulled to pieces, - it shows some small fine seeds of the same size as those of orach. The leaf and the bark are well flavoured and sweet; the leaf is like that of the ivy in shape, except that it rounds more gradually, being most curved at the part next the stalk, but in the middle contracting to a sharper and longer apex, and its edge is somewhat puckered and jagged. The timber contains little core, which is not much softer than the other part; for the rest of the wood is also soft.

3.11.1

Of the maple, as we have said, some make two kinds, some three; one they call by the general name 'maple,' another *zygia*, the third *klinotrokhos* ' ; this name, for instance, is used by the people of Stagira. The difference between *zygia* and maple proper is that the latter has white wood of finer fibre, while that of *zygia* is yellow and of compact texture. The leaf in both trees

is large, resembling that of the plane in the way in which it is divided; it is smooth/ but more delicate, less fleshy, softer, longer in proportion to its breadth, and the divisions — all tend to meet in a point, while they do not occur so much in the middle of the leaf, but rather at the tip; and for their size the leaves have not many fibres. The bark too is somewhat rougher than that of the lime, of blackish colour thick closer than that of the Aleppo pine and stiff; the roots are few shallow and compact for the most part, both those of the yellow and those of the whitewooded tree. This tree occurs chiefly in wet ground, as the people of Mount Ida say, and is rare. About its flower they did not know, but the fruit, they said, is not very oblong, but like that of Christ's thorn,- except that it is more oblong than that. But the people of Mount Olympus say that, while zygia is rather a mountain tree, the maple proper grows also in the plains; and that the form which grows in the mountains has yellow wood of a bright colour, which is of compact texture and hard, and is used even for expensive work, while that of the plains has white wood of looser make and less compact texture. And some call it gleinos instead of maple. The wood of the 'male' tree is of compacter texture and twisted; this tree, it is said, grows rather in the plain and puts forth its leaves earlier.

3.11.2

There are also two kinds of ash. Of these one is lofty and of strong growth with white wood of good fibre, softer, with less knots, and of more compact texture; the other is shorter, less vigorous in growth, rougher harder and yellower. The leaves in shape are like those of the bay, that is, the broad-leaved bay, but they contract to a sharper point, and they have a sort of jagged outline with sharp points. The whole leaf (if one may consider this as a 'leaf' because it is all shed at once) grows on a single stalk; on either side of a single fibre, as it were, the leaflets grow at a joint in pairs, which are numerous and distinct, just as in the sorb. In some leaves the joints are short and the pairs fewer

in number, but in those of the white kind the joint is long and the pairs more numerous, while the leaflets are longer narrower and leek-green in colour. Also this tree has a smooth bark, which is dry thin and red in colour. The roots are matted stout and shallow. As to the fruit, the people of Ida supposed it to have none, and no flower either; however it has a nut-like fruit in a thin pod, like the fruit of the almond, and it is somewhat bitter in taste. And it also bears certain other things like winter-buds, as does the bay, but they are more solid,' and each separate one is globular, like those of the plane; some of these occur around the fruit, some, in fact the greater number, are at a distance from it. The smooth kind grows mostly in deep ravines and damp places, the rough kind occurs also in dry- and rocky parts. Some, for instance the Macedonians, call the one 'ash' (manna-ash), the other 'horse-ash' (ash). The 'horse-ash' is a larger and more spreading tree, wherefore it is of less compact appearance. It is naturally a tree of the plains and rough, while the other belongs to the mountains and is smooth; the one which grows on the mountains is fair-coloured smooth hard and stunted, while that of the plains is colourless spreading and rough. (In general one may say of trees that grow in the plain and on the mountain respectively, that the latter are of fair colour hard and smooth, as beech elm and the rest; while those of the plain are more spreading, of less good colour and inferior, except the pear apple and wild pear, according to the people of Mount Olympus. These when they grow in the plain are better both in fruit and in wood; for on the mountain they are rough spinous and much branched, in the plain smoother larger and with sweeter and fleshier fruit. However the trees of the plain are always of larger size.)

3.12.1

Of the cornelian cherry there is a 'male' and a 'female' kind (cornel), and the latter bears a corresponding name. Both have a leaf like that of the almond, but oilier and thicker; the bark is

fibrous and thin, the stem is not very thick, but it puts out sidebranches like the chaste-tree, those of the 'female' tree, which is more shrubby, being fewer. Both kinds have branches like those of the chaste-tree, arranged in pairs opposite one another. The wood of the 'male' tree has no heart, but is hard throughout, like horn in closeness and strength; whereas that of the 'female' tree has heart-wood and is softer and goes into holes; wherefore it is useless for javelins. The height of the male tree is at most twelve cubits, the length of the longest Macedonian spear, the stem up to the point where it divides not being very tall. The people of Mount Ida in the Troad say that the 'male' tree is barren, but that the 'female' bears fruit. The fruit has a stone like an olive and is sweet to the taste and fragrant; the flower is like that of the olive, and the tree produces its flowers and fruit in the same manner, inasmuch as it has several growing from one stalk, and they are produced at almost the same time in both forms. However the people of Macedonia say that both trees bear fruit, though that of the 'female' is uneatable, and the roots are like those of the chaste-tree, strong and indestructible. This tree grows in wet ground and not only in dry places; and it comes from seed, and also can be propagated from a piece torn off.

3.12.2

The 'cedar,' some say, has two forms, the Lycian and the Phoenician; but some, as the people of Mount Ida, say that there is only one form. It resembles the arkeuthos (Phoenician cedar), differing chiefly in the leaf, that of 'cedar' being hard, sharp and spinous, while that of arkeuthos is softer: the latter tree also seems to be of taller growth. However some do not give them distinct names, but call them both 'cedar' distinguishing them however as 'the cedar' and 'prickly cedar.' Both are branching trees with many joints and twisted wood. On the other hand arkeuthos has only a small amount of close core, which, when the tree is cut, soon rots, while the trunk of 'cedar' consists

mainly of heart and does not rot. The colour of the heart in each case is red: that of the 'cedar' is fragrant, but not that of the other. The fruit of 'cedar' is yellow, as large as the myrtle-berry, fragrant, and sweet to the taste. That of arkeuthos is like it in other respects, but black, of astringent taste and practically uneatable; it remains on the tree for a year, and then, when another grows, last year's fruit falls off. According to the Arcadians it has three fruits on the tree at once, last year's, which is not yet ripe, that of the year before last which is now ripe and eatable, and it also shews the new fruit. Satyrus said that the wood-cutters gathered him specimens of both kinds which were flowerless. The bark is like that of the cypress but rougher. Both kinds have spreading shallow roots. These trees grow in rocky cold parts and seek out such districts. There are three kinds of mespile, anthedon ('oriental thorn'), sataneios (medlar) and anthedonoeides (hawthorn), as the people of mount Ida distinguish them. The fruit of the medlar is larger paler more spongy and contains softer stones; in the other kinds it is somewhat smaller, more fragrant and of more astringent taste, so that it can be stored for a longer time. The wood also of these kinds is closer and yellower, though in other respects it does not differ. The flower in all the kinds is like the almond flower, except that it is not pink, as that is, but greenish — In stature the tree is large and it has thick foliage. The leaf in the young tree is round but much divided and like celery at the tip; but the leaf of older trees is very much divided and forms angles with larger divisions; it is smooth fibrous thinner and more oblong than the celery leaf, both as a whole and in its divisions, and it has a jagged edge all round. It has a long thin stalk, and the leaves turn bright red before they are shed. The tree has many roots, which run deep; wherefore it lives a long time and is hard to kill. The wood is close and hard and does not rot. The tree grows from seed and also from a piece torn off. It is subject to a disease which causes it to become wormeaten in its old age,

and the worms are large and different to those engendered by other trees.

3.12.3

Of the sorb they make two kinds, the 'female' which bears fruit and the 'male' which is barren. There are moreover differences in the fruit of the 'female' kind; in some forms it is round, in others oblong and egg-shaped. There are also differences in taste; the round fruits are generally more fragrant and sweeter, the oval ones are often sour and less fragrant. The leaves in both grow attached to a long fibrous stalk, and project on each side in a row like the feathers of a bird's wing, the whole forming a single leaf but being divided into lobes with divisions which extend to the rib; but each pair are some distance apart, - and, when the leaves fall, these divisions do not drop separately, but the whole wing-like structure drops at once. When the leaves are older and longer, the pairs are more numerous; in the younger and shorter leaves they are fewer; but in all at the end of the leaf-stalk there is an extra leaflet, so that the total number of leaflets is an odd number. In form the leaflets resemble the leaves of the 'fine-leaved' bay, except that they are jagged and shorter and do not narrow to a sharp point but to a more rounded end. The flower is clustering and made up of a number of small white blossoms from a single knob. The fruit too is clustering, when the tree fruits well; for a number of fruits are formed from the same knob, giving an appearance like a honeycomb. The fruit gets eaten by worms on the tree before it is ripe to a greater extent than that of medlar pear or wild pear, and yet it is much more astringent than any of these. The tree itself also gets worm-eaten, and so withers away as it ages; and the worm which infests it is a peculiar one, red and hairy. This tree bears fruit when it is quite young, that is as soon as it is three years old. In autumn, when it has shed its leaves, it immediately produces its winter-bud-like knob, which is glistening and swollen as though the tree were just about to

burst into leaf, and this persists through the winter. The sorb, like the medlar, is thornless; it has smooth rather shiny bark, (except when the tree is old), which in colour is a whitish yellow; but in old trees it is rough and black. The tree is of a good size, of erect growth and with well balanced foliage; for in general it assumes a conelike shape as to its foliage, unless something interferes. The wood is hard close strong and of a good colour; the roots are not numerous and do not run deep, but they are strong and thick and indestructible. The tree grows from a root, from a piece torn off, or from seed, and seeks a cold moist position; in such a position it is tenacious of life and hard to kill: however it also grows on mountains.

3.13.1

The kerosos (bird-cherry) is peculiar in character; it is of great stature, growing as much as twenty-four cubits high; and it is of very erect growth; as to thickness, it is as much as two cubits in circumference at the base. The leaves are like those of the medlar, but very tough and thicker, so that the tree is conspicuous by its colour from a distance. The bark in smoothness colour and thickness is like that of the lime; wherefore men make their writing-cases from it, as from the bark of that tree. This bark does not grow straight nor evenly all round the tree, but runs round it in a spiral (which becomes closer as it gets higher up the tree) like the outline of the leaves. And this part of it can be stripped off by peeling, whereas with the other part — this is not possible and it has to be cut in short lengths.

3.13.2

In the same manner part is removed by being split off in flakes as thin as a leaf, while the rest can be left and protects the tree, growing about it as described. If the bark is stripped off when the tree is peeling, there is also at the time a discharge of the sap; further, when only the outside coat is stripped off, what remains turns black with a kind of mucus-like moisture; and in

the second year another coat grows to replace what is lost, but this is thinner. The wood in its fibres is like the bark, twisting spirally, - and the branches grow in the same manner from the first; and, as the tree grows, it comes to pass that the lower branches keep on perishing, while the upper ones increase.

3.13.3

However the whole tree is not much branched, but has far fewer branches than the black poplar. Its roots are numerous and shallow and not very thick; and there is a similar twisting of the root and of the bark which surrounds it. The flower is white, like that of the pear and medlar, composed of a number of small blossoms arranged like a honeycomb. The fruit is red, like that of diospyros in shape, and in size it is as large as a bean. However the stone of the diospyros fruit is hard, while that of the bird-cherry is soft. The tree grows where the lime grows, and in general where there are rivers and damp places.

3.13.4

The elder also grows chiefly by water and in shady places, but likewise in places which are not of this character. It is shrubby, with annual branches which go on growing in length till the fall of the leaf, after which they increase in thickness. The branches do not grow to a very great height, about six cubits at most. The thickness of the stem of old trees is about that of the 'helmet' of a ship; the bark is smooth thin and brittle; the wood is porous and light when dried, and has a soft heart-wood, so that the boughs are hollow right through, and men make of them their light walking-sticks. When dried it is strong and durable if it is soaked, even if it is stripped of the bark; and it strips itself of its own accord as it dries. The roots are shallow and neither numerous nor large.

3.13.6

The single leaflet is soft and oblong, like the leaf of the 'broad-leaved' bay, but larger broader and rounder at the middle and base, though the tip narrows more to a point and is

jagged all round. The whole leaf is composed of leaflets growing about a single thick fibrous stalk, as it were, to which they are attached at either side in pairs at each joint; and they are separate from one another, while one is attached to the tip of the stalk. The leaves are somewhat reddish porous and fleshy: the whole is shed in one piece; wherefore one may consider the whole structure as a 'leaf.' The young twigs too have certain crooks in them. The flower is white, made, up of a number of small white blossoms attached to the point where the stalk divides, in form like a honeycomb, and it has the heavy fragrance of lilies. The fruit is in like manner attached to a single thick stalk, but in a cluster: as it becomes quite ripe, it turns black, but when unripe it is like unripe grapes; in size the berry is a little larger than the seed of a vetch; the juice is like wine in appearance, and in it men bathe — their hands and heads when they are being initiated into the mysteries. The seeds inside the berry are like sesame.

3.13.7

The willow also grows by the water, and there are many kinds. There is that which is called the black willow because its bark is black and red, and that which is called the white from the colour of its bark. The black kind has boughs which are fairer and more serviceable for basketwork, while those of the white are more brittle. There is a form both of the black and of the white which is small and does not grow to a height, — just as there are dwarf forms of other trees, such as prickly cedar and palm. The people of Arcadia call the tree not 'willow' but helike: they believe, as was said, that it bears fruitful seed.

3.14.1

Of the elm there are two kinds, of which one is called the 'mountain elm,' the other simply the 'elm': the difference is that the latter is shrubbier, while the mountain elm grows more vigorously. The leaf is undivided and slightly jagged, longer than that of the pear, but rough rather than smooth. The tree is

large, being both tall and wide-spreading. It is not common about Ida, but rare, and likes wet ground. The wood is yellow strong fibrous and tough; for it is all heart. Men use it for expensive doors: it is easy to cut when it is green, but difficult when it is dry. The tree is thought to bear no fruit, but in the 'wallets' it produces its gum and certain creatures like gnats; and it has in autumn its peculiar 'winter-buds' which are numerous small and black, but these have not been observed at other seasons.

3.14.2

The abele and the black poplar have each but a single kind: both are of erect growth, but the black poplar is much taller and of more open growth, and is smoother, while the shape of its leaves is similar to those of the other. The wood also of both, when cut, is much the same in whiteness. Neither of these trees appears to have fruit or flower. The aspen is a tree resembling the abele both in size and in having whitish branches, but the leaf is ivy-like: while however it is otherwise without angles, its one angular projection is long and narrows to a sharp point: in colour the upper and under sides are much alike. The leaf is attached to a long thin stalk: wherefore the leaf is not set straight, but has a droop. The bark of the abele is rougher and more scaly, like that of the wild pear, and it bears no fruit.

3.14.3

The alder also has but one form: in growth it is also erect, and it has soft wood and a soft heart-wood, so that the slender boughs are hollow throughout. The leaf is like that of the pear, but larger and more fibrous. It has rough bark, which on the inner side is red: wherefore it is used for dyeing hides. It has shallow roots . . . the flower is as large as that of the bay. It grows in wet places' - and nowhere else.

3.14.4

The semyda has a leaf like that of the tree called the 'Persian nut' (walnut), but it is rather narrower: the bark is variegated

and the wood light: it is only of use for making walking-sticks and for no other purpose. The bladder-senna has a leaf near that of the willow, but is many-branched and has much foliage; and the tree altogether is a large one. The fruit is in a pod, as in leguminous plants: the pods in fact are broad rather than narrow, and the seed in them is comparatively small, and is moderately hard, but not so very hard. For its size the tree does not bear much fruit. It is uncommon to have the fruit in a pod; in fact there are few such trees.

3.15.1

The filbert is also naturally a wild tree, in that its fruit is little, if at all, inferior to that of the tree in cultivation, that it can stand winter, that it grows commonly on the mountains, and that it bears abundance of fruit in mountain regions; also because it does not make a trunk, but is shrubby with unbranched stems without knots; though some of these are long and stout. Nevertheless it also submits to cultivation. The cultivated form differs in producing better fruit and larger leaves; in both forms the leaf has a jagged edge: the leaf of the alder most closely resembles it, but is broader, and the tree itself is bigger. The filbert is always more fruitful if it has its slender boughs cut off. — There are two kinds of each sort; some have a round, others an oblong nut; that of the cultivated tree is paler, and it fruits best in damp places. The wild tree becomes cultivated by being transplanted. Its bark is smooth,' consisting of one layer, thin glossy and with peculiar white blotches on it. The wood is extremely tough, so that men make baskets even of the quite thin twigs, having stripped them of their bark, and of the stout ones when they have whittled them. Also it has a small amount of yellow heart-wood, which makes the branches hollow. Peculiar to these trees is the matter of the catkin, as we mentioned.

3.15.2

The terebinth has a 'male' and a 'female' form. The 'male' is barren, which is why it is called male'; the fruit of one of the 'female' forms is red from the first and as large as an unripe lentil; the other produces a green fruit which subsequently turns red, and, ripening at the same time as the grapes, becomes eventually black and is as large as a bean, but resinous and somewhat aromatic. About Ida and in Macedonia the tree is low shrubby and twisted, but in the Syrian Damascus, where it abounds, it is tall and handsome; indeed they say that there is a certain hill which is covered with terebinths, though nothing else grows on it. It has tough wood and strong roots which run deep, and the tree as a whole is impossible to destroy. The flower is like that of the olive, but red in colour. The leaf is made up of a number of leaflets, like bay leaves, attached in pairs to a single leaf-stalk. So far it resembles the leaf of the sorb; there is also the extra leaflet at the tip: but the leaf is more angular than that of the sorb, and the edge resembles more the leaf of the bay; the leaf is glossy all over,- as is the fruit. It bears also some hollow bag-like growths, like the elm, in which are found little creatures like gnats; and resinous sticky matter is found also in these bags; but the resin is gathered from the wood and not from these. The fruit does not discharge much resin, but it clings to the hands, and, if it is not washed after gathering, it all sticks together; if it is washed, the part which is white and unripe floats, but the black part sinks.

3.15.3

The box is not a large tree, and it has a leaf like that of the myrtle. It grows in cold rough places; for of this character is Cytora, where it is most abundant. The Macedonian Olympus is also a cold region; for there too it grows, though not to a great size. It is largest and fairest in Corsica, where the tree grows taller and stouter than anywhere else; wherefore the honey there is not sweet, as it smells of the box.

3.15.4

The krataigos is a very common tree; some call it krataigon. It has a smooth leaf like that of the medlar, but longer, and its breadth is greater than its length, while the edge is not jagged like that of the medlar. The tree does not grow very tall or thick; its wood is mottled strong and brown; it has a smooth bark like that of the medlar; it has generally a single root, which runs deep. The fruit is round and as large as that of the wild olive; as it ripens it turns brown and black; in taste and flavour it is like that of the medlar; wherefore this might seem to be a sort of wild form of that tree. There is only one form of it and it shews no variation.

3.16.1

The kermes-oak has a leaf like that of the oak, but smaller and spinous, while its bark is smoother than that of the oak. The tree itself is large, like the oak, if it has space and root-room; the wood is close and strong; it roots fairly deep and it has many roots. The fruit is like an acorn, but the kermes oak's acorn is small; the new one overtakes that of last year, for it ripens late. Wherefore some say that it bears twice. Besides the acorn it bears a kind of scarlet berry; it also has oak-mistletoe and mistletoe; so that sometimes it happens that it has four fruits on it at once, two which are its own and two others, namely those of the oak-mistletoe and of the mistletoe. It produces the oak-mistletoe on the north side and the mistletoe on the south.

3.16.2

The Arcadians have a tree which they call smilax (holm-oak), which resembles the kermes-oak, but has not spinous leaves, its leaves being softer and longer — and differing in several other ways. Nor is the wood hard and close like that of the kermes oak, but quite soft to work.

3.16.3

The tree which the Arcadians call 'cork-oak' (holm-oak) has this character: — to put it generally, it is between the kermes-oak and the oak; and some suppose it to be the 'female' kermes-oak;

wherefore, where the kermes-oak does not grow, they use this tree for their carts and such-like purposes; for instance it is so used by the peoples of Lacedaemon and Elis. The Dorians also call the tree aria. Its wood is softer and less compact than that of the kermes-oak, but harder and closer than that of the oak. When it is barked, the colour of the wood is paler than that of the kermes-oak, but redder than that of the oak. The leaves resemble those of both trees, but they are somewhat large, if we consider the tree as a kermes-oak, and somewhat small if we regard it as an oak. The fruit is smaller in size than that of the kermes-oak, and equal to the smallest acorns; it is sweeter than that of the kermes-oak, bitterer than that of the oak. Some call the fruit of the kermes oak and of the aria 'mast,' keeping the name 'acorn' for the fruit of the oak. It has a core which is more obvious than in kermes-oak. Such is the character of the 'cork-oak,'

3.16.4

The arbutus, which produces the edible fruit called memaikylon, is not a very large tree; its bark is thin and like that of the tamarisk, the leaf is between that of the kermes-oak and that of the bay. It blooms in the month Pyanepsion; the flowers grow in clusters at the end of the boughs from a single attachment; in shape each of them is like an oblong myrtle flower and it is of about the same size; it has no petals, but forms a cup like an empty eggshell, and the mouth is open: when the flower drops off, there is a hole also through the part by which it is attached, and the fallen flower is delicate and like a whorl on a spindle or a Doric karneios. The fruit takes a year to ripen, so that it comes to pass that this and the new flower are on the tree together.

3.16.5

The andrachne has a leaf like that of the arbutus and is not a very large tree; the bark is smooth and cracked,' the fruit is like that of the arbutus.

3.16.6

The leaf of the wig-tree is also like that of the last named tree, but it is a small tree. Peculiar to it is the fact that the fruit passes into down; we have not heard of such a thing in any other tree. These trees are found in a good many positions and regions.

3.17.1

Some however are more local, such as the cork-oak: this occurs in Tyrrhenia; it is a tree with a distinct trunk and few branches, and is fairly tall and of vigorous growth. The wood is strong, the bark very thick and cracked, like that of the Aleppo pine, save that the cracks are larger. The leaf is like that of the manna-ash, thick and somewhat oblong. The tree is not evergreen but deciduous. It has always an acorn-like fruit like that of the aria (holm-oak). They strip off the bark, and they say that it should all be removed, otherwise the tree deteriorates: it is renewed again in about three years.

3.17.2

The kolutea too is a local tree, occurring in the Lipari islands. It is a tree of good size, and bears its fruit, which is as large as a lentil, in pods; this fattens sheep wonderfully. It grows from seed, and also grows very well from sheep-droppings. The time for sowing it is the setting of Arcturus; and one should first soak the seed and sow it when it is already sprouting in the water. It has a leaf like telis (fenugreek). At first it grows for about three years with a single stem, and in this period men cut their walking-sticks from it; for it seems that it makes excellent ones. And, if the top is cut off during this period, it dies, for it makes no sideshoots. After this period it divides, and in the fourth year develops into a tree.

3.17.3

The tree found about Mount Ida, called koloitia, is a distinct kind and is shrubby and branching with many boughs; but it is rather rare. It has a leaf like that of the 'broad-leaved' bay, but rounder and larger, so that it looks like that of the elm but it is

more oblong; the colour on both sides is green, but the base is whitish; in this part it is very fibrous, because of its fine fibres which spring partly from the midrib, partly between the ribs (so to call them) which run out from the midrib. The bark is not smooth but like that of the vine; the wood is hard and close, the roots are shallow slender and spreading, (though sometimes they are compact), and they are very yellow. They say that this shrub has no fruit nor flower, but has its knobby winter-bud and its 'eyes'; these grow alongside of the leaves, and are very smooth glossy and white, and in shape are like a winter-bud. When the tree is cut or burnt down, it grows from the side and springs up again.

3.17.4

There are also three trees peculiar to Mount Ida, the tree called Alexandrian laurel, a sort of fig, and a vine (currant grape). The peculiarity of the laurel is that it bears fruit on its leaves, like the 'prickly myrtle' (butcher's broom): both have their fruit on the midrib of the leaf.

3.17.5

The 'fig' is shrubby and not tall, but so thick that the stem is a cubit in circumference. The wood is twisted and tough; below it is smooth and unbranched, above it has thick foliage: the colour both of leaf and bark is a dull green, the shape of the leaf is like that of the lime; it is soft and broad, and in size it also corresponds; the flower is like that of the medlar, and the tree blooms at the same time as that tree. The fruit, which they call a 'fig' is red, and as large as an olive, but it is rounder and is like the medlar in taste; the roots are thick like those of the cultivated fig, and tough. The tree does not rot, and it has a solid heart, instead of ordinary heart-wood.

3.17.6

The 'vine' (currant grape) grows about the place called Phalakrai in the district of Ida; it is shrubby with small twigs; the branches are about a cubit long, and attached to them at the

side are black berries, which are the size of a bean and sweet; inside they have a sort of soft stone; the leaf is round undivided and small.

3.18.1

Most other mountains too have certain peculiar products, whether trees shrubs or other woody plants. However we have several times remarked as to such peculiarities that they occur in all regions. Moreover the variation- between things of the same kind which we find in trees obtains also among shrubs and most other things, as has been said: for instance, we find it in buckthorn, Christ's thorn, withy, sumach, ivy, bramble, and many others.

3.18.2

Thus of buckthorn there is the black and the white form, and there is difference in the fruit, though both bear thorns. Of the withy there is a black and a white form; the flower and fruit of each respectively correspond in colour to the name; but some specimens are, as it were, intermediate, the flower being purplish, and neither wine-coloured nor whitish as in the others. The leaves in the white kind are also slenderer and smoother, as also are the branches.

3.18.3

There is variation also in the Christ's thorn . . . all these forms are fruit-bearing. Christ's thorn has its fruit in a sort of pod, resembling a leaf, which contains three or four seeds. Doctors bruise them and use them against coughs; for they have a certain viscous and oily character, like linseed. The shrub grows in wet and dry places alike, like the bramble. But it is deciduous, and not evergreen like buckthorn.

3.18.4

Of the bramble again there are several kinds, shewing very great variation; one is erect and tall, another runs along the ground and from the first bends downwards, and, when it touches the earth, it roots again; this some call the 'ground

bramble.' The 'dog's bramble' (wild rose) has a reddish fruit, like that of the pomegranate; and, like the pomegranate, it is intermediate between a shrub and a tree; but the leaf is spinous.

3.18.5

Of the sumach they recognise a 'male' and a 'female' form, the former being barren, the latter fruit-bearing. The branches are not lofty nor stout, the leaf is like that of the elm, but small more oblong and hairy. On the young shoots the leaves grow in pairs at equal distances apart, corresponding to each other on the two sides, so that they are in regular rows. Tanners use this tree for dyeing white leather. The flower is white and grows in clusters; the general form of it, with branchlets, is like that of the grape-bunch; when the flowering is over, the fruit reddens like the grape, and the appearance of it is like small lentils set close together; the form of these too is clustering. The fruit contains the drug called by the same name, which is a bony substance; it is often still found even when the fruit has been put through a sieve. The root is shallow and single, so that these trees are easily bent right over, root and all. The wood has heart-wood, and it readily perishes and gets worm-eaten. The tree occurs in all regions, but flourishes most in clayey soils.

3.18.6

The ivy also has many forms; one kind grows on the ground, another grows tall, and of the tall-growing ivies there are several kinds. However the three most important seem to be the white, the black and the helix. And of each of these there are several forms. Of the 'white' one is white only in its fruit, another in its leaves also. Again to take only white-fruited sorts, one of these has its fruit well formed close and compact like a ball; and this kind some call korymbias, but the Athenians call it the 'Acharnian' ivy. Another kind is smaller and loose in growth like the black ivy. There are also variations in the black kind, but they are not so well marked.

3.18.7

The helix presents the greatest differences; the principal difference is in the leaves, which are small angular and of more graceful proportions, while those of the ivy proper are rounder and simple; there is also difference in the length of the twigs, and further in the fact that this tree is barren. For, as to the view that the helix by natural development turns into the ivy, some insist that this is not so, the only true ivy according to these being that which was ivy from the first; (whereas if, as some say, the helix invariably turns into ivy, the difference would be merely one of age and condition, and not of kind, like the difference between the cultivated and the wild pear). However the leaf even of the fullgrown helix is very different from that of the ivy, and it happens but rarely and in a few specimens that in this plant a change in the leaf occurs as it grows older, as it does in the abele and the castor-oil plant. There are several forms of the helix, of which the three most conspicuous and important are the green 'herbaceous' kind (which is the commonest), the white, and the variegated, which some call the 'Thracian' helix. Each of these appears to present variations; of the green one form is slenderer and has more regular and also closer leaves, the other has all these characteristics in a less degree. Of the variegated kind again one sort has a larger, one a smaller leaf, and the variegation is variable. In like manner the various forms of the white helix differ in size and colour. The 'herbaceous' kind is the most vigorous and covers most space. They say that the form which is supposed to turn into ivy is clearly marked not only by its leaves, because they are larger and broader, but also by its shoots; for these are straight from the first, and this form does not bend over like the other; also because the shoots are slenderer and larger, while those of the ivy-like form are shorter and stouter. The ivy too, when it begins to seed, has its shoots upwardgrowing and erect.

3.18.8

All ivies have numerous close roots, which are tangled together woody and stout, and do not run very deep; but this is specially true of the black kind and of the roughest and wildest forms of the white. Wherefore it is mischievous to plant this against any tree; for it destroys and stanes any tree by withdrawing the moisture. This form also more than the others grows stout and becomes treelike, and in fact becomes itself an independent ivy tree, though in general it likes and seeks to be against another tree, and is, as it were, parasitic. Moreover from the first it has also this natural characteristic, that it regularly puts forth roots from the shoots between the leaves, by means of which it gets a hold of trees and walls, as if these roots were made by nature on purpose. Wherefore also by withdrawing and drinking up the moisture it starves its host, while, if it is cut off below, it is able to survive and live. There are also other not inconsiderable differences in the fruit; both in the white and in the black kind it is in some cases rather sweet, in others extremely bitter; in proof whereof birds eat one but not the other. Such are the facts about ivy.

3.18.9

The smilax is parasitic, but its stem is thorny and has, as it were, straight thorns; the leaf is ivylike small and without angles, and makes a callus at the junction with the stalk. A peculiarity of it is its conspicuous slender midrib, so to call it, which divides it in two; also the fact that the thread-like branchings do not start from this, as in other leaves, but are carried in circles round it, starting from the junction of the leaflet with the leaf. And at the joints of the stem and the spaces between the leaves there grows from the same stalk as the leaves a fine spiral tendril. The flower is white and fragrant like a lily. The fruit is like the strykhnos and the meloihron (bryony), and most of all like the berry which is called the 'wild grape' (bryony). The clusters hang down as in the ivy, but the regular setting — of the berries resembles the grape-cluster more closely; for the stalks which

bear the berries start from a single point. The fruit is red, having generally two stones, the larger ones three and the smaller one; the stone is very hard and in colour black outside. A peculiarity of the clusters is that they make a row along the sides of the stalk, and the longest cluster is at the end of the stalk, as in the buckthorn and the bramble. It is clear that the fruit is produced both at the end and at the sides.

3.18.10

The tree called the spindle-tree grows, among other places, in Lesbos, on the mountain called Ordynnos. It is as large as the pomegranate and has a leaf like that of that tree, but larger than that of the periwinkle, and soft, like the pomegranate leaf. It begins to shoot about the month Poseideon, and flowers in the spring; the flower in colour is like the gilliflower, but it has a horrible smell, like shed blood. The fruit, with its case, is like the pod of sesame; inside it is hard, but it splits easily according to its four divisions. This tree, if eaten by sheep, is fatal to them, both the leaf and the fruit, and it is especially fatal to goats unless they are purged by it; and the purging is effected by diarrhoea. So we have spoken of trees and shrubs; in what follows we must speak of the plants which remain.

4.1.1

BOOK IV. OF THE TREES AND PLANTS SPECIAL TO PARTICULAR DISTRICTS AND POSITIONS.

The Differences between trees of the same kind have already been considered. Now all grow fairer and are more vigorous in their proper positions; for wild, no less than cultivated trees, have each their own positions: some love wet and marshy ground, as black poplar abele willow, and in general those that grow by rivers; some love exposed and sunny positions; some prefer a shady place. The fir is fairest and tallest in a sunny position, and does not grow at all in a shady one; the silver-fir on the contrary is fairest in a shady place, and not so vigorous

in a sunny one.

4.1.2

Thus there is in Arcadia near the place called Krane a low-lying district sheltered from wind, into which they say that the sun never strikes; and in this district the silver-firs excel greatly in height and stoutness, though they have not such close grain nor such comely wood, but quite the reverse, — like the fir when it grows in a shady place. Wherefore men do not use these for expensive work, such as doors or other choice articles, but rather for ship-building and house-building. For excellent rafters beams and yard-arms are made from these, and also masts of great length which are not however equally strong; while masts made of trees grown in a sunny place are necessarily short but of closer grain and stronger than the others.

4.1.3

Yew, pados and joint-fir rejoice exceedingly in shade. On mountain tops and in cold positions odorous cedar grows even to a height, while silver-fir and Phoenician cedar grow, but not to a height, — for instance on the top of Mount Cyllene; and holly also grows in high and very wintry positions. These trees then we may reckon as cold-loving; all others, one may say in general, prefer a sunny position. However this too depends partly on the soil appropriate to each tree; thus they say that in Crete on the mountains of Ida and on those called the White Mountains the cypress is found on the peaks whence the snow never disappears; for this is the principal tree both in the island generally and in the mountains.

4.1.4

Again, as has been said already, both of wild and of cultivated trees some belong more to the mountains, some to the plains. And on the mountains themselves in proportion to the height some grow fairer and more vigorous in the lower regions, some about the peaks. However it is true of all trees anywhere that

with a north aspect the wood is closer and more compact and better generally; and, generally speaking, more trees grow in positions facing the north. Again trees which are close together grow and increase more in height, and so become unbranched straight and erect, and the best oar-spars are made from these, while those that grow far apart are of greater bulk and denser habit; wherefore they grow less straight and with more branches, and in general have harder wood and a closer grain.

4.1.5

Such trees exhibit nearly the same differences, whether the position be shady or sunny, windless or windy; for trees growing in a sunny or windy position are more branched shorter and less straight. Further that each tree seeks an appropriate position and climate is plain from the fact that some districts bear some trees but not others; (the latter do not grow there of their own accord, nor can they easily be made to grow), and that, even if they obtain a hold, they do not bear fruit — as was said of the date-palm the sycamore and others; for there are many trees which in many places either do not grow at all, or, if they do, do not thrive nor bear fruit, but are in general of inferior quality. And perhaps we should discuss this matter, so far as our enquiries go.

4.2.1

Thus in Egypt there are a number of trees which are peculiar to that country, the sycamore the tree called persea, the balanos, the acacia and some others.

4.2.2

Now the sycamore to a certain extent resembles the tree which bears that name in our country; its leaf is similar, its size, and its general appearance; but it bears its fruit in a quite peculiar manner, as was said at the very outset; it is borne not on the shoots or branches, but on the stem; in size it is as large as a fig, which it resembles also in appearance, but in flavour and sweetness it is like the 'immature figs,' except that it is much

sweeter and contains absolutely no seeds, and it is produced in large numbers. It cannot ripen unless it is scraped; but they scrape it with iron 'claws'; the fruits thus scraped ripen in four days. If these are removed, others and others again grow from exactly the same point, and this some say occurs three times over, others say it can happen more times than that. Again the tree is very full of sap, and its wood is useful for many purposes. There is another peculiar property which it appears to possess; when it is cut, it is at first green, but it dries in deep water; they put it at once in a hole or in pools and so season it; and it becomes dry by being soaked in the deep water, and when it is completely dry, it is fetched up and floats and is then thought to be duly seasoned; for it is now light and porous. Such are the peculiarities of the sycamore.

4.2.3

Somewhat similar appears to be the character of the tree which in Crete is called the 'Cyprian fig' (sycamore). For this also bears its fruit on the stem and on the thickest branches; but in this case there is a small leafless shoot, like a root, to which the fruit is attached. The stem is large and like the abele, but the leaf is like that of the elm. It ripens its fruit four times a year, having also four periods of growth; but it ripens no fruit unless the 'fig' is split and the juice let out. The sweet taste resembles that of the fig, and the inside of the fruit is like that of wild figs: it is as large as a plum.

4.2.4

(Like this too is the tree which the Ionians call carob; for this too bears most of its fruit on the stem, though it bears a little also on the branches, as we said. The fruit is in a pod; some call it the 'Egyptian fig' — erroneously; for it does not occur at all in Egypt, but in Syria and Ionia and also in Cnidos and Rhodes. It is evergreen and has a whitish flower and is somewhat acrid; it does not attain to a great height, and it sends out side-shoots entirely from its lower parts, while it withers above. It has on it

at the same time both last year's fruit and the new fruit; for if the one is removed after the rising of the dog-star, immediately the other is seen swelling up; for there swells up as it were another similar cluster. This then increases and flowers about the rising of Arcturus and the equinox; and thenceforward it persists through the winter to the rising of the dog-star. The likeness then consists in the fact that these trees too bear fruit on their stems, and the differences between them and the sycamore are as has been said.)

4.2.5

In Egypt there is another tree called the perseia, which in appearance is large and fair, and it most resembles the pear in leaves flowers branches and general form, but it is evergreen, while the other is deciduous. It bears abundant fruit and at every season, for the new fruit always overtakes that of last year. It ripens its fruit at the season of the etesian winds: the other fruit they gather somewhat unripe and store it. In size it is as large as a pear, but in shape it is oblong, almond-shaped, and its colour is grass-green. It has inside a stone like the plum, but much smaller and softer; the flesh is sweet and luscious and easily digested; for it does no hurt if one eats it in quantity. The tree has good roots as to length thickness and number. Moreover its wood is strong and fair in appearance, black like the nettle-tree: out of it men make their images beds tables and other such things.

4.2.6

The balanos gets its name from its fruit; its leaf is like that of the myrtle but it is longer. The tree is of a good stoutness and stature, but not of a good shape, being crooked. The perfumers use the husks of the fruit, which they bruise; for this is fragrant, though the fruit itself is useless. In size and appearance it is like the fruit of the caper; the wood is strong and useful for shipbuilding and other purposes.

4.2.7

The tree called the doum-palm is like the datepalm; the resemblance is in the stem and the leaves, but it differs in that the date-palm is a tree with a single undivided stem, while the other, as it increases, splits and becomes forked, and then each of the two branches forks again: moreover the twigs are very short and not numerous. They use the leaf, like the palm-leaf, for plaiting. It has a peculiar fruit, very different from that of the date-palm in size form and taste; for in size it is nearly big enough to fill the hand, but it is round rather than long; the colour is yellowish, the flavour sweet and palatable. It does not grow bunched together, like the fruit of the datepalm, but each fruit grows separately; it has a large and very hard stone, out of which they turn the rings for embroidered bed-hangings. The wood is very different to that of the date-palm; whereas the latter is of loose texture fibrous and porous,- that of the doum-palm is close heavy and fleshy, and when split is exceedingly compact and hard. The Persians used to esteem it highly and made the feet of their couches out of it.

4.2.8

The akanthia (acacia) is so called because the whole tree is spinous (akanthodes) except the stem; for it has spines on the branches shoots and leaves. It is of large stature, since lengths of timber for roofing of twelve cubits are cut from it. There are two kinds, the white and the black; the white is weak and easily decays, the black is stronger and less liable to decay; wherefore they use it in shipbuilding for the ribs. The tree is not very erect in growth. The fruit is in a pod, like that of leguminous plants, and the natives use it for tanning hides instead of gall. The flower is very beautiful in appearance, so that they make garlands of it, and it has medicinal properties, wherefore physicians gather it. Gum is also produced from it, which flows both when the tree is wounded and also of its own accord without any incision being made. When the tree is cut down, after the third year it immediately shoots up again; it is a

common tree, and there is a great wood of it in the Thebaid, where grow the oak, the persea in great abundance, and the olive.

4.2.9

For the olive also grows in that district, though it is not watered by the river, being more than 300 furlongs distant from it, but by brooks; for there are many springs. The oil produced is not inferior to that of our country, except that it has a less pleasing smell, because it has not a sufficient natural supply of salt.' The wood of the tree is hard in character, and, when split, is like in colour to that of the nettle-tree.

4.2.10

There is another tree, the (Egyptian) plum (sebesten), which is of great stature, and the character of its fruit is like the medlar (which it resembles in size), except that it has a round stone. It begins to flower in the month Pyanepsion, and ripens its fruit about the winter solstice, and it is evergreen.^{il} The inhabitants of the Thebaid, because of the abundance of the tree, dry the fruit; they take out the stones, bruise it, and make cakes of it.

4.2.11

There is a peculiar bush which grows about Memphis, whose peculiarity does not lie in its leaves, shoots and general form, but in the strange property which belongs to it. Its appearance is spinous and the leaf is like ferns, but, when one touches the twigs, they say that the leaves as it were wither up and collapse and then after a time come to life again and flourish. Such are the most conspicuous things peculiar to the country, to speak only of trees or shrubs. For we will speak later of the things which grow in the river and the marshes, when we come to speak of the other water plants.

4.2.12

All the trees of this kind in that country are large, both in height and stoutness; thus at Memphis there is said to be a tree of such girth that three men cannot embrace it. The wood too, when

split, is good, being of extremely close grain and in colour like the nettle-tree.

4.3.1

In Libya the lotos is most abundant and fairest; so also is the Christ's thorn, and in some parts, such as the Nasamonian district and near the temple of Zeus Ammon, the date-palm. In the Cyrenaica the cypress grows and the olives are fairest and the oil most abundant. Most special of all to this district is the silphium, and the land also bears abundant fragrant saffron-crocus. As to the lotos — the whole tree is peculiar, of good stature, as tall as a pear-tree, or nearly so; the leaf is divided and like that of the kermes-oak, and the wood is black. There are several sorts, which differ in their fruits; the fruit is as large as a bean, and in ripening like grapes it changes its colour: it grows, like myrtle-berries, close together on the shoots; to eat, that which grows among the people called the Lotus-eaters is sweet pleasant and hamaless, and even good for the stomach; but that which has no stone is pleasanter (for there is also such a sort), and they also make wine from it.

4.3.2

The tree is abundant and produces much fruit; thus the army of Ophelias, - when it was marching on Carthage, was fed, they say, on this alone for several days, when the provisions ran short. It is abundant also in the island called the island of the Lotus-eaters; this lies off the mainland at no great distance: it grows however in no less quantity, but even more abundantly on the mainland; for, as has been said, this tree is common in Libya generally as well as the Christ's thorn; for in the islands called Euesperides they use these trees as fuel. However this lotos differs from that found in the land of the Lotus-eaters.

4.3.3

The (Egyptian) Christ's thorn is more shrubby than the lotos; it has a leaf like the tree of the same name of our country, but the fruit is different; for it is not flat, but round and red, and in size

as large as the fruit of the prickly cedar or a little larger; it has a stone which is not eaten with the fruit, as in the case of the pomegranate, but the fruit is sweet, and, if one pours wine over it, they say that it becomes sweeter and that it makes the wine sweeter.

4.3.4

Some say that the lotos — is shrubby and much branched, though it has a stout stem; and that the stone in the fruit is large, while the outside is not fleshy but somewhat leathery; and that to eat it is not so much sweet as palatable; and that the wine which they make out of it does not keep more than two or three days, after which it gets sour; and so that the fruit found in the Lotus-eaters' country is sweeter, while the wood in the Cyrenaica is better; and that the country of the Lotus-eaters is hotter; and that the root is much blacker than the wood, but of less close grain, and of use for fewer purposes; for they use it only for dagger handles and tessellated work, while the wood is used for pipes and many other things.

4.3.5

In the part of Libya where no rain falls they say that, besides many other trees, there grow tall and fine date-palms; however they add that, where the date-palm is found, the soil is salt and contains water, and that at no great depth, not more than three fathoms. They say also that the water is in some places quite sweet, but in others quite close by it is brackish; that where however other things grow, the soil is dry and waterless; and that in places even the wells are a hundred fathoms deep, so that they draw water by means of a windlass worked by beasts. Wherefore it is wonderful how at any time digging to such depths was carried out. Such, they say, is the special character of the water supply which feeds the date-palms in the district also of the temple of Zeus Ammon. Further it is said that in the land where no rain falls thyme is abundant, and that there are various other peculiar plants there, and that there are found the

hare gazelle ostrich and other animals. However it is uncertain whether these do not migrate in order to find drink somewhere, (for by reason of their fleetness they are able to appear at a distant place in a short space of time), especially if they can go for several days without drinking, even as these animals, when domesticated, are only given drink every third or fourth day. While as to other animals, such as snakes lizards and the like, it is plain that they go without drink. And we are told that according to the Libyans, these animals eat the wood-louse, which is of the same kind that is found also in our country, being black, with many feet, and rolling itself into a ball; this, they say, is extremely common and is juicy by nature.

4.3.6

They say also that dew always falls abundantly in the land in which no rain falls, so that it is plain that the date-palm, as well as anything else which grows in waterless places, is kept alive by the moisture which rises from the ground, and also by the dew. For the latter is sufficient, considering the size of such trees and their natural character, which is dry and formed of dry components. And trees of that character are most abundant in, and most specially belong to such country. The character of the silphium we must discuss later.

4.4.1

In different parts of Asia also there are special trees, for the soil of the various regions produces some but not others. Thus they say that ivy and olive do not grow in Asia in the parts of Syria which are five days' journey from the sea; but that in India ivy appears on the mountain called Meros, whence, according to the tale, Dionysus came. Wherefore it is said that Alexander, when he came back from an expedition, was crowned with ivy, himself and his army. But elsewhere in Asia it is said to grow only in Media, for that country seems in a way to surround and join on to the Euxine Sea. However, when Harpalus took great pains over and over again to plant it in the gardens of Babylon,

and made a special point of it, he failed: since it could not live like the other things introduced from Hellas. The country then does not admit this plant on account of the climate, and it grudgingly admits the box and the lime; for even these give much trouble to those engaged in the gardens. It also produces some peculiar trees and shrubs. And in general the lands of the East and South appear to have peculiar plants, as they have peculiar animals; for instance. Media and Persia have, among many others, that which is called the 'Median' or 'Persian apple' (citron). This tree has a leaf like to and almost identical with that of the andrachne, but it has thorns like those of the pear or white-thorn, which however are smooth and very sharp and strong. The 'apple' is not eaten, but it is very fragrant, as also is the leaf of the tree. And if the 'apple' is placed among clothes, it keeps them from being moth-eaten. It is also useful when one has drunk deadly poison; for being given in wine it upsets the stomach and brings up the poison; also for producing sweetness of breath; for, if one boils the inner part of the 'apple' in a sauce, or squeezes it into the mouth in some other medium, and then inhales it, it makes the breath sweet. The seed is taken from the fruit and sown in spring in carefully tilled beds, and is then watered every fourth or fifth day. And, when it is growing vigorously, it is transplanted, also in spring, to a soft well-watered place, where the soil is not too fine; for such places it loves. And it bears its apples at all seasons; for when some have been gathered, the flower of others is on the tree and it is ripening others. Of the flowers, as we have said, those which have, as it were, a distaff projecting in the middle are fertile, while those that have it not are infertile. It is also sown, like date-palms, in pots with a hole in them. This tree, as has been said, grows in Persia and Media.

4.4.2

The Indian land has its so-called 'fig-tree' (banyan), which drops its roots from its branches every year, as has been said above;

and it drops them, not from the new branches, but from those of last year or even from older ones; these take hold of the earth and make, as it were, a fence about the tree, so that it becomes like a tent, in which men sometimes even live. The roots as they grow are easily distinguished from the branches, being whiter hairy crooked and leafless.- The foliage above is also abundant, and the whole tree is round and exceedingly large. They say that it extends its shade for as much as two furlongs; and the thickness of the stem is in some instances more than sixty paces, while many specimens are as much as forty paces through. The leaf is quite as large as a shield, but the fruit is very small, only as large as a chick-pea, and it resembles a fig. And this is why the Greeks named this tree a 'fig-tree.' The fruit is curiously scanty, not only relatively to the size of the tree, but absolutely. The tree also grows near the river Akesines.

4.4.3

There is also another tree which is very large and has wonderfully sweet and large fruit; it is used for food by the sages of India who wear no clothes. There is another tree whose leaf is oblong in shape, like the feathers of the ostrich; this they fasten on to their helmets, and it is about two cubits long. There is also another whose fruit is long and not straight, but crooked, and it is sweet to the taste. This causes griping in the stomach and dysentery; wherefore Alexander ordered that it should not be eaten. There is also another whose fruit is like the fruit of the cornelian cherry. There are also many more which are different to those found among the Hellenes, but they have no names. There is nothing surprising in the fact that these trees have so special a character; indeed, as some say, there is hardly a single tree or shrub or herbaceous plant, except quite a few, like those in Hellas. The ebony is also peculiar to this country; of this there are two kinds, one with good handsome wood, the other inferior. The better sort is rare, but the inferior one is common. It does not acquire its good colour by being kept, but it is

natural to it from the first. The tree is bushy, like laburnum.

4.4.4

Some say that a 'terebinth' grows there also, others that it is a tree like the terebinth; this in leaf twigs and all other respects resembles that tree, but the fruit is different, being like almonds. In fact they say that this sort of terebinth grows also in Bactria and bears nuts only as big as almonds, inasmuch as they are not large for the size of the tree; and they closely resemble almonds in appearance, except that the shell is not rough; and in palatableness and sweetness they are superior to almonds; wherefore the people of the country use them in preference to almonds.

4.4.5

The trees from which they make their clothes have a leaf like the mulberry, but the whole tree resembles the wild rose. They plant them in the plains in rows, wherefore, when seen from a distance, they look like vines. Some parts also have many date-palms. So much for what come under the heading of 'trees.'

4.4.6

These lands bear also peculiar grains, some like those of leguminous plants, some like wheat and barley. For the chick-pea lentil and other such plants found in our country do not occur; but there are others, so that they make similar mash, and one cannot, they say, tell the difference, unless one has been told. They have however barley wheat and another kind of wild barley, - which makes sweet bread and good porridge. When the horses ate this, at first it proved fatal to them, but by degrees they became accustomed to it mixed with bran and took no hurt.

4.4.7

But above all they sow the cereal called rice, of which they make their mash. This is like rice-wheat, and when bruised makes a sort of porridge, which is easily digested; in its appearance as it

grows it is like darnel, and for most of its time of growth it is in water; however it shoots up not into an ear, but as it were into a plume, like the millet and Italian millet. There was another plant which the Hellenes called lentil; this is like in appearance to 'ox-horn' (fenugreek), but it is reaped about the setting of the Pleiades.

4.4.8

Moreover this country shews differences in that part of it bears certain things which another part does not; thus the mountain country has the vine and olive and the other fruit-trees; but the olive is barren, and in its character it is as it were almost between a wild and a cultivated olive, and so it is also in its general appearance, and the leaf is broader than that of the one and narrower than that of the other. So much for the Indian land.

4.4.9

In the country called Aria there is a 'thorn' on which is found a gum resembling myrrh in appearance and smell, and this drops when the sun shines on it. There are also many other plants besides those of our land, both in the country and in its rivers. In other parts there is a white 'thorn' which branches in three, of which they make batons and sticks; its wood is sappy and of loose texture, and they call it the thorn 'of Herakles.'

4.4.10

There is another shrub as large as a cabbage, whose leaf is like that of the bay in size and shape. And if any animal should eat this, it is certain to die of it. Wherefore, whenever there were horses, they kept them under control.

4.4.11

In Gedrosia they say that there grows one tree with a leaf like that of the bay, of which if the beasts or anything else ate, they very shortly died with the same convulsive symptoms as in epilepsy.

4.4.12

And they say that another tree there is a sort of ‘thorn’ (spurge), and that this has no leaf and grows from a single root; and on each of its branches it has a very sharp spine, and if these are broken or bruised a quantity of juice flows out, which blinds animals or even a man, if any drops of it should fall on him. Also they say that in some parts grows a herb under which very small snakes lie coiled up, and that, if anyone treads on these and is bitten, he dies. They also say that, if anyone should eat of unripe dates, he chokes to death, and that this fact was not discovered at first. Now it may be that animals and plants have such properties elsewhere also.

4.4.13

Among the plants that grow in Arabia, Syria and India the aromatic plants are somewhat exceptional and distinct from the plants of other lands; for instance, frankincense, myrrh, cassia, balsam of Mecca, cinnamon and all other such plants, about which we have spoken at greater length elsewhere. So in the parts towards the east and south there are these special plants and many others besides.

4.5.1

In the northern regions it is not so, for nothing worthy of record is mentioned except the ordinary trees which love the cold and are found also in our country, as fir, oak, silver-fir, box, chestnut, lime, as well as other similar trees. There is hardly any other besides these; but of shrubs there are some which for choice — seek cold regions, as centaury and wormwood, and further those that have medicinal properties in their roots and juices, such as hellebore, squirting cucumber, scammony, and nearly all those whose roots are gathered.

4.5.2

Some of these grow in Pontus and Thrace, some about Oeta Parnassus Pelion Ossa and Telethrion, and in these parts some say that there is great abundance; so also is there in Arcadia and

Laconia, for these districts too produce medicinal plants. But of the aromatic plants none grows in these lands, except the iris in Illyria on the shores of the Adriatic; for here it is excellent and far superior to that which grows elsewhere; but in hot places and those which face the south the fragrant plants grow, as if by contrast to the medicinal plants. And the warm places have also the cypress in greater abundance; for instance, Crete Lycia, Rhodes, while the prickly cedar grows in the Thracian and the Phrygian mountains.

4.5.3

Of cultivated plants they say that those least able to thrive in cold regions are the bay and myrtle, especially the myrtle, and they give for proof that on Mount Olympus the bay is abundant, but the myrtle does not occur at all. In Pontus about Panticapaeum neither grows, though they are anxious to grow them and take special pains to do so for religious purposes. But there are many well grown fig-trees and pomegranates, which are given shelter; pears and apples are abundant in a great variety of forms and are excellent. These are springfruiting trees, except that they may fruit later here than elsewhere. Of wild trees there are oak elm manna-ash and the like (while there is no fir silverfir nor Aleppo pine, nor indeed any resinous tree). But the wood of such trees in this country is damp and much inferior to that of Sinope, so that they do not much use it except for outdoor purposes. These are the trees of Pontus, or at least of certain districts of that country.

4.5.4

In the land of Propontis myrtle and bay are found in many places on the mountains. Perhaps however some trees should be put down as special to particular places. For each district, as has been said, has different trees, differing not only in that the same trees occur but of variable quality, but also as to producing or not producing some particular tree. For instance, Tmolus and the Mysian Olympus have the hazel and chestnut in

abundance, and also the vine apple and pomegranate; while Mount Ida has some of these not at all and others only in small quantity; and in Macedonia and on the Pierian Olympus some of these occur, but not others; and in Euboea and Magnesia the sweet chestnut is common, but none of the others is found; nor yet on Pelion or the other mountains of that region.

4.5.5

Again it is only a narrow extent of country which produces wood fit for shipbuilding at all, namely in Europe the Macedonian region, and certain parts of Thrace and Italy; in Asia Cilicia Sinope and Amisus, and also the Mysian Olympus, and Mount Ida; but in these parts it is not abundant. For Syria has Syrian cedar, and they use this for their galleys.

4.5.6

The like is true of trees which love water and the riverside; in the Adriatic region they say that the plane is not found, except near the Shrine of Diomedes, and that it is scarce throughout Italy; yet there are many large rivers in both countries, in spite of which the localities do not seem to produce this tree. At any rate those which King Dionysius the Elder planted at Rhegium in the park, and which are now in the grounds of the wrestling school and are thought much of, have not been able to attain any size.

4.5.7

Some of these regions however have the plane in abundance, and others the elm and willow, others the tamarisk, such as the district of Mount Haemus. Wherefore such trees we must, as was said, take to be peculiar to their districts, whether they are wild or cultivated. However it might well be that the country should be able to produce some of these trees, if they were carefully cultivated: this we do in fact find to be the case with some plants, as with some animals.

4.6.1

However the greatest difference in the natural character itself of trees and of tree-like plants generally we must take to be that mentioned already, namely, that of plants, as of animals, some belong to the earth, some to water. Not only in swamps, lakes and rivers, but even in the sea there are some tree-like growths, and in the ocean there are even trees. In our own sea all the things that grow are small, and hardly any of them rise above the surface; but in the ocean we find the same kinds rising above the surface, and also other larger trees.

4.6.2

Those found in our own waters are as follows: most conspicuous of those which are of general occurrence are seaweed, oyster-green and the like; most obvious of those peculiar to certain parts are the sea-plants called 'fir' 'fig' 'oak' 'vine' 'palm.' Of these some are found close to land, others in the deep sea, others equally in both positions. And some have many forms, as seaweed, some but one. Thus of seaweed there is the broad-leaved kind, riband-like — and green in colour, which some call 'green-weed' and others 'girdle-weed.' This has a root which on the outside is shaggy, but the inner part is made of several coats, and it is fairly long and stout, like kromyogeteion (a kind of onion).

4.6.3

Another kind has hair-like leaves like fennel, and is not green but pale yellow; nor has it a stalk, but it is, as it were, erect in itself; this grows on oyster-shells and stones, not, like the other, attached to the bottom; but both are plants of the shore, and the hair-leaved kind grows close to land, and sometimes is merely washed over by the sea; while the other is found further out.

4.6.4

Again in the ocean about the pillars of Heracles there is a kind of marvellous size, they say, which is larger, about a palmsbreadth. This is carried into the inner sea along with the current from the outer sea, and they call it 'sea-leek' (riband-weed); and in

this sea in some parts it grows higher than a man's waist. It is said to be annual. and to come up at the end of spring, and to be at its best in summer, and to wither in autumn, while in winter it perishes and is thrown up on shore. Also, they say, all the other plants of the sea become weaker and feebler in winter. These then are, one may say, the sea-plants which are found near the shore. But the 'seaweed of ocean,' which is dived for by the sponge-fishers, belongs to the open sea.

4.6.5

In Crete there is an abundant and luxuriant growth on the rocks close to land, with which they dye not only their ribbons, but also wool and clothes. And, as long as the dye is fresh, the colour is far more beautiful than the purple dye; it occurs on the north coast in greater abundance and fairer, as do the sponges and other such things.

4.6.6

There is another kind like dog's-tooth grass; the leaf is very like, the root is jointed and long, and grows out sideways, like that of that plant; it has also a reedy stalk like the same plant, and in size it is much smaller than ordinary seaweed.

4.6.7

Another kind is the oyster-green, which has a leaf green in colour, but broad and not unlike lettuce leaves; but it is more wrinkled and as it were crumpled. It has no stalk, but from a single starting-point grow many of the kind, and again from another starting-point. These things grow on stones close to land and on oyster-shells. These are about all the smaller kinds.

4.6.8

The 'sea-oak' and 'sea-fir' both belong to the shore; they grow on stones and oyster-shells, having no roots, but being attached to them like limpets. Both have more or less fleshy leaves; but the leaf of the 'fir' grows much longer and stouter, and is not unlike the pods of pulses, but is hollow inside and contains nothing in the 'pods.' That of the 'oak' is slender and more like the

tamarisk; the colour of both is purplish. The whole shape of the 'fir' is erect, both as to the stem and the branches, but that of the 'oak' is less straight and the plant is broader. Both are found both with many stems and with one,- but the 'fir' is more apt to have a single stem. The branchlike outgrowths in the 'fir' are long straight and spreading, while in the 'oak' they are shorter less straight and closer. The whole size of either is about a cubit or rather more, but in general that of the 'fir' is the longer. The 'oak' is useful to women for dyeing wool. To the branches are attached certain creatures with shells, and below they are also found attached to the stem itself, which in some cases they completely cover; and among these are found millepedes and other such creatures, including the one which resembles a cuttlefish.

4.6.9

These plants occur close to land and are easy to observe; but some report that there is another 'sea oak' which even bears fruit and has a useful 'acorn,' and that the sponge fishers and divers told them that there were other large kinds.

4.6.10

The 'sea-vine' grows under both conditions, both close to land and in the deep sea; but the deep sea form has larger leaves branches and fruit.

4.6.11

The 'sea-fig' is leafless and not of large size, and the colour of the bark is red. The 'sea-palm' is a deep-sea plant, but with a very short stem, and the branches which spring from it are almost straight; and these under water are not set all round the stem, like the twigs which grow from the branches, but extend, as it were, quite flat in one direction, and are uniform; though occasionally they are irregular. The character of these branches or outgrowths to some extent resembles the leaves of thistle-like spinous plants, such as the sow-thistles — and the like, except that they are straight and not bent over like these,

and have their leaves eaten away by the brine; in the fact that the central stalk at least runs through the whole, they resemble these, and so does the general appearance. The colour both of the branches and of the stalks and of the plant as a whole is a deep red or scarlet. Such are the plants found in this sea. For sponges and what are called aplysiai and such-like growths are of a different character.

4.7.1

In the outer sea near the pillars of Heracles grows the 'sea-leek,' as has been said; also the well known 'plants which turn to stone, as thyma, the plants like the bay and others. And in the sea called the Red Sea a little above Coptos in Arabia there grows on the land no tree except that called the 'thirsty' acacia, and even this is scarce by reason of the heat and the lack of water; for it never rains except at intervals of four or five years, and then the rain comes down heavily and is soon over.

4.7.2

But there are plants in the sea, which they call 'bay' and 'olive' (white mangrove). In foliage the 'bay' is like the aria (holm-oak), the 'olive' like the real olive. The latter has a fruit like olives, and it also discharges a gum, from which the physicians compound a drug for stanching blood, which is extremely effective. And when there is more rain than usual, mushrooms grow in a certain place close to the sea, which are turned to stone by the sun. The sea is full of beasts, and produces sharks in great numbers, so that diving is impossible.

4.7.3

In the gulf called 'the Gulf of the Heroes,' to which the Egyptians go down, there grow a 'bay,' an 'olive,' and a 'thyme'; these however are not green, but like stones so far as they project above the sea, but in leaves and shoots they are like their green namesakes. In the 'thyme' the colour of the flower is also conspicuous, looking as though the flower had not yet completely developed. These treelike growths are about three

cubits in height.

4.7.4

Now some, referring to the occasion when there was an expedition of those returning from India sent out by Alexander, report that the plants which grow in the sea, so long as they are kept damp, have a colour like sea-weeds, but that when they are taken out and put in the sun, they shortly become like salt. They also say that rushes of stone grow close to the sea, which none could distinguish at sight from real rushes. They also report a more marvellous thing than this; they say that there are certain tree-like growths which in colour resemble an ox-horn, but whose branches are rough, and red at the tip; these break if they are doubled up, and some of them, if they are cast on a fire, become red-hot like iron, but recover when they cool and assume their original colour.

4.7.5

On the islands which get covered by the tide they say that great trees grow, as big as planes or the tallest poplars, and that it came to pass that, when the tide came up, while the other things were entirely buried, the branches of the biggest trees projected and they fastened the stem cables to them, and then, when the tide ebbed again, fastened them to the roots. And that the tree has a leaf like that of the bay, and a flower like gilliflowers in colour and smell, and a fruit the size of that of the olive, which is also very fragrant. And that it does not shed its leaves, and that the flower and the fruit form together in autumn and are shed in spring.

4.7.6

Also they say there are plants which actually grow in the sea, which are evergreen and have a fruit like lupins.

4.7.7

In Persia in the Carmanian district, where the tide is felt, there are trees of fair size like the andrachne in shape and in leaves; and they bear much fruit like in colour to almonds on the

outside, but the inside is coiled up as though the kernels were all united. These trees are all eaten away up to the middle by the sea and are held up by their roots, so that they look like a cuttle-fish. For one may see this at ebb-tide. And there is no rain at all in the district, but certain channels are left, along which they sail, and which are part of the sea. Which, some think, makes it plain that the trees derive nourishment from the sea and not from fresh water, except what they draw up with their roots from the land. And it is reasonable to suppose that this too is brackish; for the roots do not run to any depth. In general they say that the trees which grow in the sea and those which grow on the land and are overtaken by the tide are of the same kind, and that those which grow in the sea are small and look like seaweed, while those that grow — on land are large and green and have a fragrant flower and a fruit like a lupin.

4.7.8

In the island of Tylos which is situated in the Arabian gulf, they say that on the east side there is such a number of trees when the tide goes out that they make a regular fence. All these are in size as large as a fig-tree, the flower is exceedingly fragrant, and the fruit, which is not edible, is like in appearance to the lupin. They say that the island also produces the 'wool-bearing' tree (cotton-plant) in abundance. This has a leaf like that of the vine, but small, and bears no fruit; but the vessel in which the 'wool' is contained is as large as a spring apple, and closed, but when it is ripe, it unfolds and puts forth the 'wool,' of which they weave their fabrics, some of which are cheap and some very expensive.

4.7.9

This tree is also found, as was said, - in India as well as in Arabia. They say that there are other trees with a flower like the gilliflower, but scentless and in size four times as large as that flower. And that there is another tree with many leaves like the rose, and that this closes at night, but opens at sunrise, and by noon is completely unfolded; and at evening again it closes by

degrees and remains shut at night, and the natives say that it goes to sleep. Also that there are date-palms on the island and vines and other fruit-trees, including evergreen figs. Also that there is water from heaven, but that they do not use it for the fruits, but that there are many springs on the island, from which they water everything, and that this is more beneficial to the corn and the trees. Wherefore, even when it rains, they let this water over the fields, as though they were washing away the rain water. Such are the trees as so far observed which grow in the outer sea.

4.8.1

Next we must speak of plants which live in rivers marshes and lakes. Of these there are three classes, trees, plants of 'herbaceous' character, and plants growing in clumps. By 'herbaceous' I mean here such plants as the marsh celery and the like; by 'plants growing in clumps' I mean reeds, galingale, phleo, rush sedge — which are common to almost all rivers and such situations.

4.8.2

And in some such places are found brambles Christ's thorn and other trees, such as willow, abele, plane. Some of these are water plants to the extent of being submerged, while some project a little from the water; of some again the roots and a small part of the stem are under water, but the rest of the body is altogether above it. This is the case with willow alder plane lime, and all water-loving trees.

4.8.3

These too are common to almost all rivers, for they grow even in the Nile. However the plane is not abundant by rivers, while the abele is even more scarce, and the manna-ash and ash are commonest. At any rate of those that grow in Egypt the list is too long to enumerate separately; however, to speak generally, they are all edible and have sweet flavours. But they differ in sweetness, and we may distinguish also three as the most useful

for food, namely the papyrus, the plant called sari, and the plant which they call mnasion.

4.8.4

The papyrus does not grow in deep water, but only in a depth of about two cubits, and sometimes shallower. The thickness of the root is that of the wrist of a stalwart man, and the length above four cubits; it grows above the ground itself, throwing down slender matted roots into the mud, and producing above the stalks which give it its name 'papyrus'; these are three-cornered and about ten cubits long, having a plume which is useless and weak, and no fruit whatever; and these stalks the plant sends up at many points. They use the roots instead of wood, not only for burning, but also for making a great variety of articles; for the wood is abundant and good. The 'papyrus' itself- is useful for many purposes; for they make boats from it, and from the rind they weave sails mats a kind of raiment coverlets ropes and many other things. Most familiar to foreigners are the papyrus-rolls made of it; but above all the plant also is of very great use in the way of food. For all the natives chew the papyrus both raw boiled and roasted: they swallow the juice and spit out the quid. Such is the papyrus and such its uses. It grows also in Syria about the lake in which grows also sweetflag; and Antigonus made of it the cables for his ships.

4.8.5

The sari grows in the water in marshes and plains, when the river has left them; it has a hard twisted root, and from it grow what they call the saria; these are about two cubits long and as thick as a man's thumb; this stalk too is threecornered, like the papyrus, and has similar foliage. This also they chew, spitting out the quid; and smiths use the root, for it makes excellent charcoal, because the wood is hard. Mnasion is herbaceous, so that it has no use except for food.

4.8.6

Such are the plants which excel in sweetness of taste. There is also another plant which grows in the marshes and lakes, but which does not take hold of the ground; in character it is like a lily, but it is more leafy, and has its leaves opposite to one another, as it were in a double row; the colour is a deep green. Physicians use it for the complaints of women and for fractures.

4.8.7

Now these plants grow in the river, unless the stream has thrown them up on land; it sometimes happens that they are borne down the stream, and that then other plants grow from them. But the 'Egyptian bean' grows in the marshes and lakes; the length of its stalk at longest is four cubits, it is as thick as a man's finger, and resembles a pliant reed without joints. Inside it has tubes which run distinct from one another right through, like a honey-comb: on this is set the 'head,' which is like a round wasps' nest, and in each of the cells is a 'bean,' which slightly projects from it; at most there are thirty of these. The flower is twice as large as a poppy's, and the colour is like a rose, of a deep shade; the 'head' is above the water. Large leaves grow at the side of each plant, equal in size to a Thessalian hat; these have a stalk exactly like that of the plant. If one of the 'beans' is crushed, you find the bitter substance coiled up, of which the pilos is made. So much for the fruit. The root is thicker than the thickest reed, and is made up of distinct tubes, like the stalk. They eat it both raw, boiled, and roasted, and the people of the marshes make this their food. It mostly grows of its own accord; however they also sow it in the mud, having first well mixed the seed with chaff, so that it may be carried down and remain in the ground without being rotted; and so they prepare the 'bean' fields, and if the plant once takes hold it is permanent. For the root is strong and not unlike that of reeds, except that it is prickly on the surface. Wherefore the crocodile avoids it, lest he may strike his eye on it, since he has not sharp sight. This plant also grows in Syria and in parts of Cilicia, but these countries

cannot ripen it; also about Torone in Chalcidice in a certain lake of small size; and this lake ripens it perfectly and matures its fruit.

4.8.8

The plant called the lotos (Nile water-lily) grows chiefly in the plains when the land is inundated. The character of the stalk of this plant is like that of the 'Egyptian bean,' and so are the 'hat-like' leaves, except that they are smaller and slenderer. And the fruit grows on the stalk in the same way as that of the 'bean.' The flower is white, resembling in the narrowness of its petals those of the lily,' but there are many petals growing close one upon another. When the sun sets, these close and cover up the 'head,' but with sunrise they open and appear above the water. This the plant does until the 'head' is matured and the flowers have fallen off. The size of the 'head' is that of the largest poppy, and it has grooves all round it in the same way as the poppy, but the fruit is set closer in these. This is like millet. — In the Euphrates they say that the 'head' and the flowers sink and go under water in the evening till midnight, and sink to a considerable depth; for one can not even reach them by plunging one's hand in; and that after this, when dawn comes round, they rise and go on rising towards day-break, being visible above the water when the sun appears; and that then the plant opens its flower, and, after it is open, it still rises; and that it is a considerable part which projects above the water. These 'heads' the Egyptians heap together and leave to decay, and when the 'pod' has decayed, they wash the 'head' in the river and take out the 'fruit,' and, having dried and pounded it, they make loaves of it, which they use for food. The root of the lotos is called korsion, and it is round and about the size of a quince; it is enclosed in a black 'bark,' like the shell of a chestnut. The inside is white; but when it is boiled or roasted, it becomes of the colour of the yolk of an egg and is sweet to taste. The root is also eaten raw, though it is best when boiled in water or

roasted. Such are the plants found in water.

4.8.9

In sandy places which are not far from the river there grows under ground the thing called malinathalle; this is round in shape and as large as a medlar, but has no stone and no bark. It sends out leaves like those of galingale. These the people of the country collect and boil in beer made from barley, and they become extremely sweet, and all men use them as sweetmeats.

4.8.10

All the things that grow in such places may be eaten by oxen and sheep, but there is one kind of plant which grows in the lakes and marshes which is specially good for food: they graze their cattle on it when it is green, and also dry it and give it in the winter to the oxen after their work; and these keep in good condition when they have no other kind of food.

4.8.11

There is also another plant which comes up of its own accord among the corn; this, when the harvest is cleared, they crush slightly and lay during the winter on moist ground; when it shoots, they cut and dry it and give this also to the cattle and horses and beasts of burden with the fruit which forms on it. The fruit in size is as large as sesame, but round and green in colour, and exceedingly good. Such one might take to be specially remarkable plants of Egypt.

4.9.1

Every river seems to bear some peculiar plant, just as does each part of the dryland. For not even the water-chestnut grows in all rivers nor everywhere, but only in marshy rivers, and only in those whose depth is not more or not much more than five cubits, as the Strymon. (In rivers of such a depth grow also reeds and other plants.) No part of it projects from the water except just the leaves; these float as it were and conceal the 'chestnut' which is itself under water and bends down towards the bottom.

4.9.2

The leaf is broad, like that of the elm, and has a very long stalk. The stem is thickest at the top, whence spring the leaves and the fruit; below it gets thinner down to the root. It has springing from it hair-like growths, most of which are parallel to each other, but some are irregular; below, starting from the root, they are large, but, as one gets higher up the plant, they become smaller, so that those at the top are quite small and there is a great contrast between the root and the top where the fruit grows. The plant also has on the same stalk several sidegrowths; of these there are three or four, and the largest is always that which is nearer to the root, the next largest is the one next above it, and so on in proportion: this sidegrowth is like another stalk, but slenderer than the original one, though like that it has leaves and fruit. The fruit is black and extremely hard. The size and character of the root are matter for further enquiry. Such is the character of this plant. It grows from the fruit which falls, and begins to grow in spring. Some say that it is annual, others that the root persists for a time, and that from it grows the new stalk. This then is matter for enquiry. However quite peculiar to this plant is the hair-like character of the growths which spring from the stalk; for these are neither leaves nor stalk; though reeds and other things have also sidegrowths.

4.10.1

Plants peculiar to particular places must be considered separately, while a general account may be given of those which are generally distributed.' But even the latter must be classified according to locality; thus some belong to marshes, others to lakes, others to rivers, or again others may be common to all kinds of locality: we must also distinguish which occur alike in wet and in dry ground, and which only in wet ground, marking these off in a general way from those mentioned above as being most impartial.

4.10.2

Now in the lake near Orchomenos grow the following trees and woody plants: willow goat-willow water-lily reeds (both that used for making pipes and the other kind) galingale, phleas, bulrush; and also 'moon-flower' duckweed and the plant called marestalk: as for the plant called water-chickweed the greater part of it grows under water.

4.10.3

Now of these most are familiar: the goat-willow water-lily 'moon-flower' duckweed and marestalk probably grow also elsewhere, but are called by different names. Of these we must speak. The goat-willow is of shrubby habit and like the chastetree: its leaf resembles that leaf in shape, but it is soft like that of the apple,- and downy. The bloom is like that of the abele, but smaller, and it bears no fruit. It grows chiefly on the floating islands; (for here too there are floating islands, as in the marshes of Egypt, in Thesprotia, and in other lakes). When it grows under water, it is smaller. Such is the goat-willow.

4.10.4

The water-lily is in shape like the poppy. For the top of it has this character, being shaped like the pomegranate flower, but it is longer in proportion to the size of the plant. Its size in fact as a whole is that of an apple; but it is not bare, having round it white membranes, and attached to these on the outside are grass-green 'leaves,' like those of roses when they are still in bud, and of these there are four; when it is opened it shews its seeds, which are red; in shape however they are not like pomegranate seeds, but round small and not much longer than millet seeds; the taste is insipid, like that of wheat-grains. It ripens in summer and has a long stalk. The flower is like a rose-bud, but larger, almost twice as large. Now this and the leaf float on the water: but later, when the bloom is over and the fruit-case has formed, they say that it sinks deeper into the water, and finally reaches the bottom and sheds its fruit.

4.10.5

Of the plants of the lake they say that water-lily sedge and phleos bear fruit, and that that of the sedge is black, and in size like that of the water-lily. The fruit of phleos is what is called the 'plume,' and it is used as a soap-powder. It is something like a cake, soft and reddish. Moreover the 'female' plant both of phleos and sedge is barren, but useful for basket-work,- while the 'male' is useless. Duckweed 'moon-flower' and marestail require further investigation.

4.10.6

Most peculiar of these plants is the bulrush, both in being leafless and in not having so many roots as the others; for the others tend downwards quite as much as upwards, and shew their strength in that direction; and especially is this true of galingale, and also of dog's-tooth grass; wherefore these plants too and all others like them are hard to destroy. The root of galingale exceeds all the others in the diversity of characters which it shews, in that part of it is stout and fleshy, part slender and woody. So also is this plant peculiar in its way of shooting and originating; for from the trunk-like stock grows another slender root sideways, and on this again forms the fleshy part which contains the shoot from which the stalk springs. In like manner it also sends out roots downwards; wherefore of all plants it is hardest to kill, and troublesome to get rid of.

4.10.7

(Dog's-tooth grass grows in almost the same way from the joints; for the roots are jointed, and from each joint it sends a shoot upwards and a root downwards. The growth of the spinous plant called corn-thistle is similar, but it is not reedy and its root is not jointed. We have enlarged on these matters because of the resemblance.)

4.10.8

The willow and the reed (not however the reed used for pipes), galingale, bulrush, phleos, sedge grow both on land and in the water, water-lily only in the water. (As to bulrush indeed there

is a difference of opinion.) However they say that those plants which grow in the water are always finer and larger than those that grow in both positions; also that some of these plants grow also on the floating islands,- for instance galingale sedge and phleos; thus all parts of the lake contain these plants.

4.10.9

Of the plants of the lake the parts good for food are as follows: of the water-lily both the flower and the leaves are good for sheep, the young shoots for pigs, and the fruit for men. Of phleos, galingale, and sedge the part next the roots is tender, and is mostly eaten by children. The root of phleos is the only part which is edible by cattle. When there is a drought and there is no water from overhead, all the plants of the lake are dried up, but especially the reed; of this it remains to speak, since we have said almost enough about the rest.

4.11.1

Of the reed there are said to be two kinds, the one used for making pipes and the other kind. For that of the latter there is only one kind, though individual plants differ in being strong and stout, 'or on the other hand slender and weak. The strong stout one they call the 'stake-reed,' the other the 'weaving reed.' The latter they say grows on the floating islands, the stout form in the 'reed-beds'; this name they give to the places where there is a thick mass of reed with its roots entangled together. This occurs in any part of the lake where there is rich soil. It is said that the 'stake-reed' is also sometimes found in the same places as the reed used for pipes, in which places it is longer than the 'stakereed' found elsewhere, but gets worm-eaten. These then are the differences in reeds of which they tell.

4.11.2

As to the reed used for pipes, it is not true, as some say, that it only grows once in nine years and that this is its regular rule of growth; it grows in general whenever the lake is full: but, because in former days this was supposed to happen generally

once in nine years, they made the growth of the reed to correspond, taking what was really an accident to be a regular principle. As a matter of fact it grows whenever after a rainy season the water remains in the lake for at least two years,- and it is finer if the water remains longer; this is specially remembered to have happened in recent times at the time of the battle of Chaeronea. For before that period they told me that the lake was for several years deep; and, at a time later than that, when there was a severe visitation of the plague, it filled up; but, as the water did not remain but failed in winter, the reed did not grow; for they say, apparently with good reason, that, when the lake is deep, the reed increases in height, and, persisting for the next year, matures its growth; and that the reed which thus matures is suitable for making a reed mouthpiece, while that for which the water has not remained is suitable for making a 'cap.' Such then, it is said, is the reed's way of growth.

4.11.3

Also it is said to differ from other reeds, to speak generally, in a certain luxuriance of growth, being of a fuller and more fleshy character, and, one may say, 'female' in appearance. For it is said that even the leaf is broader and whiter, though the plume is smaller than that of other reeds, and some have no plume at all; these they call 'eunuch-reeds.' From these they say that the best mouthpieces are made, though many are spoiled in the making.

4.11.4

Till the time of Antigenidas, before which men played the pipe in the simple style, they say that the proper season for cutting the reeds was the month Boedromion about the rising of Arcturus; for, although the reed so cut did not become fit for use for many years after and needed a great deal of preliminary placing upon, yet the opening of the reed-tongues is well closed, which is a good thing for the purpose of accompaniment. But

when a change was made to the more elaborate style of playing, the time of cutting the reeds was also altered; for in our own time they cut them in the months Skirrophorion or Hekatombaion about the solstice or a little earlier. And they say that the reed becomes fit for use in three years and needs but little preliminary playing upon, and that the reed-tongues have ample vibration, which is essential for those who play in the elaborate style. Such, they tell us, are the proper seasons for cutting the reed used for the reed mouthpiece.

4.11.5

The manufacture is carried out in the following manner. Having collected the reed-stems they lay them in the open air during the winter, leaving on the rind; in the spring they strip this off, and, having rubbed the reeds thoroughly, put them in the sun. Later on, in the summer, they cut the sections from knot to knot into lengths and again put them for some time in the open air. They leave the upper knot on this internodal section -; and the lengths thus obtained are not less than two palmsbreadths long. Now they say that for making mouthpieces the best lengths are those of the middle of the reed, whereas the lengths towards the upper growths make very soft mouthpieces and those next to the root very hard ones. They say too that the reed-tongues made out of the same length are of the same quality, while those made from different lengths are not; also that the one from the length next to the root forms a left-hand reed-tongue, and that from the length towards the upper growths a righthand reed-tongue. Moreover, when the length is slit, the opening of the reed-tongues in either case is made towards the point at which the reed was cut; and, if the reed-tongues are made in any other manner, they are not quite of the same quality. Such then is the method of manufacture.

4.11.6

This reed grows in greatest abundance between the Kephisos and the Melas; this district is called Pelekania, and in it are

certain 'pots,' as they are called, which are deep holes in the marsh,- and in these holes they say that it grows fairest; it is also said to be found where the river called the Probatia (Sheep) comes down, which is a stream that flows from Lebadeia. But it appears to grow fairest of all near 'Sharp Bend'; this place is the mouth of the Kephisos; near it is a rich plain called Hippias. There is another region north of the Sharp Bend called Boedrias; and here too they say that the reed grows fine, and in general that it is fairest wherever there is a piece of land with deep rich alluvial soil, where also Kephisos mingles his waters with the soil, and where there is further a deep hole in the marsh; for that about the Sharp Bend and Boedrias all these conditions are found. As proof that the Kephisos has a great effect in producing the reed of good quality they have the fact that, where the river called the 'Melas' flows into the marsh, though the marsh is there deep and the bottom of good alluvial soil, it either does not grow at all or at best but of poor quality. Let this suffice for an account of the growth and character of the reed used for pipes, of the manufacture, and of its distinctive features as compared with other reeds.

4.11.7

But these are not the only kinds of reed; there are several others with distinctive characters which are easily recognised; there is one that is of compact growth in flesh and has its joints close together; another that is of open growth, with few joints; there is the hollow reed called by some the 'tubereed' inasmuch as it has hardly any wood or flesh; there is another which is solid and almost entirely filled with substance; there is another which is short, and another which is of strong growth tall and stout; there is one which is slender and has many leaves, another which has few leaves or only one. And in general there are many differences in natural character and in usefulness, each kind being useful for some particular purpose.

4.11.8

Some distinguish the various kinds by different names; commonest perhaps is the pole-reed, which is said to be of very bushy habit, and to grow chiefly by rivers and lakes. And it is said that there is a wide difference in reeds in general between those that grow on dry land and those that grow in the water. Quite distinct again is the 'archer's' reed, which some call the 'Cretan': this has few joints and is fleshier than any of the others; it can also be most freely bent, and in general, when warmed, may be turned about as one pleases.

4.11.9

The various kinds have also, as was said, great differences in the leaves, not only in number and size, but also in colour. That called the 'Laconian' reed is parti-coloured. They also differ in the position and attachment of the leaves; some have most of their leaves low down, and the reed itself grows out of a sort of a bush. Indeed some say that this may be taken as the distinctive character of those which grow in lakes, namely, that these have many leaves, and that their foliage in a manner resembles that of galingale, phleas, thryon, and sedge; but this needs further enquiry.

4.11.10

There is also a kind of reed (bush-grass) which grows on land, and which is not erect, but sends out its stem over the ground, like the dog's-tooth grass, and so makes its growth. The 'male' reed is solid: some call it eiletias.

4.11.11

The Indian reed (bamboo) is very distinct, and as it were a totally different kind; the 'male' is solid and the 'female' hollow (for in this kind too they distinguish a 'male' and a 'female' form); a number of reeds of this kind grow from one base and they do not form a bush; the leaf is not long, but resembles the willow leaf; these reeds are of great size and of good substance, so that they are used for javelins. They grow by the river Akesines. All reeds are tenacious of life, and, if cut or burnt

down, grow up again more vigorously; also their roots are stout and numerous, so that the plant is hard to destroy. The root is jointed, like that of the dogstooth grass, but this is not equally so in all kinds. However let this suffice for an account of reeds.

4.12.1

It remains to speak of the rush,- as though it belonged to this class of plants, inasmuch as we must reckon this also among water plants. Of this there are three kinds, as some distinguish, the 'sharp' rush, which is barren and is called the male'; the 'fruiting' kind which we call the 'black head' because it has black fruit; this is stouter and fleshier; and third the 'entire rush' as it is called, which is distinguished by its size, stoutness and fleshiness.

4.12.2

Now the 'black-head' grows by itself, but the 'sharp' rush and the 'entire' rush grow from the same stock, which seems extraordinary, and indeed it was strange to see it when the whole clump of rushes was brought before me; for from the same stock there were growing 'barren' rushes, which were the most numerous, and also a few 'fruiting' ones. This then is a matter for further enquiry. The 'fruiting' — ones are in general scarcer, for the 'entire rush' is more useful for wicker-work because of its fleshiness and pliancy. The 'fruiting' rush in general produces a club-like head which swells straight from the wiry stem, and then bears egg-like bodies; for attached to a single wiry base it has its very spike-like branches all round it, and on the ends of these it has its round vessels borne laterally and gaping; in each of these is the small seed, which is pointed and black, and like that of the Michaelmas daisy, except that it is less solid. It has a long root, which is stouter than that of the ordinary rush; this withers every year, and then another strikes down again from the 'head' of the plant. And it is easy to observe that some of the roots as they are let down are withered, some green. The 'head' is like that of an onion or long

onion, being, as it were, made up of several united together; it is broad, and underneath it has reddish scales. Now it is a peculiar fact about the roots of this plant that they wither every year and that the fresh growth of roots comes from the part of the plant which is above ground. Such is the character of rushes.

4.12.3

Bramble and Christ's thorn may be considered to some extent plants of the water or the waterside, as they are in some districts; but the distinctive characters of these plants are fairly clear, for we have spoken of both already.

4.12.4

The floating islands of Orchomenos — are of various sizes, the largest being about three furlongs in circumference. But in Egypt very large ones form, so that even a number of boars are found in them, and men go across to the islands to hunt them. Let this account of water-plants suffice.

4.13.1

As to the comparative shortness of life of plants and trees of the water we may say thus much as a general account, that, like the water-animals, they are shorter-lived than those of the dry land. But we must enquire into the lives of those of the dry land severally. Now the woodmen say that the wild kinds are almost without exception long-lived, and none of them is short-lived: so far they may be speaking the truth; all such plants do live far longer than others. However, just as in the case of cultivated plants, some are longer-lived than others, and we must consider which these are. Cultivated plants plainly differ as to the length of their lives, but, to speak generally, wild plants are longer-lived than cultivated ones, both taken as classes, and also when one compares the wild and cultivated forms of particular plants: thus the wild olive pear and fig are longer-lived than the corresponding cultivated trees; for the wild forms of these are stronger and of closer growth, and they do not produce such well developed fruit-pulp.

4.13.2

To the long-lived character of some plants, both cultivated and wild, witness is borne also by the tales handed down in mythology, as of the olive at Athens, the palm in Delos, and the wild olive at Olympia, from which the wreaths for the games are made; or again of the Valonia oaks at Ilium, planted on the tomb of Ilos. Again some say that Agamemnon planted the plane at Delphi, and the one at Kaphyai in Arcadia. Now how this is may perhaps be another story, but anyhow it is plain that there is a great difference between trees in this respect; the kinds that have been mentioned, and many others besides, are long-lived, while the following are admittedly short-lived — pomegranate, fig, apple: and among apples the ‘spring’ sort and the ‘sweet’ apple are shorter-lived than the ‘sour’ apple, even as the ‘stoneless’ pomegranate is shorter-lived than the other kinds. Also some kinds of vine are shortlived, especially those which bear much fruit; and it appears that trees which grow by water are shorter-lived than those which live in dry places: this is true of willow, abele, elder, and black poplar.

4.13.3

Some trees, though they grow old and decay quickly, shoot up again from the same stock, as bay apple pomegranate and most of the water-loving trees. About these one might enquire whether one should call the new growth the same tree or a new one; to take a similar case, if, after cutting down the trunk, one should, as the husbandmen do, encourage — the new shoots to grow again, or if one should cut the tree right down to the roots and burn the stump, (for these things are commonly done, and they also sometimes occur naturally); are we then here too, to call the new growth the same tree, or another one? In so far as it is always the parts of the tree which appear to alternate their periods of growth and decay and also the prunings which they themselves thus make, so far the new and the old growth might seem to be the same tree; for what difference can there be in the

one as compared with the other? On the other hand, in so far as the trunk would seem to be above all the essential part of the tree, which gives it its special character, when this changes, one might suppose that the whole tree becomes something different — unless indeed one should lay down that to have the same starting-point constitutes identity; whereas it often happens that the roots too are different and undergo a change, since some decay and others grow afresh. For if it be true, as some assert, that the reason why the vine is the longest lived of trees, is that, instead of producing new roots, it always renews itself from the existing ones, such an illustration must surely lead to an absurd conclusion, unless we assume that the stock persists, as it must do, since it is, as it were, the fundamental and essential part of a tree. However it cannot matter much for our present purpose which account is the right one. Perhaps we may say that the longest-lived tree is that which in all ways is able to persist, as does the olive by its trunk, by its power of developing sidegrowth, and by the fact that its roots are so hard to destroy. It appears that the life of the individual olive (in regard to which one should make the trunk the essential part and standard in estimating the time), lasts for about two hundred years. But if it is true of the vine, as some say, that, if the roots are partly removed, the trunk is able to survive, and the whole character of the tree remains the same and produces like fruits for any period, however long, then the vine will be the longest-lived of all trees. They say that, when the vine seems to be deteriorating, this is what one should do: — one should encourage the growth of branches and gather the fruit that year: and after that one should dig on one side of the vine and prune away all the roots on that side, and then fill the hole with brushwood and heap up the soil. In that year, they say, the vine bears very badly, but better in the next, while in the third and fourth it becomes normal again and bears many fair clusters, so that it is quite as good as when it was in its prime.

And when it goes off again, they say one should dig on the other side and apply the same treatment; and that so treated the tree lasts for ever; and this should be done at intervals of about ten years. And this is why those who adopt this treatment never cut down the vine, but the same stems remain for many generations, so that even those who planted the trees cannot remember doing so. However perhaps one should enquire of those who have had experience before accepting this statement. These examples may serve for considering which trees are long-lived and which short-lived.

4.14.1

As to diseases — they say that wild trees are not liable to diseases which destroy them, but that they get into poor condition, and that most obviously when they are smitten with hail when either they are about to bud or are just budding or are in bloom; also when either a cold or a hot wind comes at such seasons: but that from seasonable storms, even if they be violent, they take no hurt, but rather that it is good for them all to be exposed to weather: for, unless they are, they do not grow so well. Cultivated kinds however, they say, are subject to various diseases, some of which are, one may say, common to all or to most, while others are special to particular kinds. General diseases are those of being worm-eaten, of being sun-scorched, and rot. All trees, it may be said, have worms, but some less, as fig and apple, some more, as pear. Speaking generally, those least liable to be worm-eaten are those which have a bitter acrid juice, and these are also less liable to sunscorch. Moreover this occurs more commonly in young trees than in those which have come to their strength, and most of all it occurs in the fig and the vine.

4.14.2

The olive, in addition to having worms (which destroy the fig too by breeding in it), produces also a 'knot' (which some call a fungus, others a barkblister -), and it resembles the effect of

sun-scorch. Also sometimes young olives are destroyed by excessive fruitfulness. The fig is also liable to scab, and to snails which cling to it. However this does not happen to figs everywhere, but it appears that, as with animals, diseases are dependent on local conditions; for in some parts, as about Aineia, the figs do not get scab.

4.14.3

The fig is also often a victim to rot and to *krados*. It is called rot when the roots turn black, it is called *krados* when the branches do so; for some call the branches *kradoi* (instead of *kladoi*), whence the name is transferred to the disease. The wild fig does not suffer from *krados* rot or scab, nor does it get so worm-eaten in its roots' as the cultivated tree; indeed some wild figs do not even shed their early fruit- not even if they are grafted into a cultivated tree.

4.14.4

Scab chiefly occurs when there is not much rain after the rising of the Pleiad; if rain is abundant, the scab is washed off, and at such times it comes to pass that both the spring and the winter figs drop off. Of the worms found in fig-trees some have their origin in the tree, some are produced in it by the creature called the 'horned worm'; but they all turn into the 'horned worm'; — and they make a shrill noise. The fig also becomes diseased if there is heavy rain; for then the parts towards the root and the root itself become, as it were, sodden, and this they call 'bark-shedding.' The vine suffers from over-luxuriance; this, as well as sun-scorch, specially happens to it either when the young shoots are cut by winds, or when it has suffered from bad cultivation, or, thirdly, when it has been pruned upwards.'

4.14.5

The vine becomes a 'shedder,' a condition which some call 'casting of the fruit,' if the tree is snowed upon at the time when the blossom falls, or else when it becomes over lusty; what happens is that the unripe grapes drop off, and those that

remain on the tree are small. Some trees also contract disease from frost, for instance the vine; for then the eyes of the vine that was pruned early become abortive; and this also happens from excessive heat, for the vine seeks regularity in these conditions too, as in its nourishment. And in general anything is dangerous which is contrary to the normal course of things.

4.14.6

Moreover the wounds and blows inflicted by men who dig about the vines render them less able to bear the alternations of heat and cold; for then the tree is weak owing to the wounding and to the strain put upon it, and falls an easy prey — to excess of heat and cold. Indeed, as some think, most diseases may be said to be due to a blow; for that even the diseases known as ‘sun-scorch’ and ‘rot’ occur because the roots have suffered in this way. In fact they think that there are only these two diseases; but there is not general agreement on this. The spring apple and especially the sweet form of it, has the weakest constitution.

4.14.7

Some mutilations however do not cause destruction of the whole tree, but only produce barrenness; for instance, if one takes away the top of the Aleppo pine or the date-palm, the tree in both cases appears to become barren, but not to be altogether destroyed.

4.14.8

There are also diseases of the fruits themselves, which occur if the winds and rains do not come in due season. For it comes to pass that sometimes trees, figs, for example, shed their fruit when rain does or does not come, and sometimes the fruit is spoilt by being rotted and so choked off, or again ly being unduly dried up. It is worst of all for some trees, as olive and vine, if rain falls on them as they are dropping their blossom; for then the fruit, having no strength, drops also.

4.14.9

In Miletus the vines at the time of flowering are eaten by caterpillars, some of which devour the flowers, others, a different kind, the leaves; and they strip the tree; these appear if there is a south wind and sunny weather; if the heat overtakes them, the trees split.

4.14.10

About Taras the olives always shew much fruit, but most of it perishes at the time when the blossom falls.³ Such are the drawbacks special to particular regions.

4.14.11

There is also another disease incident to the olive, which is called cobweb; for this forms on the tree and destroys the fruit. Certain hot winds also scorch both olive vine-cluster and other fruits. And the fruits of some get worm-eaten, as olive, pear, apple, medlar, pomegranate. Now the worm which infests the olive, if it appears below the skin, destroys the fruit; but if it devours the stone it is beneficial. And it is prevented from appearing under the skin if there is rain after the rising of Arcturus. Worms also occur in the fruit which ripens on the tree, and these are more harmful as affecting the yield of oil. Indeed these worms seem to be altogether rotten; wherefore they appear when there is a south wind and particularly in damp places. The knips also occurs in certain trees, as the oak and fig, and it appears that it forms from the moisture which collects under the bark, which is sweet to the taste. Worms also occur in some pot-herbs, as also do caterpillars, though the origin of these is of course different.

4.14.12

Such are in general the diseases, and the plants in which they occur. Moreover there are certain affections due to season or situation which are likely to destroy the plant, but which one would not call diseases: I mean such affections as freezing and what some call 'scorching.' Also there are winds which blow in particular districts that are likely to destroy or scorch; for

instance the 'Olympian' wind of Chalcis in Euboea, when it blows cold a little before or after the winter solstice; for this wind scorches up the trees and makes them more dry and withered than they would become from the sun's heat even in a long period; wherefore its effect is called 'scorching.' In old times it occurred very frequently, and it recurred with great violence in the time of Archippus, after an interval of forty years.

4.14.13

The places which suffer most in this way are hollow places, valleys, the ground near rivers, and, in general, places which are least open to wind; the tree which suffers most is the fig, and next to that the olive. The wild olive, being stronger, suffered more than the cultivated tree, which was surprising. But the almonds were altogether unscathed, as also were the apples pears and pomegranates; wherefore this too was a surprising fact. The tree gets scorched by this wind right down to the trunk, and in general the upper are caught more and earlier than the lower parts. The effects are seen partly at the actual produce a good crop of fruit; but, I imagine, they have not been so well observed.

4.15.1

Next we must mention what trees perish when certain parts are removed. All perish alike, if the bark is stripped off all round; one may say that every tree, except the andrachne, - perishes under these circumstances; and this tree does so also, if one does violence to the flesh, and so breaks off the new growth which is forming. However one should perhaps except the cork-oak; for this, they say, is all the stronger if its bark is stripped off, that is, the outer bark and also that which lies below it next the flesh — as with the andrachne. For the bark is also stripped from the bird-cherry the vine and the lime (and from this the ropes are made), and, among smaller plants, from the mallow; but in these cases it is not the real nor the first bark

which is taken, but that which grows above that, which even of its own accord sometimes falls off because fresh bark is forming underneath.

4.15.2

In fact some trees, as andrachne and plane, have a bark which cracks. As some think, in many cases a new bark forms underneath, while the outer bark withers and cracks and in many cases falls off of its own accord; but the process is not so obvious as it is in the above mentioned cases. Wherefore, as they think, all trees are destroyed by stripping the bark, though the destruction is not in all cases equally rapid or complete. Some in fact, as fig, lime, and oak, survive for some time; indeed some say that these recover, and also the elm and date-palm, and that the bark even of the lime almost entirely closes up again, while in other trees it forms as it were a callus and acquires a peculiar new character. Men try to help the tree by plastering it with mud and tying pieces of bark reeds or something of the kind about it, so that it may not take cold nor become dried up. And they say that the bark has been known to grow again; — for instance that that of the fig-trees at the Trachinian Heraclea did so. However this does not only depend on the quality of the soil and on the climate; the other circumstances which ensue must also be favourable; for, if great cold or heat ensues, the tree perishes at once. The season also makes a difference. For if one strips the bark of a silver-fir or fir at the time when the buds are shooting during Thargelion or Skirrophorion, at which season it is separable, the tree dies at once. If it is done however in winter, the tree holds out longer; and this is especially true of the strongest trees, such as kermes-oak and oak; these it takes longer to kill. However the piece stripped off must be of a certain breadth to cause the death of the tree, especially in the case of the strongest trees; for, if one does it only a little, it is not surprising that the tree should not be killed; though some indeed say that, if it is done at

all, the tree certainly dies; this however is probably true only of the weaker kinds. For some, they say, if they are in bad barren soil, die even if the bark is not stripped all round. This then, as has been said, is a universal cause of death.

4.16.1

The process which is called topping of trees is fatal only to fir, silver-fir, Aleppo pine, and datepalm, though some add prickly cedar and cypress. These, if they are stripped of their foliage at the top and the crown is cut off, perish wholly and do not shoot again, as is the case with some, if not with all, if they are burnt. But all other trees shoot again after being lopped, and some, such as the olive, become all the fairer. However most trees perish if the stem is split; for no tree seems able to stand this, except vine, fig, pomegranate, and apple; and some perish even if they are wounded severely and deeply. Some however take no harm from this, as the fir when it is cut for tar, and those trees from which the resins are collected, as silver-fir and terebinth; though these trees are in fact then deeply wounded and mangled. Indeed they actually become fruitful instead of barren, or are made to bear plentifully instead of scantily.

4.16.2

Some trees again submit to being hewn both when they are standing and when they have been blown down, so that they rise up again and live and shoot, for instance the willow and the plane. This was known to happen in Antandros and at Philippi; a plane in Antandros having fallen and had its boughs lopped off and the axe applied to its trunk, grew again in the night when thus relieved of the weight, and the bark grew about it again. It happened that it had been hewn two thirds of the way round; it was a large tree, more than ten cubits high, and of such girth that four men could not easily have encircled it. The willow at Philippi which grew again had had its branches lopped off, but the trunk had not been hewn. A certain seer persuaded the people to offer sacrifice and take care of the tree, since what

had occurred was a good omen. Also at Stageira an abele in the school gardens which had fallen got up again.

4.16.3

Hardly any tree is destroyed by taking out the core; a proof of which is the fact that many large trees are hollow. The people of Arcadia say that the tree under these circumstances lives for a time, but that, if the tree is entirely deprived of its core, fir or silver-fir or any other tree perishes.

4.16.4

All trees alike are destroyed when the roots are cut off, whether all or most of them, if those removed are the largest and the most essential to life. Such then are the causes of death which come from the removal of a part of the tree.

4.16.5

On the other hand the destruction which oil causes is due rather to a kind of addition than to removal; for oil is hostile to all trees, and so men pour it over what remains of the roots. However oil is more potent with young trees which are just growing; for then they are weaker; wherefore men do not allow them to be touched at that time.

4.16.6

Again trees may destroy one another, by robbing them of nourishment and hindering them in other ways. Again an overgrowth of ivy is dangerous, and so is tree-medick, for this destroys almost anything. But halimon is more potent even than this, for it destroys tree-medick.

4.16.7

Again some things, though they do not cause death, enfeeble the tree as to the production of flavours and scents; thus cabbage and sweet bay have this effect on the vine. For they say that the vine scents the cabbage and is infected by it. Wherefore the vine-shoot,- whenever it comes near this plant, turns back and looks away, as though the smell were hostile to it. Indeed

Androkydes used this fact as an example to demonstrate the use of cabbage against wine, to expel the fumes of drunkenness; for, said he, even when it is alive, the vine avoids the smell. It is now clear from what has been said how the death of a tree may be caused, how many are the causes of death, and in what several ways they operate.

5.1.1

BOOK V. THE TIMBER OF VARIOUS TREES AND ITS USES.

In like manner we must endeavour to speak of timber, saying of what nature is that of each tree, what is the right season for cutting it, which kinds are hard or easy to work, and anything else that belongs to such an enquiry. Now these are the right seasons for cutting timber: — for ‘round’ timber and that whose bark is to be stripped the time is when the tree is coming into leaf. For then the bark is easily stripped (which process they call ‘peeling’) because of the moisture which forms beneath it. At a later time it is hard to strip, and the timber obtained is black and uncomely. However square logs can be cut after the time of peeling, since trimming with the axe removes the uncomeliness. In general any wood is at the best season as to strength when it has not merely ceased coming into leaf, but has even ripened its fruit; however on account of the bark-stripping it comes to pass that ‘round’ timber is in season when it is cut before it is ripe, so that, as it happens, the seasons are here reversed. Moreover the wood of the silver-fir is of a better colour at the time of the first peeling.

5.1.2

But since they strip the bark of hardly any trees except silver-fir, fir and pine, these trees are cut in the spring; for then is the time of coming into leaf. Other trees are cut sometimes after wheat-harvest, sometimes after the vintage and the rising of Arcturus, as aria (holm-oak), elm, maple, manna-ash, zygia, beech, lime, Valonia oak, and in general all those whose timber

is for underground use. The oak is cut latest of all, in early winter at the end of autumn. If it is cut at the time of peeling, it rots almost more quickly than at any other time, whether it has the bark on or not. This is especially so if it is cut during the first peeling, less so during the second, and least during the third. What is cut after the ripening of the fruit remains untouched by worms, even if it has not peeled: however worms get in under the bark and mark the surface of the stem, and such marked pieces of wood some use as seals. Oak-wood if cut in the right season does not rot and is remarkably free from worms, and its texture is hard and close like horn; for it is like the heart of a tree throughout, except that that of the kind called sea-bark oak is even at that time of poor quality.

5.1.3

Again, if the trees are cut at the time of coming into leaf, the result is the opposite of that which follows when they are cut after fruiting: for in the former case the trunks dry up and the trees do not sprout into leaf, whereas after the time of fruiting they sprout at the sides. At this season however they are harder to cut because the wood is tougher. It is also recommended to do the cutting when the moon has set, since then the wood is harder and less likely to rot. But, since the times when the fruit ripens are different for different trees, it is clear that the right moment for cutting also differs, being later for those trees which fruit later.

5.1.4

Wherefore some try to define the time for the cutting of each tree; for instance for fir and silverfir the time is, they say, when they begin to peel: for beech lime maple and zygia in autumn; for oak, as has been said, when autumn is past. Some however say that the fir is ripe for cutting in spring, when it has on it the thing called 'catkin,' and the pine when its 'cluster' is in bloom. Thus they distinguish which trees are ripe for cutting at various times; however it is clear that in all cases the wood is better

when the tree is in its prime than when it is quite young or has grown old, the wood of quite young trees being too succulent, and that of old ones too full of mineral matter.

5.1.5

Silver-fir and fir are the most useful trees and in the greatest variety of ways, and their timber is the fairest and largest. Yet they differ from one another in many respects; the fir is fleshier and has few fibres, while the silver-fir has many fibres and is not fleshy, so that in respect of each component it is the reverse of the other, having stout fibres but soft flesh of open texture. Wherefore the timber of the one is heavy, of the other light, the one being resinous, the other without resin; wherefore also it is whiter.

5.1.6

Moreover the fir has more branches, but those of the silver-fir are much tougher, or rather they are tougher than those of any other tree; — the branches of both however are of close texture, horny, and in colour brown and like resin-glutted wood. When the branches of either tree are cut, sap streams from them for a considerable time, but especially from those of the silver-fir. Moreover the wood of the silver-fir has many layers, like an onion: there is always another beneath that which is visible, and the wood is comprised of such layers throughout.

5.1.7

Wherefore, when men are shaving this wood to make oars, they endeavour to take off the several coats one by one evenly: for, if they do this, they get a strong spar, while if they do the work irregularly and do not strip off the coats evenly, they get a weak one; for the process in this case is hacking instead of stripping. The silver-fir also gives timber of the greatest lengths and of the straightest growth; wherefore yard-arms and masts are made from it. Also the vessels and fibre are more clearly seen in it than in any other tree.

5.1.8

At first it grows in height only, until it has reached — the sunshine; and so far there is no branch nor sidegrowth nor density of habit; but after that the tree proceeds to increase in bulk and density of habit, as the outgrowing branches and sidegrowths develop.

5.1.9

These are the characteristics peculiar to the silverfir. Others it shares with the fir and the other trees of this class. For instance, sometimes a tree is ‘four-cleft’ sometimes ‘two-cleft’; it is called ‘fourcleft’ when on either side of the heart-wood there are two distinct and diverse lines of fissure: in that case the blows of the axe follow these lines in cases where the hewing is stopped short on either side of the heart-wood. For the nature of the lines of fissure compels the hewing to take this course. Silver-firs or firs thus formed are said to be ‘four-cleft.’ And these are also the fairest trees for carpentry, their wood being the closest and possessing the aigis.

5.1.10

Those which are ‘two-cleft’ have one single line of fissure on either side of the heart-wood, and the lines of fissure do not correspond to each other, so that the hewing also is performed by cuts which follow the two lines of fissure, so as to reach the two sides of the heart-wood at different angles. Now such wood, they say, is the softest, but the worst for carpentry, as it warps most easily. Those trees which have only a single continuous line of fissure are said to be ‘one-cleft,’ though here too the cutting is done from either side of the heart-wood: and such wood has, they say, an open texture, and yet it is not at all apt to warp.

5.1.11

There are also differences in the bark, by observation of which they can tell at once what the timber of the tree is like as it stands. For if the timber has straight and not crooked lines of fissure, the bark also is smooth and regular, while if the timber

has the opposite character, the bark is rough and twisted; and so too is it with other points. However few trees are 'four-cleft,' and most of those which are not are 'one-cleft.' All wood, as was said before, which grows in a position facing north, is bigger, more erect, of straighter grain, tougher, and in general fairer and more abundant. Moreover of an individual tree the wood on the northward side is closer and more vigorous. But if a tree stands sideways to the north — with a draught round it, the north wind by degrees twists and contorts it, so that its core becomes twisted instead of running straight.

5.1.12

The timber of such a tree while still in one piece is strong, but, when cut, it is weak, because the grain slants across the several pieces. Carpenters call such wood 'short lengths,' because they thus cut it up for use. Again in general wood which comes from a moist, sheltered, shady or confined position is inferior both for carpentry and for fuel. Such are the differences, generally speaking, between trees of the same kind as they are affected by situation.

5.2.1

Some indeed make a distinction between regions and say that the best of the timber which comes into Hellas for the carpenter's purposes is the Macedonian, for it is smooth and of straight grain, and it contains resin: second best is that from Pontus, third that from the Rhyndakos, fourth that of the country of the Ainianes, worst is that of Parnassus and that of Euboea, for it is full of knots and rough and quickly rots. As to Arcadian timber the case is doubtful.

5.2.2

The strongest wood is that which is without knots and smooth, and it is also the fairest in appearance. Wood becomes knotty when it has been ill nourished and has suffered severely whether from winter or some such cause; for in general a knotty habit is supposed to indicate lack of nourishment. When

however, after being ill nourished, the tree recovers and becomes vigorous, the result is that the knots are absorbed by the growth which now covers them; for the tree, being now well fed and growing vigorously, recovers, and often the wood is smooth outside, though when split it is seen to have knots. And this is why they examine the core of wood that has been split; for, if this contains knots, the outward parts will also be knotty, and these knots are harder to deal with than the outer ones, and are easily recognised.

5.2.3

‘Coiling’ of the wood is also due to winter or ill nourishment. Wood is said to ‘coil’ when there is in it closer twisting than usual, made up of an unusual number of rings: this is not quite like a knot, nor is it like the ordinary curling of the wood, which runs right through it and is uniform. ‘Coiling’ is much more troublesome and difficult to deal with than knots; it seems to correspond to the so-called ‘centres’ which occur in marbles. That vigorous growth covers up the knots is plain from simple observation of the fact and also from other similar instances. For often some part of the tree itself is absorbed by the rest of the tree which has grown into it; and again, if one makes a hole in a tree and puts a stone into it or some other such thing, it becomes buried, being completely enveloped by the wood which grows all round it: this happened with the wild olive in the market-place at Megara; there was an oracle that, if this were cut open, the city would be taken and plundered, which came to pass when Demetrius took it. For, when this tree was split open, there were found greaves and certain other things of Attic workmanship hanging there, the hole in the tree having been made at the place where the things were originally hung on it as offerings. Of this tree a small part still exists, and in many other places further instances have occurred. Moreover, as has been said, such occurrences happen also with various other trees.

5.3.1

Corresponding to the individual characters of the several trees we have the following kinds of differences in the wood: — it differs in closeness, heaviness, hardness or their opposites, and in other similar ways; and these differences are common to cultivated and wild trees. So that we may speak of all trees without distinction.

5.3.2

Box and ebony seem to have the closest and heaviest wood; for their wood does not even float on water. This applies to the box-tree as a whole, and to the core of the ebony, which contains the black pigment. The nettle-tree also is very close and heavy, and so is the core of the oak, which is called 'heart of oak,' and to a still greater degree this is true of the core of laburnum; for this seems to resemble the ebony.

5.3.3

The wood of the terebinth is also very black and close-grained; at least in Syria they say that it is blacker than ebony, that in fact they use it for making their dagger handles; and by means of the lathe chisel they also make of it 'Theriklean' cups, so that no one could distinguish these from cups made of pottery; for this purpose they use, it is said, the heart-wood, but the wood has to be oiled, for then it becomes comelier and blacker.

5.3.4

There is also, they say, another tree which, as well as the black colour, has a sort of reddish variegation, so that it looks like variegated ebony, and of it are made beds and couches and other things of superior quality. This tree is very large and has handsome leaves and is like the pear.

5.3.5

These trees then, as well as the black colour, have close wood; so also have maple, zygia and in general all those that are of compact growth; so also have the olive and the wild olive, but their wood is brittle. Of wild trees which are used for

rooftimbers the wood of the silver-fir is the least compact, and among others that of the elder, fig, apple, and bay. The hardest woods are those of the oak, zygia, and aria (holm-oak); in fact men wet these to soften them for boring holes. In general, woods which are of open porous texture are soft, and of those of fleshy texture the softest is the lime. The last-named seems also to be the hottest; the proof of which is that it blunts iron tools more than any other; for they lose their edge — by reason of its heat.

5.3.6

Ivy and bay are also hot woods, and so in general are those used for making fire-sticks; and Menestor adds the wood of the mulberry. The coldest woods are those which grow in water and are of succulent character. The wood again of willow and vine is tough; wherefore men make their shields of these woods; for they close up again after a blow; but that of the willow is lighter, since it is of less compact texture; wherefore they use this for choice. The wood of the plane is fairly tough, but it is moister in character, as also is that of the elm. A proof of this is that, if it is set upright after being cut, it discharges much water. The wood of the mulberry is at once of close grain and tough.

5.3.7

The wood of the elm is the least likely to warp; wherefore they make the ‘hinges’ of doors out of elm wood; for, if these hold, the doors also keep in place; otherwise they get wrenched out of place, rhey make the ‘hinges’ by putting wood from the root above and wood ‘from the foliage’ below, thus reversing the natural position: (by wood ‘from the foliage’ joiners mean the upper wood). For, when these are fitted the one into the other, each counteracts the other, as they naturally tend in opposite directions: whei’eas, if the wood were set — as it grows, all the parts would give where the strain came.

5.3.8

(They do not finish off the doors at once; but, when they have put them together, stand them up, and then finish them off the next year, or sometimes the next year but one, if they are doing specially good work. For in summer, as the wood dries, the work comes apart, but it closes in winter. The reason is that the open fleshy texture of the wood of the silver-fir drinks in the air, which is full of moisture.)

5.3.9

Palm-wood is light easily worked and soft like cork-oak, but is superior to that wood, as it is tough, while the other is brittle. Wherefore men now make their images of palm-wood and have given up the wood of cork-oak. However the fibres do not run throughout the wood, nor do they run to a good length, nor are they all set symmetrically, but run in every direction. The wood dries while it is being planed and sawn.

5.3.10

Thyon (thyine wood), which some call *thya*, grows near the temple of Zeus Ammon and in the district of Cyrene. In appearance the tree is like the cypress alike in its branches, its leaves, its stem, and its fruit; or rather it is like a wild cypress. There is abundance of it where now the city stands, and men can still recall that some of the roofs in ancient times were made of it. For the wood is absolutely proof against decay, and the root is of very compact texture, and they make of it the most valuable articles. Images are carved from these woods, prickly cedar cypress nettle-tree box, and the small ones also from the roots of the olive, which are unbreakable and of a more or less uniformly fleshy character. The above facts illustrate certain special features of position, natural character and use.

5.4.1

Difference in weight is clearly to be determined by closeness or openness of texture, dampness or dryness, degree of glutinousness, hardness or softness. Now some woods are both

hard and heavy, as box and oak, while those that are brittle and hardest owing to their dryness, are not heavy. All wood of wild trees, as we have said before, is closer harder heavier, and in general stronger than that of the cultivated forms, and there is the same difference between the wood of 'male' and of 'female' trees, and in general between trees which bear no fruit and those which have fruit, and between those which bear inferior fruit and those whose fruit is better; on the other hand occasionally the 'male' tree is the more fruitful, for instance, it is said, the cypress, the cornelian cherry and others. However of vines it is clear that those which bear less fruit have also more frequent knots and are more solid,- and so too with apples and other cultivated trees.

5.4.2

Naturally proof against decay are cypress, prickly cedar, ebony nettle-tree, box olive, wild olive resinous fir, aria (holm-oak) oak sweet chestnut. Of these the wood of the cypress seems to last longest; at least the cypress-wood at Ephesus, of which the doors of the modern temple were made, lay stored up for four generations. And this is the only wood which takes a fine polish, wherefore they make of it valuable articles. Of the others the least liable to decay after the wood of the cypress and thyme-wood is, they say, that of the mulberry, which is also strong and easily worked: when it becomes old, this wood turns black like that of the nettle-tree.

5.4.3

Again whether a given wood is not liable to decay may depend on the purpose to which it is put and the conditions to which it is subjected: thus the elm does not decay if exposed to the air, nor the oak if it is buried or soaked in water; for it appears to be entirely proof against decay: wherefore they build vessels of it for use on rivers and on lakes, but in seawater it rots, though other woods last all the better; which is natural, as they become seasoned with the brine.

5.4.4

The beech also seems to be proof against decay in water and to be improved by being soaked. The sweet chestnut under like treatment is also proof against decay. They say that the wood of the fir is more liable to be eaten by the teredon than that of the silver-fir; for that the latter is dry, while the fir has a sweet taste, and that this is more so, the more the wood is soaked with resin; they go on to say that all woods are eaten by the teredon except the olive, wild or cultivated, and that these woods escape because of their bitter taste. Now woods which decay in sea-water are eaten by the teredon, those which decay on land by the skolex and thrips; for the teredon does not occur except in the sea. It is a creature small in size, but has a large head and teeth; the thrips resembles the skolex, and these creatures gradually bore through timber. The harm that these do is easy to remedy; for, if the wood is smeared with pitch, it does not rot in water when it is dragged down into the sea; but the harm done by the teredon cannot be undone. Of the skolekes which occur in wood some come from the decay of the wood itself, some from other skolekes which engender therein. For these produce their young in timber, as the worm called the 'horned worm' does in trees, having bored and scooped out a sort of mouse-hole by turning round and round. But it avoids wood which has a strong smell or is bitter or hard, such as boxwood, since it is unable to bore through it. They say too that the wood of the silver-fir, if barked just before the time of budding, remains in water without decaying, and that this was clearly seen at Pheneos in Arcadia, when their plain was turned into a lake since the outlet was blocked up. For at that time they made their bridges of this wood, and, as the water rose, they placed more and more atop of them, and, when the water burst its way through and disappeared, all the wood was found to be undecayed. This fact then became known by means of an accident. In the island of Tylos off the Arabian coast they say

that there is a kind of wood of which they build their ships, and that in sea-water this is almost proof against decay; for it lasts more than 200 years if it is kept under water, while, if it is kept out of water, it decays sooner, though not for some time. They also tell of another strange thing, though it has nothing to do with the question of decay: they say that there is a certain tree, of which they cut their staves, and that these are very handsome, having a variegated appearance like the tiger's skin; and that this wood is exceedingly heavy, yet when one throws it down on hard ground it breaks in pieces like pottery.

5.4.5

Moreover, the wood of the tamarisk is not weak there, as it is in our country, but is as strong as kermes-oak or any other strong wood. Now this illustrates also the difference in properties caused by country and climate. Moreover when wood, such as that of oak or fir, is soaked in brine — not all being soaked at the same depth in the sea, but some of it close to shore, some rather further out, and some at a still greater depth — in all cases the parts of the tree nearest the root (whichever tree it is) sink quicker under water, and even if they float, have a greater tendency to sink.

5.5.1

Some wood is easy to work, some difficult. Those woods which are soft are easy, and especially that of the lime; those are difficult which are hard and have many knots and a compact and twisted grain. The most difficult woods are those of aria (holm-oak) and oak, and the knotty parts of the fir and silver-fir. The softer part of any given tree is always better than the harder, since it is fleshier: and carpenters can thus at once mark the parts suitable for planks. Inferior iron tools can cut hard wood better than soft: for on soft wood tools lose their edge, as was said in speaking of the lime, while hard woods actually sharpen it: wherefore cobblers make their strops of wild pear.

5.5.2

Carpenters say that all woods have a core, but that it is most plainly seen in the silver-fir, in which one can detect a sort of bark-like character in the rings. In olive box and such woods this is not so obvious; wherefore they say that box and olive lack this tendency; for that these woods are less apt to 'draw' than any others. 'Drawing' is the closing in of the wood as the core is disturbed. For since the core remains alive, it appears, for a long time, it is always removed from any article whatever made of this wood, but especially from doors,' so that they may not warp: and that is why the wood is split.

5.5.3

It might seem strange that in 'round' timber the core does no harm and so is left undisturbed, while in wood whose texture has been interfered with, unless it is taken out altogether, it causes disturbance and warping: it were rather to be expected that it would die when exposed. Yet it is a fact that masts and yard-arms are useless, if it has been removed from the wood of which they are made. This is however an accidental exception, because the wood in question has several coats,- of which the strongest and also thinnest is the outermost, since this is the driest, while the other coats are strong and thin in proportion to their nearness to the outermost. If therefore the wood be split, the driest parts are necessarily stripped off. Whether however in the other case the object of removing the core is to secure dryness is matter for enquiry. However, when the core 'draws' it twists the wood, whether it has been split or sawn, if the sawing is improperly performed: the saw-cut should be made straight and not slantwise. Thus, if the core be represented by the line A, the cut must be made along the line BD, and not along the line BC: for in that case, they say, the core will be destroyed, while, if cut in the other way, it will live. For this reason men think that every wood has a core: for it is clear that those which do not seem to possess one nevertheless have

it, as box nettle-tree kermes-oak: a proof of this is the fact that men make of these woods the pivots of expensive doors, and accordingly the headcraftsmen specify that wood with a core shall not be used. This is also a proof that any core ‘draws,’ even those of the hardest woods, which some call the heart. In almost every wood, even in that of the silver-fir, the core is the hardest part, and the part which has the least fibrous texture: — it is least fibrous because the fibres are far apart and there is a good deal of fleshy matter between them, while it is the hardest part because the fibres and the fleshy substance are the hardest parts. Wherefore the headcraftsmen specify that the core and the parts next it are to be removed, that they may secure the closest and softest part of the wood.

5.5.4

Timber is either ‘cleft,’ ‘hewn,’ or ‘round’: it is called ‘cleft,’ when in making division they saw it down the middle, ‘hewn’ when they hew off the outer parts, while ‘round’ clearly signifies wood which has not been touched at all. Of these, ‘cleft’ wood is not at all liable to split, because the core when exposed dries and dies: but ‘hewn’ and ‘round’ wood are apt to split, and especially ‘round’ wood, because the core is included in it: no kind of timber indeed is altogether incapable of splitting. The wood of the nettle-tree and other kinds which are used for making pivots for doors are smeared with cow-dung to prevent their splitting: the object being that the moisture due to the core may be gradually dried up and evaporated. Such are the natural properties of the core.

5.6.1

For bearing weight silver-fir and fir are strong woods, when set slantwise: for they do not give like oak and other woods that contain mineral matter but make good resistance. A proof of this is that they never split like olive and oak, but decay first or fail in some other way. Palm-wood is also strong, for it bends the opposite way to other woods: they bend downwards,

palm-wood upwards. It is said that fir and silver-fir also have an upward thrust. As to the sweet chestnut, which grows tall and is used for roofing, it is said that when it is about to split, it makes a noise, so that men are forewarned: this occurred once at Antandros at the baths, and all those present rushed out. Fig-wood is also strong, but only when set upright.

5.6.2

The wood of the silver-fir may be called the strongest of all. But for the carpenter's purposes fir best takes glue because of its open texture and the straightness of its pores; for they say that it never by any chance comes apart when it is glued. Alaternus is the easiest wood for turning, and its whiteness is like that of the holly. Of the rest lime is the easiest, the whole tree, as was said, being easy to work because of the softness of the wood. In general those woods which are tough are easy to bend. The mulberry and the wild fig seem to be specially so; wherefore they make of these theatre-seats, the hoops of garlands, and, in general, things for ornament.

5.6.3

Woods which have a fair amount of moisture in them are easier to saw or split than those which are altogether dry: for the latter give, while the former resist. Wood which is too green closes up again when sawn, and the sawdust catches in the saw's teeth and clogs them; wherefore the teeth of the saw are set alternate ways, to get rid of the sawdust. Wood which is too green is also harder to bore holes in; for the auger's dust is only brought up slowly, because it is heavy; while, if the wood is dry, the air gets warmed by the boring and brings it up readily and at once. On the other hand, wood which is over dry is hard to saw because of its hardness: for it is like sawing through earthenware; wherefore they wet the auger when using it.

5.6.4

However green wood is easier to work with the axe the chisel or the plane; for the chisel gets a better hold and does not slip off.

Again softer woods are easier for the axe and for smoothing, and also a better polished surface is obtained. The cornelian cherry is also a very strong wood, and among the rest elm-wood is the strongest; wherefore, as was said, they make the 'hinges' for doors of elm-wood. Manna-ash and beech have very moist wood, for of these they make elastic bedsteads.

5.7.1

Next we must endeavour to say in a general way, distinguishing the several uses, for which purposes each kind of timber is serviceable, which is of use for ship-building, which for house-building; for these uses extend far and are important. Now silver-fir, fir and Syrian cedar are, generally speaking, useful for ship-building; for triremes and long ships are made of silver-fir, because of its lightness, and merchant ships of fir, because it does not decay; while some make triremes of it also because they are ill provided with silver-fir. The people of Syria and Phoenicia use Syrian cedar, since they cannot obtain much fir either; while the people of Cyprus use Aleppo pine, since their island provides this and it seems to be superior to their fir.

5.7.2

Most parts are made of these woods; but the keel for a trireme is made of oak, that it may stand the hauling; and for merchantmen it is made of fir. However they put an oaken keel under this when they are hauling, or for smaller vessels a keel of beech; and the sheathing is made entirely of this wood. (However oak-wood does not join well with glue on to fir or silver-fir; for the one is of close, the other of open grain, the one is uniform, the other not so; whereas things which are to be made into one piece should be of similar character, and not of opposite character, like wood and stone.)

5.7.3

The work of bentwood for vessels is made of mulberry manna-ash elm or plane; for it must be tough and strong. That made of plane-wood is the worst, since it soon decays. For

triremes some make such parts of Aleppo pine because of its lightness. The cutwater,' to which the sheathing is attached, and the catheads are made of manna-ash mulberry and elm; for these parts must be strong. Such then is the timber used in ship-building.

5.7.4

For house-building a much greater variety is used, silver-fir, fir and prickly cedar; also cypress oak and Phoenician cedar. In fact, to speak generally, any wood is here of service, unless it is altogether weak: for there are various purposes for which different woods are serviceable, just as there are in ship-building. While other woods are serviceable for special articles belonging to various crafts, such as furniture tools and the like, the wood of silver-fir is of use for almost more purposes than any other wood; for it is even used for painters' tablets. For carpentry the oldest wood is the best, provided that it has not decayed; for it is convenient for almost anyone to use. But for ship-building, where bending is necessary, one must use wood which contains more moisture (though, where glue is to be used, drier wood is convenient). For timber-work for ships is set to stand when it is newly made: then, when it has become firmly united, it is dragged down to the water, and then it closes up and becomes watertight,- unless all the moisture has been dried out of it, in which case it will not take the glue, or will not take it so well.

5.7.5

But we must consider for what purposes each several wood is serviceable. Silver-fir and fir, as has been said, are suitable both for ship-building house-building and also for other kinds of work, but silver fir is of use for more purposes than fir. Aleppo pine is used for both kinds of building, but especially for ship-building, yet it soon rots. Oak is used for house-building, for ship-building, and also for underground work; lime for the deck-planks of long ships, for boxes, and for the manufacture of

measures; its bark is also useful for ropes and writing-cases, for these are sometimes made of it.

5.7.6

Maple and zygia are used for making beds and the yokes of beasts of burden: yew for the ornamental work attached — to chests and footstools and the like: kermes-oak for the axles of wheelbarrows and the cross-bars of lyres and psalteries: beech for making waggons and cheap carts: elm for making doors and weasel-traps, and to some extent it is also used for waggon work; pedos for waggon-axles and the stocks of ploughs: andrachne is used for women for parts of the loom: Phoenician cedar for carpenters' work and for work which is either to be exposed to the air or buried underground, because it does not decay. Similarly the sweet chestnut is used, and it is even less likely to decay if it is used for underground work. Box is used for some purposes; however that which grows on Mount Olympus ' is useless, because only short pieces can be obtained and the wood is full of knots. Terebinth is not used, except the fruit and the resin. Alaternus is only useful for feeding sheep; for it is always leafy. Hybrid arbutus is used for making stakes and for burning: holly and Judas-tree for walking-sticks: some also use bay for these; for of this they make light sticks and sticks for old men. Willow is used for shields hampers baskets and the like. We might in like manner add the several uses of the other woods.

5.7.7

Distinction is also made between woods according as they are serviceable for one or other of the carpenter's tools: thus hammers and gimlets are best made of wild olive, but box elm and manna-ash are also used, while large mallets are made of Aleppo pine. In like manner there is a regular practice about each of the other tools. Such are the differences as to the uses of various woods.

5.8.1

Each kind of timber, as was said before, differs according to the place where it grows; in one place nettle-tree, in another the cedar is remarkably fine, for instance in Syria; for in Syria and on its mountains the cedars grow to a surpassing height. ^{5.8.2} For it is told how the Romans once made an expedition to that island with twenty-five ships, wishing to found a city there; and so great was the size of the trees that, as they sailed into certain bays and creeks, they got into difficulties through breaking their masts. And in general it is said that the whole island is thickly wooded and, as it were, one wild forest; wherefore the Romans gave up the idea of founding their city: however some of them made an excursion into the island and cleared away a large quantity of trees from a small area, enough to make a raft with fifty sails; but this broke up in the open sea. Corsica then, whether because of its uncultivated condition or because of its soil and climate, is very superior in trees to other countries.

^{5.8.3}

The country of the Latins is all well watered; the lowland part contains bay, myrtle, and wonderful beech: they cut timbers of it of such a size that they run the whole length of the keel of a Tyrrhenian vessel. The hill country produces fir and silver-fir. The district called by Circe's name is, it is said, a lofty promontory, but very thickly wooded, producing oak, bay in abundance, and myrtle. There, according to the natives, dwelt Circe, and they shew Elpenor's tomb, on which grow myrtles like those used for garlands, though other kinds of myrtle are large trees. Further it is said that the district is a recent addition to the land, and that once this piece of land was an island, but now the sea has been silted up by certain streams and it has become united to the coast, and the size of the 'island' is about eighty furlongs in circumference. There is then much difference in trees, as has been said repeatedly, which is due to the individual character of particular districts.

^{5.9.1}

Next we must state in like manner and endeavour to determine the properties of each kind of timber in relation to making fire. The best charcoal is made from the closest wood, such as *aria* (holm-oak) oak, *arbutus*; for these are the most solid, so that they last longest and are the strongest; wherefore these are used in silver-mines for the first smelting of the ore. Worst of the woods mentioned is oak, since it contains most mineral matter, and the wood of older trees is inferior to that of the younger, and for the same reason that of really old trees is specially bad. For it is very dry, wherefore it sputters as it burns; whereas wood for charcoal should contain sap.

5.9.2

The best charcoal comes from trees in their prime, and especially from trees which have been topped: for these contain in the right proportion the qualities of closeness admixture of mineral matter and moisture. Again better charcoal comes from trees- in a sunny dry position with a north aspect than from those grown in a shady damp position facing south. Or, if the wood used contains a good deal of moisture, it should be of close texture; for such wood contains more sap. And, for the same reason, that which is of closer texture either from its own natural character or because it was grown in a drier spot, is, whatever the kind of tree, better. But different kinds of charcoal are used for different purposes: for some uses men require it to be soft; thus in iron-mines they use that which is made of sweet chestnut when the iron has been already smelted, and in silver-mines they use charcoal of pine-wood: and these kinds are also used by the crafts. Smiths require charcoal of fir rather than of oak: it is indeed not so strong, but it blows up better into a flame, as it is less apt to smoulder: and the flame from these woods is fiercer. In general the flame is fiercer not only from these but from any wood which is of open texture and light, or which is dry': while that from wood which is of close texture or green is more sluggish and dull. The fiercest flame of all is given

by brushwood; but charcoal cannot be made from it at all, since it has not the necessary substance.

5.9.3

They cut and require for the charcoal-heap straight smooth billets: for they must be laid as close as possible for the smouldering process. When they have covered the kiln, they kindle the heap by degrees, stirring it with poles. Such is the wood required for the charcoal-heap.

5.9.4

In general damp wood makes an evil smoke, and for this reason green wood does so: I mean the damp woods which grow in marshy ground, such as plane willow and black poplar: for even vine-wood, when it is damp, gives an evil smoke. So does palm-wood of its own nature, and some have supposed it to give the most evil smoke of all: whence Chaeremon speaks of 'Veins issuing underground from roots of palm with its malodorous smoke. Most pungent is the smoke of fig-wood, whether wild or cultivated, and of any tree which has a curdling juice; the reason lies in the sap; when such wood has been barked and soaked in running water and then dried, it gives as little smoke as any other, and sends up a very soft flame, since its natural moisture also has been removed. The cinders and ashes of such wood are also pungent, and especially, they say, those of almond-wood.

5.9.5

For the crafts requiring a furnace and for other crafts various woods are serviceable according to circumstances. For kindling fig and olive are best: fig, because it is tough and of open texture, so that it easily catches fire and does not let it through, olive, because it is of close texture and oily.

5.9.6

Fire-sticks are made from many kinds of wood, but best, according to Menestor, from ivy: for that flares up most quickly and freely. They say also that a very good fire-stick is made of

the wood which some call traveller's joy; this is a tree like the vine or the 'wild vine' which, like these, climbs up trees. The stationary piece should be made of one of these, the drill of bay; for the active and passive parts of the apparatus should not be of the same wood, but different in their natural properties to start with, one being of active, the other of passive character. Nevertheless they are sometimes made of the same wood, and some suppose that it makes no difference. They are made in fact of buckthorn kemiesoak, lime and almost any wood except olive; which seems surprising, as olive-wood is rather hard and oily; however it is plainly its moisture which makes it less suitable for kindling. The wood of the buckthorn is also good, and it makes a satisfactory stationary piece; for, besides being dry and free from sap it is necessary that this should also be of rather open texture, that the friction may be effectual; while the drill should be one which gets little worn by use. And that is why one made of bay is best; for, as it is not worn by use, it is effective through its biting quality. All fire-sticks take fire quicker and better in a north than in a south wind, and better in an exposed spot than in one which is shut in.

5.9.7

Some woods, such as prickly cedar, exude moisture, and, generally speaking, so do those whose sap is of an oily character; and this is why statues are sometimes said to 'sweat'; for they are made of such woods. That which seems call the menses of Eileithuia, and for the appearance of which they make atonement, forms on the wood of the silver-fir when some moisture gathers on it: the formation is round in shape, and in size about as large as a pear, or a little larger or smaller. Olive-wood is more apt than other woods to produce shoots even when lying idle or made into manufactured articles; this it often does, if it obtains moisture and lies in a damp place; thus the socket of a door-hinge has been known to shoot, and also an oar which was standing in damp earth in an earthenware vessel.

BOOK VI OF UNDER-SHRUBS.

6.1.1

Of the classification of under-shrubs: the wild kinds: the chief distinction that between spinous and spineless.

WE have spoken already of trees and shrubs, and next we must speak of under-shrubs and herbaceous plants and of any other natural classes which are included with these; for instance, cereals come under herbaceous plants.

6.1.3

But first let us tell of under-shrubs, for this class comes near those mentioned above because of its woody character. Now it may be said that with all plants the wild kinds are more abundant than the cultivated, and this is certainly true of the undershrubs. For the cultivated kinds of this class J are not numerous, and consist almost entirely of coronary plants, as rose gilliflower carnation sweet marjoram martagon lily, to which may be added tufted thyme bergamot-mint calamint southernwood. For all these are woody and have small leaves; wherefore they are classed as under-shrubs. This class covers also pot-herbs, such as cabbage rue and others like them. Of these it is perhaps more appropriate to speak under their proper designation, that is, when we come to make mention of coronary plants and pot-herbs. Now let us first speak of the wild kinds. Of these are several classes and subdivisions, which we must distinguish by the characteristics of each sub-division as well as by those of each class taken as a whole.

6.1.4

The most important difference distinguishing class from class which one could find is that between the spineless and the spinous kinds. Again under each of these two heads there are many differences distinguishing kinds and forms, of which we must endeavour to speak severally.

6.1.5

Of spinous kinds some just consist of spines, as asparagus and skorpios; for these have no leaves except their spines. Then there are the spinousleaved plants, as thistle eryngo safflower; these and the like have their spines on the leaves, whence their name. Others again have leaves as well as their spines, as rest-harrow caltrop and pheos? which some call stoibe. Caltrop is also spinous-fruited, having spines on the fruit-vessel. Wherefore this peculiarity marks it off from almost all other plants; though many trees and shrubs have spines on the shoots, as wild pear pomegranate Christ's thorn bramble rose caper. Such are the general distinctions which may be made among spinous plants.

6.1.6

With spineless plants it is not possible to make such 'generic' distinctions; for the variation of the leaves in size and shape is endless, and the differences are not clearly marked; but we must try to distinguish on another principle. There are many classes of such plants and they differ widely, as rock-rose, bryony, madder, privet, kneoron, marjoram, savory sphakos (sage), elelispakos (salvia), horehound, konyza, balm, and others like these; and in addition to these we have the plants with a ferula-like stem or with a stem composed of fibre, as fennel, horse-fennel, narthekia (ferula), narthex (ferula) and the plant called by some wolf's bane, and others like these. All these, as well as any other ferula-like plants, may be placed in the class of undershrubs.

6.2.1

The various forms and the differences between the above mentioned plants are in some cases more, in some less easy to distinguish. Of rock-rose they distinguish two kinds, 'male' and 'female' in that the one is larger, tougher, more glossy, and has a crimson flower; both however are like the wild rose, save that the flower is smaller and scentless.

6.2.2

There are also two kinds of kneoron, one white, the other black. The white has a leathery oblong leaf, somewhat like that of the olive; the leaf of the black is like that of the tamarisk and fleshy; the white grows more on the ground and is scented, while the black is scentless. In both the root, which runs deep, is large (and the branches which divide at the ground level are numerous thick and woody), and the root is also very woody. It is also very tough, wherefore it is used for binding and to put round things, like the withy. It grows and flowers after the autumnal equinox, and remains in flower a long time.

6.2.3

Of marjoram the black form is barren, the white bears fruit. There is a black and a white thyme, and it flowers very freely: it is in bloom about the summer solstice. It is from this flower that the bee gets the honey, and by it beekeepers say that it is made known whether they have a good yield of honey or not; for, if the thyme flowers abundantly, they have a good yield, but the bloom is injured or even destroyed if it is rained upon.

6.2.4

Savory, and still more marjoram, has a conspicuous fruitful seed, but in thyme it is riot easy to find, being somehow mixed up with the flower; for men sow the flower and plants come up from it. This plant is sought and obtained by those in Athens who wish to export such herbs. But it has a peculiarity as compared both with similar plants and with most others, namely the kind of region which it affects; they say that it can not be grown or become established where a breeze from the sea does not reach. This is why it does not grow in Arcadia, while savory marjoram and such plants are common in many parts. (A similar peculiarity is found in the olive; for it appears that it likewise will not grow more than three hundred furlongs from the sea.)

6.2.5

The difference between sphakos (sage) and elelisphakos (salvia) is like that between cultivated and wild; for the leaf of sphakos is smoother smaller and less succulent, while that of elelisphakos is rougher.

6.2.6

There are also two kinds of horehound: one has a narrow leaf with a more jagged edge, and the notches are very conspicuous and deep, and this is the plant used by druggists for certain purposes; the other has a rounder leaf, which, like that of sphakos, is not at all succulent; the notches are less conspicuous and the edge less jagged.

6.2.7

Of konyza there is a 'male' and a 'female' kind, the differences between them being such as are usual between forms so distinguished; the 'female' has slenderer leaves, is more compact, and a smaller plant; the 'male' is larger, has thicker stalks, is more branched, has larger glossier leaves, and moreover the flower is more conspicuous. Both bear fruit; the plant as a whole is late in growing and in blooming; it blooms about the rising of Arcturus and is full grown after his setting. The smell of the 'male' plant is strong, but that of the 'female' more pungent; wherefore both of them are of use against wild beasts.

6.2.8

These plants then and others like them have, as it were, different forms. Again there are some which have but one form both among those already mentioned and others as well; for there are numerous plants of this class.

6.2.9

The class of ferula-like plants (for this too belongs to the under-shrubs) comprises many kinds: here we must first speak of the characteristic which is common to all, including ferula itself (narthex) and narthekia, whether they both belong to the same kind and differ only in size, or whether, as some say, they

are distinct. The obvious character of both is alike, except as to size; for narthex grows very tall, while narthekia is a small plant. Each of them has a single stalk, which is jointed; from this spring the leaves and some small stalks; the leaves come alternately by which I mean that they do not spring from the same part of the joint, but in alternating rows. For a considerable distance they embrace the stalk, like the leaves of the reed, but they turn back from it more owing to their softness and their size; for the leaf is large soft and much divided, so that it is almost hair-like; the largest leaves are the lowest ones next the ground, and so on in proportion. The flower is quince-yellow and inconspicuous, the fruit like dill, but larger. The plant divides at the top and has some small branches, on which grow the flower and the fruit. It also bears flowers and fruit on the side-stalks all the way up, like dill. The stalk only lasts a year, and the growth takes place in spring, the leaves growing first and then the stem, as with other plants. It roots deep and has but a single root. Such is the ferula.

6.2.10

Of the others some to a certain extent resemble ferula, that is, in having a hollow stem; for instance deadly nightshade, hemlock, hellebore, asphodel: while some have a stem more or less, as it were, consisting of fibre, as fennel, aconite, and others like these. The fruit of deadly nightshade is peculiar in being black and like a grape and like wine in taste.

6.2.11

Of certain specially important spineless under-shrubs silphium and magydaris belonging to ferula-like plants.

6.3.1

Most important and peculiar in their characters are the silphium and papyrus of Egypt. These too come under the class of ferula-like plants; of these we have spoken of the papyrus already under the head of plants living in water; of the other we have now to speak.

6.3.2

The silphium has a great deal of thick root; its stalk is like ferula in size, and is nearly as thick; the leaf, which they call inaspeton, is like celery: it has a broad fruit, which is leaf-like, as it were, and is called the phyllon. The stalk lasts only a year, like that of ferula. Now in spring it sends up this maspelon, which purges sheep and greatly fattens them, and makes their flesh wonderfully delicious; after that it sends up a stalk, which is eaten, it is said, in all ways, boiled and roast, and this too, they say, purges the body in forty days. It has two kinds of juice, one from the stalk and one from the root; wherefore the one is called 'stalk-juice,' the other 'root-juice.' The root has a black bark, which is stripped off. They have regulations, like those in use in mines, for cutting the root, in accordance with which they fix carefully the proper amount to be cut, having regard to previous cuttings and the supply of the plant. For it is not allowed to cut it wrong nor to cut more than the appointed amount; for, if the juice is kept and not used, it goes bad and decays. When they are conveying it to Peiraeus, they deal with it thus: having put it in vessels and mixed meal with it, they shake it for a considerable time, and from this process it gets its colour, and this treatment makes it thenceforward keep without decaying. Such are the facts in regard to the cutting and treatment. The plant is found over a wide tract of Libya, for a distance, they say, of more than four thousand furlongs, but it is most abundant near the Syrtis, starting from the Euesperides islands. It is a peculiarity of it that it avoids cultivated ground, and, as the land is brought under cultivation and tamed, it retires, plainly shewing that it needs no tendance but is a wild thing. The people of Cyrene say that the silphium appeared seven years before they founded their city; now they had lived there for about three hundred years before the archonship at Athens of Simonides.

6.3.3

Such is their account. Others however say that the root of the silphium grows to the length of a cubit or a little longer, and in the middle of this is a head, which is the highest part and almost comes above ground, and is called the 'milk', from this then presently grows the stalk, and from that the magydaris which is also called the phyllon; but it is really the seed, and, when a strong south wind blows after the setting of the dog-star, it is scattered abroad and the silphium grows from it. The root and the stalk grow in the same year; nor is this a singular feature unless they mean that it grows immediately after the dispersal of the seed since the same thing occurs with other plants also.

6.3.4

There is this singular statement, which is inconsistent with what was said above, that, it is said, it is necessary to dig the ground every year, and that, if it be left alone, it bears the seed and the stalk, but these are inferior and so is the root; on the other hand, that with digging they are improved because the soil is changed. (This is inconsistent with the statement that silphium avoids cultivated land.) They add that the roots are cut up into vinegar and eaten fresh, and that the leaf is of a golden colour. We have also the inconsistent statement that sheep are not purged by eating the leaves; for they say that in spring and in winter they are driven into the hill-country, where they feed on this and on another plant which is like southernwood; both these plants appear to be heating and not to cause purging, but, on the contrary, to have a drying effect and promote digestion. It is also said that, if a sheep which is sick or in bad condition comes to that district, it is quickly cured or else dies, but usually it recovers. Which of these accounts is true is matter for enquiry.

6.3.5

The plant called magydaris is distinct from silphium, being of later growth and less pungent, and it does not produce the characteristic juice; experts can also easily distinguish it by its

appearance. It grows in Syria and not in Cyrene, and they say that it is also abundant on Mount Parnassus, and some call it silphium. Whether however, like silphium, it avoids cultivated ground is matter for enquiry, as also whether it has any resemblance or likeness in leaf and stalk, and, in general, whether it produces a juice. In these examples we may consider the class of ferula-like plants [and, in general, that of spinous plants.]

6.4.1

Taking next the class of spinous plants (for we must next speak of them), we have already distinguished those which are altogether spinous and those which have spinous leaves, and now we must speak of each of these classes separately, and also, in the third place, of those which have leaves as well as their spines, such as pheos and caltrop. Moreover caper has the peculiarity of possessing not only spines on its stems but also a spinous leaf. Of the classes thus distinguished that with spinous leaves is the largest, while that which is altogether spinous is about the smallest. It is indeed, as was said, a very small class, and it would not be easy to find examples of such plants besides asparagus and skorprios. Both of these flower after the autumnal equinox. Skorprios produces its flower in the fleshy swelling below the top of the spinous twig; at first it is white, but afterwards it becomes purplish. Asparagus produces alongside of the spines a small knob, and from this grows the flower, which is of small size. Skorprios has a single root which runs deep; asparagus roots very deep and its roots are numerous and matted, the upper part of them being in one piece, and from this the actual shoots spring. The stalk comes up from the plant in spring and is edible; afterwards, as the season advances, it acquires its rough and spinous character; the bloom appears not only on this stalk, but on those of previous years, for the stalk is not annual. Such is the character of plants which are altogether spinous.

6.4.2

Of those which have spinous leaves the largest class, one may say, consists of those plants which are thistle-like, by which I mean that the swollen part, that part which contains the flower, or, it may be, the fruit, is in all cases a thistle-head, or has that appearance. However there are differences in the 'head' itself, in size shape colour number of spines and in other respects. For, apart from quite a few plants, such as soap-wort sow-thistle and possibly some others, nearly all the rest have this character (even sow-thistle has a spinous character, but its seed-process is different). The list includes all the following: akorna, milk-thistle, khalkeios, safflower, polykanthos, distaff-thistle, onopyxos, ixine, chamaeleon (the last-named, however, has not spinous leaves, though golden thistle, which is also called 'meadow-thistle,' has), and so on, for there are many more. These differ from one another not only in the aforesaid ways, but in that some of them have many stalks and side-growths, like the pinethistle, while some have a single stalk and no sidegrowths, like the safflower, and some again have out-growths above from the top of the plant, like the globe-thistle. Again some grow directly the first rains come, others at a later time, some again in summer, as the plant which some call yellow starthistle, and ixine. So too the flowering-time differs: golden thistle blooms late and is in bloom for a long time.

6.4.3

Pine-thistle has but one kind, but there are different kinds of safflower, the wild and the cultivated. Again of the wild kind there are two forms, one very like the cultivated except that the stalk is straighter; wherefore in ancient times women sometimes used it to make distaffs. It has a fruit which is black large and bitter. The other is leafy, and its stalks are like those of the sow-thistle, so that to some extent it comes to have a prostrate stem; for on account of the softness of the stalks it

bends down towards the ground; and it has a small fruit, which is bearded. All the forms produce abundant seed, but it is larger and more crowded in the wild forms. This kind has also a peculiarity as compared with other wild plants; these are usually coarser and more spinous than the cultivated forms, but in this plant the wild form is softer and smoother.

6.4.4

The akorna resembles in a general way in appearance the cultivated safflower, but has a yellowish colour and a sticky juice. There is also a plant called distaff-thistle, which is whiter than these. A peculiarity of the leaf of this is that, if it is stripped off and applied to the flesh, the contact makes the juice blood-coloured, wherefore some call this kind of spinous plant 'blood-wort'; also it has an abominable smell, like that of blood; it matures its fruit late, towards autumn. Indeed, generally speaking, all plants like the thistle-tribe are late fruiting. All these plants grow both from seed and from the root, so that there is but a short period between the beginning of growth and the maturing of the seed.

6.4.5

Golden thistle has not only this peculiarity, that it has a root which is edible, whether boiled or raw, but the root is best when the plant is in flower, and, as it becomes hard, it produces a juice. The flowering time is also peculiar, about the solstice.

6.4.6

The root of the sow-thistle is also fleshy and edible; but the swollen part is elongated and not thistle-like; and, alone of the spinous-leaved plants, it has this peculiarity, in which it is the reverse of the chamaeleon, (for that plant, though it has not spinous leaves, has a thistle-like flower-head). The flower of the sow-thistle, as it ages, turns into down, as do that of the dandelion the tamarisk and other plants like these. In its growth there is a succession up to the summer, part forming flowers, part flowering, and part producing seed; this has little moisture

in it and has a sharp point. The leaf, as it dries, becomes flaccid and no longer pricks.

6.4.7

Ixine does not grow in many places, and it has leaves on the root. From the middle of the root grows the seed-bearing thistle-head, which is like an apple and well hidden by the leaves; this on its head produces its gum, which is pleasant to the taste, and this is the 'thorn-mastich.' These plants and others like them are found almost everywhere.

6.4.8

But the plant called *kaktos* (cardoon) grows only in Sicily, and not in Hellas. It is a plant quite different from any other; for it sends up straight from the root stems which creep on the ground, and its leaf is broad and spinous: these stems are called *kaktoi*; they are edible, if peeled, and are slightly bitter, and men preserve them in brine.

6.4.9

There is another kind which sends up an erect stem, called the *pternix*. This too is edible, but cannot be preserved. The fruit-vessel, which contains the seed, is in shape like a thistle-head: and when the downy seeds are taken off, this too is edible and resembles the 'brain' of the palm; and it is called *skalias*. Such are the different characteristics in the light of which we may observe the spinousleaved plants.

6.5.1

Examples of plants which have leaves as well as spines are *pheos* *rest-harrow* *star-thistle* *caltrop* 'horse-pheos' (spurge) *butcher's broom* . . . , and it has a fleshy leaf: it is much divided and has many roots, but is not deep-rooting. It grows at the rising of the *Pleiad*, the first seed-time, and then puts forth its leaf; for it is not annual, but lives longer than one year.

6.5.2

Caper, as was said, is quite distinct from these; it has a spinous leaf and a spinous stem, whereas pheos and 'horse-pheos' have no spines on their leaves; it has a single root, is low-growing, and has a creeping stem; it grows and flowers in summer, and the leaf remains green till the rising of the Pleiad. It rejoices in sandy light soils, and it is said that it is unwilling to grow on cultivated land, and that though it grows near towns and in good soil, and not, like silphium, in mountain country. This account however is not altogether accurate.

6.5.3

A peculiarity of caltrop is that it is spinousfruited. There are two kinds; one has a leaf like that of chick-pea, the other has spinous leaves. Both are low-growing and much divided, but the spinous-leaved form grows later and is found near enclosures. The seed of the early kind is like that of sesame, that of the late kind is round and blackish and enclosed in a pod. These may serve as examples of plants which have spines as well as leaves.

6.5.4

Rest-harrow has spines on the shoots; the leaf, which is annual, is like that of rue, and grows right along the stem, so that the general appearance is that of a garland, the leaves being set at intervals alternately along it; the flower is irregular, and the fruit contained in a pod, which is not divided into compartments. It grows in sticky rich soil and especially in sown and cultivated land; wherefore it is an enemy to husbandmen, and it is hard to kill; for, when it gets hold of a piece of ground, it immediately pushes its roots down deep, and every year it sends up new growths at the sides and the next year it roots these again. Wherefore it has to be dragged up entire; this is done when the ground has been moistened, and then it is easier to destroy. But, if but a small piece is left, it shoots again from this. It begins to grow in summer and completes its growth in autumn. Let these examples serve for a survey of the wild forms of under-shrubs.

6.6.1

The cultivated kinds need but a brief survey; these come under the class of coronary plants. Of coronary plants we must endeavour to give a general account, so that the whole class may be included. This group has a somewhat peculiar position, since it overlaps partly the under-shrubs, partly the herbaceous plants; wherefore the latter must also be included and we must mention them as occasion serves, taking first the under-shrubs.

6.6.2

These may be divided into two groups according to their uses. Of some only the flower is serviceable; and of these some are sweet-scented, as gilliflower, some scentless, as carnation and wallflower. Of others again the branches leaves and in fact the whole growth are sweet-scented, as with tufted thyme calamint bergamot-mint and the rest. Both groups however belong to the under-shrubs. And of the first-mentioned, those valued for their flowers, the growth is in most cases that of undershrubs, (in some annual merely, in others of longer duration) except in the violet; for this is altogether without branches, its leaves grow close to the root, and it is always in leaf; while, as some say, it is able to bear flowers continuously, if it is tended in a certain way. This may be considered a peculiar characteristic of this plant.

6.6.3

Of the others, or rather of all the group, the general appearance is in each case plain to all; any peculiarities that they may exhibit we must mention, for instance, if some appear to have but a single form, while others have various forms.

6.6.4

Thus those of woody character, as tufted thyme bergamot-mint calamint, have but one form, unless one counts wild and cultivated, scented and scentless plants, as belonging to distinct forms; and again there are with these plants differences of culture of position and of climate. Some also of the group valued

for their flowers have each but one form, for instance, the black ion (violet); for this does not appear to have different forms like the white ion (gilliflower) in which the colour evidently varies; as does still more that of the lilies, if it be true, as some say, that there is a crimson kind.

6.6.5

Among roses there are many differences, in the number of petals, in roughness, in beauty of colour, and in sweetness of scent. Most have five petals, but some have twelve or twenty, and some a great many more than these; for there are some, they say, which are even called 'hundredpetalled.' Most of such roses grow near Philippi; for the people of that place get them on Mount Pangaeus, where they are abundant, and plant them. However the inner petals are very small, (the way in which they are produced being such that some are outside, some inside). Some kinds are not fragrant nor of large size. Among those which have large flowers those in which the part below the flower is rough are the more fragrant. In general, as has been said, good colour and scent depend upon locality; for even bushes which are growing in the same soil shew some variation in the presence or absence of a sweet scent. Sweetest scented of all are the roses of Cyrene, wherefore the perfume made from these is the sweetest. (Indeed it may be said generally that the scents of the gilliflowers also and of the other flowers of that place are the purest, and especially the scent of the saffron-crocus (a plant which seems to vary in this respect more than any other). Roses can be grown from seed, which is to be found below the flower in the 'apple' and is like that of safflower or pine-thistle, but it has a sort of fluff, so that it is not unlike the seeds which have a pappus. As however the plant comes slowly from seed, they make cuttings of the stem, as has been said, and plant them. If the bush is burnt or cut over, it bears better flowers; for, if left to itself, it grows luxuriantly and makes too much wood. Also it has to be often transplanted; for

then, they say, the roses are improved. The wild kinds are rougher both in stem and in leaf, and have also smaller flowers of a duller colour.

6.6.6

The black ion (violet) differs from the white ion (gilliflower) not only in other respects but in the plant itself, in that in the former the leaves are broad, lie close to the ground, and are fleshy, and there is much root.

6.6.7

Krina (lilies) shew the variation in colour which has been already mentioned. The plant has in general a single stem, but occasionally divides into two, which may be due to differences in position and climate. On each stem grows sometimes one flower, but sometimes more; (for it is the top of the stem which produces the flower) but this sort is less common. There is an ample root, which is fleshy and round. If the fruit is taken off, it germinates and produces a fresh plant, but of smaller size; the plant also produces a sort of tear-like exudation, which men also plant, as we have said.

6.6.8

The narcissus or leirion (for some call it by the one name, some by the other) has its ground-leaves like those of the asphodel, but much broader, like those of the krinon (lily); its stem is leafless and grass-green and bears the flower at the top; the fruit is in a kind of membrane-like vessel, and is very large, black in colour, and oblong in shape. This as it falls germinates of its own accord; however men collect and set the seed, and also plant the root, which is fleshy round and large. The plant blooms very late, after the setting of Arcturus about the equinox.

6.6.9

The saffron-crocus is herbaceous in character, like the above-mentioned plants, but has a narrow leaf; indeed the leaves are, as it were, hair-like; it blooms very late, and grows

either late or early, according as one looks at the season; for it blooms after the rising of the Pleiad and only for a few days. It pushes up the flower at once with the leaf, or even seems to do so earlier. The root is large and fleshy, and the whole plant vigorous; it loves even to be trodden on and grows fairer when the root is crushed into the ground by the foot: wherefore it is fairest along the roads and in well-worn places. It is propagated from the root.

6.6.10

These are the ways then in which the above plants are grown. All the above-mentioned flowers are grown from seed, as gilliflower, carnation, spikelavender, wall-flower, martagon-lily; these plants themselves, as well as their roots, are woody. Drop-wort is also grown from seed; for that too is a plant grown for its flower. These and other plants like them may serve as examples of plants grown for their flowers.

6.7.1

All the others flower and bear seed, though they do not all appear to do so, since in some cases the fruit is not obvious. Indeed in some the flower too is inconspicuous, but, because these grow slowly and with some difficulty, men propagate them rather by off-shoots, as was said at the beginning. However some contend that they have no fruit: and there are men who have actually tried with the following plants; they have, they say, themselves often dried and rubbed out and sown the apparent fruit of thyme calamint bergamot-mint and green mint (for even that they have tried) and there was no germination from such sowing. However, the account given above is the truer, and the character of the wild forms testifies to this; for there is also a wild thyme (Attic thyme), which they bring from the mountains and plant at Sicyon, or from Hymettus and plant at Athens; and in other districts the mountains and hills are quite covered with it, for instance in Thrace. There is also a wild bergamot-mint, and wild forms of

the other plants mentioned, having a more pungent smell. Thyme is sometimes quite like cultivated thyme.' Now it is plain that these wild forms possess this means of reproducing themselves.

6.7.2

Southernwood actually grows more readily from seed than from a root or a piece torn off (though it grows even from seed with difficulty); however it can be propagated by layering in pots in summertime, like the 'gardens of Adonis'; it is indeed very sensitive to cold and generally delicate even where the sun shines brightly; but, when it is established and has grown, it becomes tall and strong and tree-like, like rue, except that the latter is much more woody drier and less succulent.

6.7.3

Sweet marjoram grows in either way, from pieces torn off or from seed; it produces a quantity of seed, which is fragrant with a delicate scent; it can also be transplanted. Southernwood also produces much seed, which has some scent. This plant has straight roots which run deep; it has, as it were, its single stout root, from which the others spring; while sweet marjoram thyme bergamot-mint and calamint have surface roots which are much divided and matted; in all these plants the roots are woody, but especially in southernwood, because of its size and because it is so dry. The growth of the shoots of thyme is peculiar. If it has a stake, or is planted against a wall, it can send them out to any length; so also if it is let grow downwards; indeed it is most vigorous when grown into a pit. It is not possible to distinguish different forms of the cultivated kind, as has been said, but they say that of the wild kind (Attic thyme) there is more than one form; for that of the kind which grows on the mountains one form is like savory and very pungent, while the other is fragrant and more delicate.

6.7.4

The season for planting most of these is autumn, and then men hasten to plant them as early as possible; however some are planted also in spring. All of them love shade, water, and especially dung; however thyme is patient of drought and, in general, needs moisture less than the others. These plants especially delight in the dung of beasts of burden; and it is said that they should often be transplanted, for that it improves them, while bergamot-mint, as has been said, actually degenerates if it is not transplanted.

6.8.1

Of the flowers the first to appear is the gilliflower; where the air is mild, it appears as soon as winter comes, but, where it is more severe, later, sometimes in spring. Along with the gilliflower, or a little later, appears the flower called the wild wallflower. These, of all the flowers that the garlandmakers use, far outrun the others. After these come pheasant's eye and polyanthus narcissus (and, among wild plants, the kind of anemone which is called the 'mountain anemone') and the 'head' of pursetassels; for this too some interweave in their garlands. After these come dropwort violet, and of wild plants, gold-flower, the meadow kind of anemone corn-flag hyacinthos (squill), and pretty well all the mountain flowers that are used. The rose comes last of these, and is the first of the spring flowers to come to an end, as it is the first to appear, for its time of blooming is short. So too is that of the rest of the wild plants mentioned, except hyacinthos the wild kind (squill), and also the cultivated (larkspur); this lasts on, and so does the gilliflower, and for a still longer time the wallflower, while the violet, as has been said, blooms throughout the year, if it receives tendance. So too dropwort (for that too is one of the plants valued for their flowers, though it is herbaceous in character) if one pinches off and removes the flower instead of letting it go to seed, and if, further, it has a sunny position. The flower is clustering and white, like that of the wild These then are, we may say, the plants of spring.

6.8.2

The following belong rather to summer: rosecampion carnation krinon (lily) spike-lavender and the Phrygian sweet marjoram]; also the plant called 'regret,' of which there are two kinds, one with a flower like that of larkspur, the other not coloured but white, which is used at funerals; and this one lasts longer. The iris also blooms in summer, and the plant called soap-wort, which has a beautiful flower but is scentless. In autumn bloom the other kind of narcissus, the crocus, both the scentless mountain form and the cultivated one (saffron crocus); for these bloom directly the first rains come. The fruit of the cotoneaster and the flower of the smilax, both of them wild plants, are also used in garlands.

6.8.3

Such are the seasons at which each appears; and, to speak generally, there is no interval of time nor flowerless period, but even winter produces flowers, for all that it seems to be unproductive by reason of the cold, since the autumn flowers continue into winter, and to a much greater extent if the season be mild. For all things, one may say, or at least most of them, extend beyond their proper season, and all the more if the place be sunny; so that there is a continuous succession. These then are the periods and seasons at which the various flowers are produced.

6.8.4

The life of the gilliflower is at most three years; as it ages it degenerates and produces paler flowers. A rose-bush lives five years, after which its prime is past, unless it is pruned by burning; with this plant too the flowers become inferior as it ages. Position and a suitable climate contribute most to the fragrance of roses, gilliflowers and other flowers. Thus in Egypt, while all other flowers and sweet herbs are scentless, the myrtles are marvellously fragrant. In that country it is said that the roses gilliflowers and other flowers are as much as two

months ahead of those in our country, and also that they last a longer, or at least not a shorter, time than those of our country.

6.8.5

And, as has been said, the particular season according to its character, makes a great difference to the fragrance, not only by reason of rains and droughts, but also according as rain, wind, and in general, the changes of climate occur or do not occur at the fitting moment. Also it appears that in general roses gilliflowers and the rest bloom well on the mountains, but many of them have there an inferior scent. Concerning coronary plants and under-shrubs in general these examples and others like them suffice for our enquiry.

7.1.1

BOOK VII. OF HERBACEOUS PLANTS, OTHER THAN CORONARY PLANTS: POT-HERBS AND SIMILAR WILD HERBS.

Next we have to tell of herbaceous plants: for this class remains of those which we distinguished at the outset, and it includes to some extent the classes of pot-herbs and of cereals. And first we must speak of the class of pot-herbs, beginning with the cultivated kinds, since it happens that these are better known than the wild kinds.

7.1.2

There are three seed-times for all things grown in gardens, at which men sow the various herbs, distinguishing by the season. One is the 'winter' seed-time, another the 'summer' and the third is that which falls between these, coming after the winter solstice. These terms however are given in regard not to the sowing, but to the growth and use of each kind; for the actual sowing takes place, one might almost say, at the opposite seasons. Thus, the 'winter' period begins after the summer solstice in the month Metageitnion, in which they sow cabbage radish turnip, and what are called 'secondary crops' that is to say, beet, lettuce, rocket, monk's rhubarb, mustard, coriander,

dill, cress; and this is also called the 'first' period of cultivation. The second period begins after the winter solstice in the month Gamelion, in which they scatter or plant the seed of leeks, celery, long, orach. The third period, which is called the 'summer' period, begins in the month Munychion: in this are sown cucumber gourd blite basil purslane savory. Moreover they make several sowings of the same herb at each season, as of radish basil and the others. And at all the periods are sown the 'secondary crops.'

7.1.3

Not all herbs germinate within the same time, but some are quicker, others slower, namely those which germinate with difficulty. The speediest are basil, blite, rocket, and of those sown for winter use, radish; for these germinate in about three days. Lettuce takes four or five, cucumber and gourd about five or six, or, as some say, seven; however, cucumber is earlier and quicker than the others. Purslane takes a longer time, dill four days, cress and mustard five. Beet in summer takes six days, in winter ten, orach takes eight, and turnip ten. Leek and long onion do not take the same time, but the former nineteen to twenty days, the latter ten to twelve. Coriander germinates with difficulty; indeed fresh seed will not come up at all unless it is moistened. Savory and marjoram take more than thirty days; but celery germinates with the greatest difficulty of all; for those who make the time comparatively short say forty days, and others fifty, and that too, at whichever period it is sown, for some sow it as a 'secondary crop' at all the periods.

7.1.4

Generally speaking, those herbs which are sown at more than one season do not mature faster in the summer. Howbeit it is strange if the season and the state of the atmosphere do not contribute at all to quicker growth, and if, when there is an unfavourable cold season and the atmosphere is cloudy, these conditions do not tend to make growth slower, seeing that,

when stormy or fair weather follows the sowing, germination is slower or quicker accordingly. And there is another thing which makes a difference as to the raising of the various herbs; germination begins earlier in sunny places which have an even temperature.

7.1.5

As a matter of fact, to speak roundly, the causes of such differences must be found in several different circumstances, in the seeds themselves, in the ground, in the state of the atmosphere, and in the season at which each is sown, according as it is stormy or fair. However it is a point for consideration with which herbs the time of sowing makes a difference and with which it makes none; thus it is said that radish germinates on the third day whether it be sown in summer or in winter, while beet, as has been said, behaves differently according to the season. Anyway such are and are said to be the seasons of germination in each case.

7.1.6

Another thing which makes a difference as to the rapidity with which the seeds germinate is their age; for some herbs come up quicker from fresh seed, as leek, long onion, cucumber, gourd; (some even soak the seed of cucumber first in milk or water, to make it germinate quicker). Some come up quicker from old seed, as celery beet cress savory coriander marjoram (unless indeed they are raised from fresh seed in the manner which we have mentioned). There is, they say, a singular feature about beet; the seed does not all germinate at once, but some of it not for some time, some even in the next or in the third year; wherefore it is said that little comes up from much seed.

7.1.7

Any of the seeds, if they are ripe when they fall, last till their own proper season and do not sprout till then. And in this they are consistent; for we note that the same thing happens with the seed of wild plants, unless it is destroyed. However all

mature their fruits in the summer, though sooner and quicker, generally speaking, when they are sown earlier. The season also makes a difference; things sown in the hot season push up their shoots and go to seed sooner, as radish and turnip. Some however bear their fruit not in the same year but in the next, as celery leek long onion, which plants also last a longer time, and are not annual; for most herbs wither with the ripening of their seed.

7.1.8

Generally speaking, all those that push up shoots and mature their fruit reach their perfection of form in having side-shoots branching from the main stem except those which have but a single stem, as leek, long onion, onion, garlic.

7.1.9

All these herbs are lovers of water and of dung, and especially the weaker ones, which require more attention or in some cases more feeding.

7.2.1

All these herbs are propagated from seed, and some also by a piece torn off, a shoot, or a piece of root. Cabbage is propagated by a piece torn off, since it is essential in this case to take a piece which has root attached to it. From cuttings are grown rue marjoram basil; for slips of this too men plant when it has grown to the height of a span or more, cutting off half the plant. By root are planted garlic onion purse-tassels cuckoo-pint and in general such bulbous plants. Such propagation is also possible in cases where the roots persist for more than a year, though the shoots last but for a year. And it is plain that all these herbs can be grown from seed; for even rue can (which some deny), though the process is slow, and so cuttings are also taken.

7.2.2

Of those which are propagated by a piece of root the root is long-lived, though the plant itself may be annual; wherefore the

roots of such plants make offsets and so increase; and this is true not only of plants cultivated in the garden, but also of wild plants, as we have said, for instance of purse-tassels long onion squill and so forth. Some plants even which are not bulbous but longer-lived make offsets, as celery and beet; for these send out roots from which grow leaves and stems. Long onion and leek also make offsets, sending out a 'head' below, like the bulb of purse-tassels, from which the leaves spring; but this only takes place when the stem has withered and the seed has been removed. But, as the 'heads' of such plants are not useful, they do not collect them for storing dry; wherefore also they do not plant these. It may be that somehow these are akin and closely allied to onion, wherefore what has been said is not surprising. However in all those plants, both wild and cultivated alike, which have an annual stem, but yet live longer than a year, there is an outgrowth of the roots, just as there is in under-shrubs and shrubby plants: while in onions garlic and purse-tassels even a number, as it were, of such roots is formed. In fact, they are reproduced in three ways, as has been said; from seed in all cases and from the stem and root in those specified.

7.2.3

Almost all shoot again if the stem is broken (except those which are stemless), but most obviously basil lettuce cabbage, which are, as it were, broken for a practical reason. Indeed they say that the stems of lettuce which thus grow again are sweeter, for that the original stem has a taste like fig-juice and is bitter, as being not properly ripened. Some however say that the later stems have the taste of fig-juice more than the original one, but that, so long as they are tender, they appear sweeter. Be that as it may, it is admitted that in the case of cabbage the stem is sweeter if it should have grown again after being broken, provided that the leaves are stripped off before the plant runs to stalk.

7.2.4

In most cases the roots persist, but they do not in all cases produce fresh growth. Thus radish and turnip persist till summer, if earth is thrown on them, and they increase in size; and some gardeners do this deliberately; but they do not make fresh growth nor send out leaves, even if one removes the earth heaped over them. And this may also be observed in other plants. However, most pot-herbs have the single stout root which runs deep; for even in those which produce these side-roots of equal stoutness, as celery and beet, the side-growth comes, as it were, from the middle root and it is not separate to start with; but to this single root are attached the small out-growths, both in radish and in turnip. These instances are familiar to all because of the use which is made of these plants.

7.2.5

The beet has a single long stout straight root like that of the radish, and has stout out-growths, sometimes two, sometimes three, sometimes only one, and the small ones are attached to these. The root is fleshy and sweet and pleasant to the taste, wherefore some even eat it raw. The 'bark' is not thick and cannot be detached, like that of the radish, but rather resembles that of alexanders. In like manner the root of orach is single and runs deep, and other roots are attached to it.

7.2.6

Monk's rhubarb however has a single root in a truer sense than any of the others, for it has no stout out-growths of root, but only the slender ones; its root also runs deeper than that of the others, being more than a foot and a half long. The wild sort however has a shorter root, and has several stems and branches, and its shape, as a whole, when fully grown resembles that of beet. Cultivated monk's rhubarb moreover is longer lived than the wild form, and, in general, we may say, than any other pot-herb, for, they say, it may live any time. It has a fleshy

root, full of moisture, wherefore, if pulled up, it will live some time.

7.2.7

Basil has the single stout root, the one which runs deep, and the others at the sides are slender and fairly long.

7.2.8

Some herbs, as blite, have not the single straight root, but a number of roots which start directly from the top and are of a good stoutness and longer than those of orach.

7.2.9

The roots of basil are woodier than those of any of the other herbs, as also is its stem; for those of blite orach and the like are less woody. In general we may say that the roots of any of these herbs are either woody or fleshy. Examples of fleshy roots are beet, celery, alexanders, monk's rhubarb, radish, turnip, and especially all 'heavy-headed' kinds, for the roots of these do not wither up altogether even when they are dried. Examples of those with woody roots are basil, blite, orach, rocket, dill, coriander, and in general, those with fibrous stems; for in dill and coriander, which have a single root, the root is woody and not long, and the slender side-roots from it are not numerous; but both plants have several stems and branches; wherefore in neither of these plants does the part above ground correspond to the part which is below.

7.2.10

The following have short roots: lettuce and purslane, in which both the straight main root and the side ones are short. Lettuce may be said to have no such side-roots, but only the slender ones, and may be called in the strictest sense a plant of a single root. In general all summer herbs have short roots: we may include cucumber gourd and bottlegourd, both because of the season to which they belong and perhaps still more because of their character, which corresponds to the season. However the transplanted lettuce has a shorter root than one that is raised

from seed, since it is more apt to send out side-growths; also the wild kind has a shorter root than the cultivated, and the part above ground has more stems.

7.3.1

All, except one, of these herbs produce all their bloom at once, but basil has a succession of flowers, the lower part of the plant flowering first, and then, when that bloom is over, the upper part. Wherefore its season of bloom is a long one, like that of the bean, and among herbaceous plants that of the plant called heliotropion, and also other wild plants. Cucumber also has a long period of bloom, for this plant has a second growth. The flowers are in some cases whitish, in others quince-yellow, in others somewhat reddish; but the flower is never of a bright colour.

7.3.2

The seeds too differ in shape; most are round, but some are oblong; some again are broad and leaflike, as those of orach, for the seed of this is like that of silphium; others again are narrow or marked in lines, as those of cummin. They also vary in colour, some being black, some the colour of wood, some paler. The seeds of all are either in pods or naked, or have an integument or have a pappus. Radish mustard and turnip have their seeds in pods; coriander fennel dill and cummin, have naked seeds; those of blite beet orach and basil are enclosed in an integument; those of lettuce have a pappus on them.

7.3.3

All have numerous fruits and numerous shoots, but cummin has the most fruits of all. And there is another peculiarity told of this plant: they say that one must curse and abuse it, while sowing, if the crop is to be fair and abundant.

7.3.4

Nearly all of these, except cummin, are hard to dry for keeping, unlike corn; for this, when once it is ripened, quickly dries and is shed, and the herbs whose seed have an integument are

harder still to dry, especially basil. All however, when dried, produce more fruit: wherefore it is the custom to gather the seed early and dry it. All of them are prolific and produce many seeds, but basil produces most of all.

7.3.5

Examples of those which produce their fruit at the top of the stem are basil leek onion: of those which produce it rather at the sides, radish turnip and the like; of those which produce it in both ways, blite and orach; both of these produce it at the side as well as at the top; in fact blite has its seed in clusters, closely attached to each branch. Some push up their shoots fairly soon from old seed, but seed from plants in their prime is the most rapid; for these plants too have a time when they are at their best. The beauty of the plant also corresponds in proportion, provided that equal care in other respects is shewn in cultivation.

7.3.6

It likewise appears that, if a quantity of seed is sown in the same place, the resulting crop comes up and germinates better; thus they tie up seed of leek and celery in a piece of cloth before sowing, and then there is a large crop.

7.3.7

The position also contributes to growth; at least, when celery is transplanted, they suggest that one should hammer in a peg of whatever size one wishes to make the celery; and also that one should sow the seed in a piece of cloth after hammering in a peg and filling the hole with dung and soil.

7.3.8

Some things again come to resemble in their shape even the position in which they grow: thus the bottle-gourd becomes like in shape to the vessel in which it has been placed.

7.3.9

Moreover differences in taste are acquired in some cases when the seed has been treated specially beforehand; for instance, the seed of the cucumber produces a fruit with different taste if it is soaked in milk before sowing. But such matters belong perhaps more properly to the subject of cultivation.

7.4.1

Of some herbs there are several kinds, but of others only one, as basil monk's rhubarb blite cress rocket orach coriander dill rue; of each of these they say that there is but one kind. But of others there is more than one, as radish cabbage beet cucumber gourd cummin garlic lettuce. Differences are marked in the leaves, the root, the colour, the taste, and so forth.

7.4.2

Thus of radish they recognise these various kinds the Corinthian, that of Cleonae, the Leiothasian, amoreia, the Boeotian. The Corinthian is said to be the strongest in growth, and it has an exposed root; for it pushes upwards, and not downwards like the others. The Leiothasian is called by some the Thracian radish, and it stands the winter best. The Boeotian is said to be the sweetest and to be round in shape, not of a long shape like that of Cleonae. Those kinds whose leaves are smooth are sweeter and pleasanter to the taste, those whose leaves are rough have a somewhat sharp taste. Besides the above-mentioned kinds there is yet another, whose leaves resemble those of rocket. These then are the different kinds of radish.

7.4.3

Of the turnip all do not agree that there are several kinds, but some say that the only difference is between the 'male' and the 'female,' and that both forms come from the same seed. In order to produce 'female' plants it is said that the seed should be sown thinly, for that, if it is sown thick, the result is all 'male' plants; and that the same result follows if the seed is sown in poor soil. Wherefore, when they are shifting plants for seeding, they plant

the seedlings wide apart. Good and inferior seed can be easily distinguished by their appearance; the seed of a good plant is fine, that of a poor one coarse. Both this plant and radish like exposure to winter; for it is supposed that this makes them sweeter and that they are thus made to grow roots rather than leaves. With a south wind and warm weather they run up quickly. It needs explanation that both plants should thus adapt themselves in special ways.

7.4.4

Of cabbage three kinds are distinguished, the curly-leaved, the smooth-leaved, and thirdly, the wild form. The wild form has a small round leaf, it has many branches and many leaves, and further a sharp medicinal taste; wherefore physicians use it for the stomach. Between the other two kinds there seem also to be differences, inasmuch as one of them bears no seed or only inferior seed. In general the curly-leaved kind has a better flavour than the smooth and it has larger leaves.

7.4.5

So too with beet; the white kind has a better flavour than the black and produces fewer seeds; some call it 'Sicilian' beet.

7.4.6

So too with lettuce; the white kind is sweeter and tenderer. Of this plant there are three other kinds, the flat-stalked, the round-stalked, and the Laconian; the last-named has a leaf like the golden thistle, but is erect and strong-growing and has no side-shoots from the main stem. Of the 'flat' kinds some have such flat stalks that some, they say, use them to make a garden trellis. The third kind, which has much milky juice and small leaves and a whiter stem, is like a wild plant.

7.4.7

In celery the differences between the various kinds lie in the leaves and stem; one kind is close and curly and has rough leaves, the other is more open in growth and flatter, but has a larger stalk. Again there are kinds with stems white, red or

particoloured; and in general all such forms resemble more the wild kind.

7.4.8

As to cucumber and gourd, it is said that there are various forms of the former, but of the latter, just as in radish and turnip, the differences are only between better and inferior individuals. Of the cucumber there are three forms, the Laconian the cudgel-shaped and the Boeotian. Of these the Laconian is better with moisture, the others without it.

7.4.9

There are also various kinds of onion and of garlic; those of the onion are the more numerous, for instance, those called after their localities Sardinian, Cnidian, Samothracian; and again the 'annual' the 'divided' (shallot) and that of Ascalon. Of these the annual kind is small but very sweet, while the divided and the Ascalonian differ plainly as to their character as well as in respect of their cultivation. For the 'divided' kind they leave untended in winter with its foliage, but in spring they strip off the outside leaves and tend the plant in other ways; when the leaves are stripped off, others grow, and at the same time division takes place under ground, which is the reason of the name 'divided.' Some indeed say that all kinds should be thus treated, in order that the force of the plant may be directed downwards and it may not go to seed. The Ascalonian kind has a somewhat peculiar character; it is the only kind which does not divide and which does not, as it were, reproduce itself from the root; moreover in the plant itself there is no power of increasing and multiplying; wherefore many do not plant these, but raise them from seed; and the sowing is made late, towards the spring; and then, when the seed has germinated, they transplant. And the plant arrives at maturity so fast that it is taken up with the others or even earlier; whereas, if it is left a longer time in the ground, it rots. If planted on the other hand, it sends up a stem and merely produces seed, and then shrivels

up and withers. Such then is the character of these.

7.4.10

Some also shew differences in colour; thus at Issus are found plants which in other respects resemble the others, but which are extremely white in colour; and they bear, 'it is said, onions like those of Sardis. Most distinct however is the character of the Cretan kind, which resembles to some extent that of Ascalon, if indeed it be not the same. For in Crete there is a kind which when sown produces a root, but when planted produces a stem and seed but has no 'head'; and it is sweet in flavour. This kind in fact has just the contrary character to the others; for they all grow better and faster when they are planted. All are planted after the rising of Arcturus while the earth is still warm, so that the rains may come upon them after planting. They are planted either entire or else in sections made by cutting at the 'head.' The growth which results is not uniform; from the lower part comes an onion, from the upper only foliage; while, if the plant is divided vertically, no growth at all takes place. The kind called horn- onion has no head/ but has as it were a long neck, at the top of which comes the new growth; it is often cut, like the leek; wherefore it is raised from seed and not planted. Such then, one may say, are the forms of the onion.

7.4.11

Garlic is planted a little before or after the solstice, when it divides into cloves. There are different kinds distinguished as late or early, for there is one kind which matures in sixty days. There are also differences as to size. There is one kind which excels in size, especially that variety which is called Cyprian, which is not cooked but used for salads, and, when it is pounded up, it increases wondrously in bulk, making a foaming dressing. There is a further difference, in that some kinds cannot be divided into cloves. The sweetness of taste and smell and the vigour depend on the position and on cultivation, as with other herbs. Garlic reaches maturity from seed, but slowly, for in the

first year it acquires a 'head' which is only as large as that of the leek, but in the next year it divides into cloves, and in the third is fully grown, and is not inferior, indeed some say it is superior, to the garlic which has been planted. The growth of the root in garlic and onion is not the same; in garlic, when the clove has swollen, the whole of it becomes convex; then it increases and divides again into the cloves, and becomes several plants instead of one by the maturing of the 'head,' while the onion puts out another and another growth straight from the root, as do purse-tassels and squill and all such plants. For both onions and garlic multiply if they are not removed but left alone. They say also that garlic produces garlic heads on the stalk, and that the onion in like manner produces onions. Let this suffice for an account of their ways of growth.

7.5.1

All the pot-herbs are lovers of water and of dung, except rue, which does not at all like dung; this is true of the winter no less than of the summer herbs, and of the tender no less than of the strong ones. The dung which is most commended is that which is mixed with litter, while that of beasts of burden is held to be bad, because it is most apt to lose its moisture. Dung which is mixed with the seed is most in request, but some cast the manure on while they are sowing, and they also use fresh human dung as a liquid manure. The winter crops like moisture more than the summer ones, and the weak more than the strong, as well as those which specially need feeding. Onion and long onion also love moisture, though some say that they do not require it, if at the outset it has been applied twice or thrice. Fresh cold water is the best, and the worst is that which is brackish and thick: wherefore the water from irrigation ditches is not good, for it brings with it seeds of weeds. Rain-water is good, for it also appears to destroy the pests which devour the young plants. Some however say that rain-water is not good for melons nor for onions. Most herbs are watered in early morning

or at evening, so that they may not be dried up; but basil is watered even at noon, for it is said that it grows more quickly if it is watered at first with warm water. In general water seems to be extremely beneficial, especially if it is mixed with dung; for, they say, pot-herbs often are hungry, and experienced gardeners can recognise when this is so.

7.5.2

All herbs grow finer and larger if transplanted; for even the size of leeks and radishes depends on transplantation. Transplanting is done especially in view of collecting seed: and, while most herbs bear it well, as long onion leek cabbage cucumber celery turnip lettuce, others bear it less well. All however make better growth and are larger if the seed is planted rather than scattered.

7.5.3

As for pests, radish is attacked by spiders, cabbage by caterpillars and grubs, while in lettuce, leek, and many other herbs occur 'leek-cutters.' These are destroyed by collecting green fodder, or when they have been caught somewhere in a mass of dung, the pest being fond of dung emerges, and, having entered the heap, remains dormant there; wherefore it is then easy to catch, which otherwise it is not. To protect radishes against spiders it is of use to sow vetch among the crop; to prevent the spiders from being engendered they say that there is no specific. Basil turns pale about the rising of the dog-star, and coriander becomes mildewed. In these instances we may observe the accidents which occur to pot-herbs.

7.5.4

Of seeds some have more vitality than others as to keeping; among the more vigorous ones are coriander beet leek cress mustard rocket savory, and in general those of pungent taste; among the less vigorous are long onion which will not keep orach basil gourd cucumber; and in general the summer herbs keep less well than the winter ones. No seed will keep more

than four years so as still to be of use for sowing; though it is better in the second year, in some cases it does not deteriorate in three years, but after that time deterioration begins.

7.5.5

However for cooking purposes seed will keep a longer time, except that such seed must necessarily become less vigorous by reason of evaporation and destruction by worms. The chief cause of loss is vermin; for vermin occur in all the seeds, even those which are pungent, though least in the gourd tribe; such seeds however, as they lose their moisture, become bitter in taste and inferior for use. Let this suffice for an account of the seeds and in general of herbs cultivated in gardens.

7.6.1

We must now endeavour to speak in the same way of the wild kinds and of those which are called uncultivated herbs. Some of these have the same names as the cultivated kinds; for all these kinds exist also in a wild form, and most of them resemble the cultivated kinds in appearance, except that in the wild forms the leaves and also the stalks are smaller and rougher, and in particular these forms are more pungent and stronger in taste, for instance, savory marjoram cabbage and rue; the wild monk's rhubarb (dock) indeed, though it has a pleasanter taste than the cultivated, yet has a sharper flavour; and this is the chief difference. Moreover all the wild kinds are less juicy than the cultivated, and perhaps this is the very reason why most of them are more pungent and stronger.

7.6.2

A peculiarity of 'wild cabbage' as compared with the others is that its stems are rounder and smoother than in the cultivated kind, and, while in the latter the attachment of the leaf is flat, in the wild kind it is rounder, and the leaf itself has less angles; in other cases the wild form is the rougher both in stem and leaf. The wild turnip has a long root, like that of the radish, and a short stem.

7.6.3

The wild lettuce has a shorter leaf than the cultivated kind, and, as the plant matures, it becomes spinous; the stem is also shorter, while the juice is pungent and medicinal. It grows in fields; they extract its juice at the time of wheat-harvest, and it is said that it purges away dropsy and takes away dimness of sight and removes ulcers on the eye; for which purpose it is administered in human milk.

7.6.4

‘Horse-celery’ (alexanders), ‘marsh-celery’ and ‘mountain-celery’ (parsley) differ both from one another and from the cultivated kind; ‘marsh-celery,’ which grows by irrigation-ditches and in marshes, has scanty leaves, and is not of close habit, yet it somewhat resembles the cultivated kind in smell taste and appearance. ‘Horse-celery’ has a leaf like that of the marsh kind, but is of close habit and has a big stalk, and its root is as thick as a radish and black; the fruit is also black, and in size is larger than the seed of a vetch. They say that both kinds are serviceable in cases of strangury and for those suffering from stone, being administered in sweet white wine. Both kinds grow equally everywhere. There is also a sort of gum which exudes from the plant, like myrrh, and some say that it is myrrh.

7.6.5

‘Mountain-celery’ (parsley) exhibits even greater differences; its leaf is like that of hemlock, the root is slender, and the fruit like that of dill, but smaller; it is given in dry wine for diseases of women.

7.6.6

In some cases however the wild kinds are not in the least like the cultivated in taste and properties; thus the wild and the cultivated cucumber are quite different, and their resemblance is due only to their general look, as, among coronary plants, there is resemblance between the wild and the cultivated kinds

of gilliflower; for the leaves are alike. We have then described the differences which these plants present.

7.7.1

Next we must speak of the differences found in the herbs called ‘uncultivated,’ and in general in any herbaceous plants which are not edible. For we give the name of ‘pot-herbs’ to those which are cultivated for our own use, but in a wider sense the term includes these also; wherefore we must speak of them too. Under the name ‘pot-herbs’ are included also such plants as chicory, dandelion, khondrylla, cat’s ear, groundsel, and in general all those that are called ‘chicory-like’ because of the resemblance in the leaves; for to a certain extent the leaves of all these are like those of chicory; and we may add kaukalis, chervil, green mint. Some include under the name countless others, as wild chervil and all plants that resemble it, and goat’s beard, which some call home (‘hair’), which has a long sweet root and leaves like those of the crocus, but longer, and a short stem, on which is set the sheath; this is large, and on the top is the large mass of grey pappus, from which it gets its name of ‘goat’s beard.’

7.7.2

In like manner all those may be included which have a similar appearance, but juices suitable for food whether raw or cooked; for some need the action of fire, as malakhe (cheese-flower) beet, monk’s rhubarb, nettle, and bachelor’s buttons; while garden nightshade is also eaten raw, and some in former times considered it worth growing in gardens. There are also many more, including the plant which has become proverbial for its bitterness, blue pimpernel, which has a leaf like basil. All these are either annual or have annual stems; for some of them wither away altogether in one season, while of others the roots persist for a longer time, and to this class belong the majority.

7.7.3

Some of these plants grow from roots and also from seed unless in some cases they come up spontaneously. The growth alike of these and of others takes place in some cases with the first rains after the equinox, for instance, dandelion ribgrass and the plant which some call buprestis; in other cases after the rising of the Pleiad, for instance, chicory and most of the plants of that class. Some produce their flower immediately at the time of making growth, as lesser celandine, some not long after, as anemone, while some as soon as spring comes send up both their stems and flower, as chicory and the plants which resemble it, and those spinous plants which come under the head of pot-herbs.

7.7.4

There is much difference in the flowers, of which we have spoken already; for such difference is a thing common to all; and some are altogether flowerless, as stonecrop. Those which produce their flower with the stem quickly shed the flower; except that dandelion, when the first flower is past its prime, produces another and yet another, and continues to do so right through the winter and spring up to the summer. Groundsel also blooms for a long time; the others however do not do this; for instance the crocus does not, neither the scented (saffron crocus) nor the white nor the spinous kind, which last are scentless.

7.8.1

A distinction which is found in all herbaceous plants alike is the following: some have straight and fibrous stems, some prostrate stems, as malakhe (cheese-flower), wild chervil, 'wild cucumber' (squirting cucumber); while heliotropion has this character to an even greater extent, and so, among spinous plants, have caltrop, caper and several others; for in these too the above-mentioned distinction is even more marked. Some again have clasping stems, but if they have nothing on which to throw themselves, their stems become prostrate, as epetine bedstraw and in general those which have a slender soft long stem;

wherefore these in general grow in the midst of other plants. This point of difference too is common not only to all herbaceous plants and under-shrubs, but also to shrubby ones; for helix (ivy) has a clasping stem, and, still more, smilax.

7.8.2

Again of herbaceous plants too some have several stems, some only one; and of the latter some have no side-shoots along the stem, while others have side-shoots, for instance, among cultivated plants radish and some others. Those with prostrate stems have generally more than one, while those with erect stems have but one or a few. Of these those with smooth stems have no side-shoots, as onion leek garlic the wild, as well as the cultivated forms; and of these again some have straight, some crooked stems.

7.8.3

There is also the following point of difference in herbaceous plants: some have their leaves on the ground, some on the stem, some have both characters. The following have ground leaves: crowsfoot, the anthemion whose flowers have no petals (wild camomile), alkanet grass, anemone, hawk's beard, plantain, dandelion; the following have leaves on the stem ox-tongue, the anthemion which has petalled flowers, trefoil, gilliflower; while chicory has both kinds of leaves; for this plant produces, as well as leaves, a certain number of flowers on the stems at the points where the side-shoots are attached. Similar too are some of the plants with spinous leaves, but not those that are altogether spinous, as sow-thistle.

7.9.1

Again some are barren, while others bear fruit, and, speaking generally, of herbaceous plants some get as far as producing leaves only, others have a stem and flower, but no fruit; some again have fruit as the completion of their development, while some bear fruit even though they have no flower, as is the case with some trees.

7.9.2

The leaves of herbaceous plants again differ in hardly fewer, nay, even in more, ways than those of trees, and further, they present differences as compared with these, the chief being perhaps that some are attached by a leaf-stalk, some are attached directly, some attached with cauline appendages. And in some herbaceous plants the stalk is the first part to grow, but in most the leaves, which almost at the outset grow to their largest and are best for eating; whereas the leaves of trees always push out first a sort of stalk.

7.9.3

There is also much difference as to the flowers between herbaceous plants and trees; for in trees most of the flowers are white, while some are slightly reddish, others are greenish or greenish-yellow, but none of them have distinct gay colours; while in herbaceous plants the flowers shew many and various colours, both simple and in combination, and further, some of them are scented, others not. Again trees produce all their bloom at once, while some herbaceous plants have a succession of flowers, as we said of basil; wherefore it is in flower for a long period, as are many other herbs, such as heliotropion and chicory.

7.9.4

There are also many differences in the roots, and, in a way, the differences in these are more obvious; some are woody, some fleshy and fibrous, just as in the cultivated kinds, as are those of corn and most kinds of grass. Again the roots themselves exhibit in each case very many differences in colour smell taste and size; some are white, some black, some red, as those of alkanet and madder; some are yellowish, or the colour of wood. Again there are roots which are sweet, bitter, pungent, fragrant, evil-smelling; and some are medicinal, as has been said elsewhere.

7.9.5

There are also differences between those with fleshy roots; the roots of some are round, of some oblong and acorn-shaped, as those of asphodel and crocus; some consist of several layers, as those of purse-tassels squill and others which belong to that class, onion long onion and others like these. Some are smooth loose and soft throughout, and, as it were, without 'bark,' as those of cuckoo-pint, while some have a 'bark' attached to the fleshy part, as those of cyclamen and turnip. And not all those that are fragrant or sweet or pleasant to the taste are also edible, any more than all those that are bitter are uneatable; any (whether sweet or bitter) that are harmless to the body after being eaten are edible; for some that are sweet are deadly and dangerous to health, while some are beneficial even if they are bitter or have an evil smell. The same may be said of the leaves and stalks, as in the case of wormwood and centaury. There are also differences in the time of growth and of flowering, the season being variously the beginning or middle of winter, or again spring, summer, or autumn. So too is there in like manner a difference in the fruits, which in some of these plants are edible and juicy, as well as the leaves seeds and roots. And in these cases there are further differences in the taste (of those which are edible and juicy), which may be sharp, pungent, sweet, dry, or exhibit other similar differences, either altogether or in degree. These are examples of the differences which we find.

7.10.1

Each plant having its proper season for growth, flowering and maturing of the fruit, nothing grows before its proper season either of those grown from a root or of those grown from seed, but each awaits its proper season and is not affected even by rain. For some are plants which belong properly to summer as to their growth and their flowering, as golden thistle and squirting cucumber, as was said of shrubby plants and of konyza caper and the rest; for of these too none blooms or grows before

its proper season. Wherefore in this respect too these plants would seem to differ from trees. For trees make their growth all at once or nearly so, or at all events we may say that they do so all at one season; but the plants of which we are now speaking have their times of growing and still more of flowering at many or rather at all seasons; so that, if one will consider it, both the growing and the flowering are almost continuous throughout the year; for one continually succeeds to another, so that all seasons are covered; thus after the dandelion will come the crocus anemone groundsel and the other plants of winter, and after these those of spring summer and autumn. Some again, as was said, because they do not produce all their bloom at once, cover a longer season; for there are some that thus bloom, for instance dandelion bugloss chicory plantain, and others; but because of this continuity and overlapping it does not seem easy in some cases to define which first make growth and which are late in growing, unless one were to lay down that the 'year' begins when a certain season begins. Further in these plants it is not easy to define in each case the time of first growth and the season when, the fruits being matured, it makes a fresh start in reproduction. This seems chiefly to occur after the autumnal equinox; for by that time most of the seeds are matured, like most of the fruits of trees: moreover a change then takes place in the seed itself as well as in the season. But in the case of any seeds which are still immature and unripe and so are overtaken by winter, the period of first growth, the flowering of the new plant, and the period of maturity are proportionally later. Wherefore it comes to pass that some bloom at the solstice, some at the rising of the dog-star, and some after the rising of Arcturus and the autumnal equinox.

7.10.2

But these matters seem to require a wider investigation in order to determine when the process begins. However it is clear from what has been said that these plants present at least as many

differences as trees. For some again of this class are evergreen, as hylotrichon heliotropion and maidenhair.

7.10.3

Of the classes into which herbaceous plants may be divided, as those having a spike and chicory-like plants.

7.11.1

Having then made these explanations we must now give a separate account of each plant, discussing the differences (in those plants in which they occur) and saying how they arise except those peculiarities which belong to the character of individual kinds. I mean for instance the plants which have a spike, those which may be classed with wild chervil, and those which have a single stem, or any other such class in which one can find some such general characteristics obvious to the senses either in leaves flowers roots or fruits; (for the classification is to be made by the visible parts, as well as by the roots).

7.11.2

An example of the plants which have a spike is the plant which some call (dog's eye ' (rib-grass), which comprises several forms; we have also foxbrush,' stelephuros (plantain) which some call (lamb's tongue 'and some' quail-plant ' ; and somewhat similar to this is thryallis. These are simple plants and uniform in character, having a spike which is not pointed nor bearded; while in 'fox-brush' it is soft and somewhat downy, in that it actually resembles the brush of a fox, whence also it has obtained its name. Similar to this is stelephuros (plantain), except that it does not, like that plant, flower here and there, but all up the spike like wheat. The bloom of both is downy like that of corn, and the plants in their general appearance resemble wheat, but have broader leaves. Of the other plants which have a spike a similar account may be given.

7.11.3

The chicory-like plants all have annual leaves and have root-leaves, and they begin to grow after the Pleiad, except dandelion; but in their stems and roots they exhibit great differences; for in some these are simpler and fewer, but the stem of chicory is large and has many side-shoots; also it is tough and hard to break, wherefore it is used for withes; it makes side-growths from the root, and also has long roots, wherefore it is hard to kill; for, when the top is taken off to use as a vegetable, what remains starts growing again. Moreover different parts of it flower at different times, and the flowering goes on till autumn, since the stem appears to be hard. Also it bears a pod, which contains the seed, at the top of the stem.

7.11.4

Cat's ear is smoother and has a more cultivated appearance, and is also sweeter and not like khondrylla; for the latter is altogether uneatable and unfit for food, and its root contains a quantity of pungent juice.

7.11.5

Dandelion is also unfit for food and bitter: it flowers early and quickly waxes old and the flower turns to pappus; but then another flower forms, and yet another, and this goes on right through the winter and spring up to the summer; and the flower is yellow.

7.11.6

The like may be said of pikris: for this plant too blooms in spring, and like dandelion it flowers throughout the winter, and it flowers also to some extent in summer; in taste it is bitter, whence its name. These are the special points of difference about these plants; now we must endeavour, as was said/ to set forth the special points of the other classes in like manner.

7.12.1

There is a large class of these which have fleshy or bulbous roots: these exhibit differences both as compared with other plants and with one another both in roots leaves stems and

their other prominent features. Of the roots, as has been said already, some are in layers, some fleshy, some have a 'bark,' some not; and again some are round, some oblong, some edible and some not fit for food. Among edible roots are not only purse-tassels and others which resemble them, but also the roots of asphodel and squill, though not of all kinds of the latter, but only of the kind called 'Epimenides' squill (French sparrow-grass) which gets its name from its use; this kind has narrower leaves and is smoother than the others.

7.12.2

The root of cuckoo-pint is also edible, and so are the leaves, if they are first boiled down in vinegar; they are sweet, and are good for fractures. To increase the root, having first stripped off the leaves (and the leaf is very large), they dig it up and invert it in order that it may not shoot, but may draw all the nourishment into itself. This some do also with purse-tassels, when they lay them by. However the root of edder-wort (for a kind of cuckoo-pint is so called because of its variegated stem) is not good for food, but is used for drugs.

7.12.3

But the root of the plant called corn-flag is sweet, and, if cooked and pounded up and mixed with the flour, makes the bread sweet and wholesome. It is round and without 'bark,' and has small offsets like the long onion. Many of them are found in moles' runs; for this animal likes them and collects them.

7.12.4

The root of theseion is bitter to the taste, but when pounded purges the bowels. There are also certain others of these roots which have medicinal properties, but of many the roots are neither medicinal nor edible. Such are the differences in the roots.

7.13.1

In the leaves the differences are in size and shape. Asphodel has a long leaf, which is somewhat narrow and tough, while that of

squill is broad and tears easily; corn-flag, which is called by some xiphos ('sword'), has a sword-like leaf, whence its name, and iris one more like a reed. That of cuckoopint, in addition to being broad, is concave and like that of cucumber; that of the narcissus is narrow substantial and glossy, those of purse-tassels and plants of that character are quite narrow, and that of crocus narrower still.

7.13.2

Some have not a stem at all, nor a flower, as the edible cuckoo-pint; some have only the flower-stem, as narcissus and crocus; some however have a stem, as squill purse-tassels iris and corn-flag; but asphodel has the tallest of all for the antherikos (asphodelstalk) is very tall: that of iris is smaller, but tougher, though in general it is like the asphodel-stalk. Asphodel also produces much fruit, and its fruit is woody: in shape it is triangular and in colour black; it is found in the round vessel which is below the flower, and it falls out in summer when this splits open. It does not produce all its flowers at once; in which respect it resembles squill, but the flowering begins at the bottom. In the stalk of asphodel forms a grub which changes into another creature like a hornet, and then, when the stem withers, eats its way out and flies away. A peculiarity of the plant as compared with others which have a smooth stem appears to be that, though it is slender, it has outgrowths at the top. It provides many things useful for food: the stalk is edible when fried, the seed when roasted, and above all the root when cut up with figs; in fact, as Hesiod says, the plant is extremely profitable.

7.13.3

Now all bulbous plants are tenacious of life, but especially squill; for this even lives when hung up and continues to do so for a very long time; it is even able to keep other things that are stored, for instance the pomegranate,' if the stalk of the fruit is set in it; and some cuttings strike more quickly if set in it; and it

is said that, if planted before the entrance door of a house, it wards off mischief which threatens it. All these bulbs grow in masses, as do onions and garlic; for they make offsets from the root, and some plainly are also increased by seed, as the asphodel polyanthus narcissus corn-flag and purse-tassels.

7.13.4

However it is said to be a peculiarity of pursatassels that all the seeds do not germinate at once, but some in the same year, some the next year; a like account is given of aigilops and trefoil. If then this is true, it is not peculiar to this plant. Nor perhaps is the following characteristic, which is not found in many plants and is marvellous wherever it does occur and it is found in squill and narcissus: namely that, whereas in most plants, whether those originally planted or those which are produced from them in season, the leaf comes up first and then presently the stem, in these plants the stem comes up first.

7.13.5

In the case of narcissus it is only the flower-stem which comes up, and it immediately pushes up the flower. But in squill it is the stem proper which thus appears, and presently the flower appears emerging from and sitting on it. And it makes three flowerings, of which the first appears to mark the first seed-time, the second the middle one, and the third the last one; for, according as these flowerings have occurred, so the crops usually turn out. But, when the flower-stem has waxed old, then the growth of the leaves follows many days later. So also is it with narcissus, except that it has no second stem besides the flower-stem, as we said, nor any visible fruit; but the flower itself perishes with the stem, and when it has withered, then the plant puts up its leaves.

7.13.6

These two plants then, as compared with the other bulbous plants are peculiar; and, as compared with those which bloom before the leaves and stems appear (as the autumn squill seems

to do, and other plants with conspicuous flowers, as well as, among trees, the almond especially, if not alone), there is the distinction that, while these two put forth their leaves along with the flowers or immediately afterwards (so that about some the matter is uncertain) in the case of these two the flower appears, as it were, from a different starting-point, there being a considerable number of days in between, and the growth of the leaves not beginning till, in the case of one of them, the flower, and in the case of the other, the whole stem has withered. Squill produces its leaves before the flower, narcissus afterwards; but the latter produces much more abundant foliage, and the individual root is small rather than large, resembling purse-tassels in shape, except that it is not formed of scales. About these matters then there is doubt.

7.13.7

Of purse-tassels it is plain that there are several kinds; for they differ in size colour shape and taste. In some places they are so sweet as to be eaten raw, as in the Tauric Chersonese. But the greatest and most distinct difference is shown by the 'wool-bearing' purse-tassels; for there is such a kind, and it grows on the sea-shore, and has the wool beneath the outer tunic, so that it is between the edible inside and the outside: of it are woven felt shoes and other articles of apparel. Wherefore this kind is woolly and distinct from the Indian kind, which is hairy.

7.13.8

There are also several kinds of plants of the same class as purse-tassels such as snowdrop, starflower, opition, kyix, and to a certain extent Barbary nut. These belong to this class only in having round roots; for in colour they are white, and the bulbs are not formed of scales. A peculiarity of Barbary nut is that the lower end of the root grows first, and this is called . . . ; it grows during winter, but, when spring appears, it decreases, while the upper part, which is edible, grows, Such are the

differences in these plants.

7.14.1

There are also the following peculiarities in herbaceous plants, for instance that which we find in wet-proof (maidenhair); the leaf does not even get wet when it is watered, nor does it catch the dew, because the dew does not rest on it; whence its name. There are two kinds, the white 'wet-proof' (English maidenhair), and the black (maidenhair); and both are useful to prevent the falling off of the hair of the head, for which purpose they are pounded up and mixed with olive-oil. They grow especially in damp places. Some think that trikhomanes (English maidenhair) is also useful in cases of strangury. Its stem is like that of the black kind, but it has small leaves, which are close set and grow in opposite pairs; there is no root below, and the plant loves shady places.

7.14.2

Of those plants which do not flower all at once anemone has the peculiarity that, while in all others the lower part flowers first, in this plant it is the upper part which does so; the outer circle of the flower is white, and the centre green; and the fruit falls off, as in spinous plants, leaving the attachment bare. There are several forms of it.

7.14.3

Bedstraw has the peculiarity that it sticks to clothes owing to its roughness, and it is hard to pull away; indeed it is in this rough part that the flower is contained: it does not project nor show, but matures within itself and produces seed; so that its habit is like that of weasels and sharks; for, as these animals likewise produce eggs in themselves and then bear their young alive, so this plant keeps its flower within itself, matures it and produces fruit.

7.15.1

As to these plants whose flowering time is dependent on the heavenly bodies, as the plant called heliotropion, golden thistle

(for this also blooms at the solstice), and also ‘swallow-plant’ (greater celandine) for this blooms when the Swallow-wind blows the reason in these cases would seem to be partly in their nature and partly accidental.

7.15.2

Such peculiarities are common in other plants also; thus it is the nature of the house-leek to remain always moist and green, its leaf being fleshy smooth and oblong. It grows on flat shores, on the earthy tops of walls, and especially on tiled roofs, when there is on them a sandy accumulation of earth.

7.15.3

Possibly one might mention many other eccentricities. But, as has been repeatedly said, we must only observe the peculiarities and differences which one plant has as compared with others. Some plants are found in several forms which have almost the same name, for instance the lotos; for of this there are many forms differing in leaves stems flowers and fruit, including the plant called melilotos; there are also forms differing in the virtues for which they are used as food, and again in their fondness for different localities. So too is it with many other plants.

7.15.4

Others are found in fewer forms, as strykhnos, which is a general name covering plants that are quite distinct; one is edible and like a cultivated plant, having a berry-like fruit, and there are two others, of which the one is said to induce sleep, the other to cause madness, or, if it is administered in a larger dose, death. The same thing may be observed in other plants which are widely different. Now about the other herbaceous plants enough has been said; but concerning corn and corn-like plants we must speak next; for this subject still lies before us.

8.1.1

BOOK VIII. OF HERBACEOUS PLANTS: CEREALS, PULSES, AND SUMMER CROPS.

Let the above suffice for an account of the other herbaceous plants; let us now discuss corn and corn-like plants in the same manner as those already treated; for this class of herbaceous plants we reserved.

8.1.2

There are two principal classes; there are the corn-like plants such as wheat, barley, one-seeded wheat, rice-wheat, and the others which resemble either of the first two; and again there are the leguminous plants, as bean, chick-pea, pea, and in general those to which the name of pulses is given. Besides these there is a third class, which includes millet, Italian millet, sesame, and in general the plants which belong to the summer seed-time, which lack any common designation.

8.1.3

There is only one single way of propagating these; they grow from seed, except that some may grow rarely and scantily from a root. There are two seasons for sowing most of them; the first and most important is about the setting of the Pleiad; this rule we find even Hesiod following with most authorities; wherefore some call it simply 'the seed-time.' Another time is at the beginning of spring after the winter equinox. However different crops are sown at the two seasons. For some of them love to be sown early, some late because they cannot bear the winters, while some will do not amiss at either season, both towards winter and towards spring.

8.1.4

Crops sown early are wheat and barley, and of these the latter is sown the earlier; also rice-wheat one-seeded wheat olyra, and others which resemble wheat. For all of these the time of sowing is about the same. Of leguminous plants bean and okhros? it may be said, are specially sown at this time; for these on account of their weakness like to be well rooted before the winter. Lupin is also sown early; in fact they say it should be sown straight from the threshing-floor.

8.1.5

Those which are sown late are certain special varieties of these very kinds, as a certain kind of wheat, and of barley the kind which is called 'three months barley' because it takes that time to mature; and among leguminous plants lentil tare pea. However some of these plants are sown at both seasons, as vetch and chick-pea; some also sow beans late, if they have missed the first seed-time. To speak generally, some crops are sown early because of their robustness, since they can stand the winters, some because of their weakness, so that their growth may be secured in the fine weather. These then are the two seasons; the third is that of the summer crops of which we spoke, which are sown millet Italian millet sesame, and also erysimon and horminon. Such then are the times for each.

8.1.6

Some are quicker in coming up, some slower. Barley and wheat generally come up on the seventh day, but barley is the earlier. Pulses take four or five days, except beans; for they, like some kinds of corn, require a longer time; in some places they take as much as fifteen days, or even twenty. This crop indeed is the slowest to start of all, and if after the sowing there is a long spell of wet weather, it is extremely slow. Whether the sprouting of crops sown at the spring seed-time is quicker because of the season is matter for enquiry.

8.1.7

These times of sprouting or germination must be taken generally; for at some times and places germination takes fewer days, as with barley in Egypt, where it is said to come up on the third or fourth day; while elsewhere it takes longer than the period mentioned, which is not surprising when both soil and climate are different, when one makes the sowing earlier or later, and when the crop is subjected to different influences afterwards. For open light soil with a favourable climate produces quick and easy growth, while soil that is sticky and

heavy tends to slow growth, and that of a specially dry district to slower growth still.

8.1.8

Moreover the time of growth is affected, according as storms supervene, or droughts, or fine weather or again rain; for these conditions make wide differences. So too it makes a difference if the ground has been well tilled and given dung, or if neither of these things has been done: for the soil makes a difference even as to the early or late germination of each crop. In Hellas some are used to sow everything earlier because of the coldness of the soil, for instance the Phocians; the object being that the winter may not overtake the crop while it is still tender.

8.2.1

In germinating some of these plants produce their root and their leaves from the same point, some separately, from either end of the seed. Wheat barley one-seeded wheat, and in general all the cereals produce them from either end, in a manner corresponding to the position of the seed in the ear, the root growing from the stout lower part, the shoot from the upper part; but the part corresponding to the root and that corresponding to the stem form a single continuous whole. Beans and other leguminous plants do not grow in the same manner, but they produce the root and the stem from the same point, namely the point at which the seed is attached to the pod, which, it is plain, is a sort of starting point of fresh growth. In some cases there is also a formation resembling the penis, as in beans chick-peas and especially in lupins; from this the root grows downwards, the leaf and the stem upwards.

8.2.2

There are then these different ways of germinating; but a point in which all these plants agree is that they all send out their roots at the place where the seed is attached to the pod or ear, whereas the contrary is the case with the seeds of certain trees, as almond hazel acorn and the like. And in all these plants the

root begins to grow a little before the stem; whereas in certain trees the bud first begins to grow within the seed itself, and, as it increases in size, the seeds split for all such seeds are in a manner in two halves, and those of leguminous plants again all plainly have two valves and are double and then the root is immediately thrust out; but in cereals, since the seeds are in one piece, this does not occur, but the root grows a little before the bud.

8.2.3

Barley and wheat come up with a single leaf, but peas beans and chick-peas with several. All the leguminous plants have a single woody root, and also slender side-roots springing from this. The chick-pea is about the deepest rooting of these, and sometimes it has side-roots; but wheat barley and the other cereals have a number of fine roots, wherefore they are matted together. Again all such plants have many branches and many stems. And there is a sort of contrast between these two classes; the leguminous plants, which have a single root, have many side-growths above from the stem all except beans; while the cereals, which have many roots, send up many shoots, but these have no side-shoots except such sorts of wheat as are called sitanias and krithanias ('barley-wheat').

8.2.4

During winter cereals remain in the blade, but, as the season begins to smile, they send up a stem from the midst and it becomes jointed. And it comes to pass that the ear also at once appears in the third, or in some cases in the fourth joint, though it is not distinctly seen in the mass of growth (the whole stem contains more joints than three or four), so that it must be formed at the same time that the straw grows or but a little later; though it does not become conspicuous till it has first swollen and formed in the sheath, and by that time its size makes its development visible.

8.2.5

Four or five days after being set free wheat and barley flower and remain in bloom for a like number of days; those who put the period at the longest say that the bloom is shed in seven days. On the other hand the flowering period of leguminous plants lasts a long time; that of vetch and chick-pea is longer than that of most, but that of the bean is far longer than that of any of them; they say that it is in bloom for forty days; some however give this period absolutely, others say that at different times different parts are in flower, since the whole plant does not flower at once. For plants with an ear bloom all at once, but plants with pods and all leguminous plants bloom part at a time; the lower part blooms first, and, when this bloom has fallen, the part next above it, and so on up to the top. Wherefore, at the time when some of the vetches are gathered, the lower seeds have already fallen, while the upper ones are still quite green.

8.2.6

After the flowering is over wheat and barley develop and mature in about forty days; one-seeded wheat and other such plants take about the same time. So too, they say, does the bean, which blooms and matures in a like number of days: but the others take fewer, and fewest of all the chick-pea, since, as some say, it takes only forty days from the time when it is sown to that when it is mature; and in any case it is clear that the plant as a whole develops very rapidly. Millet sesame Italian millet and the summer crops in general, it is fairly well agreed, take the same number of days, that is, forty; though some say that they take less.

8.2.7

Again as to the development of the plant there are differences according to soil and climate. Some soils seem to produce the crop in fewer days; for instance, Egypt may be given as a specially conspicuous example; in that country barley is reaped in six months and wheat in seven: while in Hellas the barley harvest is in the seventh month, or in most parts in the eighth,

and wheat requires an even longer time. However even in Egypt the whole harvest is not gathered at such an early date, but only what is required for the first-fruits; for they gather new grain for the meal required in certain sacrifices in the sixth month, and that too in the regions high up the Nile, above Memphis.

8.2.8

It is said also that in the Messenian district in Sicily at the place called Mylae the late sown crops mature rapidly; thus the sowing of pulses goes on for six months, but he that made the last sowing gathers his crop at the same time as the first: also that the soil is exceedingly good, so that it yields thirty-fold; and there are also wonderful pastures and forest-land. They tell of an even more wonderful thing in Melos; there they reap thirty or forty days after sowing; wherefore it is a saying of the islanders that one should continue sowing till one sees a swathe. However it is said that pulses in their country do not grow like this, nor are they abundant. Yet they say that the soil is wonderfully productive; for it is good both for corn and olives, and fairly good for vines.

8.2.9

However what occurs in Chalkia, an island belonging to the Rhodians, goes even beyond this and is more extraordinary than all the instances given; there they say that there is a place which is so early and so fertile that, when the barley is sown after reaping the crop with the other crops, they then sow again, and then reap the crop thus sown at the same time as the remaining crops; this then, if it be true, marks a difference greater than we find anywhere else. For it is less surprising that there should be a difference in crops transferred to another region, as they say occurs when they are transferred from Cilicia to Cappadocia or in general beyond the Taurus; for these regions are obviously very dissimilar. But that one particular land should produce two crops in the time that other lands to which it is close take to produce one, is very remarkable;

wherefore Chalkia exhibits the greatest difference.

8.2.10

The crops grown in other regions show not much, if any, dissimilarity as to time; those grown at Athens are only about thirty days or not much more before those of the Hellespont region. Now, if the sowing should turn out to be also earlier, that would shift the season back; if it is at the same time, it is plain that the difference of time would be greater.

8.2.11

Again the particular district makes a considerable difference, even as between places which are not far apart; thus the crops of Salamis are far earlier than those of the rest of Attica, and so in general are those of places by the sea; and this applies to other fruits as well as these: for instance, those of the place called Akte in the Peloponnese and of Phalykos in the Megarid are early; but here something is contributed by the fact that the soil is light and crumbling. Such are the facts in regard to growth and development.

8.3.1

There are also differences between the whole classes which we have mentioned, namely cereals leguminous plants and summer crops, as well as between the several members of the same class. Cereals have the leaf of a reed, while of leguminous plants some have a round leaf, as beans and most others, some a more oblong leaf, as pea, lathyros okhros and the like. Some again have fibrous leaves, others leaves without veins and fibres. Again sesame and erysimon have leaves quite distinct from these.

8.3.2

Again the stem of cereals is jointed and hollow; wherefore it is called the 'reed,' while that of the bean is hollow, and that of the other leguminous plants is more woody, that of chick-pea woodiest of all; of the summer crops that of millet and Italian millet is reed-like, that of sesame and erysimon is more like the

stem of ferula. Some again have erect stems, as wheat barley and in general the cereals and summer crops; some have rather a crooked stem, as chick-pea vetch lentil; some a creeping stem as okhros pea lathyros; while calavance, if long stakes are set by it, climbs them and becomes fruitful, whereas otherwise the plant is unhealthy and liable to rust; the bean, most of all leguminous plants, if not alone among them, has an erect stem.

8.3.3

The flowers also shew differences in character and in position (of which matters we have to some extent treated in our general account); thus some are 'downy,' as those of corn and of any plant that has an 'ear'; others are 'leafy,' as those of leguminous plants, and in most cases they are irregular flowers; for most of these have such flowers. Those of millet and Italian millet are also 'downy' those of sesame and erysimon 'leafy.' Another difference is that in some cases the flowers are round the fruit; thus those of corn and millet are round the ear; while in leguminous plants the fruit comes as it were from the flower itself, or at least from the same starting-point. Another difference is that some produce all their flowers at once, others in succession. And there are other differences akin to these.

8.3.4

In like manner there are differences in the fruits; some have an 'ear', leguminous plants a pod, and millet-like plants a 'plume' which is the name given to an inflorescence such as reeds have. Again, generally speaking, some have their seeds in a vessel, some in pods, some naked; and further some bear their fruit at the top, some at the sides; and there are other differences which bear on this enquiry.

8.3.5

In general the leguminous plants produce more fruit and are more prolific, and the summer crops millet and sesame are even more so than these, while among the leguminous plants themselves lentil is the most prolific. Generally speaking, those

that have small seeds are more prolific, as cummin among pot-herbs, which are all prolific of seed. The seeds of cereals are more robust as to standing winter and conditions of climate generally, while those of leguminous plants are stronger as to providing food. However it may be that in this respect the other animals are affected in the opposite way to men.

8.4.1

There are then these differences between the various classes; and as between plants of the same class there are plainly differences due to the unlikeness in the various parts. Thus among cereals wheat as compared with barley has a narrower leaf, and a smoother stem of closer texture tougher and less brittle. Again the seed of wheat has several coats/ that of barley is naked, that plant having its seeds specially naked. Also one-seeded wheat rice-wheat and all such plants have their seed in several coats, and above all, it may be said, is this true of oats. Also the 'reed' of wheat is taller than that of barley, and wheat has its ear less distant from the 'leaf.'

8.4.2

Further the husk of wheat is distinct from that of barley, being less dry and softer. Barley also differs from wheat in this respect; it has grains in a regular row, whereas those of wheat are not in a row, but the ear is as it were quite simple in form.

8.4.3

Such then are the differences as between one whole kind and another. But in each of these kinds again, for instance in barley and wheat, there are many sub-divisions differing both in the actual fruits, in the ear, and in the other characteristic features; and again in capacities and properties. Of barley different sorts have respectively two, three, four, and five rows of seeds; the largest number known is six, for there is a kind which bears that number. And those which have more rows have generally the grains set closer together. Another great difference is that of having side-shoots, as we said of the Indian kind. Again in

barley the ears are in some kinds large and of looser make, in some smaller and set closer; in some kinds the ear is some way from the 'leaf,' in some it is nearer to it, as in the kind called 'Achillean.' Again of the grains themselves some are rounder and smaller, some more oblong and larger and set at wider intervals on the ear. Moreover some are white, some black or reddish, and the latter are thought to produce much meal and to be more robust than the white as to bearing winter wind or conditions of climate generally.

8.4.4

There are also many kinds of wheat which take their names simply from the places where they grow, as Libyan Pontic Thracian Assyrian Egyptian Sicilian. They show differences in colour size form and individual character, and also as regards their capacities in general and especially their value as food. Some again get their distinctive names for other reasons, as kankhrydias stlengys 'Alexandrian'; all of which must be distinguished by the above-mentioned characters. Again, if one takes such differences as the following, they are quite characteristic thus some are early, some late, some are vigorous and prolific, some are small and produce little, some have a large, some a small ear. The ears of some remain a long time in the sheath, of some it remains but a short time, as that of the Libyan kind. Again some have a slender, some a stout haulm; the Libyan kind has this characteristic also, and that of kankhrydias is also stout. Again the grain of some has few coats, of some many, for instance the Thracian. Some kinds have a single reed,' some more than one, and in the latter class the number varies.

8.4.5

So too must we distinguish any differences like these or those mentioned above which are found in the several capacities; for these would seem to be the most essential differences. In this connexion we may distinguish kinds which mature in three or in two months, and those, if there be such, which take a less

number of days; for instance, they say that in the region of Aineia there is a kind which ripens and attains perfection within forty days from the time of sowing; they say too that this grain is strong and heavy, not light like that which takes three months; wherefore they give it even to the servants, for it also does not contain much bran. Now this kind is the rarest and the quickest to mature. But there is also a kind which takes two months; this was brought to Achaia from Sicily; it is not however prolific nor fertile, though as food it is light and sweet. There is another such kind which grows in Euboea and especially in the region of Karystos. There are several kinds that take three months, and these, wherever they are found, are light and not prolific; their growth consists of a single 'reed,' and in general they are not robust. Lightest of all we may say is the Politic wheat; the Sicilian is heavier than most of those imported into Hellas, but heavier still than this is the Boeotian; in proof of which it is said that the athletes in Boeotia consume scarcely three pints, while, when they come to Athens, they easily manage five. The Laconian kind is also light. The reason for these differences is to be found in the respective soils and in the climate; for in Asia not far from Bactra they say that in a certain place the corn is so vigorous that the grains grow as large as an olive-stone, while in the country called that of the Pissatoi it is so strong that, if a man eats too much of it, he bursts, which was actually the fate of many of the Macedonians. There is one curious thing about the corn of Pontus, which is an exception to the rule as to the lightness of crops raised in three months; for there the hard crops are those of the spring, the soft ones those of the winter; for soft kinds are exceedingly light. Two sowings, as it appears, are made of all corn alike, one in winter and one in spring, at which time they also plant the seed of the pulses.

8.4.6

Some kinds are free from darnel, as the Pontic and the Egyptian; the Sicilian is also fairly free from it, and that of Akragas is especially immune from darnel.

8.4.7

Peculiar however to the Sicilian is the plant called melampyron? which is harmless and not, like the darnel, injurious and productive of headache. However such peculiarities, as was said, must be ascribed to the soil, and to a certain extent to the different characters of different kinds.

8.5.1

In pulses we cannot find such differences to the same extent, whether for the want of equally careful enquiry or because there is actually less diversity in these plants. For, apart from chickpea lentil and to a certain extent bean and vetch (in so far as in these we find differences of colour and taste), among the rest no distinct forms are recognised. Chick-peas however differ in size colour taste and shape; thus there are the varieties called 'rams,' 'vetch-like' chick-peas and the intermediate forms. In all pulses the white are the sweeter, and this applies to vetch lentil chick-pea bean and sesame, of which also there is a white form.

8.5.2

However it is more possible to recognise the differences in such points as these: all these plants have pods, but whereas the pods in some kinds have no divisions, but the seeds as it were touch one another, as in vetch pea and most kinds, in some there are divisions, as in lupin and still more in sesame, in which the divisions are of a peculiar kind. Again some have long, some round pods, as chick-pea. And the number of seeds follows in proportion, since they are fewer in the small pods, as in those of chick-pea and lentil.

8.5.3

Possibly these differences correspond to those which we mentioned in the case of cereals as to the ears and the actual fruits; for what are called 'pods' also fairly correspond to the

shape of the seeds, some being flat, as those of lentil and tare, some more or less cylindrical, as those of vetch and pea: for in the case of either pair of plants the seeds correspond in shape. However one might discover and distinguish many such differences, of which some are common to a whole kind, others special to particular varieties.

8.5.4

In all cases the seeds are attached to the pods and have a sort of starting-point, which in some cases projects, as in bean and chick-pea, in some is hollow, as in lupin and some others, and in some is not thus conspicuous but smaller and, as it were, only indicated; this is plain from observation; it is from this point that the seeds germinate and take root when they are sown, as was said: but to start with they are themselves nourished by being so attached to the pod until they are matured. This is clear both from what is said now and from what was said before. Enough then about the points of difference.

8.6.1

It is expedient to sow all these, if possible, at the early seed-time; however some plant the seed even in dry ground, and especially wheat and barley, on the theory that they are most likely to hold their own at a time when the ground is not infested with birds or other creatures. For it appears that in general the first sowing is better, and worst that which is made in half-soaked ground; for then the seeds perish and become 'milky'; moreover many weeds come up at that time. After the sowing however it is beneficial for all that rain should fall on them, except in the case of some which appear to germinate then with more difficulty, as seems to be the case with beans, and among summer crops with sesame cummin and cress.

8.6.2

As to sowing thickly or scantily one should have regard to the soil as well as to other considerations; for a fat good soil can bear more than one which is sandy and light. However there is a

saying that the same soil can take at one time more, at another less seed; and in general the former condition is taken as an unfavourable omen, for then they say at once that the soil is hungry; however this is perhaps a rather foolish saying. If a man should have regard to the kind of the seed and especially to the actual situation, considering the aspect in respect of winds and sun, as well as the soil itself, he would more properly gauge the differences.

8.6.3

Similarly manuring for the sown crops should be done with regard to the soil; and it is better to turn up fallow land in winter than in spring. And there are some places in which deep ploughing is not expedient, as in Syria; wherefore they use small plough-shares. In other parts to work the ground too much is injurious, as in Sicily: wherefore many settlers in the country, it appears, make a mistake. From every point of view therefore the soil must be considered.

8.6.4

The seeds are also classified according as each suits a particular soil; in wintry lands wheat is sown rather than barley, and in general they say that corn rather than leguminous plants should be sown in barren soils which are only disturbed at long intervals; and such soils bear wheat better than barley. Moreover wheat welcomes abundant rain more than barley, and bears better on land which is not manured. In like manner they distinguish among wheats themselves which suits which kind of soil, namely which grows best in good fat soil and which in crumbling light soil, and so on with other kinds of soil.

8.6.5

More abundant rain is beneficial to all crops when they have come into leaf and formed the flower; however it is harmful to wheats and barleys and other cereals when they are actually in flower; for it destroys the flower. But to pulses it is harmless, except to chick-peas; for these, if the salt is washed off them,

perish from rot or from being eaten by caterpillars. However the black and the red chick-pea are stronger than the white, and it is beneficial, they say, to sow this crop late in moist soil. The bean likes especially to receive rain when it is in flower; wherefore men are unwilling, as we said, to sow it late, because it flowers for a long time; but after it has shed its flowers, it needs very little water, since its time of maturity is now near. But, when cereals have matured, it appears that water actually injures them, and barley more than wheat.

8.6.6

In Egypt Babylon and Bactra, where the country receives no rain, or but little, the dews are sufficient nourishment; and so is it also in the regions about Cyrene and the Euesperides. However to all, generally speaking, it is the spring rains which are the most seasonable; and that is why Sicily is rich in corn; for there is abundance of soft rain in spring and little of it in winter. A light soil requires plenty of rain, but little at a time; while that which is fat can indeed bear both an abundance of rain and a drought; (for a droughty country seawinds and breezes seem to be helpful, and various breezes of this kind prevail in various countries, as has been said already). Yet in general drought suits corn better than excessive rain; for heavy showers, apart from the harm which they do in other ways, often actually destroy the seed, or at least cause a luxuriant growth of leafage, so that the grain is choked and becomes abortive.

8.7.1

Now, while it is not the nature of any other of these seeds to degenerate and change into something else, they say that wheat and barley change into darnel, and especially wheat; and that this occurs with heavy rains and especially in well-watered and rainy districts. But that darnel is not a plant of the spring, like other weeds (for some endeavour to make this out) is clear from the following consideration: it springs up and becomes

noticeable directly winter comes; and it is distinguished in many ways; the foliage is narrow abundant and glossy, and this gloss is the most marked of these differences; (the leaves of aigilops are indeed also abundant, but this character does not shew itself in them till spring). This then is peculiar to the seeds of wheat and barley, and also to those of flax; for that too, they say, turns into darnel.

8.7.2

A peculiarity of chick-pea as compared with other leguminous plants is that which has been mentioned as to its flowering; and also the fact that it is the quickest to mature its fruit, being very strong and woody; and again there is the fact that in general it does not reinvigorate the ground, since it exhausts it; but it destroys weeds, and above all and soonest caltrop. And in general it is not every kind of soil which suits it; the soil should be black and fat. Of the other leguminous plants the bean best reinvigorates the ground, even if it is sown thick and produces much fruit.

8.7.3

All those crops sown at the summer seed-time need little water, and they say also that spring water is better for them than rain water; and Italian millet and millet need less water, for, if they have too much, they shed their leaves. Millet is the robuster plant, Italian millet is sweeter and less robust. Sesame and lupin are not eaten green by any animal; whether the same is true of erysimon and horminon is matter for enquiry; for these too are bitter. Erysimon is like sesame and is oily; horminon is like cummin and black, and is sown at the same time as sesame. These matters then require investigation.

8.7.4

In good soils to prevent the crop running wildly to leaf they graze and cut down the young corn, for instance in Thessaly. And the result is that, however often they graze it, the crop is not impaired; while if they cut it down not more than once, the

wheat changes in character and becomes tall and weak what they call long-shafted corn, and, if seed of this is sown, it does not recover its character. This the Thessalians tell of as having occurred in a few cases. At Babylon however they cut it down twice always and as it were systematically, and after that they let the sheep on to it; for in that case it makes its straw, but otherwise it runs wildly to leaf; and, if the ground is ill cultivated, it produces fifty fold, if it is carefully cultivated, a hundred fold. And the 'cultivation' consists in letting the water lie on it as long as possible, so that it may make much silt; for the soil being fat and close must be made open. And at Babylon the ground does not produce weeds and grasses, as it does in Egypt. Such are the things which depend on the quality of the soil.

8.7.5

Wheat and barley also in many places grow from the root in the next year, or in the same year from crops cut down for fodder, since a second haulm shoots up. The like happens also if the plant has been nipped by winter; for it shoots again when rain comes; but such plants produce an ear which is imperfect and under-sized. There is also new growth the next year from plants which are roughly treated or trodden down so that hardly anything remains visible, as happens when an army has marched over the field; the ears in such cases too are undersized and are called 'lambs.' But no kind of leguminous plant can do anything of the kind, or at least not to the same extent. In these various ways may new growth occur.

8.7.6

For growth and nourishment the climate is the most important factor, and in general the character of the season as a whole; for when rain, fair weather and storms occur opportunely, all crops bear well and are fruitful, even if they be in soil which is impregnated with salt or poor. Wherefore there is an apt proverbial saying that it is the year which bears and not the

field.

8.7.7

But the soil also makes much difference, according as it is fat or light, well watered or parched, and it also makes quite as much difference what sort of air and of winds prevails in that region; for some soils, though light and poor, produce a good crop because the land has a fair aspect in regard to sea breezes. But, as has been repeatedly said already, the same breeze has not this effect in all places; some places are suited by a west, some by a north, some by a south wind.

8.7.8

Again the working of the soil and above all that which is done before the sowing has an important effect; for when the soil is well worked it bears easily. Also dung is helpful by warming and ripening the soil, for manured land gets the start by as much as twenty days of that which has not been manured. However manure is not good for all crops; and further it is beneficial not only to corn and the like but to most other things, except fern, which they say it destroys if it is put on. (Fern is also destroyed if sheep lie on it, and, as some say, lucerne is destroyed by their dung and urine.)

8.8.1

There is a particular kind of soil which best suits each kind of seed, whether we compare one class with another or those of the same class; and attempts are made to distinguish these. Foreign seeds change into the native sorts in about three years. It is well that they should be imported from a warm climate to one that is rather less warm, or from a cold one to one that is rather less cold. Those imported from a wintry climate, if they be those of early crops, are late in coming into ear, so that they get destroyed by drought unless rain late in the season saves them. Wherefore they say that one should take good heed not to mix foreign with native seeds, unless they come from a similar place, since they do not agree with the soil as to the time of

being sown and of germinating, and accordingly need different cultivation; and so that one should take good heed to the differences of soil, the properties of the seed, and further the seasons appropriate to each.

8.8.2

When however there is a good season, the grain also is fuller. For instance at Athens the barley produces more meal than anywhere else, since it is an excellent land for that crop; and this is so, not merely when a very large crop is sown, but when the weather has been favourable for it. And in Phocis about Elateia the wheats produce half as much meal again as elsewhere; while at Soli in Cilicia this is true of both wheat and barley; and in other parts there are other crops for which the soil is severally well adapted. Wherefore grain turns out better or worse because of the soil as well as because of cultivation; for in some places it changes into the cultivated from the wild form, or the reverse, like trees; and in general it changes according to the soil in which it is grown, just as some trees, when transplanted, forthwith deteriorate.

8.8.3

But no kind can change altogether into another, except one-seeded wheat and rice-wheat, as we said in our previous discussions, and darnel which comes from degenerate wheat and barley: at least, if this is not the true account, darnel loves chiefly to appear among wheat, as does the Pontic melampyros and the seed of purse-tassels, even as other seeds appear in other crops; thus aigilops seems to grow for choice among barley, and among lentils the rough hard kind of arakos, while among tares occurs the axe-weed, which resembles an axe-head in appearance. Indeed in the case of nearly every crop there is a plant which grows up with it and mingles with it, whether this is due to the soil, which is a reasonable explanation, or to some other cause. Some plants of this character evidently attach themselves to more than one kind of crop, but, because they are

specially vigorous in some one particular crop, they are thought to be peculiar to that one, as ‘vetch-strangler’ (dodder) to vetches and bedstraw to lentils. But the former gains the mastery over the vetches especially because of the weakness of that plant; and bedstraw is specially luxuriant among lentils; to some extent it resembles dodder, in that it overspreads the whole plant and holds it fast as it were in coils, for it is thus that dodder strangles the plant, and this is the origin of its name (‘vetch-strangler’).

8.8.4

The plant which springs up straight from the roots of cummin and the plant called broom-rape which similarly attaches itself to ox-horn (fenugreek) are somewhat more peculiar in their habits. Broom-rape has a single stem, and is not unlike . . ., but is much shorter and has on the top a sort of head, while its root is more or less round; and there is no other plant which it starves except fenugreek. These plants grow in light and not in fat soils; thus in Euboea they do not occur at Lelanton, but only about Kanethos and in districts of like character.

8.8.5

The reason then why these plants, which attach themselves to more than one kind, grow stronger when attached to the plants specified, is that the latter are not robust.

8.8.6

The terms ‘cookable’ and ‘uncookable’ are only applied to pulses, but it is not unreasonable to suppose that conditions like those indicated, if not identical with them, occur also in cereals, though they are not so obvious, since these plants are not put to the same use. Indeed it is said that these terms are not applied even to all pulses alike, but chiefly to beans and lentils, either because these are specially subject to these conditions, or because the use to which they are put makes them more conspicuous. At all events the conditions occur for a variety of reasons; for in many parts there are places which regularly

produce seeds that are 'cookable,' while others again produce seeds that are 'uncookable'; in general however it is light soils which tend to produce the former. Now it is a certain condition of the climate which causes this variation; a proof of which is the fact that the same piece of land, tilled in the same manner, produces sometimes seeds that are 'cookable,' sometimes seeds that are 'uncookable.' In the district of Philippi, if the beans, while being winnowed, are caught by the prevailing wind of the country, they become 'uncookable,' having previously been 'cookable.' These facts prove that for various reasons, of districts which are close together, have the same aspect and shew no difference of soil, some bear 'cookable' some 'uncookable' seeds, and that sometimes when there is only the breadth of a furrow between them.

8.9.1

Wheat exhausts the land more than any other crop, and next to it barley; wherefore the former requires good soil, while barley will bear even on somewhat crumbling soils; and of leguminous plants chick-pea is the most exhausting, although this crop is in the ground only a very short time. Beans, as was said, are in other ways not a burdensome crop to the ground, they even seem to manure it, because the plant is of loose growth and rots easily; wherefore the people of Macedonia and Thessaly turn over the ground when it is in flower.

8.9.2

Of the plants which resemble wheat or barley such as zeia (rice-wheat) one-seeded wheat olyra (rice-wheat) oats aigilops zeia is the strongest and most exhausts the ground; for it has many roots which run deep and many stems; but its fruit is the lightest and is welcome to all animals. Of the rest oats is the most exhausting; for this too has many roots and many stems. Olyra is a more delicate plant and not so robust as these. But one-seeded wheat is the crop which is of all the least burdensome to the soil; for it has but a single slender stem;

wherefore also it requires a light soil and not, like zeia, one that is fat and good. These last two, zeia and one-seeded wheat, are also those which are likeliest to wheat, while aigilops and oats are as it were wild and uncultivated things.

8.9.3

Aigilops also greatly exhausts the land, having many roots and many stems; while darnel is a plant which has become altogether wild. Of the crops sown at the summer seed-time sesame seems to be most severe on the land and to exhaust it most; yet millet has more numerous and stouter stems and more roots. Moreover there is a difference between crops which are called 'light' in relation to the soil and those called 'light' in regard to human use. For some, such as leguminous plants and millet, are light in one sense but not in the other; and, as was said, what is light for men is not necessarily so for the other animals. Now enough of these matters.

8.10.1

As to diseases of seeds some are common to all, as rust, some are peculiar to certain kinds; thus chick-pea is alone subject to rot and to being eaten by caterpillars and by spiders; and some seeds are eaten by other small creatures. Some again are liable to canker and mildew, as cummin. But creatures which do not come from the plant itself but from without do not do so much harm; thus the kantharis is a visitor among wheat, the phalangion in vetches, and other pests in other crops.

8.10.2

Generally speaking, cereals are more liable to rust than pulses, and among these barley is more liable to it than wheat; while of barleys some kinds are more liable than others, and most of all, it may be said, the kind called 'Achillea.' Moreover the position and character of the land make no small difference in this respect; for lands which are exposed to the wind and elevated are not liable to rust, or less so, while those that lie low and are not exposed to wind are more so. And rust occurs chiefly at the

full moon. Again wheat and barley are destroyed by winds, if they are caught by them either when in flower, or when the flower has just fallen and they are weak; and this applies specially to barley, indeed it occurs when the grain is already ripening, if the winds are violent and last a long time; for they dry up and parch the grain, which some call being 'wind-bitten.' Also a hot sun after cloudy weather destroys both, and wheat more than barley, so that the ear is not even conspicuous, since it is empty.

8.10.3

Wheat is also destroyed by grubs; sometimes they eat the roots, as soon as they appear, sometimes they do their work when by reason of drought the ear cannot be formed; for at such times the grub is engendered, and eats the haulm as it is becoming unrolled; it eats right up to the ear and then, having consumed it, perishes. And, if it has entirely eaten it, the wheat itself perishes; if however it has only eaten one side of the haulm and the plant has succeeded in forming the ear, half the ear withers away, but the other half remains sound. However it is not everywhere that the wheat is so affected; for instance this does not occur in Thessaly, but only in certain regions, as in Libya and at Lelanton in Euboea.

8.10.4

Grubs occur also in okhros, lathyros and peas, whenever these crops get too much rain and then hot weather supervenes; and caterpillars occur in chick-peas under the same conditions. All these pests perish, when they have exhausted their food, whether the fruit in which they occur be green or dry, just as wood-worms do and the grubs found in beans and other plants, as was said of the pests found in growing trees and in felled timber. But the creature called 'horned worm'⁶ is an exception. Now in regard to all these pests the position makes a great difference, as might be expected. For the climate, it need hardly be said, makes a difference according as it is hot or cold, moist

or dry; and it was the climate which gave rise to these pests; wherefore they are not always found even in places in which they ordinarily occur.

8.11.1

The seeds have not all the same capacity for germination and for keeping well. Some germinate and mature very quickly, and keep excellently, as Italian millet and millet. Some germinate well, but soon rot, as beans, and especially those that are 'cookable'; so do tare and calavance; also barley perishes sooner than wheat; and dusty grain and that which is kept in plastered store-rooms perishes sooner than that which is kept in unplastered rooms.

8.11.2

Again, as seeds decay, they engender special creatures, except chick-pea, which alone engenders none. As they rot, all produce a grub; but, as they get worm-eaten, each produces a special creature. Chick-pea and vetch keep best of all, and better still than these lupin; but this, as it were, is like a wild kind.

8.11.3

It appears that soil and climate make a difference as to whether the seed gets worm-eaten or not; at least they say that at Apollonia on the Ionian Sea beans do not get eaten in this way at all, and therefore they are put away and stored; and about Cyzicus they keep an even longer time. It also makes a great difference to keeping that the seed should be gathered dry, for then there is less moisture in it. However the seeds of leguminous plants are gathered with a certain amount of moisture in them, because then they can be collected in greater quantity and more easily; for otherwise they are soon shed and get shrivelled up and split; and wheat and one kind of barley are gathered before they are dry, because then they are better for meal.

8.11.4

Wherefore the grain of wheat and barley is put into heaps, and it seems to ripen in a heap rather than to lose substance. (However corn does not get worm-eaten when it is reaped after exposure to rain.) Also corn lasts better than other things if it is left standing, and so does lupin to an even greater extent; indeed this crop is not even gathered till rain has fallen, because, if it is gathered, the seed springs out and is lost.

8.11.5

For propagation and sowing generally seeds one year old seem to be the best; those two or three years old are inferior, while those kept a still longer time are infertile, though they are still available as food. For each kind has a definite period of life in regard to reproduction. However these seeds too differ in their capacity according to the place in which they are stored. For instance, in Cappadocia at a place called Petra they say that seed remains even for forty years fertile and fit for sowing, while as food it is available for sixty or seventy years; for that it does not get worm-eaten at all, like clothes and other stored-up articles. For that the region is, apart from this, elevated and always exposed to fair winds and breezes which prevail alike from the east, the west, and the south. They say that in Media also and other elevated countries the seed when stored keeps for a long time. And it is plain that chick-pea, lupin, vetch, millet and the like will keep a far longer time than these seeds, as they do even in districts of Hellas. However these peculiarities, as has been said, are due to the particular region.

8.11.6

There appears to be a kind of earth in some places, which when sprinkled over the seed helps to make wheat keep, for instance, the earth found at Olynthos and at Kerinthos in Euboea; this makes the grain inferior for food, but fuller in appearance; the earth is sprinkled in the proportion of one pint to twenty-four of grain.

8.11.7

All seeds if exposed to fire perish and become infertile. Yet they say that at Babylon the grains of barley and wheat jump on the threshing-floor like corn which is being parched. However it is plain that it is some particular kind of warmth which produces this effect: or else the jumping is simply another effect of heat. Such behaviour would appear to be common to most, if not to all kinds.

8.11.8

Some even of those kinds which seem to be more or less wild have peculiarities as to their germination and growth, for instance, lupin and aigilops. For lupin, although it is very robust, unless it is planted immediately after leaving the threshing-floor, turns out of poor growth, as was said, and refuses altogether to be buried in the ground; wherefore they sow it without first ploughing the land. And often if the seed has fallen amid thick undergrowth or herbage, it thrusts this aside, fastens on to the earth with its root and grows vigorously. It seeks sandy and poor soil for choice, and will not grow at all in cultivated soil.

8.11.9

Aigilops has the opposite character; it grows better in tilled soil; and in some places where at first it would not grow, if the ground is tilled, it grows and yields a large crop, and in general it likes good soil. A peculiarity mentioned in regard to it as compared with other cereal seeds is that one seed in two does not germinate for a year. Wherefore those who wish to destroy it entirely, (since it is naturally hard to destroy), leave the fields unsown for two years, and, when it springs up, send in the sheep several times till they have grazed it down, and this is a way of completely destroying it. At the same time this testifies to the fact that the seed does not all germinate at once.

9.1.1

BOOK IX. OF THE JUICES OF PLANTS, AND OF THE MEDICINAL PROPERTIES OR HERBS.

Moisture belongs to plants as such and some call it the 'sap' to give it a general name; and it plainly has special qualities in each plant. This moisture is attended by a taste, in some cases more, in some less, while in some it would seem to have none, so weak and watery is it. Now all plants have most moisture at the time of making growth, but it is strongest and most shows its character when the plant has ceased to grow and to bear fruit. Again in some plants the juice has a special colour; in some it is white, as in those which have a milky juice; in some blood-red, as in centaury and the spinous plant which is called distaff-thistle; in some green: and in some of other colours. And these qualities are more obvious in annual plants and those with annual stems than in trees.

9.1.2

Again in some plants the juice is merely thick, as in those in which it is of milky character; but in some it is of gummy character, as in silver-fir, fir, terebinth, Aleppo pine, almond, kerosos (bird-cherry), bullace, Phoenician cedar prickly cedar, acacia, elm. For this last also produces a gum, though it does not exude from the bark, but is found in the 'bag' of the leaves; there are also the juices from which come frankincense and myrrh; for these too are gums; so too are balsam of Mecca khalbane and any others of the kind that there may be, such as, they say, the Indian akantha, from which comes something resembling myrrh; and a similar substance forms on mastich and the spinous plant called urine (pine-thistle), whence mastic-gum is made.

9.1.3

All these have a fragrant odour, as in general have those which contain a viscous substance and are fatty; while those that are not fatty have no scent, as gum and the juice which exudes from the almond. The pine-thistle of Crete has also a gum, and so has the plant called tragacanth; this was formerly supposed to grow only in Crete, but now it is well known to grow also in Achaia in

the Peloponnese and elsewhere in Hellas and in Asia in the Median country. In all these plants the gum occurs in the stems the trunks and the branches, but in some plants it is found in the roots, as in alexanders, scammony and many other medicinal plants. In some it is found in the stem and also in the root; for of some plants they tap the stem and the roots as well, as is done with silphium.

9.1.4

Now the juice of alexanders is like myrrh, and some, having heard that myrrh comes from it, have supposed that, if myrrh is sown, alexanders comes up from it; for, as was said, this plant can be grown from an exudation, like the krinonia (lily) and other plants. The juice of silphium is pungent like the plant itself; for what is called the 'juice' of silphium is a gum. Scammony and similar plants, as was said, have medicinal properties.

9.1.5

In all the plants mentioned the juice either forms naturally, or when incisions are made, or in both ways, but it is obvious that men only make incisions in plants whose juice is of use and is specially sought after. Now there is no use in the gum which exudes from the almond, wherefore men do not tap it. However it is plain that in plants whose gum forms naturally the flow of juice is greater. The incisions and the clotting of the juice do not take place at the same season in all cases; but the juice of the vine clots best they say if the incision is made a little before budding begins, less well in the autumn or at the beginning of winter; (although in regard to production of fruit these seasons are the best in the case of most vines). However with terebinth fir or any other tree which produces resin the best time is after the period of budding; yet in general these trees are not cut every year, but at longer intervals. The frankincense and myrrh trees they say should be cut at the rising of the Dogstar and on the hottest days, and so also the Syrian balsam ' (balsam of

Mecca).

9.1.6

The cutting of these is also a more delicate matter and is done on a smaller scale; for the flow of juice is less. In those plants whose stem and root are both cut the stem is cut first, as also with silphium; and the juices so obtained are called respectively stalk-juice and root-juice, of which the latter is the better, for it is clear transparent and less liquid. The stalk-juice is more liquid, and for this reason they sprinkle meal over it to make it clot. The Libyans know the season for cutting, for it is they that gather the silphium. So also do the rootdiggers and those that collect medicinal juices, for these too tap the stems earlier. And in general all those who collect whether roots or juices observe the season which is appropriate in each case. And this remark applies generally.

9.2.1

Resin is made in the following manner: in fir it is done by removing the resinous wood after the tree has been tapped; for then the juice flows into the hole so made in greater abundance; in silver-fir and Aleppo pine it is done by tapping the wood, after tasting it. For there is no fixed rule for all alike; thus with terebinth they tap both the stem and the branches; but the juice which runs into the stem is always more abundant and better than that which flows into the branches.

9.2.2

There are also differences in the resin obtained from different trees. The best is that of terebinth; for it sets firm, is the most fragrant, and has the most delicate smell; but the yield is not abundant. Next comes that of silver-fir and Aleppo pine, for these are more delicate than that of the fir. But that of the fir is the most abundant, the grossest and the most pitchlike, because this tree has the greatest amount of resinous wood. It is carried about in baskets in a liquid state, and so acquires the more solid form which we know. However they say that in Syria pitch is

extracted even from the terebinth by burning; for there is in that land a mountain which, as we said before, is all covered with great terebinths.

9.2.3

Some say the same of Aleppo pine and also of Phoenician cedar; but this must be taken as only indicating what can be done, the practice not being common; for the people of Macedonia do not extract pitch by burning even from fir, except from the ‘male’ kind (they call the kind which bears no fruit the ‘male’); the ‘female’ kind they only treat in this way when they have found roots containing pitch; for all firs have resinous wood extending to the roots. The finest and purest pitch is that obtained from trees growing in a sunny position and facing north; that obtained from trees growing in shade is coarser and muddy; (in exceedingly shady places the fir does not even grow at all).

9.2.4

Again the yield may be either good or bad as to amount and as to quality; thus, when there is a moderate winter, it is abundant and good and whiter in colour, but, when there is a severe winter, it is scanty and of inferior quality. And it is these conditions, and not the tree’s capacity for bearing fruit, which determine the amount and quality of pitch.

9.2.5

The people of Mount Ida distinguish different kinds of fir, calling one ‘that of Mount Ida’ (Corsican pine), another the ‘fir of the seashore’ (Aleppo pine); and they say that the pitch obtained from the former is more abundant blacker sweeter and generally more fragrant in the raw state, but that boiling down reduces the amount; for that it contains a larger proportion of watery matter, wherefore it is less substantial; but that derived from the ‘fir of the seashore’ is browner and thicker in the raw state, so that the amount is less reduced by boiling down; that the ‘fir of Mount Ida’ however contains more resinous wood. And, speaking generally, they say that from an equal amount of

resinous wood more pitch is obtained and in a more liquid state in wet weather than during a drought, and from a wintry and shady position than from one that is sunny and enjoys fair weather. Such is the account given by the peoples of Mount Ida and of Macedonia respectively.

9.2.6

The holes for the pitch fill up, so that the pitch can be again removed/ in good firs in a year, in those of more moderate quality in two years, in poor trees in three. The filling-up is composed of the pitch; it is not caused by closing up of the wood; for the wood cannot close up and become one again, but the effect which takes the time mentioned is due to the formation of the pitch. However it is clearly inevitable that there should be some new growth of the wood too, seeing that the resinous wood is removed and burnt when the discharge of pitch takes place. So much for this account.

9.2.7

The people of Mount Ida however say that, when they bark the stem, and they bark the side towards the sun to a height of two or three cubits from the ground, the flow of pitch takes place in that part, and in about a year the wood becomes full of pitch; and that, when they have hewn this part out, pitch forms again in the next year, and in the third year in like manner; after which that the tree, because it has been cut away underneath, is rotted by the winds and falls; and that then they take out its heart, for that is especially full of pitch, and that they also extract pitch from the roots; for that these too, as we said, are full of pitch in all firs.

9.2.8

Now it is plainly to be expected that they should, as was said, repeatedly thus treat a good tree, but an inferior one at longer intervals, and that, if the tree is husbanded, the supply should hold out longer, while, if they remove all the pitch, it will not hold out so long; it appears as a matter of fact that the tree will

stand about three such removals of its substance. However firs do not produce both fruit and pitch at once; they begin to bear fruit when they are quite young, but they only produce pitch much later, when they are older.

9.3.1

This is the manner in which they make pitch by fire: having prepared a level piece of ground, which they make like a threshing-floor with a slope for the pitch to run towards the middle, and having made it smooth, they cleave the logs and place them in an arrangement like that used by charcoal-burners, except that there is no pit; but the billets are set upright against one another, so that the pile goes on growing in height according to the number used. And they say that the erection is complete, when the pile is 180 cubits in circumference, and fifty, or at most- sixty, in height; or again when it is a hundred cubits in circumference and a hundred in height, if the wood happens to be rich in pitch. Having then thus arranged the pile and having covered it in with timber they throw on earth and completely cover it, so that the fire may not by any means show through; for, if this happens, the pitch is ruined. Then they kindle the pile where the passage is left, and then, having filled that part up too with the timber and piled on earth, they mount a ladder and watch wherever they see the smoke pushing its way out, and keep on piling on the earth, so that the fire may not even shew itself. And a conduit is prepared for the pitch right through the pile, so that it may flow into a hole about fifteen cubits off, and the pitch as it flows out is now cold to the touch. The pile burns for nearly two days and nights; for on the second day before sunset it has burnt itself out and the pile has fallen in; for this occurs if the pitch is no longer flowing. All this time they keep watch and do not go to rest, in case the fire should come through; and they offer sacrifice and keep holiday, praying that the pitch may be abundant and good. Such is the manner in which the people of

Macedonia make pitch by fire.

9.3.2

They say that in Asia in the Syrian region they do not extract the pitch by cutting out of the tree the wood containing it, but use fire to the tree itself, applying an instrument fashioned on purpose, with which they set fire to it. And then, when they have melted out the pitch at once place, they shift the instrument to another. But they have a limit and indications when to stop, chiefly of course the fact that the pitch ceases to flow. They also, as was said before, use fire to get pitch out of the terebinth; for the places where this tree grows do not produce the fir. Such are the facts about resin and pitch.

9.4.1

As to frankincense myrrh balsam of Mecca and similar plants it has been said that the gum is produced both by incision and naturally. Now we must endeavour to say what is the natural character of these trees and to mention any peculiarities as to the origin of the gum or its collection or anything else. So too concerning the other fragrant plants; most of these come from places in the south and east.

9.4.2

Now frankincense myrrh cassia and also cinnamon are found in the Arabian peninsula about Saba Hadramyta Kitibaina and Mamali. The trees of frankincense and myrrh grow partly in the mountains, partly on private estates at the foot of the mountains; wherefore some are under cultivation, others not; the mountains, they say, are lofty, forest-covered and subject to snow, and rivers from them flow down to the plain. The frankincense-tree, it is said, is not tall, about five cubits high, and it is much branched; it has a leaf like that of the pear, but much smaller and very grassy in colour, like rue; the bark is altogether smooth like that of bay.

9.4.3

The myrrh-tree is said to be still smaller in stature and more bushy; it is said to have a tough stem, which is contorted near the ground, and is stouter than a man's leg; and to have a smooth bark like that of andrachne. Others who say that they have seen it agree pretty closely about the size; neither of these trees, they say, is large, but that which bears myrrh is the smaller and of lower growth; however they say that, while the frankincense-tree has a leaf like that of bay and smooth bark, that which bears myrrh is spinous and not smooth, and has a leaf like that of the elm, except that it is tiurly and spinous at the tip like that of kermes-oak.

9.4.4

These said that on the coasting voyage which they made from the bay of the Heroes they landed to look for water on the mountains and so saw these trees and the manner of collecting their gums. They reported that with both trees incisions had been made both in the stems and in the branches, but that, while the stems looked as if they had been cut with an axe, in the branches the incisions were slighter; also that in some cases the gum was dropping, but that in others it remained sticking to the tree; and that in some places mats woven of palm-leaves were put underneath, while in some the ground underneath was merely made level and clean; and that the frankincense on the mats was clear and transparent, that collected on the ground less so; and that that which remained sticking to the trees they scraped off with iron tools, wherefore sometimes pieces of bark remained in it. The whole range, they said, belongs to the portion of the Sabaeans; for it is under their sway, and they are honest in their dealings with one another. Wherefore no one keeps watch; so that these sailors greedily took, they said, and put on board their ships some of the frankincense and myrrh, since there was no one about, and sailed away. They also reported another thing which they said they had been told, that the myrrh and frankincense are

collected from all parts into the temple of the sun; and that this temple is the most sacred thing which the Sabaeans of that region possess, and it is guarded by certain Arabians in arms. And that when they have brought it, each man piles up his own contribution of frankincense and the myrrh in like manner, and leaves it with those on guard; and on the pile he puts a tablet on which is stated the number of measures which it contains, and the price for which each measure should be sold; and that, when the merchants come, they look at the tablets, and whichever pile pleases them, they measure, and put down the price on the spot whence they have taken the wares, and then the priest comes and, having taken the third part of the price for the god, leaves the rest of it where it was, and this remains safe for the owners until they come and claim it.

9.4.5

Others report that the tree which produces the frankincense is like mastich, and its fruit is like the fruit of that tree, but the leaf is reddish: also that the frankincense derived from young trees is whiter and less fragrant, while that derived from those which have passed their prime is yellower and more fragrant; also that the tree which produces myrrh is like the terebinth, but rougher and more thorny; that the leaf is somewhat rounder, and that, if one chews it, it resembles that of the terebinth in taste; also that of myrrh-trees too those that are past their prime give more fragrant myrrh.

9.4.6

Both trees, it is said, grow in the same region; the soil is clayey and caked, and spring waters are scarce. Now these reports are contradictory to that which says that the country is subject to snow and rain and sends forth rivers. However others make the statement that the tree is like the terebinth; in fact some say that it is the same tree; for that logs of it were brought to Antigonus by the Arabs who brought the frankincense down to the sea, and that these did not differ at all from logs of

terebinth. However these informants were guilty of a further more important piece of ignorance; for they believed that the frankincense and the myrrh were produced by the same tree. Wherefore the account derived from those who sailed from the city of Heroes is more to be believed; in fact the frankincense-tree which grows above Sardes in a certain sacred precinct has a leaf like that of bay, if we may judge at all by this; and the frankincense derived both from its stem and its branches is like in appearance and in smell, when it is burnt as incense, to other frankincense. This is the only tree which can never be cultivated.

9.4.7

Some say that the frankincense-tree is more abundant in Arabia, but finer in the adjacent islands over which the Arabians bear rule; for there it is said that they mould the gum on the trees to any shape that they please. And perhaps this is not incredible, since it is possible to make any kind of incision that they like. Some of the lumps of gum are very large, so that one is large enough in bulk to fill the hand and in weight is more than a third of a pound. All frankincense is gathered in the rough and is like bark in appearance. Myrrh is either 'fluid' (myrrh-oil) or 'solid' (agglutinated). That of better quality is tested by its taste, and of this they select that which is of uniform colour. Now of frankincense and myrrh these are about all the facts that have come to our notice at present.

9.5.1

Of cinnamon and cassia the following account is given: both are shrubs, it is said, and not of large size, but of the same size as bushes of chaste-tree, with many branches and woody. When they cut down the whole cinnamon-tree, they divide it into five parts; of these the first is that which grows next the branches and this is the best: this is cut in lengths a span long or a little longer; next comes the second kind, which is cut in shorter lengths; then come the third and the fourth, and last the least

valuable wood, which grows next the root; for this has least bark, and it is the bark and not the wood which is serviceable; wherefore the part which grows high up the tree is the best, since it has the most bark. Such is the account given by some.

9.5.2

Others say that cinnamon is shrubby or rather like an under-shrub; and that there are two kinds, one black, the other white. And there is also a tale told about it; they say that it grows in deep glens, and that in these there are numerous snakes which have a deadly bite; against these they protect their hands and feet before they go down into the glens, and then, when they have brought up the cinnamon, they divide it in three parts and draw lots for it with the sun; and whatever portion falls to the lot of the sun they leave behind; and they say that, as soon as they leave the spot, they see this take fire. Now this is sheer fable.

9.5.3

Cassia, they say, has stouter branches, which are very fibrous and difficult to strip of the bark; and it is the bark of this tree also which is serviceable. When then they cut off the branches, they chop them up into lengths of about two fingers' breadth or rather more, and these they sew up in raw hide; and then from the leather and the decaying wood little worms are engendered, which devour the wood but do not touch the bark, because it is bitter and has a pungent odour. This is all the information forthcoming about cinnamon and cassia.

9.6.1

Balsam of Mecca grows in the valley of Syria. They say that there are only two parks in which it grows, one of about four acres, the other much smaller. The tree is as tall as a good-sized pomegranate and is much branched; it has a leaf like that of rue, but it is pale; and it is evergreen; the fruit is like that of the terebinth, in size shape and colour, and this too is very fragrant, indeed more so than the gum.

9.6.2

The gum, they say, is collected by making incisions, which is done with bent pieces of iron at the time of the Dog-star, when there is scorching heat; and the incisions are made both in the trunks and in the upper parts of the tree. The collecting goes on throughout the summer; but the quantity which flows is not large; in a day a single man can collect a shell-full; the fragrance is exceeding great and rich, so that that which comes from a small amount is perceived for a wide distance. However it does not reach us in a pure state; what is collected is mixed with other things; for it mixes freely with other things; and what is known in Hellas is generally mixed with something else. The boughs are also very fragrant. In fact it is on account of these boughs, they say, that the tree is pruned (as well as for a different reason), since the boughs cut off can be sold for a good price. In fact the culture of the trees has the same motive as the irrigation (for they are constantly irrigated). And the cutting of the boughs seems likewise to be partly the reason why the trees do not grow tall; for, since they are often cut about, they send out branches instead of putting out all their energy in one direction.

9.6.3

Balsam is said not to grow wild anywhere. From the larger park are obtained twelve vessels containing each about three pints, from the other only two such vessels; the pure gum sells for twice its weight in silver, the mixed sort at a price proportionate to its purity. Balsam then appears to be of exceptional fragrance.

9.7.1

Sweet-flag and ginger-grass grow beyond the Libanus between that range and another small range, in the depression thus formed; and not, as some say, between Libanus and Anti-Libanus>. For Anti-Libanus is a long way from Libanus, and between them is a wide fair plain called 'The Valley.' But, where

the sweet-flag and gingergrass grow, there is a large lake, and they grow near it in the dried up marshes, covering an extent of more than thirty furlongs. They have no fragrance when they are green, but only when they are dried, and in appearance they do not differ from ordinary reeds and rushes; but, as you approach the spot, immediately a sweet smell strikes you. However it is not true, as some say, that the fragrance is wafted to ships approaching the country; for indeed this district is more than 150 furlongs from the sea. However it is said that in Arabia the breezes wafted from the land are fragrant.

9.7.2

Such then are the plants in Syria which have remarkable fragrance. For that of khalbane is more oppressive and somewhat medicinal; for this perfume also is produced in Syria from the plant called allheal. As to all the other fragrant plants used for aromatic odours, they come partly from India whence they are sent over sea, and partly from Arabia, for instance, komakon as well as cinnamon and cassia. The fruit called komakon is said to be distinct from this; the komakon of which we are speaking is a perfume which they mix with the choicest unguents. Cardamom and Nepaul cardamom some say come from Media; others say that these come from India, as well as spikenard and most, if not all, of the other species.

9.7.3

Now this is a general list of the plants used for perfumes: cassia, cinnamon, cardamom, spikenard, natron, balsam of Mecca, aspalathos, storax, iris, narte, kostos, all-heal, saffron-crocus, myrrh, kypeiron, gingergrass, sweet-flag, sweet marjoram, lotos, dill. Of these it is the roots, bark, branches, wood, seeds, gum or flowers which in different cases yield the perfume. Some of them grow in many places, but the most excellent and most fragrant all come from Asia and sunny regions. From Europe itself comes none of them except the iris.

9.7.4

This is best in Illyria, not in the part near the sea, but in that which is further inland and lies more to the north. In different districts it varies in quality; no special attention is needed, except to scrape the roots clean and dry them.

9.7.5

As for the roots which grow in Thrace, such as one which has a smell like spikenard and certain others, their fragrance is but slight and feeble. Let this suffice for an account of sweet-smelling plants.

9.8.1

Now we must endeavour to speak in like manner of those juices which have not been mentioned already, I mean, such as are medicinal or have other properties; and at the same time we must speak of roots; for some of the juices are derived from roots, and apart from that roots have in themselves divers properties of all kinds; and in general we must discuss medicinal things of all kinds, as fruit, extracted juice, leaves, roots, 'herbs'; for the herb-diggers call some medicinal things by this name.

9.8.2

The properties of 'roots' are numerous and they have numerous uses; but those which have medicinal virtues are especially sought after, as being the most useful; and they differ in not all being applied to the same purposes and in not all having their virtue in the same parts of them. To speak generally, most 'roots' have it in themselves; or else it is found in the fruits or the juices of the plant; and in some cases in the leaves as well, and it is to the virtues of the leaves in most cases that the herb-diggers refer, when they speak, as has just been said, of 'herbs.'

9.8.3

The collection of the juice from plants from which it is collected is mostly done in summer, in some cases at the beginning of that season, in others when it is well advanced. The digging of roots is done in some cases at the time of wheat-harvest or a

little earlier, but the greater part of it in autumn after the rising of Arcturus when the plants have shed their leaves, and, in the case of those whose fruit is serviceable, when they have lost their fruit. The collection of juice is made either from the stalks, as with tithymallos (spurge) wild lettuce and the majority of plants, or from the roots, or thirdly from the head, as in the case of the poppy; for this is the only plant which is so treated and this is its peculiarity. In some plants the juice collects of its own accord in the form of a sort of gum, as with tragacanth; for incision of this plant cannot be made; but in most it is obtained by incision. In some cases the juice is collected straight into vessels, for instance that of tithymallos (spurge) or mekonion (for the plant has both names) and in general the juice of specially juicy plants is so collected. But that of those which do not yield abundant juice is taken with a piece of wool, as also that of wild lettuce.

9.8.4

In some cases there can be no collection of juice, but there is a sort of extraction of it, for instance in the case of plants which are cut down or bruised; they then pour water over them and strain off the fluid, keeping the sediment; but it is plain that in these cases the juice obtained is dry and less copious. In most 'roots' the juice thus extracted is less powerful than that of the fruit, but in hemlock it is stronger and it causes an easier and speedier death even when administered in a quite small pill; and it is also more effective for other uses. That of thapsia is also powerful, while all the rest are less so. Such then is a general account of the various ways of obtaining the juices of plants.

9.8.5

As to cutting of the roots there is no such diversity of practice, except as to the season, which may be summer or autumn, and as to the particular roots selected. Thus in hellebore the slender lower roots are taken, for they say that the thick upper part

which forms a sort of head is useless, and that it is only given to dogs when it is desired to purge them. And in certain other plants also such differences are mentioned.

9.8.6

Further we may add statements made by druggists and herb-diggers, which in some cases may be to the point, but in others contain exaggeration. Thus they enjoin that in cutting some roots one should stand to windward, for instance, in cutting thapsia among others, and that one should first anoint oneself with oil, for that one's body will swell up if one stands the other way. Also that the fruit of the wild rose must be gathered standing to windward, since otherwise there is danger to the eyes. Also that some roots should be gathered at night, others by day, and some before the sun strikes on them, for instance those of the plant called honeysuckle.

9.8.7

These and similar remarks may well seem to be not off the point, for the properties of these plants are hurtful; they take hold, it is said, like fire and burn; for hellebore too soon makes the head heavy, and men cannot go on digging it up for long; wherefore they first eat garlic and take a draught of neat wine therewith. On the other hand the following ideas may be considered far-fetched and irrelevant; for instance they say that the peony, which some call glykyside, should be dug up at night, for, if a man does it in the day-time and is observed by a woodpecker while he is gathering the fruit, he risks the loss of his eyesight; and, if he is cutting the root at the time, he gets a prolapsed rectum.

9.8.8

It is also said that, while cutting feverwort one must beware of the buzzard-hawk, if one wishes to come off unhurt; and other reasons for caution are also given. That one should be bidden to pray while cutting is not perhaps unreasonable, but the additions made to this injunction are absurd; for instance as to

cutting the kind of all-heal which is called that of Asklepios; for then it is said that one should put in the ground in its place an offering made of all kinds of fruits and a cake; and that, when one is cutting gladwyn, one should put in its place to pay for it cakes of meal from spring-sown wheat, and that one should cut it with a two-edged sword, first making a circle round it three times, and that the piece first cut must be held up in the air while the rest is being cut.

9.8.9

And many similar notions are mentioned. Thus it is said that one should draw three circles round mandrake with a sword, and cut it with one's face towards the west; and at the cutting of the second piece one should dance round the plant and say as many things as possible about the mysteries of love. (This seems to be like the direction given about cummin, that one should utter curses at the time of sowing.) One should also, it is said, draw a circle round the black hellebore and cut it standing towards the east and saying prayers, and one should look out for an eagle both on the right and on the left; for that there is danger to those that cut, if your eagle should come near, that they may die within the year. These notions then seem to be irrelevant, as has been said. There are however no methods of root-cutting besides those which we have mentioned.

9.9.1

As was said, of some plants the root, fruit and juice are all serviceable, as of all-heal among others; of some the root and the juice, as of scammony, cyclamen, thapsia and others, such as mandrake; for the leaf of this, they say, used with meal, is useful for wounds, and the root for erysipelas, when scraped and steeped in vinegar, and also for gout, for sleeplessness, and for love potions. It is administered in wine or vinegar; they cut little balls of it, as of radishes, and making a string of them hang them up in the smoke over must.

9.9.2

Of hellebore both root and fruit are useful for the same purposes, if it is true, as is said, that the people of Anticyra use the fruit as a purge; this fruit contains the well-known drug called sesamodes.

9.9.3

Various parts of all-heal are also useful, and not all for the same purposes; the fruit is used in cases of miscarriage and for disorders of the bladder, while the juice, which is called khalbane, is used in cases of miscarriage and also for sprains and such-like troubles; also for the ears, and to strengthen the voice. The root is used in childbirth, for diseases of women, and for flatulence in beasts of burden. It is also useful in making the iris-perfume because of its fragrance; but the seed is stronger than the root. It grows in Syria and is cut at the time of wheat-harvest.

9.9.4

Of cyclamen the root is used for suppurating boils; also as a pessary for women and, mixed with honey, for dressing wounds; the juice for purgings of the head, for which purpose it is mixed with honey and poured in; it also conduces to drunkenness, if one is given a draught of wine in which it has been steeped. They say also that the root is a good charm for inducing rapid delivery and as a love potion; when they have dug it up, they burn it, and then, having steeped the ashes in wine, make little balls like those made of wine-lees which we use as soap.

9.9.5

Of 'wild cucumber'(squirting cucumber) the root is used for white leprosy and for mange in sheep, while the extracted juice makes the drug called 'the driver' (elaterion). It is collected in autumn, for then it is best.

9.9.6

Of germander the leaves pounded up in olive-oil are used for fractures and wounds and for spreading sores; the fruit purges

bile, and is good also for the eyes; for ulcers in the eye they pound up the leaf in olive-oil before applying it. It has leaves like the oak, but its entire growth is only about a palm high; and it is sweet both to smell and taste.

9.9.7

Now that all parts are not serviceable for the same purpose is perhaps not strange; it is more surprising that part of the same 'root' should purge upwards and another part downwards, as is the case with thapsia and iskhas which some call apios (spurge) and with libanotis; for it is not strange that on the other hand the same parts should purge both upwards and downwards, as is the case with 'driver.'

9.9.8

Thapsia has a leaf like fennel, but broader, a stalk like that of ferula, and a white root.

9.9.9

Iskhas (or apios) has a leaf like rue and short, three or four prostrate stems, and a root like that of asphodel, except that it is composed of scales; it loves mountain districts with a gravelly soil. It is collected in spring. Now this account applies only to the above-mentioned plants.

9.10.1

The white and the black hellebore appear to have nothing in common except the name. But accounts differ as to the appearance of the plants; some say that the two are alike and differ only in colour, the root of the one being white, of the other black; some however say that the leaf of the 'black' is like that of bay, that of the white like that of the leek, but that the roots are alike except for their respective colours. Now those who say that the two plants are alike describe the appearance as follows: the stem is like that of asphodel and very short; the leaf has broad divisions, and is extremely like that of ferula, but is long; it is closely attached to the root and creeps on the ground; the plant has numerous roots, to wit, the slender roots which

are serviceable.

9.10.2

Also they say that the black is fatal to horses oxen and pigs, wherefore none of these animals eat it; while the white is eaten by sheep, and from this circumstance the virtue of the plant was first observed, since it purges them; it is at its prime in autumn, and past its prime when spring comes. However the people of Mount Oeta gather it for the meetings of the Amphictyons; for it grows there in greatest abundance and best, though at only one place in the district of Oeta, namely about Pyra.

9.10.3

(The seed of rupture-wort is mixed with the potion given to promote easy vomiting; this plant is a small herb).

9.10.4

The black kind of hellebore grows everywhere; it is found in Boeotia, in Euboea and in many other places; but best is that from Mount Helicon, which mountain is in general rich in medicinal herbs. The white occurs in few places; the best and that which is most used comes from one of four places, Oeta, Pontus, Elea, and Malea. They say that that of Elea grows in the vineyards and makes the wine so diuretic that those who drink it become quite emaciated.

9.10.5

But best of all these and better than that found anywhere else is that of Mount Oeta, while that of Parnassus and that of Aetolia (for the plant is common in these parts too and men buy and sell it, not knowing the difference) are tough and exceeding harsh. These plants then, while resembling the best form in appearance, differ in their virtues.

9.10.6

Some call the black the 'hellebore of Melampus,' saying that he first cut and discovered it. Men also purify horses and sheep with it, at the same time chanting an incantation; and they put

it to several other uses.

9.11.1

There are also several kinds of all-heal tithymallos (spurge) and other herbs. To begin with, one plant called all-heal is the one found in Syria, of which we have recently spoken. Then come the three other kinds, known as that of Chaeronea, that of Asclepios, and that of Heracles. That of Chaeronea has a leaf like monk's rhubarb, but larger and rougher, a golden flower, and a small root; and it specially loves rich ground; they use it for the bites of snakes, spiders, vipers and other reptiles, administering it in wine or anointing the place with it mixed with olive-oil. In treating a snake-bite they use a plaster of it, and also give a draught of it mixed with vinegar; and they also say that it is good for sores when mixed with wine and olive-oil, and for tumours when mixed with honey.

9.11.2

The kind called after Asclepios has a white and very stout root about a span long and a thick bark which is crusted with salt; its stem is jointed all the way up, its leaf like that of thapsia, but thicker; it is said that it is good to scrape and drink it against bites of reptiles, to take it in a posset of honey for disorders of the spleen, when the blood collects about it, and against headache to pound it up in olive-oil and anoint the head; that it is of use also in other obscure troubles, and against stomachache, if scraped and taken in wine. It is said also to be able to prevent long periods of sickness. Again for running sores one may sprinkle it on in hot wine, first washing the place, while for dry sores one may soak it in wine and apply a plaster.

9.11.3

The kind named after Herakles has a large broad leaf, three spans each way, a root as thick as a man's finger, forking in two or three; in taste it is somewhat bitter, in smell like pure frankincense; it is good to drink it against epilepsy, mixed with the rennet of a seal in the proportion of one to four, or in sweet

wine against pain in the stomach; it may be used dry for running sores, and mixed with honey for dry ones. Such are the special features about these plants and their respective virtues.

9.11.4

There are also other kinds of all-heal, of which one has a fine leaf, the other not; the properties of both kinds are the same; namely they are used as a pessary for women, and a plaster may be made of them mixed with meal for spreading sores as well as for ordinary sores.

9.11.5

Of the various plants called strykhnos. As to strykhnos again and tithymallos (spurge) there is in either case more than one form of the plant denoted by the name. Of the plants called strykhnos one induces sleep, the other (thorn-apple) causes madness. The first-mentioned has a root which becomes red like blood as it dries, but when first dug up it is white; its fruit is a deeper orange than saffron, its leaf like that of tithymallos or the sweet apple; and it is itself rough, and about a foot high. The 'bark' of the root of this they bruise severely, and soaking it in neat wine give it as a draught, and it induces sleep. It grows in water-courses and on tombs.

9.11.6

The kind which produces madness (which some call thryoron and some peritton) has a white hollow root about a cubit long. Of this three twentieths of an ounce in weight is given, if the patient is to become merely sportive and to think himself a fine fellow; twice this dose if he is to go mad outright and have delusions; thrice the dose, if he is to be permanently insane; (and then they say that the juice of centaury is mixed with it); four times the dose is given, if the man is to be killed. The leaf is like that of rocket, but larger, the stem about a fathom long; the 'head' is like that of a long onion, but larger and rougher. And it also resembles the fruit of the plane-tree.

9.11.7

Of the various plants called tithymallos (spurge) that which is called sea-spurge has a round scarlet leaf; the stem (and the size of the plant generally) is about a span long, and the fruit is white. It is gathered when the grape is just turning, and the dried fruit is given in a draught, the dose being the twenty-fourth part of a pint.

9.11.8

That which is called the 'male' has a leaf like the olive, and the height of the whole plant is a cubit. Of this they collect the juice at the time of vintage, and, after preparing it, use it as occasion demands; and it purges chiefly downwards.

9.11.9

The kind of tithymallos called 'myrtle-like' is white; it has a leaf like the myrtle, but spinous at the tip; it puts out earthward twigs about a span long, and these bear the fruit not all at the same time but in alternate years, so that from the same root grow fruits partly this and partly next year. It loves hill-country. The fruit of it is called a 'nut.' They gather it when the barley is ripening and dry and clean it; (it is the actual fruit which they clean); they wash it in water and, after drying it again, give it in a draught, mixing with it two parts of 'black poppy'; and the whole dose amounts to about an eighth of a pint. It purges phlegm downwards. If they administer the 'nut' itself, they first pound it up in sweet wine, or give it in parched sesame to bite up. These plants then have leaves juices or fruits which are serviceable.

9.11.10

Of the plants called libanotis, (for there are two) one is barren, the other fruitful, the latter having both fruit and leaves that are serviceable, the former only a serviceable root. The fruit is called kakhry. This plant has a leaf like marsh celery, but much larger, a stem a cubit long or more, a large stout white root, which smells like frankincense, and a white rough elongated fruit. It grows chiefly wherever there is parched and rocky soil;

the root is serviceable for sores, and for diseases of women when given in a draught of dry black wine. The fruit is good for strangury, for the ears, for ulcers on the eye, for ophthalmia and for producing milk in women.

9.11.11

The barren kind has a leaf like that of the bitter lettuce, but rougher and paler; the root is short. It grows where there is abundance of heather. The root can purge both upwards and downwards, the upper part being used for the former, that nearer the ground for the latter purpose. Also, if it is put among clothes, it prevents moth. It is gathered at the time of wheat-harvest.

9.12.1

Of chamaeleon there is the white kind and the dark; the properties of the roots are different, and the roots also differ in appearance. In the one case the root is white, stout and sweet, and it has a heavy smell; they say that when cooked it is serviceable against flux; it is chopped up like radishes and the pieces strung on a rush; it is also good against the broad maw-worm; the patient first eats a bunch of raisins and then drinks about an eighth of a pint of this scraped up in a draught of dry wine. It is fatal to dogs and pigs; to kill a dog it is well mixed up in a meal paste with oil and water, to kill a pig it is mixed with 'mountain cabbage' (spurge). It is given to a woman in sweet winelees or sweet wine. And if one wishes to discover whether a man that is sick will recover, they say that he should be washed with this for three days, and, if he survives the experience, he will recover. It grows everywhere alike and has a leaf like the golden thistle, but larger; the plant itself has a large thistle-like head close to the ground; some actually call it the thistle.

9.12.2

The dark kind resembles the other in leaf, which is like that of the golden thistle but smaller and smoother; the plant itself is in

general appearance like a sunshade; the root is stout and black, and when broken is yellowish. It likes cold uncultivated soil: it has the property of expelling leprosy; for this it is given pounded up in vinegar, or else scrapings of it are made into a plaster; and it is also used for the white leprosy. This plant is also fatal to dogs.

9.12.3

There are several kinds of wild poppy: the one called the horned poppy is black: the leaf of this is like that of the black mullein, but it is not so black; the stem grows about a cubit high, the root is stout and shallow, the fruit is twisted like a little horn: it is gathered at the time of wheat harvest. It has the property of purging the belly, and the leaf is used for removing ulcers on sheep's eyes. It grows by the sea, wherever there is rocky ground.

9.12.4

Another kind of poppy is that called rhoias, which is like wild chicory, wherefore it is even eaten: it grows in cultivated fields and especially among barley. It has a red flower, and a head as large as a man's finger-nail. It is gathered before the barley-harvest, when it is still somewhat green. It purges downwards.

9.12.5

Another kind of poppy is called Herakleia: it has a leaf like soap-wort, with which they bleach linen: the root is slender and does not run deep, and the fruit is white. The root of this plant purges upwards: and some use it in a posset of mead for epileptics. These kinds then are distinct plants, though they come under one name.

9.13.1

The differences between roots are shown in their tastes and in their smells: some are pungent, some bitter, some sweet: some again have a pleasant, others a disagreeable smell. The plant called yellow water-lily is sweet: it grows in lakes and marshy

places, as in the district of Orchomenus, at Marathon and in parts of Crete: the Boeotians, who eat the fruit, call it madondis. It has a large leaf which lies on the water: and it is said that it acts as a styptic if it is pounded up and put on the wound: it is also serviceable in the form of a draught for dysentery.

9.13.2

'Scythian root'(liquorice) is also sweet; some indeed call it simply 'sweet-root.' It is found about Lake Maeotis: it is useful against asthma or a dry cough and in general for troubles in the chest: also, administered in honey, for wounds: also it has the property of quenching thirst, if one holds it in the mouth: wherefore they say that the Scythians, with the help of this and mares' milk cheese can go eleven or twelve days without drinking.

9.13.3

[Birthwort is fragrant to the smell but in taste is very bitter: in colour it is black. The best grows on the mountains: it has a leaf like alsine, but rounder: it is useful for many purposes, and is best for sores on the head and other sores, also for bites of reptiles, for inducing sleep and for disorders of the womb. It is directed that it should be applied as a plaster, steeped in water, and for the other purposes should be given shredded into honey and olive-oil: for snake-bites it should be taken in sour wine and also used as a plaster on the bite: to induce sleep it should be scraped up and administered in black dry wine; in cases of prolapsed uterus a lotion of it mixed with water should be applied.]

9.13.4

These then are sweet: other roots are bitter, and some unpleasant to the taste. Of those that are sweet there are some that cause mental derangement, as the plant like the golden thistle which grows near Tegea: of this Pandeios the sculptor ate, and went mad while he was working in the temple. Others have fatal effects, as that which grows near the mines in the

fields of Thrace: this however is inoffensive and quite sweet to the taste, and the death which it causes is easy and like falling asleep. There are also differences in colour, not merely as to being black or white or yellow, but some are quite winecoloured and some are red, as the root of madder.

9.13.5

The root of pentaphyllon or pentapetes (cinquefoil) (for the plant bears both names) is red when it is dug up, but as it dries it becomes black and square: its leaf is like a vine-leaf, and it is small and like it in colour: it grows and fades along with the vine. It only has five leaves in all, whence its name: it sends out long slender stems on the ground, and it has joints.

9.13.6

Madder has a leaf like ivy, but it is rounder: it grows along the ground like dog's-tooth grass and loves shady spots. It has diuretic properties, wherefore it is used for pains in the loins or hip-disease.

9.13.7

Some roots are of peculiar shape, as that of the plant called 'scorpion-plant' (leopard's bane) and that of polypody. For the former is like a scorpion and is also useful against the sting of that creature and for certain other purposes. The root of polypody is rough and has suckers like the tentacles of the polyp. It purges downwards: and, if one wears it as an amulet, they say that one does not get a polypus. It has a leaf like the great fern, and it grows on rocks.

9.14.1

Some roots keep a longer, some a shorter time. Hellebore retains its usefulness for as much as thirty years, birthwort five or six, the black chamaeleon for forty, feverwort (whose root is thick and compact) for ten or twelve. Sulphur-wort keeps five or six years, the root of the 'wild vine' (bryony) for a year, if it be kept in the shade and not damaged: otherwise it rots and becomes spongy. Others keep for various periods. But, to speak

generally, of all plants used as drugs the 'driver' keeps longest, and, the older it is, the better it is. At least a certain physician, who was no boaster nor liar, said that he had some which was 200 years old and of marvellous virtue, and that it was a present to him from some one. The cause of its keeping so long is its moisture: for to secure this, as soon as they have cut it, they put it among ashes without drying it, and not even so does it become dry, but up to fifty years it will put the lamp out if it is brought near it. And they say that alone of all drugs, or to a greater degree than any, it effects a thorough purge upwards: this then is a virtue peculiar to it.

9.14.2

Those roots which contain any sweetness become worm-eaten in course of time, but those that are pungent are not so affected, though their virtues diminish as they become flabby and waste away. No creature coming from without touches a pungent root, but the sphondyle attacks them all; this then is a peculiarity of this creature.

9.14.3

Any root, they say, deteriorates if one lets the fruit grow to maturity and ripen: and so in like manner does the fruit, if you drain the root of its juice: and in general roots with medicinal properties do not have the juice of their roots taken, and only those whose seeds are medicinal are thus treated. But some say that they use the roots for choice, because the fruit is too powerful for the human body to be able to bear it. However this does not appear to be true as a universal rule, seeing that the people of Anticyra administer doses of the drug *sesamodes* made from hellebore, which is so called because its fruit is like sesame.

9.15.1

The places outside Hellas which specially produce medicinal herbs seem to be the parts of Tyrrhenia and Latium (where they say that Circe dwelt), and still more parts of Egypt, as Homer

says: for thence he says that Helen brought things of virtue which Polydamna, the Egyptian wife of Thon, gave her; there the grain-bearing earth produces most drugs, many that are good, and many baneful. Among these he says was nepenthes, the famous drug which cures sorrow and passion, so that it causes forgetfulness and indifference to ills. So these lands seem to have been pointed out, as it were, by the poets. For Aeschylus too in his elegies speaks of Tyrrhenia as rich in drugs, for he tells of the Tyrrhenian stock, a nation that makes drugs.

9.15.2

It seems that almost all places take their share in producing drugs, but that they differ in the extent to which they do so; for the regions of the North, South, and East have herbs of marvellous virtue. Thus in Ethiopia there is a certain deadly root with which they smear their arrows. And in Scythia there is this and there are also others, some of which kill at once those who eat them, some after an interval, shorter or longer, so that in the latter case men have a lingering death. In India there are many other kinds, but the most extraordinary, if they tell the truth, are these: there is one which has the power to make the blood disperse and as it were to put it to flight, and another which collects it and draws it to itself; these they say were discovered as remedies for the bites of deadly serpents.

9.15.3

In Thrace it is said there are fairly numerous other kinds, but that about the most powerful is 'blood-stancher,' which stops and prevents the flow of blood, some say if the vein is merely pricked, others even if it is deeply cut into. These then of the places outside Hellas are those that are most productive of drugs.

9.15.4

Of places in Hellas those most productive of drugs are Pelion in Thessaly, Telethron in Euboea, Parnassus, and also Arcadia and Laconia, for both these states produce medicinal herbs;

wherefore the Arcadians are accustomed, instead of drinking medicine, to drink milk in spring when the juices of such plants are at their best, for then the milk has most medicinal virtue. It is cows' milk that they drink, since it appears that the cow eats more than any other animal and is more impartial as to what she eats.

9.15.5

Both kinds of hellebore, the white and the black, grow in their country, and also carrot, a saffron coloured plant like bay, and a plant which the Arcadians call 'wild cabbage' (spurge) but some physicians kerdis; also a plant called by some marsh mallow, also birthwort, hartwort, alexanders, sulphurwort, Herakleia, and both kinds of strychnos, that which has a scarlet and that which has a black fruit.

9.15.6

There also grow there the 'wild cucumber' (squirting cucumber), of which the drug 'driver' is compounded, and the tithymallos (spurge) of which hippophaes is made; this is best about Tegea, and that kind is much sought after; it grows there in considerable abundance, but in greatest abundance and best about Kleitoria.

9.15.7

All-heal grows in great abundance and best in the rocky ground about Psophis, moly about Pheneos and on Mount Kyllene. They say that this plant is like the moly mentioned by Homer, that it has a round root like an onion and a leaf like squill, and that it is used against spells and magic arts, but that it is not, as Homer says, difficult to dig up.

9.15.8

Hemlock is best about Susa and in the coldest spots. Most of these plants occur also in Laconia, for this too is a land rich in medicinal herbs. In Achaia tragacanth is abundant and is as good as that of Crete, it is believed, and even fairer in appearance. Daukon again is excellent in the country about

Patrai; this is by nature healing, and it has a black root. Most of these grow also on Mount Parnassus and about Telethron. So these plants are common to several lands.

9.16.1

But dittany is peculiar to Crete. This plant is marvellous in virtue and is useful for many purposes, but especially for women in child-birth. Its leaf is like pennyroyal, to which it also bears some resemblance in taste; but the twigs are slenderer. They use the leaves, not the twigs nor the fruit: and the leaf is useful for many other purposes, but above all, as was said, against difficult labour in women; for it is said that either it makes labour quite easy or at least it confessedly makes the pains to cease: it is given as a draught in water. It is a scarce plant: for the region which bears it is not extensive, and the goats graze it down because they are fond of it. The story of the arrows is also said to be true, that, if goats eat it when they have been shot, it rids them of the arrow. Such then is dittany and such its properties.

9.16.2

'False dittany' is like it in leaf, but has smaller twigs, and in virtue is far inferior. For it is of service in the same ways, but is feebler and not nearly so powerful. The virtue of dittany is perceived directly it is taken into the mouth: for a small piece of it has a very warming effect. The bunches of it are put in the hollow stem of ferula or a reed, so that it may not exhale its virtue: for, if it does so, it is less effective. Some say that dittany and 'false dittany' are essentially the same plant, but that the latter is an inferior form produced by growing in places with richer soil; just as many other things become inferior in their properties for the same cause. For dittany loves rough ground.

9.16.3

There is also another plant called 'dittany,' though it has nothing in common with these except the name. This has neither the same appearance nor the same virtue; for its leaf is

like bergamot mint and its twigs are larger, and further its use and virtue are differently shewn. The true plant is, as was said, marvellous, and is also peculiar to the island of Crete. Indeed some say that the plants of Crete are superior in leaves, boughs, and in general all the parts above ground to those of other places; while those of Parnassus are superior to most of those found elsewhere.

9.16.4

Wolf's-bane grows in Crete and in Zakynthos, but is most abundant and best at Herakleia in Pontus. It has a leaf like chicory, a root like in shape and colour to a prawn, and in this root resides its deadly property, whereas they say that the leaf and the fruit produce no effects. The fruit is that of a herb, not that of a shrub or tree. It is a lowgrowing herb and shows no special feature, but is like corn, except that the seed is not in an ear. It grows everywhere and not only at Akonai, from whence it gets its name (this is a village of the Mariandynoi): and it specially likes rocky ground. Neither sheep nor any other animals eat it. In order to be effective it is said that it must be compounded in a certain manner, and that not everyone can do this: and so that physicians, not knowing how to compound it, use it as a septic and for other purposes: and that, if drunk mixed in wine or a honey-posset, it produces no sensation: but that it can be so compounded as to prove fatal at a certain moment which may be in two three or six months, or in a year, or even in two years: and that the longer the time the more painful the death, since the body then wastes away, while, if it acts at once, death is quite painless. And it is said that no antidote which can counteract it has been discovered, like the natural antidotes to other poisonous herbs of which we are told: though the country-folk can sometimes save a man with honey and wine and such like things, only however occasionally and with difficulty.

9.16.5

(On the other hand they say that for meadow saffron the antidote has been found: for that there is another root which counteracts that herb: and that it has a leaf like hellebore or the madonna lily: and that this is generally known. Wherefore they say that slaves often take meadow-saffron when greatly provoked, and then themselves have recourse to the antidote and effect a cure, seeing that the poison does not cause a speedy and easy death, but one that is lingering and slow, unless indeed, merely because the cure is so easy, the antidote has not been properly prepared. At least they say that though death may ensue at once, sometimes it only occurs after a considerable interval, which in some cases extends to a year, and that in these latter cases the dose given has incurable effects: and that these facts have been most carefully ascertained among the Tyrrhenians of Herakleia. Now it is not surprising that in some circumstances the effects of the poison should be incurable, and yet in others that a cure should be possible, this being also the case with other deadly poisons.)

9.16.6

To return, wolf's-bane, as has been said, is useless to those who do not understand it; in fact it is said that it is not lawful even to have it in one's possession, under pain of death; also that the length of time which it takes to produce its effects depends on the time when it is gathered; for that the time which it takes to kill is equal to that which has elapsed since it was gathered.

9.16.7

Thrasyas of Mantinea had discovered, as he said, a poison which produces an easy and painless end; he used the juices of hemlock poppy and other such herbs, so compounded as to make a dose of conveniently small size, weighing only somewhat less than a quarter of an ounce. For the effects of this compound there is absolutely no cure, and it will keep any length of time without losing its virtue at all. He used to gather his hemlock, not just anywhere, but at Susa or some other cold

and shady spot; and so too with the other ingredients; he also used to compound many other poisons, using many ingredients. His pupil Alexias was also clever and no less skilful than his master, being also versed in the science of medicine generally.

9.16.8

Now these things seem to have been ascertained far better in recent than in former times. And many things go to shew that the method of using the various drugs makes a difference; thus the people of Ceos formerly did not use hemlock in the way described, but just shredded it up for use, as did other people; but now not one of them would think of shredding it, but they first strip off the outside and take off the husk, since this is what causes the difficulty, as it is not easily assimilated; then they bruise it in the mortar, and, after putting it through a fine sieve, sprinkle it on water and so drink it; and then death is made swift and easy.

9.17.1

The virtues of all drugs become weaker to those who are accustomed to them, and in some cases become entirely ineffective. Thus some eat enough hellebore to consume whole bundles and yet suffer no hurt; this is what Thrasyas did, who, as it appeared, was very cunning in the use of herbs. And it appears that shepherds sometimes do the like; wherefore the shepherd who came before the vendor of drugs (at whom men marvelled because he ate one or two roots) and himself consumed the whole bundle, destroyed the vendor's reputation: it was said that both this man and others did this every day.

9.17.2

For it seems that some poisons become poisonous because they are unfamiliar, or perhaps it is a more accurate way of putting it to say that familiarity makes poisons non-poisonous; for, when the constitution has accepted them and prevails over them, they cease to be poisons, as Thrasyas also remarked; for he said that the same thing was a poison to one and not to another;

thus he distinguished between different constitutions, as he thought was right; and he was clever at observing the differences. Also, besides the constitution, it is plain that use has something to do with it. At least Eudemus, the vendor of drugs, who had a high reputation in his business, after making a wager that he would experience no effect before sunset, drank a quite moderate dose, and it proved too strong for his power of resistance: while the Chian Eudemus took a draught of hellebore and was not purged. And on one occasion he said that in a single day he took two and twenty draughts in the market-place as he sat at his stall, and did not leave the place till it was evening, and then he went home and had a bath and dined, and was not sick. However this man was able to hold out because he had provided himself with an antidote; for he said that after the seventh dose he took a draught of tart vinegar with pumice-stone dust in it, and later on took a draught of the same in wine in like manner; and that the virtue of the pumice-stone dust is so great that, if one puts it into a boiling pot of wine, it causes it to cease to boil, not merely for the moment, but altogether, clearly because it has a drying effect and it catches the vapour and passes it off. It was then by this antidote that Eudemus was able to contain himself in spite of the large quantity of hellebore which he took.

9.17.3

However many things go to show that use makes much difference; thus some say that the sheep of some places do not eat wormwood; yet those of Pontus not only eat it but become fatter and fairer and, as some say, have no bile. But these things may be said to belong to a different enquiry.

9.18.1

Herbs and shrubs, as has been said, have many virtues which are shown in their effects not only on living bodies but on lifeless ones. Thus they say that there is a kind of akantha (gum arabic) which thickens water, when it is put in it; and that so

also does the root of marsh-mallow if one shreds it and puts it in and stands the water in the open air. Marsh-mallow has a leaf like mallow, but larger and rougher; the stems are soft, the flower yellow, the fruit like that of mallow, the root fibrous and white, with a taste like that of the stem of mallow. They use it for fractures and for coughs in sweet wine, and for sores in olive-oil.

9.18.2

They say that there is another kind which, if cooked with meat, combines with it and as it were sets it hard; and there are others that attract things to them, like the magnet or amber. So much for effects produced on lifeless things.

9.18.3

Wolf's bane, which some call 'scorpion-plant' because it has a root like a scorpion, kills that animal if it is shredded over him; while if one then sprinkles him with white hellebore, they say that he comes to life again. It is also fatal to oxen, sheep, beasts of burden and in general to any fourfooted animal, and kills them the same day if the root or leaf is put on the genitals; and it is also useful as a draught against a scorpion's sting. It has a leaf like cyclamen, and a root, as was said, like a scorpion. It grows like dog's-tooth grass, and is jointed, and it loves shady places. Now if what has been told already about the scorpion be true, then other similar tales are not incredible. (Indeed fabulous tales are not composed without some reason). And in relation to our own persons, apart from their effects in regard to health disease and death, it is said that herbs have also other properties affecting not only the bodily but also the mental powers.

9.19.1

As to those which affect the mind, strychnos, as was said before, is said to upset the mental powers and make one mad; while the root of onotheras (oleander) administered in wine makes the temper gentler and more cheerful. This plant has a leaf like the

almond, but smaller, and the flower is red like a rose. The plant itself (which loves hilly country) forms a large bush; the root is red and large, and, if this is dried, it gives off a fragrance like wine. And this does not seem surprising, since there is a sort of 'bouquet' given off by a thing which has the peculiar quality of wine.

9.19.2

On the other hand what is said of amulets and charms in general for the body or the house is somewhat foolish and incredible. Thus they say that tripolion according to Hesiod and Musaeus is useful for every good purpose, wherefore they dig it up by night, camping on the spot. So too what is said of good or fair fame as affected by plants is quite as foolish or more so: for they say that the plant called snapdragon produces fair fame. This plant is like bedstraw but it has no root: and the fruit has what resembles a calf's nostrils. The man who anoints himself with this they say wins fair fame. And they say that the same result follows, if he crowns himself with the flower of gold-flower, sprinkling it with unguent from a vessel of unfired gold. The flower of gold-flower is like gold, the leaf is white. The stem also is white and hard, the root is slender and does not run deep. Men use it in wine against the bites of serpents, and to make a plaster for burns after burning it and mixing the ashes with honey. Such tales then, as was said before, proceed from men who desire to glorify their own crafts.

9.19.3

Now since the natural qualities of roots, fruits and juices have many virtues of all sorts, some having the same virtue and causing the same result, while others have opposite virtues, one might raise a question which is perhaps equally perplexing in regard to other matters, to wit, whether those that produce the same effect do so in virtue of some single virtue which is common to them all, or whether the same result may not come about also from different causes. Let us be content to put the

question thus: but now we must proceed to speak of the natural qualities or virtues of any other plants that we can mention.

9.20.1

Pepper is a fruit, and there are two kinds: one is round like bitter vetch, having a case and flesh like the berries of bay, and it is reddish; the other is elongated and black and has seeds like those of poppy: and this kind is much stronger than the other. Both however are heating: wherefore these, as well as frankincense, are used as antidotes for poisoning by hemlock.

9.20.2

The 'Cnidian berry' is round, red in colour, larger than that of pepper, and far stronger in its heating power; wherefore, when it is given as a pill (for it is given to open the bowels) they knead it up in a piece of bread or dough: otherwise it burns the throat.

9.20.3

The root of sulphur-wort is also heating, wherefore they make of it an ointment to produce a sweat, as with other things so used. This root is also given for the spleen: but neither its seed nor its juice is of use: it grows in Arcadia. Daukon of excellent quality grows in the district of Patrai in Achaia, and is heating by nature: it has a black root.

9.20.4

The root of the 'wild vine' (bryony) is also heating and pungent: wherefore it is useful as a depilatory and to remove freckles: and the fruit is used for smoothing hides. It is cut at any season, but especially in autumn. The root of edderwort given in milk is useful for stopping a cough. It has a variegated snake-like stem: the seed is not used.

9.20.5

The root of thapsia has emetic properties: and, if one retains it, it purges both upwards and downwards. It is also able to remove bruises: and it restores other contusions to a pale colour. Its juice is stronger and purges both upwards and downwards: the

seed is riot used. It grows especially in Attica, but also in other places: the cattle of the country do not touch it, but imported cattle feed on it and perish of diarrhoea. Polypody springs up after rain, and produces no seed.

9.20.6

The wood of ebony is in appearance like box, but when barked it becomes black: it is useful against ophthalmia, and is rubbed on a whetstone for that use. Birthwort is a stout plant and is bitter to the taste: it is black in colour and fragrant; the leaf is round. However there is not much of the plant above ground. It grows especially on mountains, and then it is best. Many uses of it for various purposes are enumerated; it is best for bruises on the head, good also for other wounds, against snake-bites, to produce sleep, for the womb as a pessary: for some purposes it is soaked with water and applied as a plaster, for others it is scraped into honey and olive oil: against snake-bites it is drunk in sour wine and also sprinkled over the bite; to induce sleep it is given pounded up in black dry wine: in cases of prolapsed uterus it is used in water as a lotion. This plant then seems to have a surpassing variety of usefulness.

9.20.7

Of scammony, as though by contrast, only the juice is useful and no other part. Of male-fern no part but the root is useful and it has a sweet astringent taste. It expels the flat worm. It has no seed nor juice: and they say it is ripe for cutting in autumn. (This worm naturally infests certain races: speaking generally the following are liable to it the Egyptians, the Arabians, the Armenians, the Matadides, the Syrians, the Cilicians: the Thracians have it not, nor the Phrygians. Among the Hellenes those Thebans who frequent wrestling-schools and the Boeotians generally are liable to it: but not the Athenians.) Of all drugs, to speak generally, those are better which come from places that are wintry, face the north and are dry: wherefore of those which grow in Euboea best, they say, are the drugs of

Aigai or Telethrion, these places being dry, while Telethrion is also shady.

9.20.8

Thus we have spoken of drugs, those that are medicinal and those that have virtues of whatsoever kind, whether in the root itself, or in the juice, or in any other of their parts, and in general of all the shrubby or herbaceous plants which have such virtues, as well as their tastes, whether they be fragrant or without fragrance, with the differences between them, which are equally part of their essential character.